Cambridge Assessment International Education

Cambridge IGCSE™

PHYSICS **0625/13**

Paper 1 Multiple Choice (Core) **October/November 2025**

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

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INSTRUCTIONS

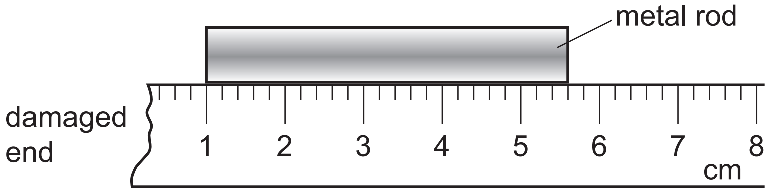
- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m / s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.

1 A girl uses a ruler to measure the length of a metal rod. The end of the ruler is damaged, so she places one end of the rod at the 1 cm mark, as shown.



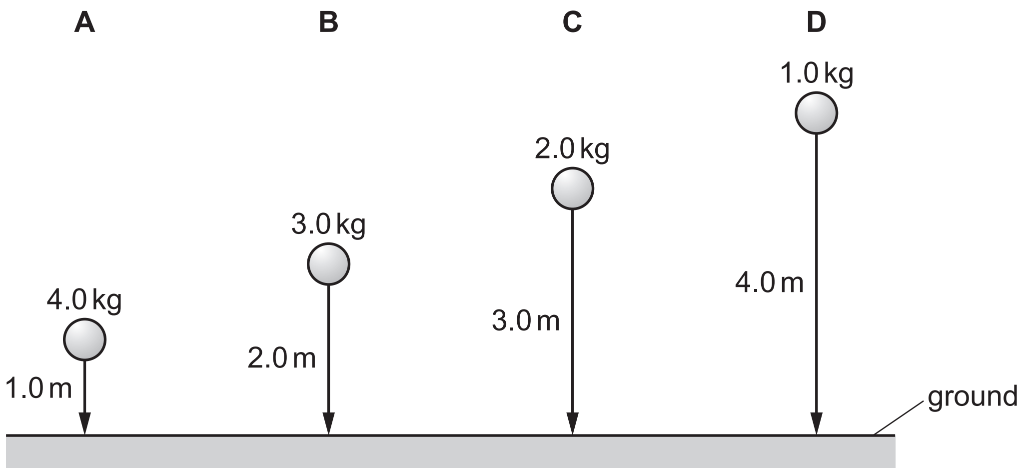
How long is the metal rod?

A 43 mm **B** 46 mm **C** 53 mm **D** 56 mm

2 Four balls with different masses are dropped from the heights shown.

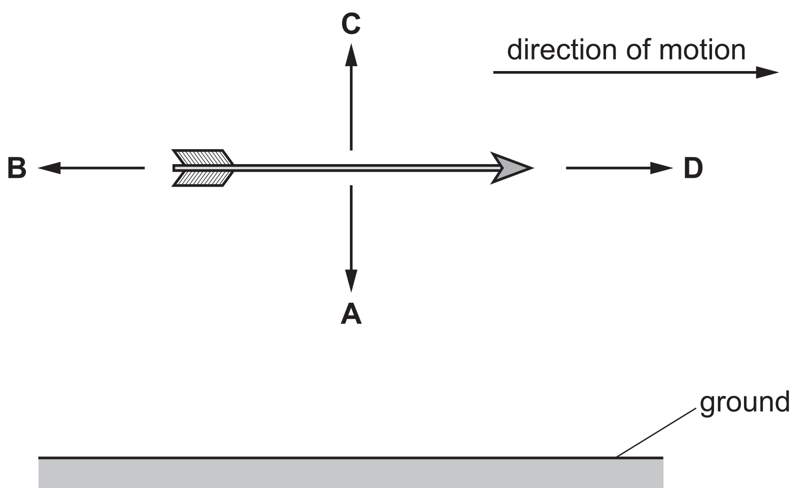
Air resistance can be ignored.

Which ball has the greatest average speed?



3 An arrow travels horizontally in a straight line at constant speed.

In which direction does the weight of the arrow act?



4 Which piece of equipment is useful in determining the density of an irregularly-shaped stone?

A stop-watch

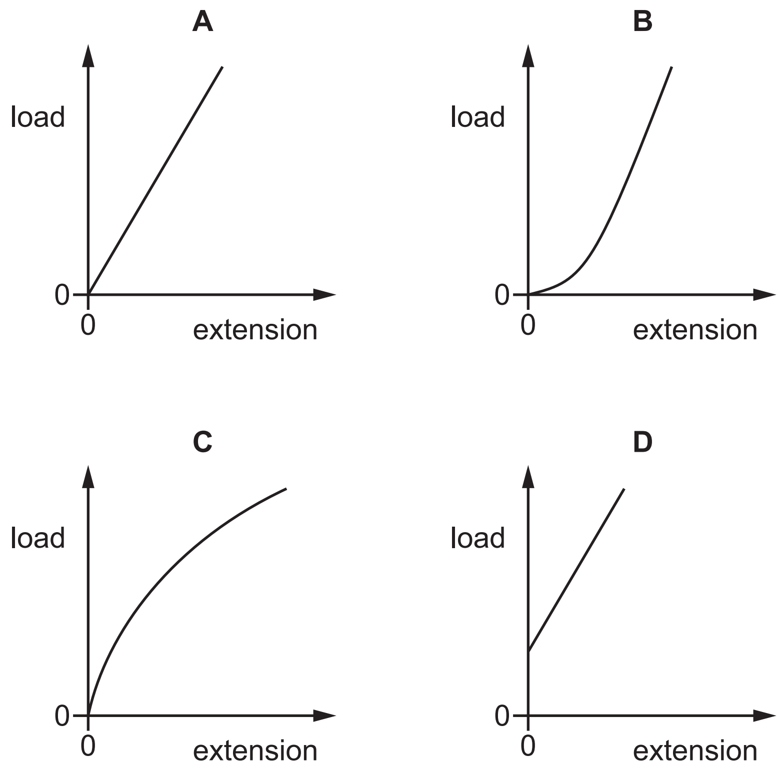
B metre ruler

C measuring cylinder

D voltmeter

5 Four load–extension graphs are shown for a spring.

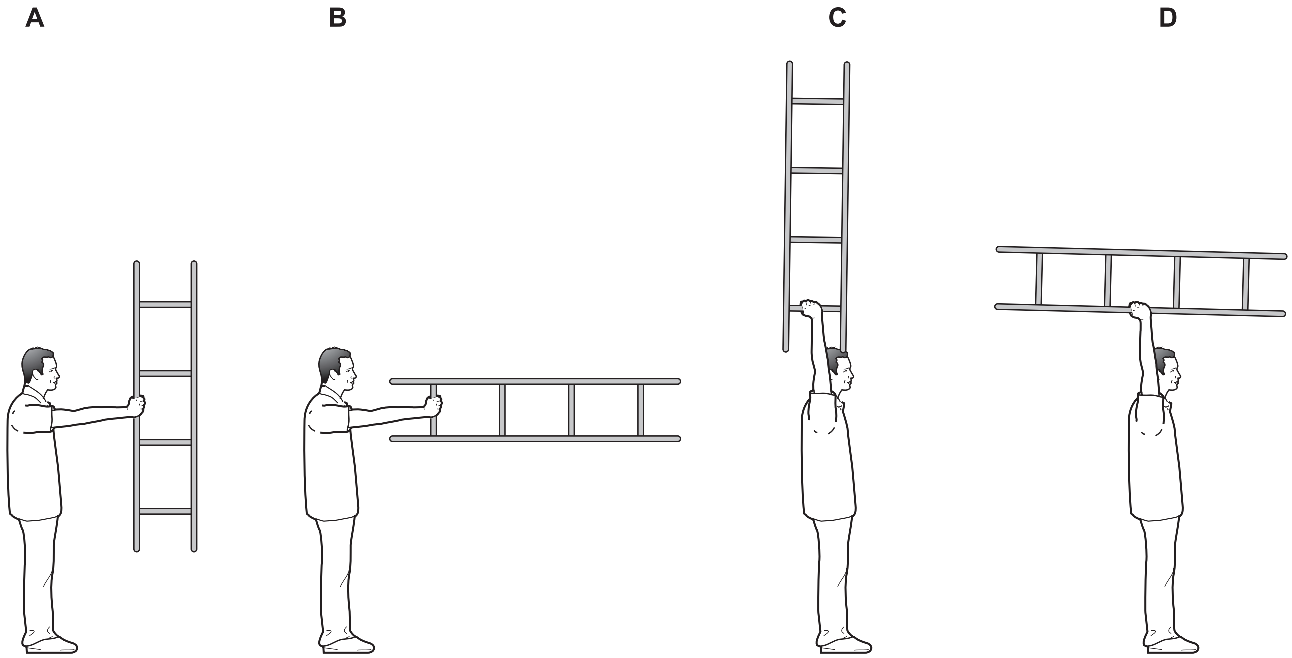
Which graph shows that the load is proportional to the extension?



6 A man holds a ladder in four different positions, pivoting around his shoulder.

The weight of the ladder causes a moment about the man's shoulder.

In which position is the moment greatest?



7 A child has four blocks of wood in the shapes of the letters A, M, T and X.

The blocks have equal heights and thicknesses. At their widest parts, their widths are equal.

The child stands the four blocks the right way up on a flat table.

Which block is most likely to fall over when the block is slightly knocked?

A letter A

B letter M

C letter T

D letter X

8 When an object falls freely due to gravity, energy that was stored as gravitational potential energy is transferred to the kinetic energy store.

How is the energy transferred in this process?

A by electrical work

B by electromagnetic waves

C by heating

D by mechanical work

9 A woman pushes a mower across a lawn.

Which quantities does the woman use to calculate the work she does?

A distance, force and time

B distance and force only

C distance and time only

D force and time only

10 A steel knife needs a cutting edge with a small surface area.

What is the reason for this?

A A small surface area increases the pressure.

B A small surface area increases the density of the steel.

C A small surface area prevents the cutting edge from rusting.

D A small surface area prevents the knife from becoming magnetised.

11 When a hot gas cools, its internal energy decreases.

What causes this?

A a decrease in the average kinetic energy of the gas particles

B a decrease in the gravitational potential energy of the gas particles

C an increase in the average speed of the gas particles

D an increase in the average distance of separation of the gas particles

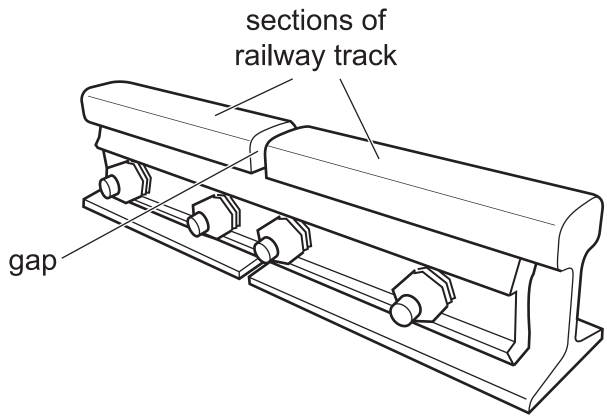
12 The temperature of an object is 37 °C.

What is the temperature of the object in kelvin?

A –236 K **B** 7.4 K **C** 236 K **D** 310 K

13 The diagram shows part of a railway track.

Each section of the railway track is separated by a gap.



Why are these gaps necessary?

A The tracks contract on cold days.

B The tracks contract on warm days.

C The tracks expand on cold days.

D The tracks expand on warm days.

14 A test-tube contains a volume of 1.0 cm^3 of liquid water at a temperature of 100 °C. The liquid water boils to form a volume of 1600 cm^3 of steam.

What is the reason for the large increase in volume?

A Steam particles are bigger than water particles.

B The average distance between the particles is much greater in the steam.

C The particles do **not** move until the water turns into a gas.

D There are more steam particles than there were water particles.

15 When a liquid is exposed to the air, the more-energetic particles escape from the surface of the liquid.

What is the name of this process?

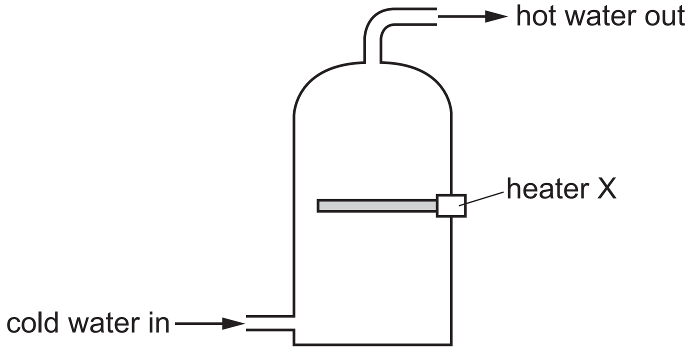
A boiling

B condensation

C convection

D evaporation

16 The diagram shows a hot-water tank. The tank has a heater, X, halfway up the tank.



Heater X is switched on.

Which row states the main processes by which thermal energy is transferred to the water above the heater and to the water below the heater?

	thermal energy transfer to the water above the heater	thermal energy transfer to the water below the heater
A	conduction	conduction
B	conduction	convection
C	convection	conduction
D	convection	convection

17 The handle of a metal cooking pan is made from wood.

What is a reason for making the handle from wood?

A to increase thermal energy transfer by conduction

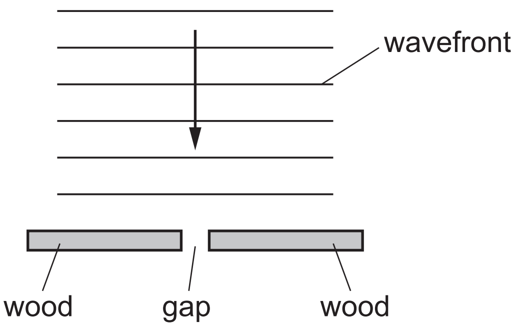
B to increase thermal energy transfer by convection

C to reduce thermal energy transfer by conduction

D to reduce thermal energy transfer by convection

18 A student observes waves travelling across the surface of water. The diagram shows two pieces of wood resting in shallow water of constant depth.

Straight parallel wavefronts approach the pieces of wood, as shown.



The gap between the pieces of wood is 2.0 cm wide.

The wavefronts are 3.0 cm apart.

What is the appearance of the wavefronts after they pass through the gap?

A semicircular and 2.0 cm apart

B semicircular and 3.0 cm apart

C straight and 2.0 cm apart

D straight and 3.0 cm apart

19 P-waves and S-waves are produced in the ground during an earthquake.

Which statement about these waves is correct?

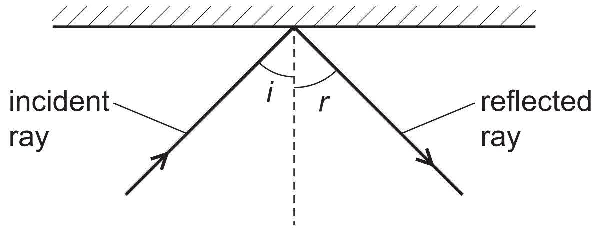
A Only P-waves are transverse waves.

B Only P-waves are longitudinal waves.

C Both P-waves and S-waves are longitudinal waves.

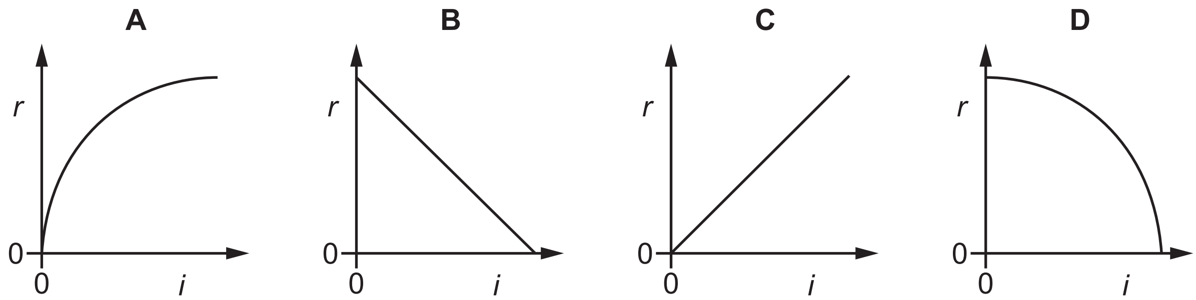
D Both P-waves and S-waves are transverse waves.

20 A ray of light is incident on a plane mirror. A student measures the angle of incidence *i* and the angle of reflection *r*.



The student varies the angle of incidence and then plots a graph of *r* against *i*.

What does the graph look like?



21 A ray of light passes from air into a block of glass. The critical angle is 42°.

Which statement is correct?

A The angle of incidence is the angle between the ray of light and the surface of the glass block.

B The angle of incidence is always bigger than 42°.

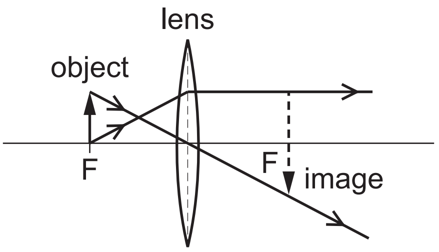
C The angle of refraction **cannot** be less than 42°.

D The angle of refraction is smaller than the angle of incidence.

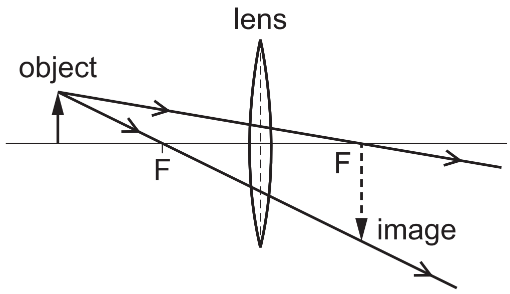
22 A converging lens is used to produce a real image of a 2 cm high object. The image is 4 cm high.

Which ray diagram shows how the lens forms the image?

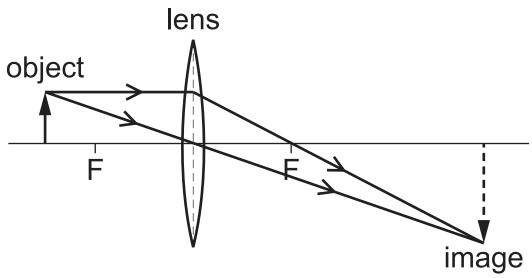
A



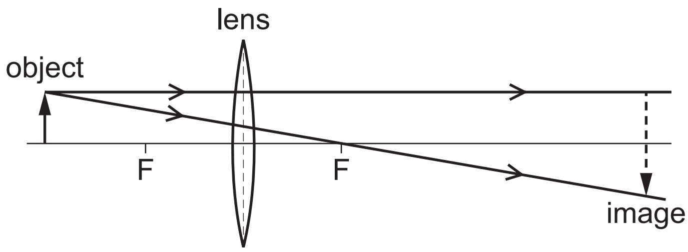
B



C



D



23 White light can be split into different colours by passing it through a prism.

What is the name of this process?

A diffraction

B dispersion

C reflection

D total internal reflection

24 A sound wave with a frequency of 25 000 Hz travels through air with a speed of 340 m / s.

Why can a human **not** hear the sound?

A The frequency is too low.

B The frequency is too high.

C The speed is too low.

D The speed is too high.

25 A drummer hits a drum harder to make a louder sound without changing the pitch of the sound.

Which statement describes the change in properties of the sound as the drummer hits the drum harder?

A Only the amplitude of the sound increases.

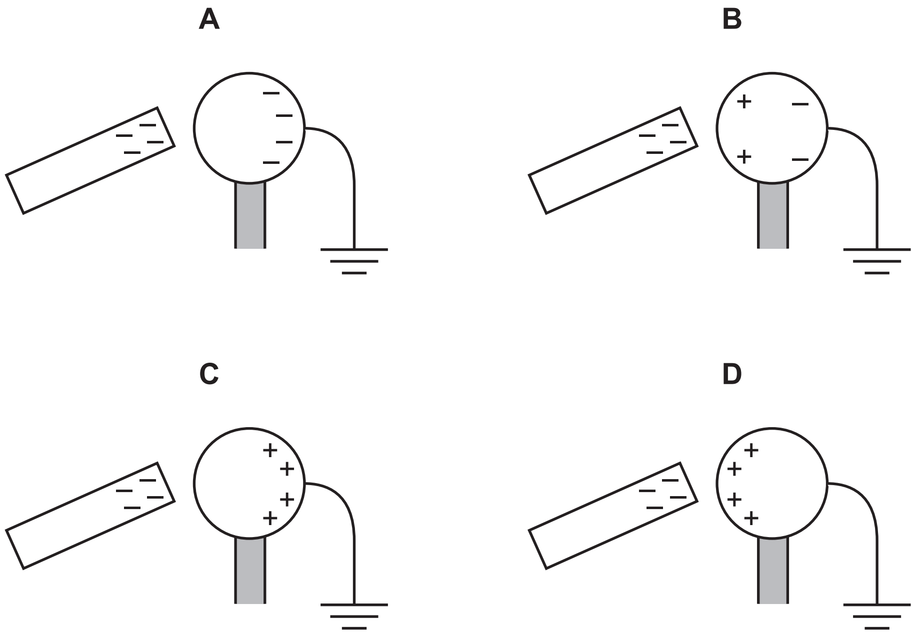
B Only the frequency of the sound decreases.

C Both the amplitude and the frequency of the sound increase.

D Both the amplitude and the frequency of the sound decrease.

26 A metal sphere is attached to the top of an insulated rod. A negatively charged rod is held close to one side of the metal sphere. The other side of the metal sphere is earthed.

Which diagram shows the distribution of charge on the metal sphere?



27 A student writes three statements about electric current.

1 Electric current is a flow of charge.

2 Electric current in a circuit is measured using a voltmeter.

3 The direction of an alternating current (a.c.) continually changes.

Which statements are correct?

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

28 Electromotive force (e.m.f.) is defined in terms of the energy supplied in driving which physical quantity around a complete circuit?

A charge

B resistance

C potential difference (p.d.)

D power

29 A student measures the resistance of a wire.

Which two measurements must the student make?

A length and current

B potential difference (p.d.) and current

C cross-sectional area and power

D potential difference (p.d.) and diameter

30 There is a current of 2.0 A in a resistor of resistance 8.0 Ω.

What is the potential difference (p.d.) across the resistor?

A 0.25 V **B** 4.0 V **C** 8.0 V **D** 16 V

31 There is a current of 2.0 A in a resistor for a time of 30 s. The potential difference (p.d.) across the resistor is 12 V.

How much energy is transferred in the resistor?

A 1.25 J **B** 5.0 J **C** 180 J **D** 720 J

32 The circuit symbol for an ammeter is a circle with a letter A in it. The symbol for a fuse is a rectangle with a central line.

Which pair of circuit symbols is also one involving a circle followed by one involving a rectangle?

A a lamp and a cell

B a motor and a thermistor

C a switch and a heater

D a voltmeter and a battery

33 Which component is **not** needed when using an appliance that is double-insulated?

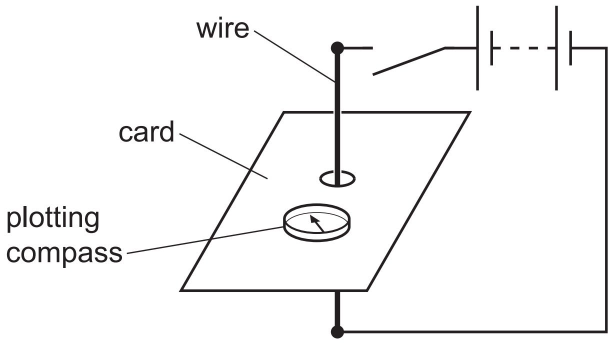
A earth wire

B fuse

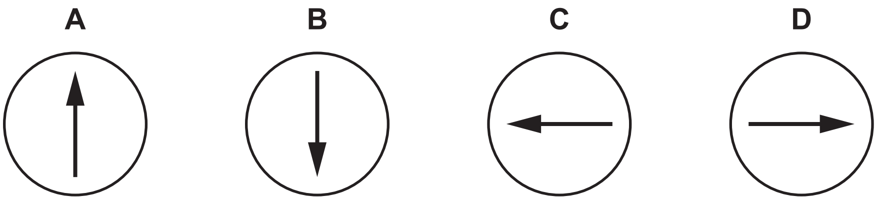
C live wire

D neutral wire

34 A vertical wire passes through a hole in a piece of card. A plotting compass is placed on the card. There is **no** current in the circuit.



Which diagram shows the plotting compass when the switch is closed?



35 Which description of the structure of an atom is correct?

A a negatively charged nucleus with negatively charged electrons around it

B a negatively charged nucleus with positively charged electrons around it

C a positively charged nucleus with negatively charged electrons around it

D a positively charged nucleus with positively charged electrons around it

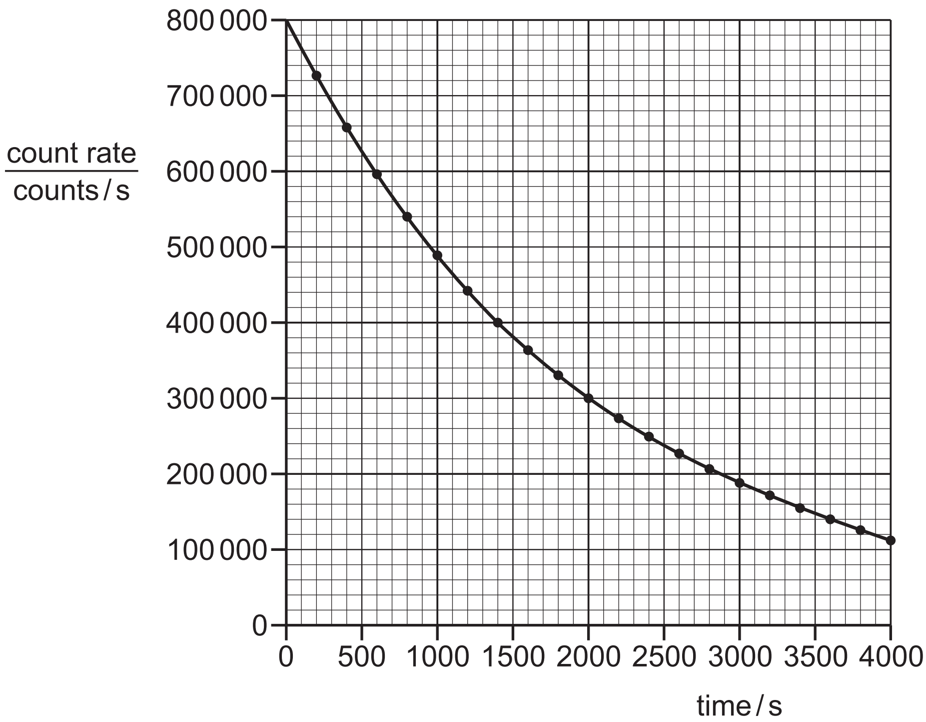
36 Alpha, beta and gamma emissions are compared.

Alpha radiation is the ......1...... ionising as it has the ......2...... electric charge.

Which words correctly complete the sentence?

	1	2
A	least	largest
B	least	smallest
C	most	largest
D	most	smallest

37 A student collects data for a new radioactive substance.



Using the graph, what is the approximate half-life of the substance?

A 1400 s **B** 2000 s **C** 100 000 s **D** 800 000 s

38 Four rocky planets orbit the Sun.

Which list shows these planets in order of increasing distance from the Sun?

A Earth → Mars → Mercury → Venus

B Venus → Earth → Mars → Mercury

C Mercury → Venus → Earth → Mars

D Mars → Mercury → Venus → Earth

39 In which given region of the electromagnetic spectrum does the Sun radiate the least of its energy?

A infrared

B microwave

C ultraviolet

D visible

40 Light from distant galaxies is redshifted when it is observed from Earth.

Which property of the light increases when light is redshifted?

A amplitude

B frequency

C speed

D wavelength

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PHYSICS **0625/12**

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45 minutes

You must answer on the multiple choice answer sheet.

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You will need: Multiple choice answer sheet
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INSTRUCTIONS

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- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = $9.8\,\text{m/s}^2$).

INFORMATION

- The total mark for this paper is 40.
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This document has **16** pages. Any blank pages are indicated.

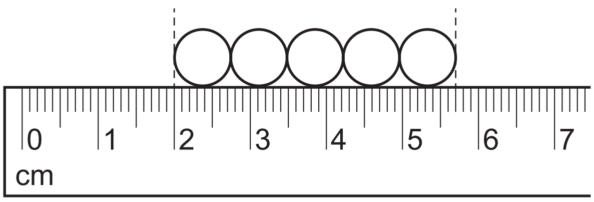
1 An athlete runs around a circular track.

Readings on the stop-watch used to time the athlete are shown.

How long did it take the athlete to run the third lap?

A 1 min 19 s **B** 1 min 22 s **C** 1 min 37 s **D** 1 min 59 s

2 The diagram shows a ruler being used to measure the length of a line of ball-bearings.



Which expression gives the average diameter of a ball-bearing?

A $\dfrac{(5.7 + 2.0)}{5}$

B $\dfrac{(5.7 - 2.0)}{5}$

C $(5.7 + 2.0) \times 5$

D $(5.7 - 2.0) \times 5$

3 Which statement about acceleration is correct?

A It is related to the changing speed of an object.

B It is the distance an object travels in one second.

C It is the force acting on an object divided by the distance it travels in one second.

D It is the force acting on an object when it is near to the Earth.

4 An object with a mass of 6.0 kg rests on the surface of a planet.

On this planet, g = 20 N / kg.

What is the weight of the object on this planet?

A 0.30 N **B** 0.60 N **C** 60 N **D** 120 N

5 Which quantity is represented by the symbol ρ?

A density

B momentum

C pressure

D resistance

6 A person is standing in a lift. The lift is moving downwards at constant speed.

Which statement about the force exerted by the lift floor on the person's feet is correct?

A It is zero.

B It is less than the person's weight.

C It is equal to the person's weight.

D It is greater than the person's weight.

7 What is the unit of the moment of a force?

A N **B** N / kg **C** N / m **D** N m

8 A stone has a gravitational energy store equal to 46 J.

When the stone falls in air, it does 21 J of work against air resistance.

What is the gain in kinetic energy stored in the stone during this fall?

A 21 J **B** 25 J **C** 46 J **D** 67 J

9 Which physical quantity is transferred when work is done?

A distance

B energy

C force

D temperature

10 Which unit is a unit of pressure?

A kg/m^2 **B** kg/m^3 **C** N m **D** N/m^2

11 A kettle of boiling water converts water to steam at 100 °C.

How does the mass and volume of the steam produced compare with the mass and volume of the water that is lost through boiling?

	mass of steam	volume of steam
A	same as water	same as water
B	same as water	greater than water
C	less than water	same as water
D	less than water	greater than water

12 Which row lists two properties of a solid?

	property 1	property 2
A	particles vibrate in fixed positions	has a fixed volume
B	particles move around freely	has no fixed volume
C	particles vibrate and can change position	has a fixed volume
D	particles vibrate in fixed positions	has no fixed volume

13 Air is trapped in a cylinder by a piston.

The piston is pushed inwards. The volume of the air reduces.

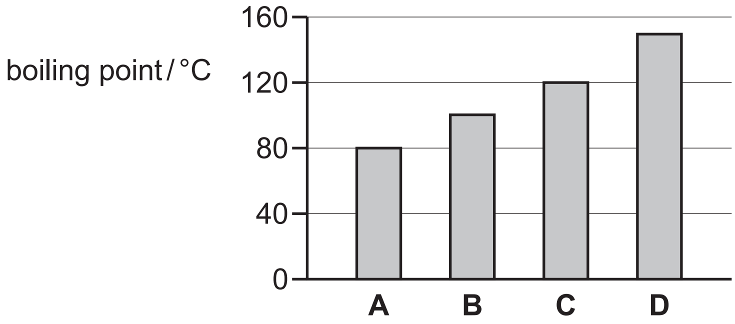
The temperature of the air stays the same.

Which row describes how the average speed of the air molecules and the average distance between them changes?

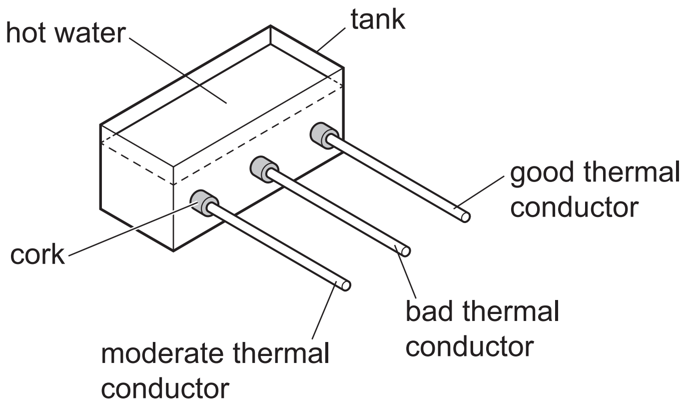
	average speed of molecules	average distance between molecules
A	increases	decreases
B	increases	unchanged
C	unchanged	decreases
D	unchanged	increases

14 The chart shows the boiling points for four different liquids at standard atmospheric pressure.

Which liquid is water?



15 Rods of the same shape and size are inserted through corks into an empty tank. Each rod is covered with a layer of solid wax that has a low melting point. Hot water is added to the tank, as shown. After a period of time, some wax melts.



On which of the rods will the wax melt first?

A all at the same time

B good thermal conductor

C bad thermal conductor

D moderate thermal conductor

16 Which method of thermal transfer occurs when the density of some of a liquid decreases and the liquid moves upwards?

A conduction

B convection

C evaporation

D radiation

17 Which surface is the best reflector of infrared radiation?

A dull and black

B dull and white

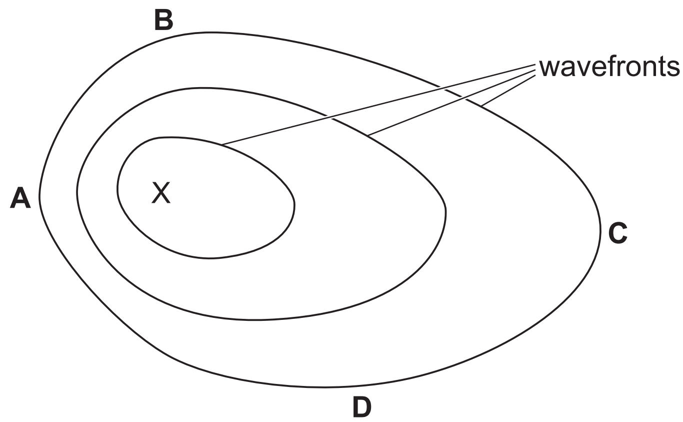
C shiny and black

D shiny and white

18 Waves travel more quickly on the surface of water when the water is deep.

A stone is dropped at point X into a pool of varying depth. The diagram shows the first three wavefronts on the surface of the pool.

The region between X and which labelled point is likely to be the deepest?



19 The table gives information about seismic waves.

Which row is correct?

	seismic S-waves (secondary)	seismic P-waves (primary)
A	longitudinal	longitudinal
B	longitudinal	transverse
C	transverse	longitudinal
D	transverse	transverse

20 Light reflects from a plane mirror.

The angle between the incident ray and the reflected ray is 68°.

What is the angle of incidence?

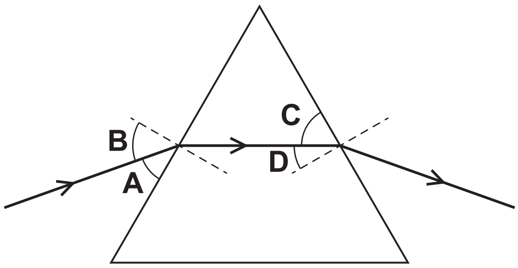
A 11° **B** 22° **C** 34° **D** 56°

21 The diagram shows a ray of red light travelling through a transparent prism.

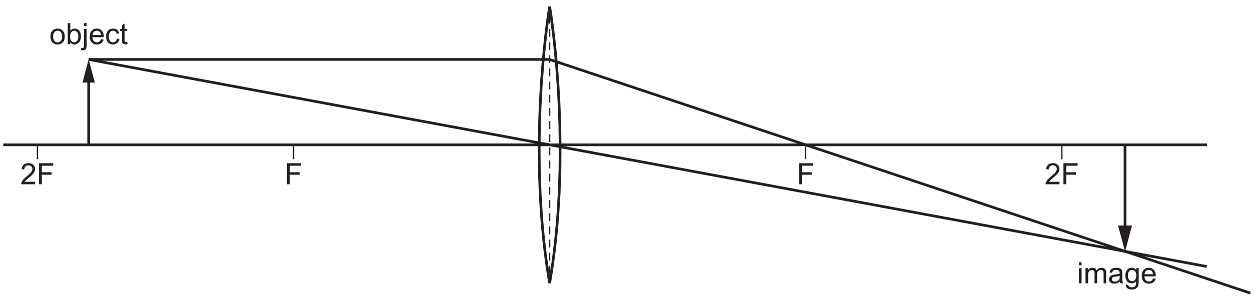
Normals to the prism surfaces are marked at the points where the ray meets the prism surfaces.

The direction of the incident ray is changed until total internal reflection just occurs at one of the surfaces.

Which angle **must** now be equal to the critical angle?



22 The diagram shows an image being formed by a converging lens.



Which description of the image formed is correct?

A enlarged and inverted

B enlarged and upright

C diminished and inverted

D diminished and upright

23 Which list shows the colours present in the visible spectrum in order of increasing wavelength?

A green → yellow → orange → red

B red → green → orange → yellow

C red → orange → yellow → green

D yellow → green → orange → red

24 Which parts of the electromagnetic spectrum complete the list of applications?

Mobile phones use ......1...... .

Security scanners use ......2...... .

Intruder alarms use ......3...... .

	1	2	3
A	radio waves	X-rays	radio waves
B	microwaves	microwaves	infrared
C	microwaves	X-rays	radio waves
D	microwaves	X-rays	infrared

25 A student places a loudspeaker inside each of four sealed jars.

From which jar will the student **not** hear a sound from the loudspeaker?

A sealed jar with air inside

B sealed jar with carbon dioxide inside

C sealed jar with a vacuum inside

D sealed jar with water inside

26 Soft iron and steel are magnetic materials.

What are permanent magnets and temporary magnets made of?

	permanent magnet	temporary magnet
A	soft iron	soft iron
B	soft iron	steel
C	steel	soft iron
D	steel	steel

27 A negatively charged plastic rod is brought near to an isolated, uncharged metal sphere and held there.

What happens when the metal sphere is earthed?

A Electrons flow from the metal sphere to earth.

B Electrons flow from earth to the metal sphere.

C Positive charge flows from the metal sphere to earth.

D Positive charge flows from earth to the metal sphere.

28 Which statement describes a direct current in a circuit?

A Charges change their direction of flow.

B Charges flow in many directions.

C Charges flow in only one direction.

D Charges only flow away from a cell.

29 Potential difference (p.d.) is the .... work done by a unit charge passing through a component.

Which word completes the sentence?

A electrical

B magnetic

C mechanical

D thermal

30 There is a current I in a resistor for a time t. The potential difference (p.d.) across the resistor is V.

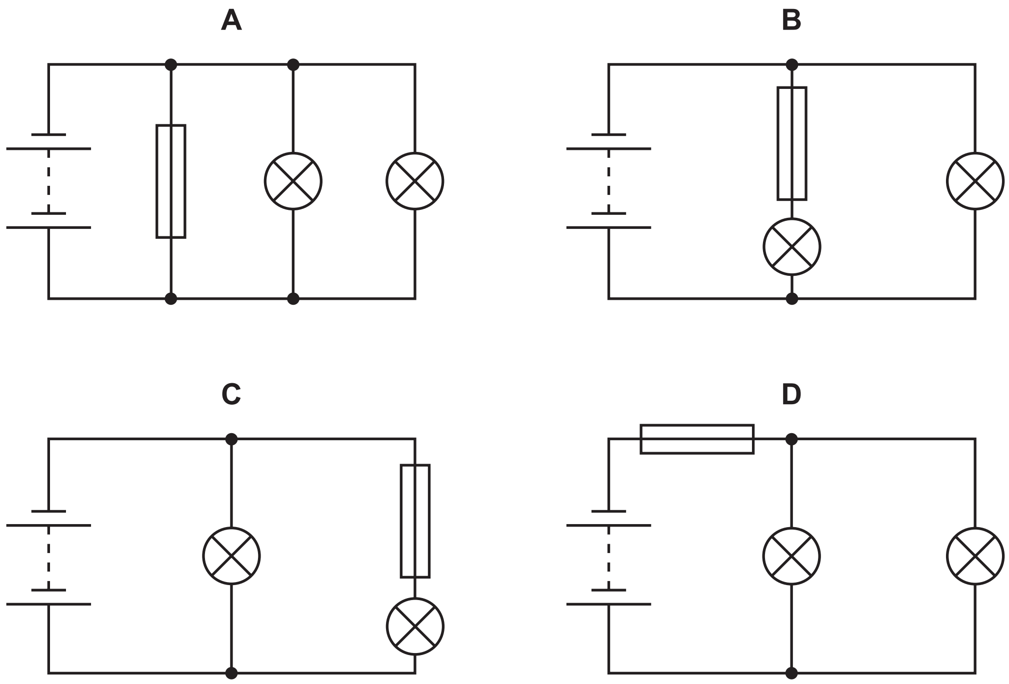
Which equation gives the energy E transferred by the resistor?

A $E = \frac{IV}{t}$ **B** $E = \frac{It}{V}$ **C** $E = \frac{Vt}{I}$ **D** $E = IVt$

31 A student constructs four circuits, each containing a fuse.

The fuse blows in one circuit and both lamps in the circuit go out.

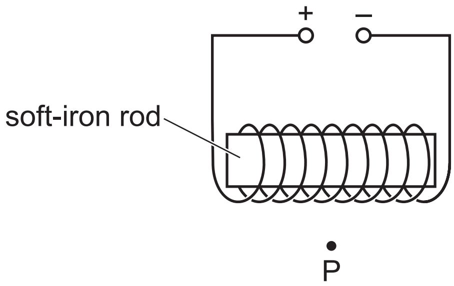
In which circuit does the fuse blow and both lamps go out?



32 The diagram shows a coil of insulated wire wrapped around a soft-iron rod.

The wire is connected to a d.c. power supply as indicated.

The apparatus is in a region which is totally shielded from the Earth's magnetic field.



A small compass needle is placed at point P.

In which direction does the N pole of the compass needle point?

A towards the bottom of the page

B towards the left of the page

C towards the right of the page

D towards the top of the page

33 Which components are used to construct an efficient simple transformer?

A two insulated copper wire coils wound on a copper core

B two insulated copper wire coils wound on an iron core

C two insulated iron wire coils wound on a copper core

D two insulated iron wire coils wound on an iron core

34 An isotope of carbon has the nuclide notation ${}^{14}_{6}C$.

What does the nucleus of one atom of this isotope contain?

	number of protons	number of neutrons
A	6	6
B	6	8
C	8	6
D	8	8

35 When the count rate from a radioactive source is measured, background radiation is taken into account.

Which option shows only examples of nuclear background radiation?

A alpha-particles from radon gas and gamma radiation from space

B alpha-particles from radon gas and infrared radiation from the Sun

C gamma radiation from the Sun and microwaves from satellites

D microwaves from satellites and X-rays from hospitals

36 Strontium-89 is a radioactive beta-emitter used to treat bone cancers.

A solution of strontium chloride is injected into patients and accumulates in their bones.

Which statement about this treatment is **not** correct?

A Beta radiation is a form of electromagnetic radiation.

B Beta-particles are a form of ionising radiation.

C The strontium-89 atoms change to atoms of another element when they decay.

D Beta-particles are negatively charged.

37 An isotope of strontium decays by β emission. It takes 87 hours for its activity to fall to $\frac{1}{8}$ of its original value.

What is the half-life of the isotope?

A 11 hours

B 29 hours

C 44 hours

D 260 hours

38 What causes the phases of the Moon?

A the Earth orbiting the Sun

B the Earth turning on its axis

C the Moon orbiting the Earth

D the Moon orbiting the Sun

39 The Sun is a star that radiates energy towards the Earth.

Which regions of the electromagnetic spectrum make up most of the Sun's radiation received on the Earth?

A gamma, X-ray and ultraviolet

B X-ray, ultraviolet and visible

C ultraviolet, visible and infrared

D visible, infrared and microwaves

40 Light received from distant galaxies is observed to have a longer wavelength than the light originally emitted. This is called redshift.

Which conclusion do astronomers form about distant galaxies from this observation?

A They are all moving around each other.

B They are all moving away from each other.

C They are all moving in circular orbits.

D They are all moving towards each other.

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Cambridge Assessment International Education

Cambridge IGCSE™

PHYSICS **0625/11**

Paper 1 Multiple Choice (Core) **October/November 2025**

45 minutes

You must answer on the multiple choice answer sheet.

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You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m / s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.

IB25 11_0625_11/3RP

1 Which measuring instrument is used to determine the volume of water in a cup?

A clock

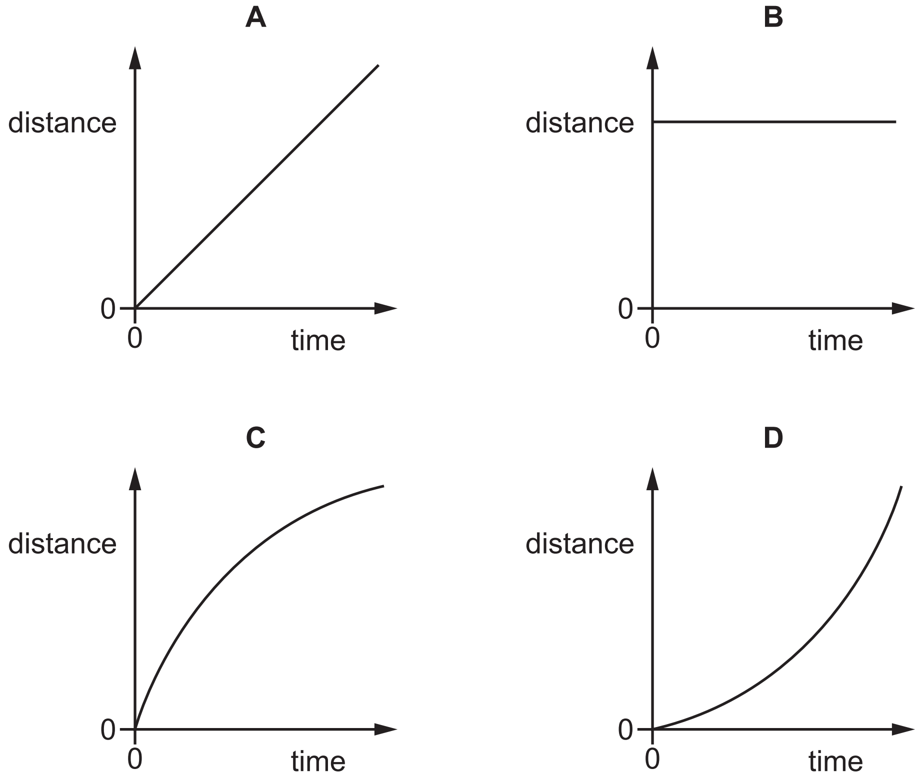
B measuring cylinder

C ruler

D thermometer

2 The diagrams show distance–time graphs for four objects.

Which object is moving with increasing speed?



3 A space station orbits above the Earth. There is an astronaut in the space station.

In the space station, the acceleration of free fall is $7.5\,m/s^2$.

On Earth, the acceleration of free fall is $9.8\,m/s^2$.

Which row about the astronaut's mass and weight in the space station is correct?

	mass of astronaut	weight of astronaut
A	same as on the Earth	less than on the Earth
B	same as on the Earth	greater than on the Earth
C	lower than on the Earth	less than on the Earth
D	lower than on the Earth	greater than on the Earth

4 A student is asked to predict whether a solid floats in a liquid.

Which information does the student require?

A the density of the liquid and the mass of the solid

B the density of the solid and the density of the liquid

C the density of the solid and the mass of the liquid

D the mass of the solid and the mass of the liquid

5 Which property of an object **never** changes when a force acts on the object?

A mass

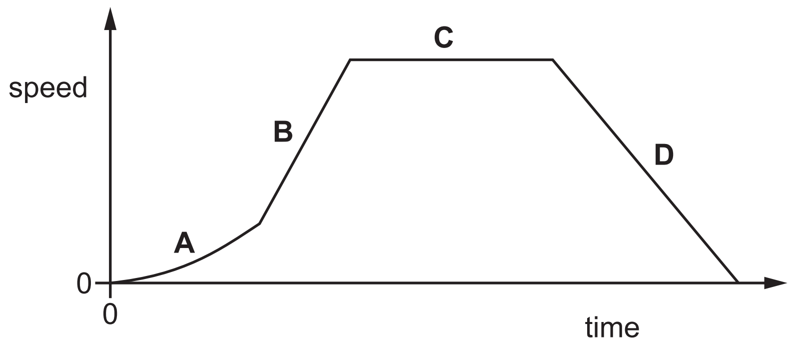
B shape

C size

D speed

6 A car is travelling along a straight horizontal road. The speed–time graph is shown.

In which labelled part of the journey is the resultant force on the car zero?



7 Which list contains only energy stores?

A electrostatic, kinetic, light

B electrostatic, kinetic, nuclear

C electrostatic, light, nuclear

D kinetic, light, nuclear

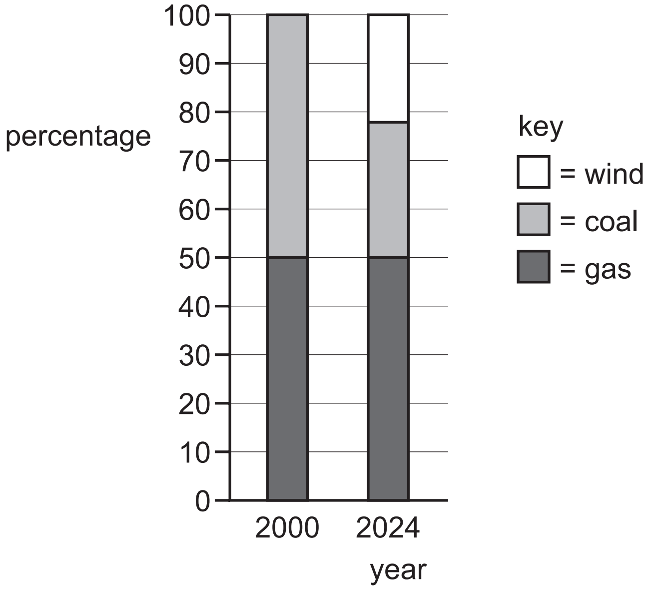
8 What needs to be known to calculate the work done by a force acting on an object?

	the size of the force	the distance the force moves the object	the time for which the force acts
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✓	✗	✗

key
✓ = needed
✗ = **not** needed

9 Different energy resources are used to produce electricity on a small island.

The chart shows the percentages of each energy resource used in 2000 and in 2024.



Which row correctly describes any changes in the percentage for 2024 compared with 2000?

	percentage of energy resource used	
	coal	renewable
A	decrease	no change
B	no change	increase
C	decrease	increase
D	no change	no change

10 Which situation is **not** explained by the equation $P = \frac{F}{A}$?

A using a longer spanner than normal to undo a tight nut

B hammering a nail into a piece of wood

C tractors using wide tyres in a muddy field

D a sharp kitchen knife cutting vegetables more easily than a blunt one

11 A substance is heated. The substance changes state from a liquid to a gas.

When the gas is cooled, it first becomes a liquid and finally a solid.

What is the order of changes of state for this substance?

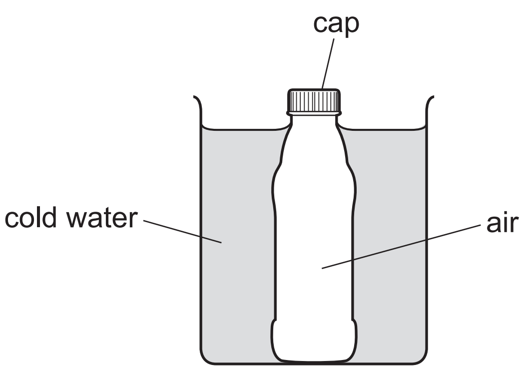
A condensation → melting → freezing

B vaporisation → condensation → melting

C vaporisation → condensation → freezing

D melting → vaporisation → freezing

12 A glass bottle containing warm air is sealed with a screw cap and then cooled in cold water.



The contraction of the glass bottle can be ignored.

What remains the same during the cooling?

A the air pressure inside the bottle

B the energy of the air particles in the bottle

C the force on the cap made by the air particles in the bottle

D the volume of air in the bottle

13 A gas is stored in a container of constant volume.

The temperature of the gas is increased.

What happens to the particles of the gas?

A The distance between the particles increases.

B The mass of the particles increases.

C The number of particles increases.

D The speed of the particles increases.

14 One end of a shiny metal rod is heated, and the other end quickly gets hot.

Which statement describes why the other end quickly gets hot?

A Metals are good thermal conductors.

B Metals are poor thermal conductors.

C Shiny surfaces are good emitters of infrared radiation.

D Shiny surfaces are poor emitters of infrared radiation.

15 Vacuum flasks usually have silvered walls that help to keep the contents of the flask hot.

Why are the walls silvered?

A to absorb thermal energy from the air around the flask

B to increase the rate of convection inside the flask

C to reduce energy loss to the surroundings by conduction

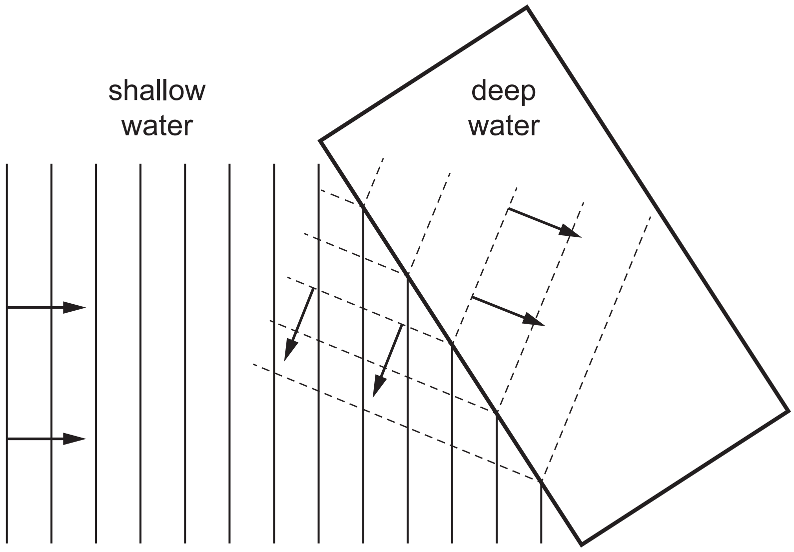
D to reflect thermal radiation back into the flask

16 The temperature of the air in the atmosphere around us increases.

Which row describes the change in the density of the air and which form of thermal energy transfer occurs as a result of this change?

	what happens to the density of the air	thermal energy transfer that occurs as a result
A	decreases	conduction
B	decreases	convection
C	increases	conduction
D	increases	convection

17 A student draws a diagram to show two different properties of a water wave. The arrows show the wave directions.



Which two wave properties does the diagram show?

A refraction and diffraction

B reflection and dispersion

C reflection and diffraction

D reflection and refraction

18 A seismic wave consists of vibrations that are parallel to the direction of propagation.

What is the type of wave and what is the name of the seismic wave?

	type of wave	name of seismic wave
A	longitudinal	P-wave
B	longitudinal	S-wave
C	transverse	P-wave
D	transverse	S-wave

19 A person stands 1.0 m in front of a plane mirror. The mirror is moved away from the person at a speed of 1.0 m / s.

Which statement is correct?

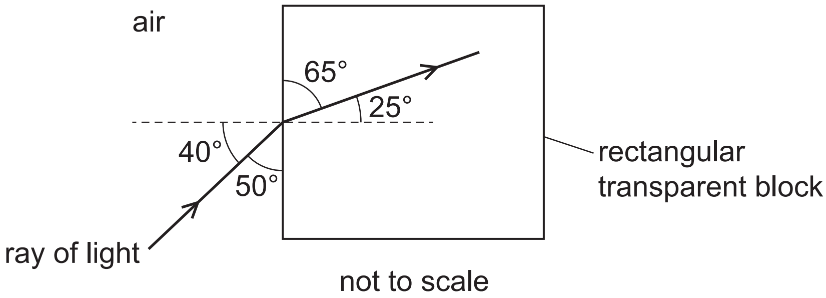
A The image moves away from the person at a speed of 1.0 m / s.

B The image moves away from the person at a speed of 2.0 m / s.

C The image moves towards the person at a speed of 1.0 m / s.

D The image moves towards the person at a speed of 2.0 m / s.

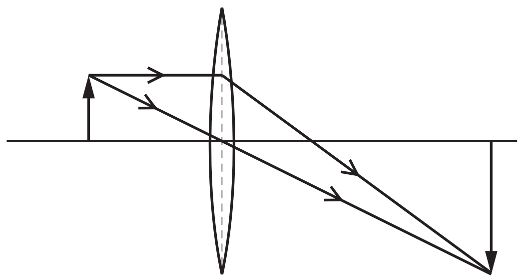
20 A ray of light passes from air into a rectangular transparent block.



What is the angle of refraction?

A 25° **B** 40° **C** 50° **D** 65°

21 The diagram shows the formation of an image by a thin converging lens.



Which terms describe the image?

A diminished and upright

B real and inverted

C virtual and diminished

D virtual and inverted

22 Valuable objects are often marked with a special ink. In visible light, the ink **cannot** be seen.

The ink can be seen when exposed to the right type of electromagnetic waves.

Which type of electromagnetic waves are used for this type of security marking?

A infrared

B microwaves

C ultraviolet

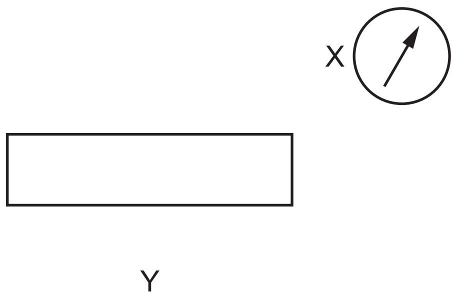
D X-rays

23 A siren is emitting a sound. As time passes, the sound becomes louder and higher pitched.

What is happening to the amplitude and to the frequency of the emitted sound wave?

	amplitude	frequency
A	decreasing	decreasing
B	decreasing	increasing
C	increasing	decreasing
D	increasing	increasing

24 A plotting compass is placed at position X near to a bar magnet. The diagram shows the direction in which the compass points.



Which diagram shows the direction in which the plotting compass placed at position Y points?

A **B** **C** **D**

25 A student rubs a plastic rod with a cloth.

Electrons move from the cloth to the rod.

Which statement about the rod is correct?

A It is negatively charged and is attracted by a negatively charged object.

B It is negatively charged and is attracted by a positively charged object.

C It is positively charged and is attracted by a negatively charged object.

D It is positively charged and is attracted by a positively charged object.

26 Which expression is used to calculate electrical resistance?

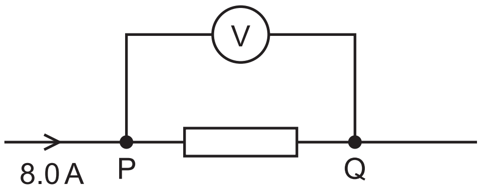
A charge ÷ potential difference

B current ÷ potential difference

C potential difference ÷ charge

D potential difference ÷ current

27 The diagram shows part of an electric circuit. The reading on the voltmeter is 16 V. The current in the resistor is 8.0 A.



One coulomb of charge flows from P to Q through the resistor.

How much energy is transferred in the resistor?

A 2.0 J **B** 8.0 J **C** 16 J **D** 128 J

28 What is the combined resistance of a 3.0 Ω resistor and a 6.0 Ω resistor connected in parallel?

A 2.0 Ω **B** 4.5 Ω **C** 9.0 Ω **D** 18 Ω

29 A 5.0 Ω resistor is connected in series with a 10 Ω resistor.

There is a constant current in the 5.0 Ω resistor.

Which statement about the two resistors is correct?

A The current is greater in the 5.0 Ω resistor.

B The current is greater in the 10 Ω resistor.

C The potential difference (p.d.) is greater across the 5.0 Ω resistor.

D The potential difference (p.d.) is greater across the 10 Ω resistor.

30 Which statement explains where the switch for an electric lamp is connected?

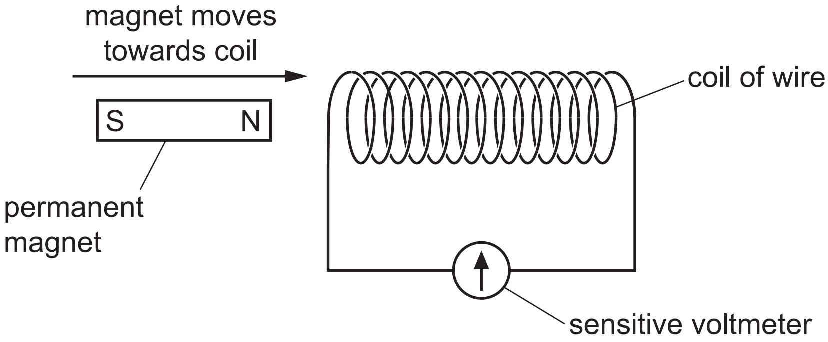
A The switch is in **either** the live wire **or** in the neutral wire as both wires carry the current.

B The switch is in the earth wire to prevent the fuse from blowing when the lamp is in use.

C The switch is in the live wire so that the lamp is disconnected from the mains supply when the switch is off.

D The switch is in the neutral wire to prevent current leakage when the lamp is switched off.

31 The diagram shows a magnet moving towards a coil of wire.



When the magnet moves towards the coil of wire, an electromotive force (e.m.f.) is induced.

Students are asked for one change that will increase the magnitude of the induced e.m.f.

Three changes are proposed.

1 Move the magnet more quickly.

2 Use a stronger magnet.

3 Increase the number of turns on the coil.

Which changes, on their own, will increase the magnitude of the induced e.m.f.?

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

32 What is used to identify the pattern and direction of the magnetic field due to the current in a straight wire?

A an ammeter

B a compass

C iron filings

D a voltmeter

33 Which description of the structure of an atom is correct?

A negatively charged electrons surrounding an uncharged nucleus

B negatively charged electrons surrounding a positively charged nucleus

C positively charged electrons surrounding an uncharged nucleus

D positively charged electrons surrounding a negatively charged nucleus

34 A radioactive source is placed 20 mm in front of a detector.

The count rate from the source is 2000 counts / minute.

With a thin sheet of paper between the source and the detector, the count rate is 2000 counts / minute.

With a sheet of aluminium 5 mm thick between the source and the detector, the count rate is 1200 counts / minute.

Which radioactive emissions are coming from the source?

A alpha, beta and gamma

B alpha only

C beta and gamma only

D beta only

35 Which type of radiation is measured using a detector connected to a counter?

A visible light

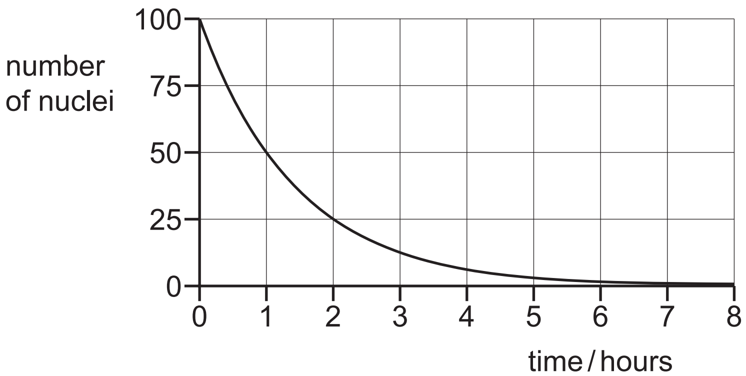
B ionising nuclear

C microwave

D thermal

36 A sample of a radioactive isotope contains 100 nuclei of the isotope.

The graph shows the number of nuclei of that isotope that remain in the sample as time passes.



What is the half-life of the radioactive isotope?

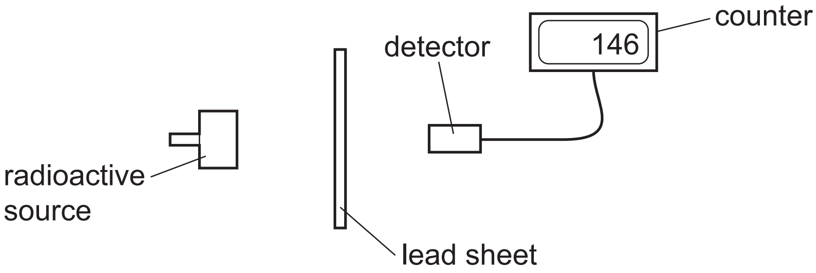
A 1 hour

B 4 hours

C 6 nuclei

D 50 nuclei

37 A teacher does an experiment to measure radiation passing through a lead sheet.



Three students suggest how the teacher can reduce the risk to her health.

student 1 Move the radioactive source with her bare hands.

student 2 Stay as close as possible to the radioactive source during the experiment.

student 3 Minimise the time of the experiment.

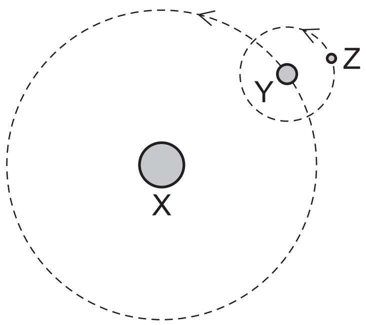
Which students are correct?

A 1, 2 and 3 **B** 1 and 2 only **C** 2 only **D** 3 only

38 The diagram represents three objects, X, Y and Z, in space.

Object Y orbits object X.

Object Z orbits object Y.



Which types of object are X, Y and Z?

	object X	object Y	object Z
A	planet	moon	star
B	planet	star	moon
C	star	planet	moon
D	star	moon	planet

39 Which statement about the Sun is correct?

A The Sun consists of mainly carbon and nitrogen.

B The Sun is the largest planet in the Solar System.

C The Sun is a medium-sized star.

D The Sun radiates most of its energy in the gamma-ray region of the electromagnetic spectrum.

40 What is the name of the galaxy that contains the Sun?

A Andromeda

B Milky Way

C Red Shift

D Universe

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Cambridge IGCSE™

PHYSICS **0625/13**

Paper 1 Multiple Choice (Core) **October/November 2025**

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

Question	Answer	Marks
1	B	**1**
2	D	**1**
3	A	**1**
4	C	**1**
5	A	**1**
6	B	**1**
7	C	**1**
8	D	**1**
9	B	**1**
10	A	**1**
11	A	**1**
12	D	**1**
13	D	**1**
14	B	**1**
15	D	**1**
16	C	**1**
17	C	**1**
18	B	**1**
19	B	**1**
20	C	**1**
21	D	**1**
22	C	**1**
23	B	**1**
24	B	**1**
25	A	**1**
26	D	**1**
27	C	**1**
28	A	**1**

Question	Answer	Marks
29	B	**1**
30	D	**1**
31	D	**1**
32	B	**1**
33	A	**1**
34	C	**1**
35	C	**1**
36	C	**1**
37	A	**1**
38	C	**1**
39	B	**1**
40	D	**1**

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PHYSICS **0625/12**

Paper 1 Multiple Choice (Core) **October/November 2025**

MARK SCHEME

Maximum Mark: 40

Published

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Question	Answer	Marks
1	A	**1**
2	B	**1**
3	A	**1**
4	D	**1**
5	A	**1**
6	C	**1**
7	D	**1**
8	B	**1**
9	B	**1**
10	D	**1**
11	B	**1**
12	A	**1**
13	C	**1**
14	B	**1**
15	B	**1**
16	B	**1**
17	D	**1**
18	C	**1**
19	C	**1**
20	C	**1**
21	D	**1**
22	A	**1**
23	A	**1**
24	D	**1**
25	C	**1**
26	C	**1**
27	A	**1**
28	C	**1**

Question	Answer	Marks
29	A	**1**
30	D	**1**
31	D	**1**
32	C	**1**
33	B	**1**
34	B	**1**
35	A	**1**
36	A	**1**
37	B	**1**
38	C	**1**
39	C	**1**
40	B	**1**

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PHYSICS **0625/11**

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4	B	**1**
5	A	**1**
6	C	**1**
7	B	**1**
8	B	**1**
9	C	**1**
10	A	**1**
11	C	**1**
12	D	**1**
13	D	**1**
14	A	**1**
15	D	**1**
16	B	**1**
17	D	**1**
18	A	**1**
19	B	**1**
20	A	**1**
21	B	**1**
22	C	**1**
23	D	**1**
24	D	**1**
25	B	**1**
26	D	**1**
27	C	**1**
28	A	**1**

Question	Answer	Marks
29	D	**1**
30	C	**1**
31	A	**1**
32	B	**1**
33	B	**1**
34	C	**1**
35	B	**1**
36	A	**1**
37	D	**1**
38	C	**1**
39	C	**1**
40	B	**1**

Cambridge Assessment International Education

Cambridge IGCSE™

PHYSICS **0625/13**

Paper 1 Multiple Choice (Core) **October/November 2024**

45 minutes

You must answer on the multiple choice answer sheet.

* 3 4 5 0 0 5 7 6 4 7 *

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
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- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m / s^2).

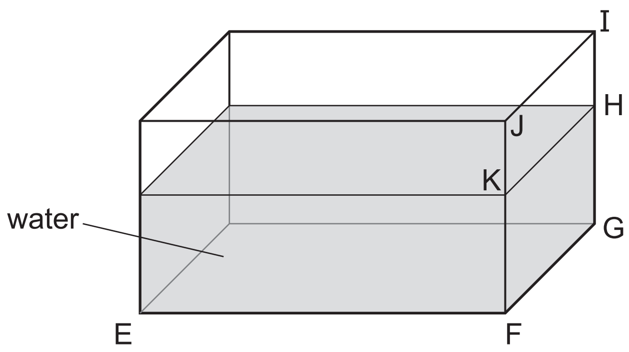
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.

1 A student uses a ruler to find the volume of water in a tank.

She measures the lengths EF and FG.



What other length does she need to measure?

A FJ **B** FK **C** HI **D** IJ

2 Which statement is correct?

A speed = distance travelled × time taken

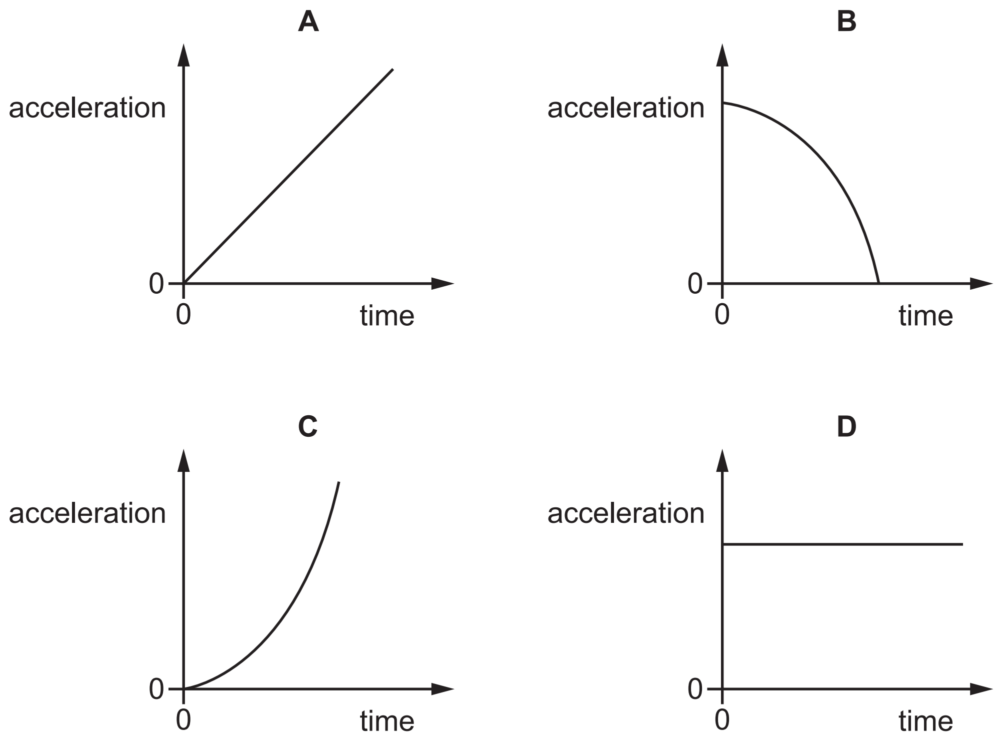
B speed = velocity in a given direction

C velocity = $\dfrac{\text{time taken}}{\text{distance travelled}}$

D velocity = speed in a given direction

3 A stone falls freely from the top of a cliff. Air resistance may be ignored.

Which graph shows how the acceleration of the stone varies with time as it falls?



4 Which statement about weight is correct?

A It is a measure of the quantity of matter in an object.

B It is the gravitational force on an object that has mass.

C It is equivalent to the acceleration of free fall.

D It is the gravitational force per unit mass.

5 A student uses the equation $\rho = \dfrac{m}{V}$.

Which row shows the correct meaning of these symbols?

	ρ	m	V
A	density	mass	velocity
B	density	mass	volume
C	pressure	mass	volume
D	pressure	moment	velocity

6 What is the point at which the weight of an object appears to act?

A the centre of gravity of the object

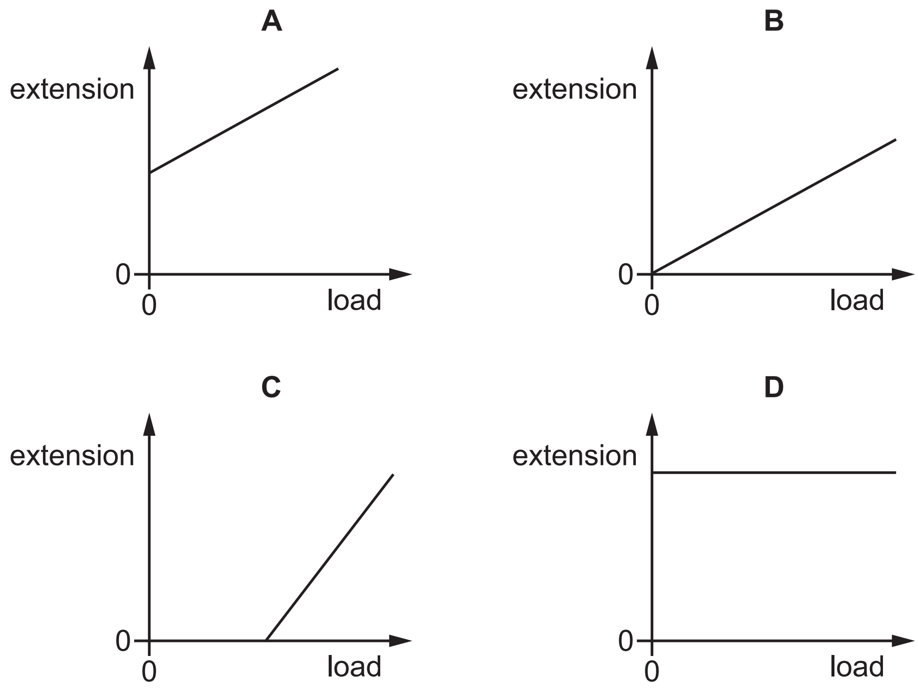
B the density of the object

C the moment of a force about a point

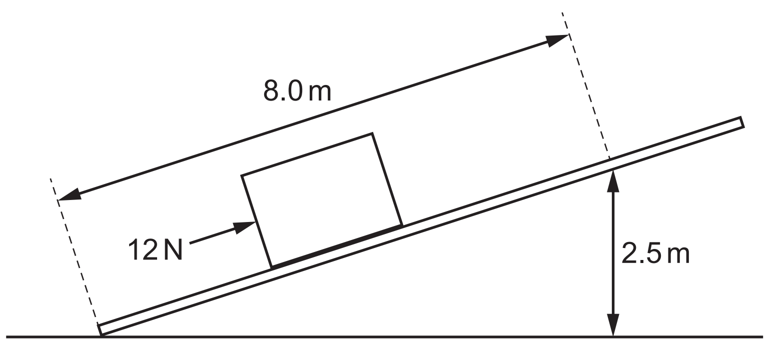
D the pressure on the object due to its depth

7 A spring is stretched by adding loads to it.

Which diagram shows the extension–load graph for the spring?



8 The diagram shows a force of 12 N being used to push a box up a slope. The box is moved 8.0 m along the slope. This lifts the box through a vertical height of 2.5 m.



How much work is done pushing the box from the bottom to the top of the slope?

A 1.5 J **B** 4.8 J **C** 30 J **D** 96 J

9 Which energy resource is non-renewable?

A biofuel

B energy from the Sun

C nuclear fuel

D water behind a dam

10 What is the relationship between the power of a motor, the force exerted by the motor, the distance moved by the force and the time taken?

A power = $\dfrac{\text{distance moved} \times \text{time taken}}{\text{force}}$

B power = force $\times$ distance moved $\times$ time taken

C power = $\dfrac{\text{force} \times \text{time taken}}{\text{distance moved}}$

D power = $\dfrac{\text{force} \times \text{distance moved}}{\text{time taken}}$

11 How is pressure defined?

A area per unit force

B force per unit area

C mass per unit area

D mass per unit volume

12 A fixed mass of gas is trapped in a cylinder with a piston. The volume of the gas is slowly reduced at constant temperature without any particles of gas escaping.

Which statement is correct?

A The force exerted by the gas on the piston will decrease because the particles move more quickly.

B The force exerted by the gas on the piston will decrease because the particles move more slowly.

C The force exerted by the gas on the piston will increase because the particles are hitting the piston harder.

D The force exerted by the gas on the piston will increase because the particles are hitting the piston more frequently.

13 The air temperature rises from 10 °C to 40 °C.

What is the change in temperature, expressed in kelvin?

A 30 K **B** 50 K **C** 243 K **D** 303 K

14 Which effect is **not** due to thermal expansion?

A convection currents in air

B expansion of a bubble rising through a fizzy drink

C expansion of a liquid in a thermometer

D lengthening of overhead power lines in the summer

15 On a hot summer day, the level of the water in a pond falls.

Which statement explains this?

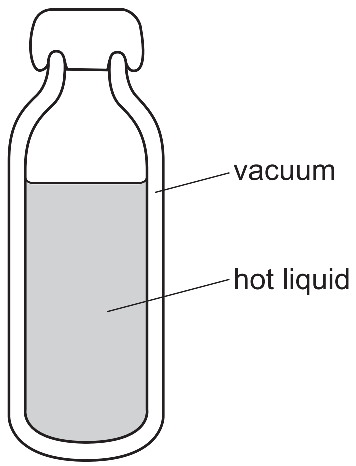
A The least energetic water particles escape from the surface and do not return.

B The least energetic water particles escape from the surface and then return.

C The most energetic water particles escape from the surface and do not return.

D The most energetic water particles escape from the surface and then return.

16 The diagram shows a vacuum flask used to keep liquid hot.



How does thermal energy pass through the vacuum?

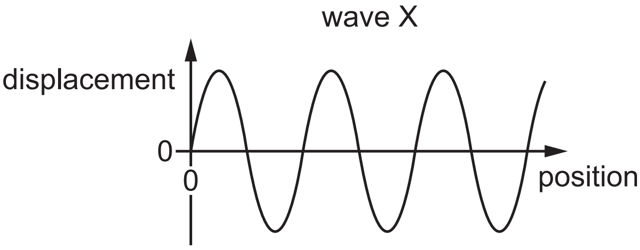
A conduction and convection only

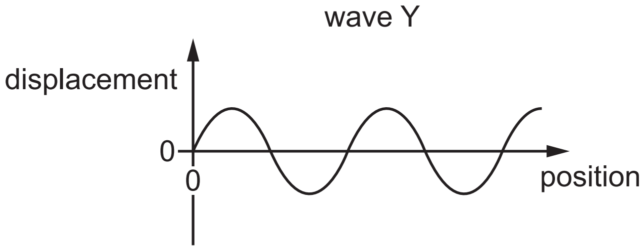
B conduction only

C convection only

D radiation only

17 The diagrams show two transverse waves X and Y and are to the same scale.





Which row is correct?

	wave with the largest amplitude	wave with the largest wavelength
A	X	X
B	X	Y
C	Y	X
D	Y	Y

18 Which row shows an example of a transverse wave and an example of a longitudinal wave?

	transverse	longitudinal
A	light	radio
B	radio	sound
C	sound	water
D	water	light

19 A ray of light is reflected from a plane mirror.

Which angle is the angle of reflection?

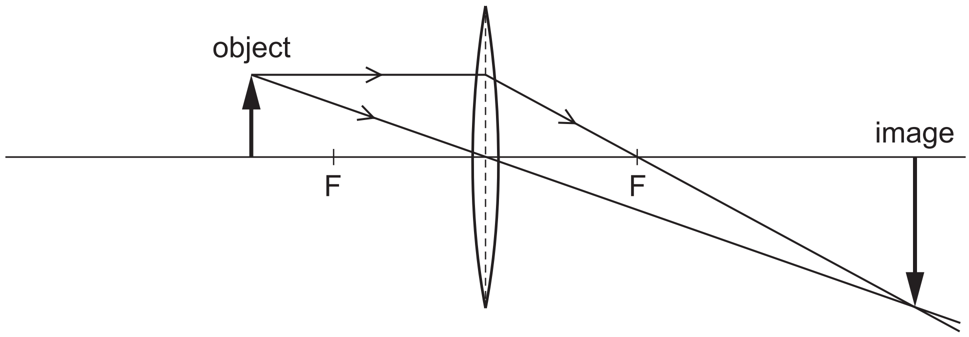
A the angle between the incident ray and the normal

B the angle between the incident ray and the reflected ray

C the angle between the reflected ray and the normal

D the angle between the reflected ray and the surface

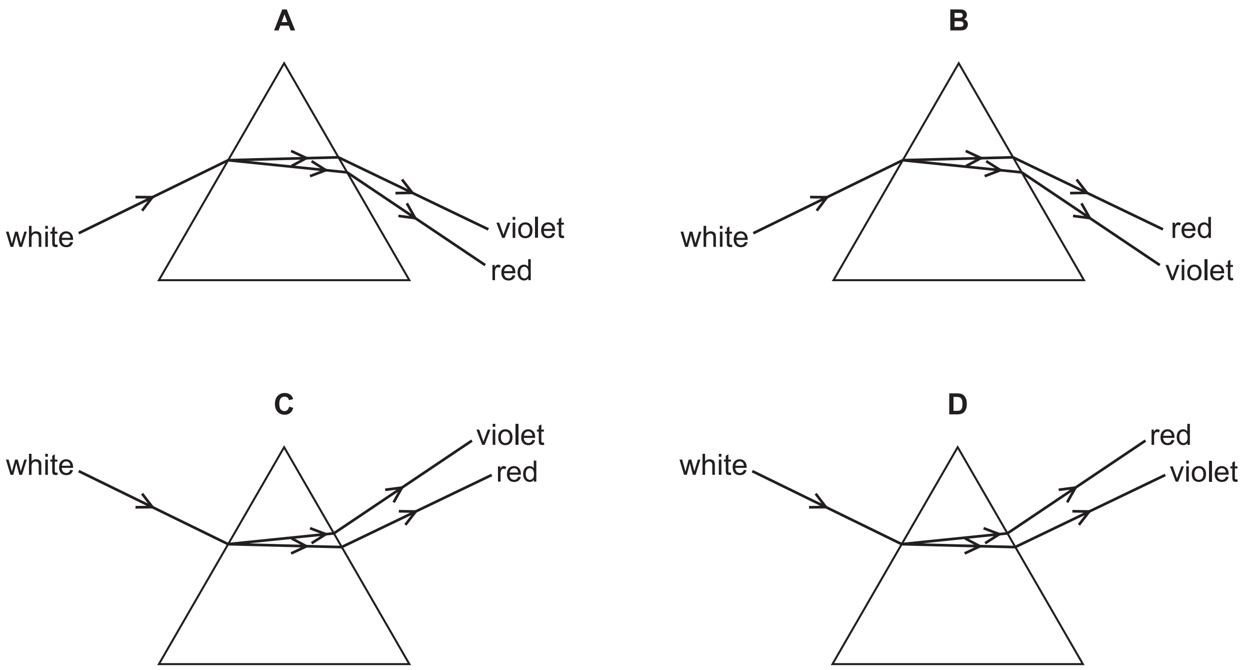
20 A thin converging lens forms an image.



What is the nature of this image and can it be formed on a screen?

	nature of image	can be formed on a screen
A	virtual	no
B	virtual	yes
C	real	no
D	real	yes

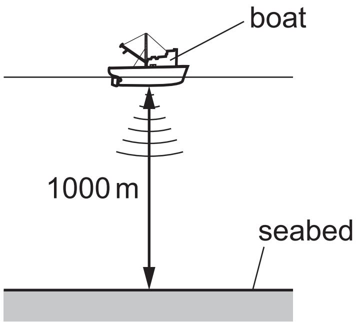
21 Which diagram shows the dispersion of white light by a prism?



22 Which row correctly compares microwaves and ultraviolet?

	frequency of microwaves compared to ultraviolet	uses
A	higher	microwaves are used to heat water and ultraviolet is used to sterilise water
B	higher	microwaves are used to sterilise water and ultraviolet is used to heat water
C	lower	microwaves are used to heat water and ultraviolet is used to sterilise water
D	lower	microwaves are used to sterilise water and ultraviolet is used to heat water

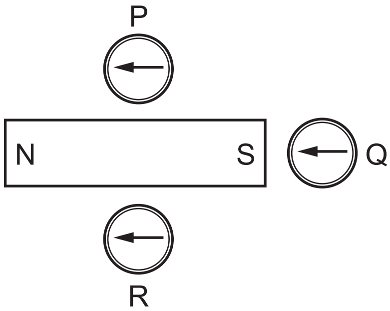
23 A pulse of sound is produced at the bottom of a boat. The sound travels through the water and is reflected from the seabed. The reflected sound reaches the boat 1.3 s after the sound was produced. The seabed is 1000 m below the boat.



Using this information, what is the speed of sound in the water?

A 770 m / s **B** 1300 m / s **C** 1500 m / s **D** 2600 m / s

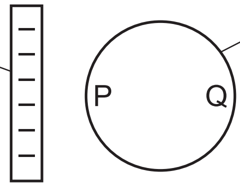
24 The diagram shows three small compasses near to a bar magnet.



Which compasses show the direction of the magnetic field due to the bar magnet?

A P and Q **B** P only **C** Q and R **D** Q only

25 A negatively charged rod is brought close to an isolated, uncharged metal sphere.



What are the charges on sides P and Q of the sphere?

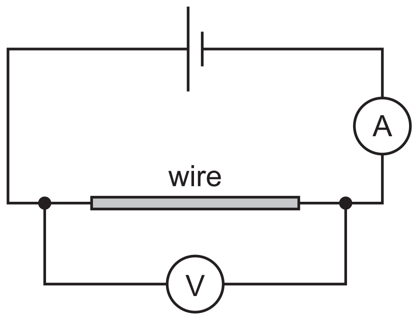
A P and Q are both negatively charged.

B P and Q are both positively charged.

C P is negatively charged and Q is positively charged.

D P is positively charged and Q is negatively charged.

26 The diagram shows a simple circuit.



The ammeter reading is 2.0 A and the voltmeter reading is 6.0 V.

What is the resistance of the wire?

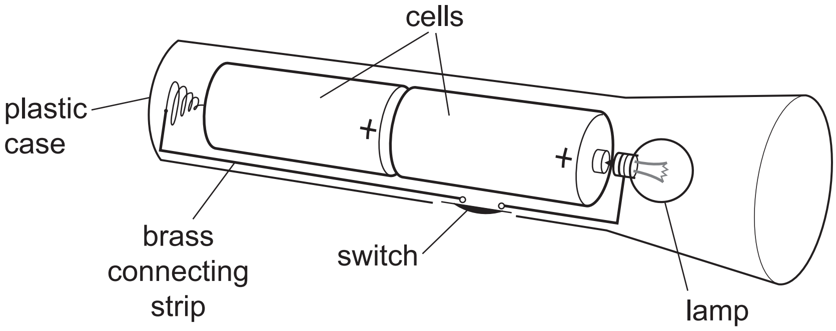
A 0.33 Ω **B** 3.0 Ω **C** 8.0 Ω **D** 12 Ω

27 A lamp is rated 240 V, 0.40 A. The lamp operates at normal brightness for 1.0 min.

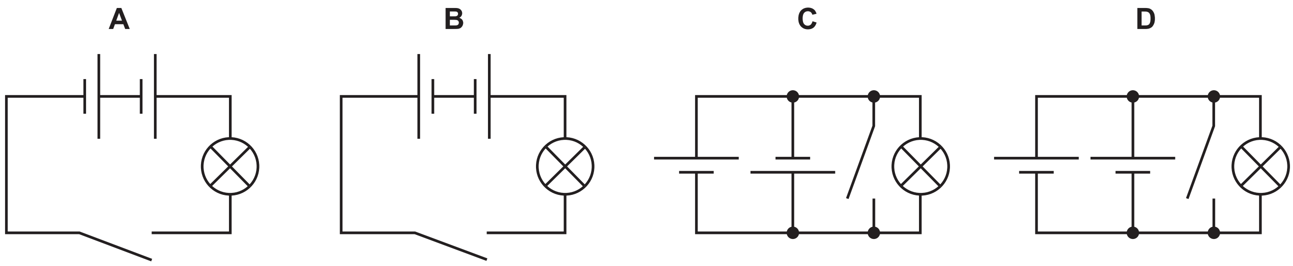
How much energy does the lamp transfer?

A 60 J **B** 96 J **C** 600 J **D** 5800 J

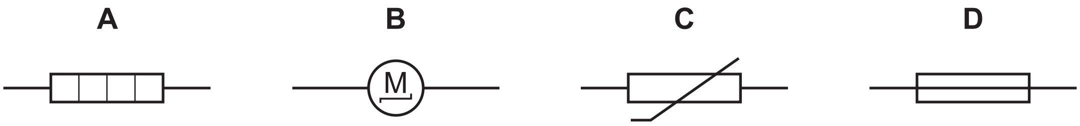
28 The diagram shows a torch containing two cells, a switch and a lamp.



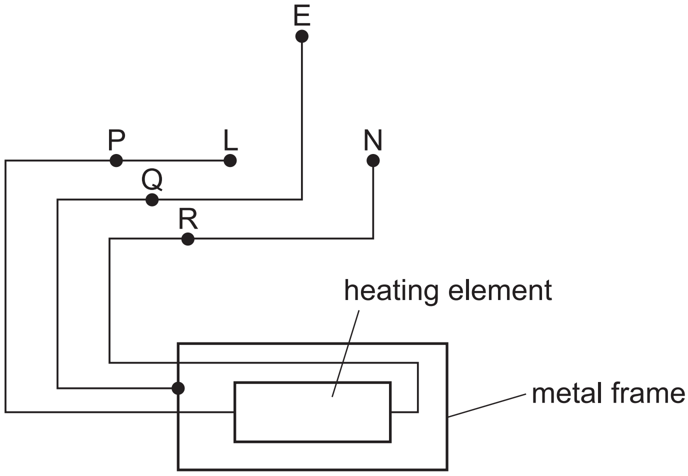
Which circuit diagram is correct for the torch?



29 Which symbol represents a device that contains a current-carrying coil in a magnetic field?



30 The diagram shows a three-core cable connecting a three-pin socket to an electric fire with a metal frame.



The earth, live and neutral connections of the socket are labelled E, L and N respectively.

A switch is connected into the circuit.

Where is a switch connected?

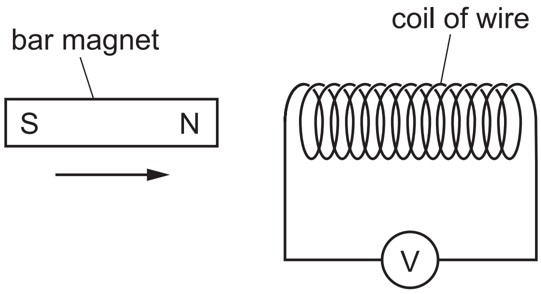
A at either of P or R

B at P only

C at Q only

D at R only

31 A student investigates electromagnetic induction. She moves the N pole of a magnet quickly towards a coil of wire. There is a reading on the sensitive voltmeter.



What can she do to get a greater reading on the sensitive voltmeter?

A Hold the bar magnet stationary inside the coil.

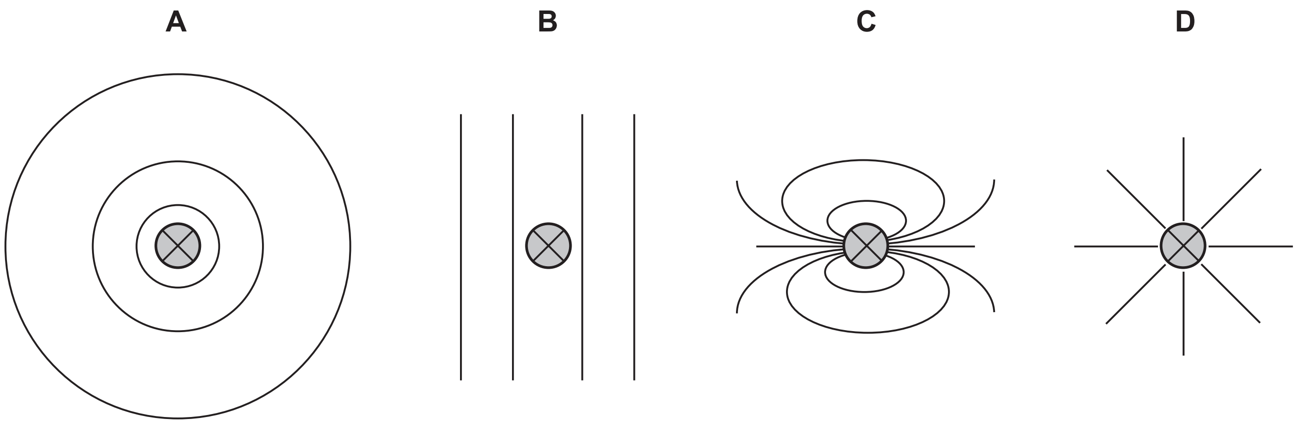
B Move the bar magnet slowly away from the coil.

C Use a coil of wire with fewer turns on it.

D Use a stronger bar magnet.

32 The direction of the current in a straight wire X is into the paper.

Which diagram shows the shape of the magnetic field pattern around the wire?



33 What does a step-up transformer increase?

A current

B energy

C power

D voltage

34 When measuring the emissions from a radioactive rock brought into the laboratory, a teacher mentions that background radiation must be taken into account.

What is this background radiation?

A ionising radiation in the laboratory when the radioactive rock is not present

B ionising radiation from the radioactive rock brought into the laboratory

C infrared radiation from the Sun

D infrared radiation from warm objects in the laboratory

35 How do the ionising effect and the penetrating ability of α-particles compare with those of β-particles and γ-rays?

	ionising effect	penetrating ability
A	higher	lower
B	higher	higher
C	lower	lower
D	lower	higher

36 Half-life is ......1...... for the ......2...... a sample of a radioactive isotope to halve.

Which words correctly complete gaps 1 and 2?

	1	2
A	half of the time taken	nucleon number of
B	half of the time taken	number of nuclei in
C	the time taken	nucleon number of
D	the time taken	number of nuclei in

37 Radioactive materials must be handled in a safe way.

What is **not** a safety procedure?

A storing radioactive materials in cardboard boxes

B monitoring exposure time to radioactive materials

C using tongs to pick up the radioactive source

D wearing protective clothing

38 The Solar System contains eight major planets.

What is the correct order of three of the major planets, starting with the planet furthest from the Sun?

A Mercury → Venus → Earth

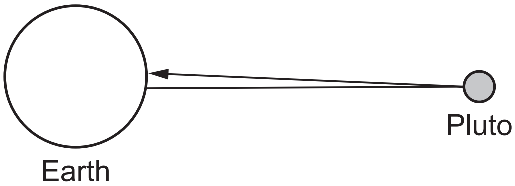
B Neptune → Jupiter → Saturn

C Neptune → Mars → Earth

D Jupiter → Saturn → Uranus

39 The diagram shows a radio signal from the Earth being reflected by Pluto and returning to the Earth.

The distance between the Earth and Pluto is 0.00050 light-years.



What is the time taken for the radio signal to return to the Earth after being sent?

A 2000 years

B 1000 years

C 0.0010 years

D 0.00050 years

40 The Milky Way is a ......1...... made up of about a hundred ......2...... stars.

Which words correctly complete gaps 1 and 2?

	1	2
A	galaxy	billion
B	universe	billion
C	galaxy	thousand
D	universe	thousand

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Cambridge IGCSE™

PHYSICS **0625/12**

Paper 1 Multiple Choice (Core) **October/November 2024**

45 minutes

You must answer on the multiple choice answer sheet.

3707671878

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

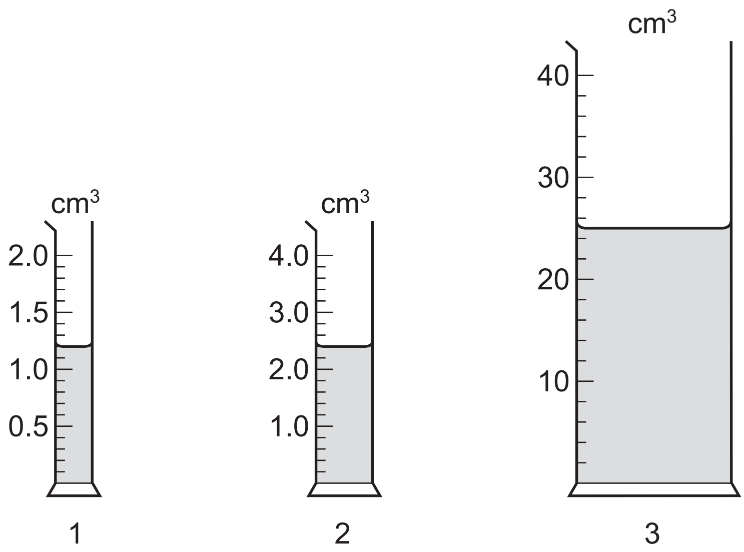
- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m / s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.

1 A student measures the volumes of three liquids using three different measuring cylinders.



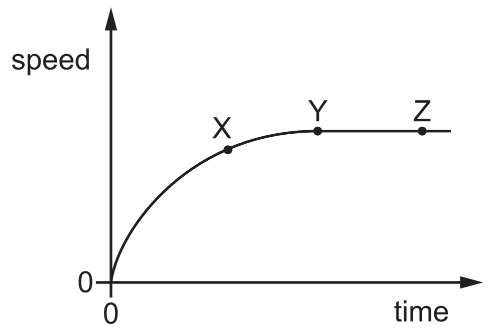
The table shows the volumes recorded by the student.

measuring cylinder	volume / cm^3
1	1.2
2	2.2
3	25

Which readings are correctly recorded?

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 1 only

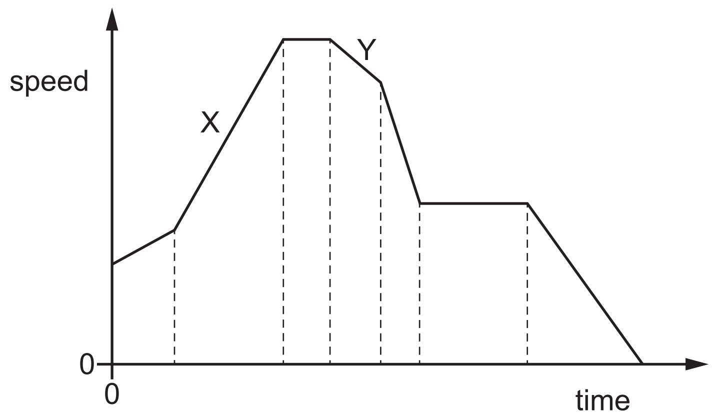
2 The diagram shows how the speed of a falling object changes with time.



Which row describes the motion of the object between X and Y, and between Y and Z?

	between X and Y	between Y and Z
A	accelerating	at rest
B	accelerating	constant speed
C	decelerating	at rest
D	decelerating	constant speed

3 The speed–time graph represents a journey.



How does the graph show that the distance travelled in section X of the journey is greater than the distance travelled in section Y?

A The area under section X of the graph is greater than the area under section Y.

B The gradient of section X of the graph is greater than the gradient of section Y.

C The speed at the end of section X of the journey is greater than the speed at the end of section Y.

D The time for section X of the journey is greater than the time for section Y.

4 Which row contains two correct statements about the mass and the weight of an object?

	mass of an object	weight of an object
A	is measured using a measuring cylinder	is measured using a balance
B	is the gravitational force exerted on the object	is the quantity of matter in the object
C	is measured in newtons	is measured in kilograms
D	is the same everywhere	can vary from place to place

5 Which equipment is used to find the density of an irregularly shaped stone?

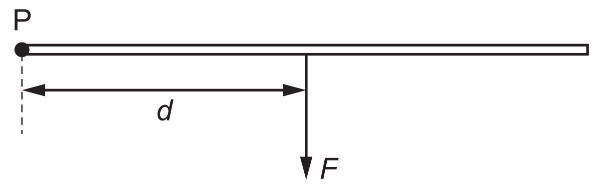
A force meter and ruler

B measuring cylinder, ruler and water

C measuring cylinder, top pan balance and water

D top pan balance and ruler

6 The diagram shows a force *F* acting at 90° to a bar at a distance *d* from the point P.



Which pair of changes causes the greatest increase in the moment of the force *F* about the point P?

A halving *F* and halving *d*

B halving *F* and doubling *d*

C doubling *F* and halving *d*

D doubling *F* and doubling *d*

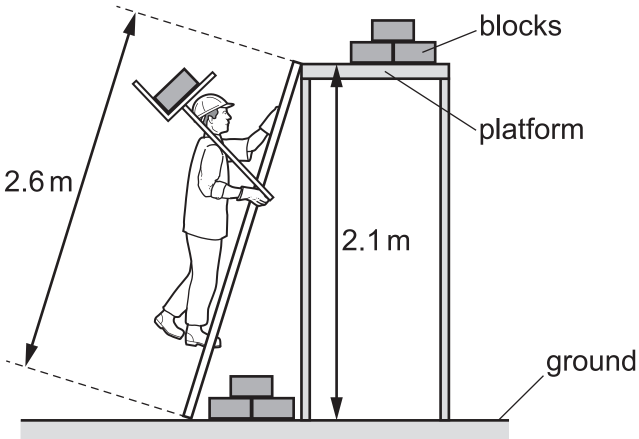
7 A child of weight 420 N is sitting on a swing with her feet on the ground.

The child experiences an upward force from the ground of 130 N and an upward force from the swing of 290 N.

What is the resultant force on the child?

A 0 N **B** 260 N **C** 580 N **D** 840 N

8 A builder lifts one concrete block from the ground onto a platform.



The weight of one block is 170 N.

What is the useful work done against gravity on one block?

A 360 J **B** 440 J **C** 360 W **D** 440 W

9 Students are asked for examples of water being used to store energy.

Three examples are listed.

1 energy stored in water waves

2 energy stored in tides

3 energy stored in water behind dams

Which examples describe water being used to store energy?

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

10 Which statement about power is correct?

A Power is measured in joules.

B Power = energy × time.

C Power is the rate of energy transfer.

D Greater power always gives higher efficiency.

11 In which position does a person exert least pressure on the ground?

A kneeling on the ground

B lying flat on the ground

C sitting on the ground

D standing on the ground

12 Which name is given to the change of state when steam at 100 °C changes to water at 100 °C?

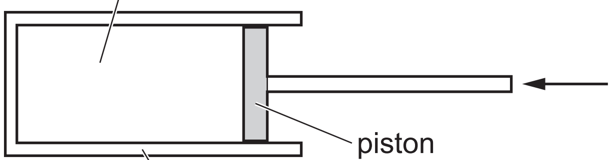
A boiling

B condensation

C evaporation

D melting

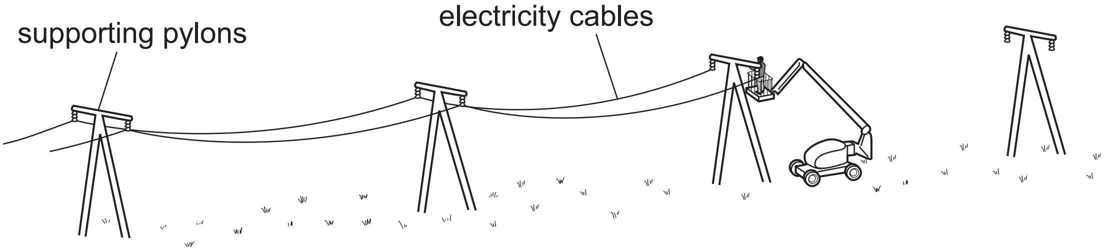
13 A fixed mass of air is trapped inside a cylinder fitted with a moveable piston. The piston is slowly pushed in.



What happens to the pressure and the volume of the trapped air?

	pressure	volume
A	decreases	decreases
B	decreases	does not change
C	increases	decreases
D	increases	increases

14 The diagram shows electricity cables being put up on a warm day. The cables are suspended between the supporting pylons, as shown.



Why are the cables **not** tightened to make them higher above the ground?

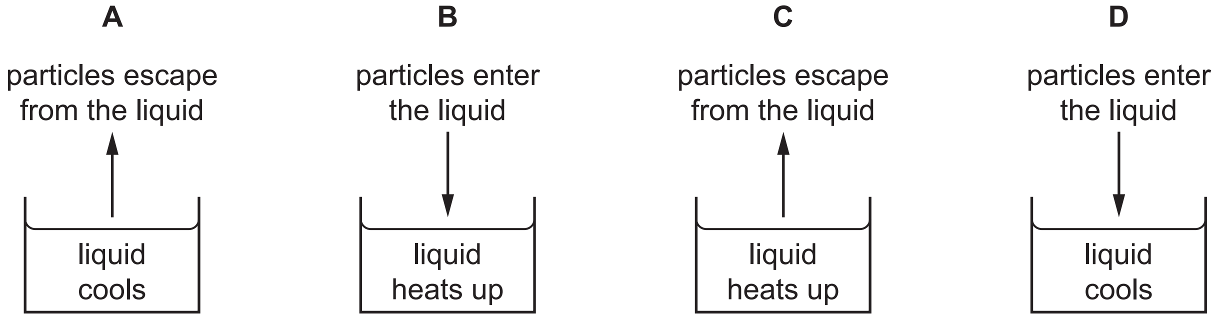
A They will contract on cold days.

B They will contract on very warm days.

C They will expand on cold days.

D They will expand on very warm days.

15 Which diagram shows the overall movement of particles and the temperature change when a liquid evaporates?



16 There is a vacuum between the double walls of a vacuum flask.

Which types of thermal transfer are reduced by the vacuum?

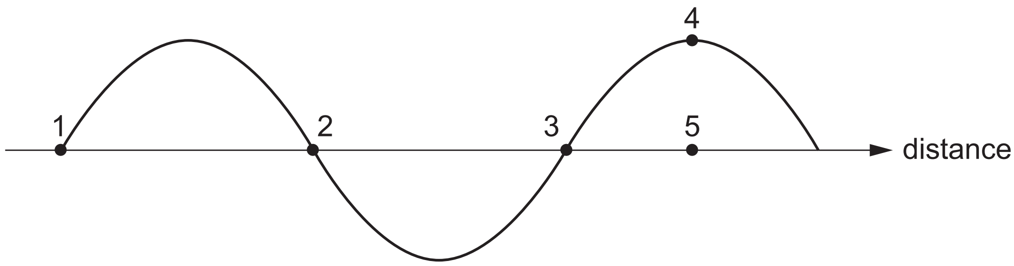
A conduction, convection and radiation

B conduction and convection only

C conduction and radiation only

D convection and radiation only

17 The diagram shows a transverse wave.



Which distance is equal to one wavelength?

A the distance between points 1 and 2

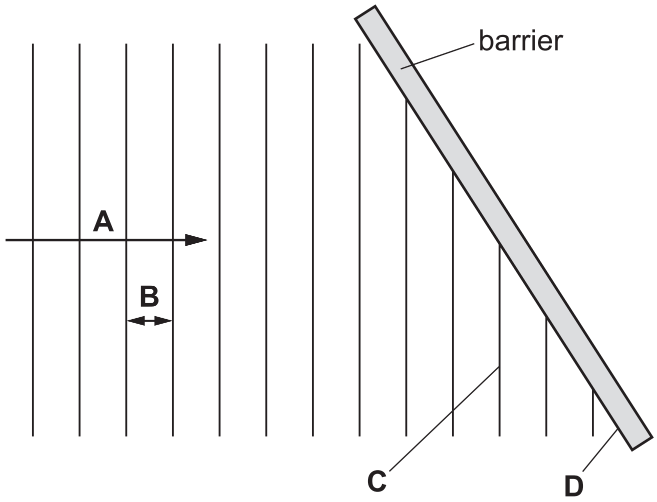
B the distance between points 1 and 3

C the distance between points 2 and 3

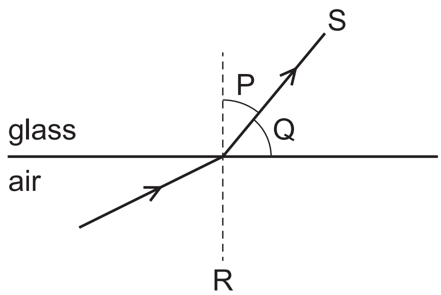
D the distance between points 4 and 5

18 The diagram shows a wave before it reflects from a barrier.

Which labelled section of the diagram represents a wavefront?



19 The diagram shows a ray of light as it passes from air into glass.

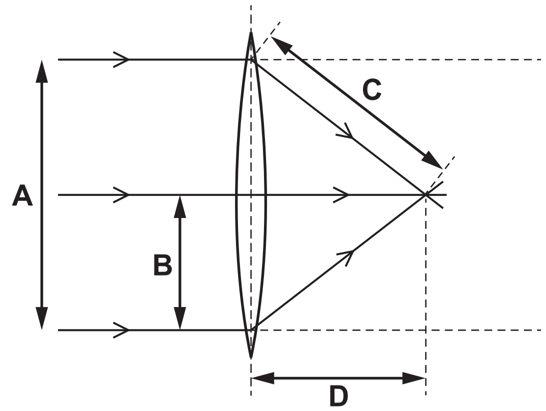


Which row shows the angle of refraction and the normal?

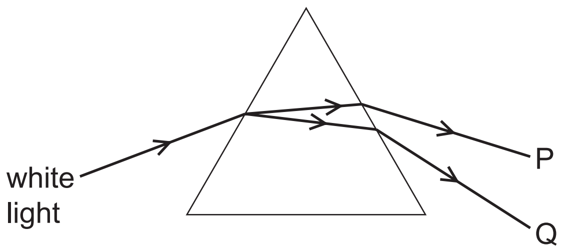
	angle of refraction	normal
A	P	R
B	P	S
C	Q	R
D	Q	S

20 The diagram shows rays of light passing through a converging lens.

Which labelled arrow represents the focal length of the lens?



21 The diagram shows a narrow beam of white light being dispersed by a glass prism to produce a visible spectrum.



Which statement describes what happens to the frequency and wavelength of the light in observing from P to Q across the spectrum?

A The frequency and the wavelength both increase.

B The frequency and the wavelength both decrease.

C The frequency decreases but the wavelength increases.

D The frequency increases but the wavelength decreases.

22 The early Universe was filled with gamma radiation. Since then, the radiation has shifted to the microwave region of the electromagnetic spectrum.

How has this change affected the wavelength and speed of the radiation?

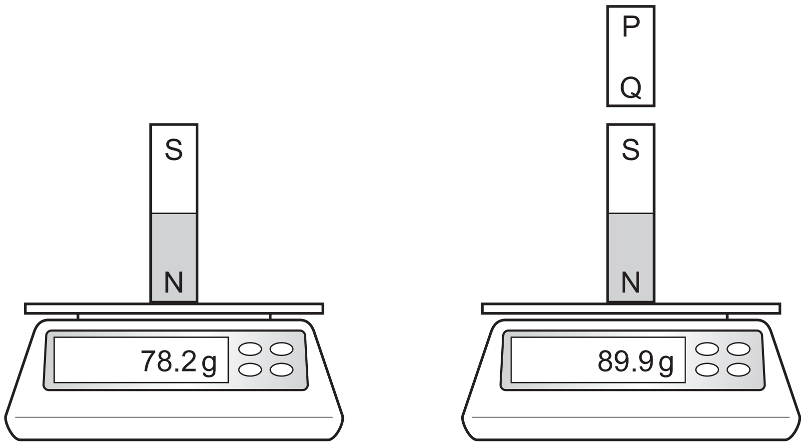
	wavelength	speed
A	decreased	decreased
B	decreased	stayed the same
C	increased	decreased
D	increased	stayed the same

23 Sound waves may cause an echo.

What happens to sound waves to cause an echo and what is the nature of sound waves?

	what an echo is caused by	nature of sound waves
A	reflection	longitudinal
B	reflection	transverse
C	refraction	longitudinal
D	refraction	transverse

24 The diagram shows what happens when a bar PQ is brought near to a permanent magnet which is standing on a balance.



What could bar PQ be?

A an iron bar or a permanent magnet with P a N pole

B an iron bar or a permanent magnet with Q a N pole

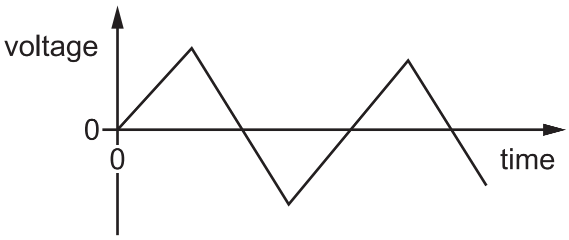
C an iron bar only

D a permanent magnet only

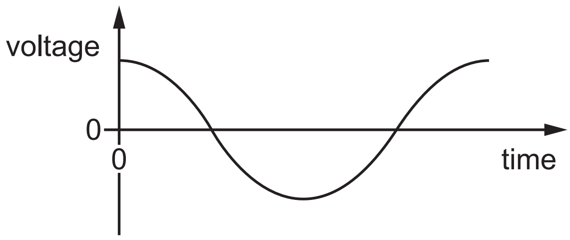
25 Direct current (d.c.) can be represented on a voltage–time graph.

Which graph shows the correct waveform for a d.c. supply?

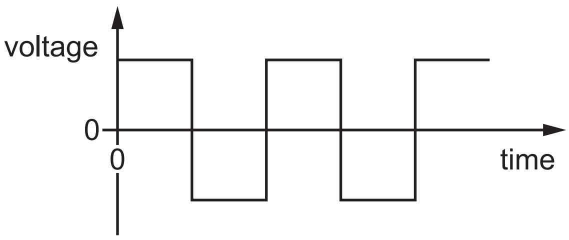
A



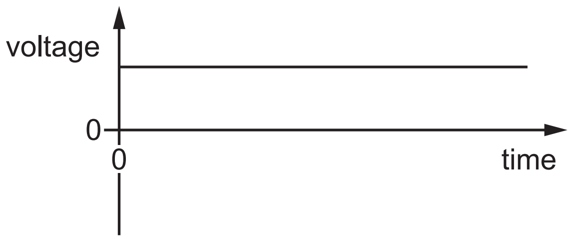
B



C



D



26 A resistor with a potential difference (p.d.) of 100 V across it carries a current of 5.0 mA.

What is the resistance of the resistor?

A 0.50 Ω **B** 20 Ω **C** 500 Ω **D** 20 000 Ω

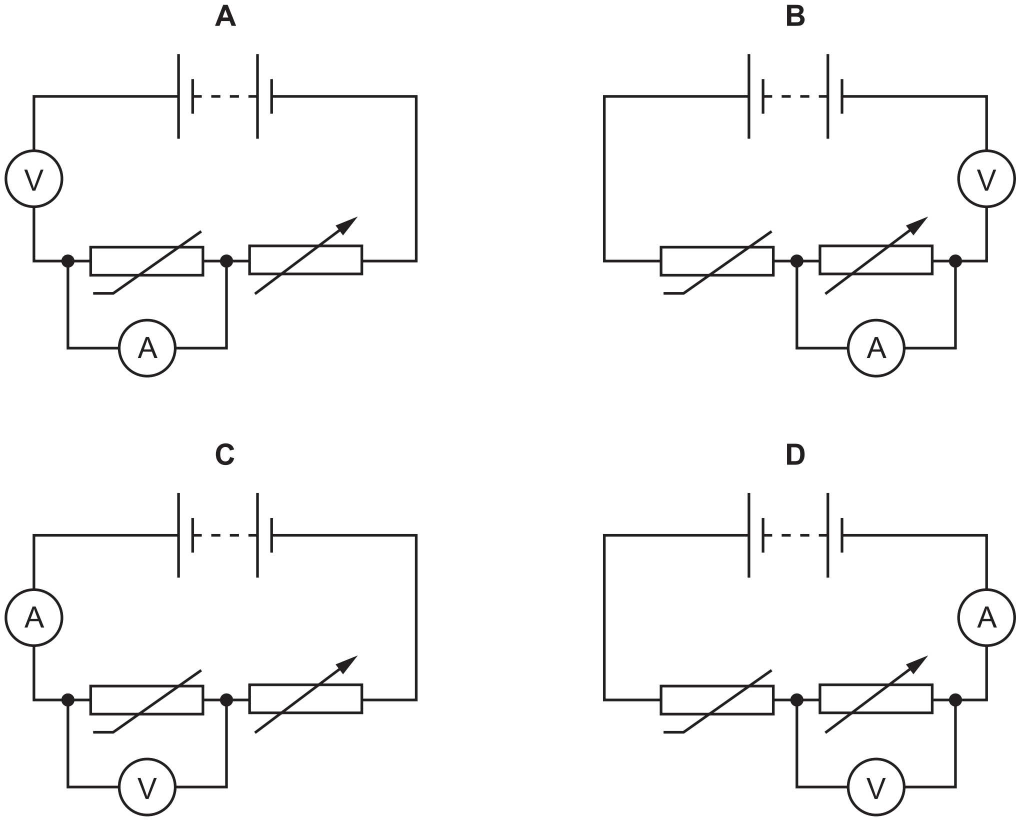
27 The diagram shows the information label on an electric kettle.

Model CIB xxx001	
Voltage:	110 V a.c.
Power:	1500 W
Frequency:	50 Hz

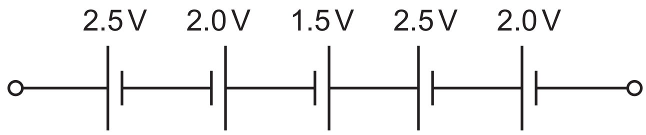
What is the value of the current when the kettle is heating water?

A 0.073 A **B** 2.2 A **C** 3.7 A **D** 14 A

28 Which circuit enables the resistance of the thermistor to be determined?



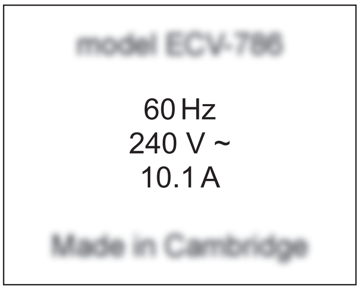
29 A student makes a battery by connecting five cells, as shown.



What is the electromotive force (e.m.f.) of the battery?

A 1.0 V **B** 2.1 V **C** 3.5 V **D** 10.5 V

30 The diagram shows the safety label on an electric oven.



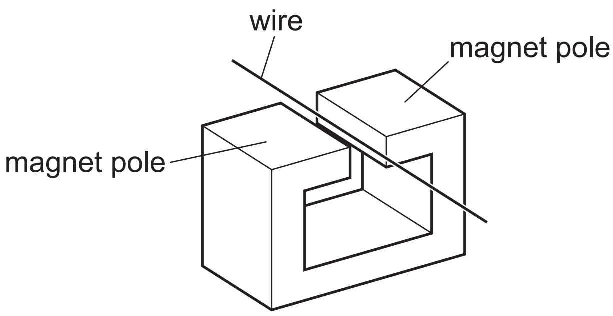
The oven is connected to a mains electric circuit. A fuse is placed in the circuit to protect the cabling and prevent overheating if there is fault.

Where should the fuse be placed and which rating should the fuse have?

	position of fuse	fuse rating / A
A	live wire	10
B	live wire	13
C	neutral wire	10
D	neutral wire	13

31 A straight wire is stationary between the poles of a magnet.

It lies perpendicular to the magnetic field.



Which action does **not** induce an e.m.f. in the wire?

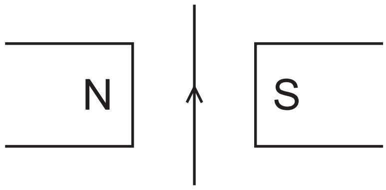
A moving the magnet up towards the top of the page and the wire down towards the bottom of the page

B moving the magnet up towards the top of the page only

C moving the wire and the magnet up towards the top of the page at the same speed

D moving the wire down towards the bottom of the page only

32 The diagram shows a wire hanging vertically between the poles of a magnet.



There is a current in the wire in the direction shown. The wire moves into the plane of the page away from the observer.

In which direction does the wire move when the current is reversed?

A The wire moves into the plane of the page, away from the observer.

B The wire moves to the left.

C The wire moves to the right.

D The wire moves out of the plane of the page, towards the observer.

33 A current-carrying coil in a magnetic field experiences a turning effect.

Students are asked how the turning effect can be increased.

Three suggestions are listed.

1 decreasing the number of turns on the coil

2 increasing the current

3 increasing the strength of the magnetic field

Which suggestions will increase the turning effect?

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

34 A student suggests three sources of naturally occurring background radiation.

1 cosmic rays

2 medical X-rays

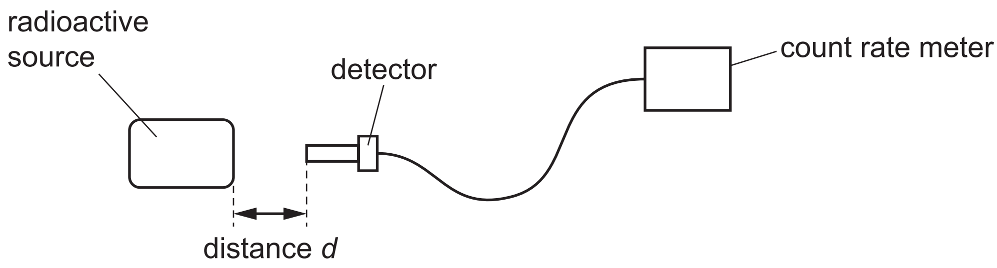
3 radioactive emissions from radon gas from the ground

Which suggestions are correct?

A 1 and 3 **B** 1 only **C** 2 and 3 **D** 2 only

35 A student measures the rate at which ionising radiation is emitted from a radioactive substance.

He places a detector at different distances from the radioactive source.



The table shows how the count rate from the source varies with distance *d*.

distance *d* / cm	0	2	4	6
count rate / counts per minute	1250	115	0	0

Which type of ionising radiation is being emitted by the substance?

A α-particles

B β-particles

C γ-rays

D X-rays

36 Radioactive decay results in the emission of α-particles, β-particles and γ-radiation.

Which types of emission result in a nucleus changing to that of a different element?

A α-particle emission and β-particle emission

B α-particle emission and γ-radiation emission

C β-particle emission and γ-radiation emission

D γ-radiation emission only

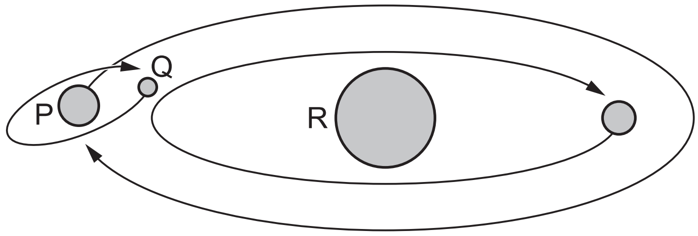
37 A radioactive isotope has a half-life of 120 minutes.

It emits radiation at a rate of 100 particles per second.

How long does it take for the rate of emission to fall to 25 particles per second?

A 30 minutes **B** 45 minutes **C** 90 minutes **D** 240 minutes

38 The diagram shows part of a solar system.



Which row correctly identifies the bodies P, Q and R?

	P	Q	R
A	moon	planet	star
B	planet	moon	star
C	planet	star	moon
D	star	planet	moon

39 Which statement about the Sun is **not** correct?

A The Sun is one of the biggest stars.

B The Sun emits ultraviolet radiation.

C The Sun emits infrared radiation.

D Helium is present in the Sun.

40 Which statement about the Milky Way galaxy is correct?

A It contains stars that are about the same distance from the Earth as the Sun.

B It is a collection of about 1000 stars that can be seen from the Earth.

C It is the galaxy that contains the Sun and the Earth.

D Its diameter is about 1 light-year.

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PHYSICS **0625/11**

Paper 1 Multiple Choice (Core) **October/November 2024**

45 minutes

You must answer on the multiple choice answer sheet.

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You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m / s^2).

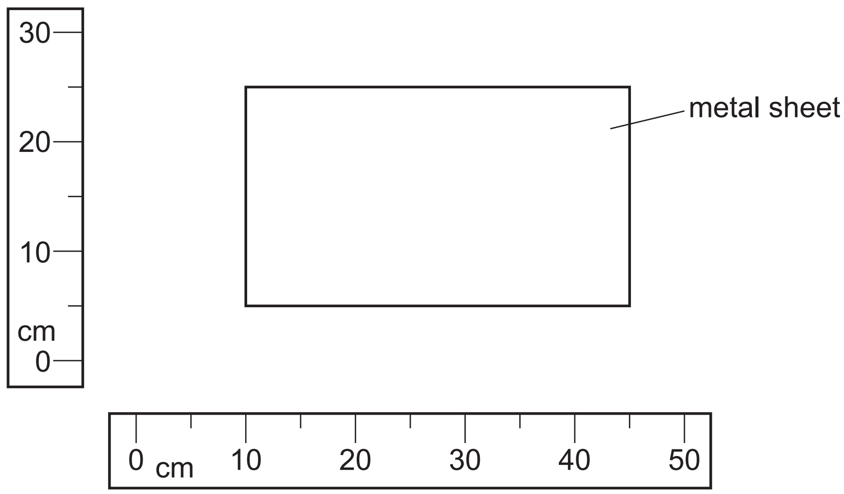
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.

IB24 11_0625_11/RP

1 The diagram shows a rectangular metal sheet close to two rulers.

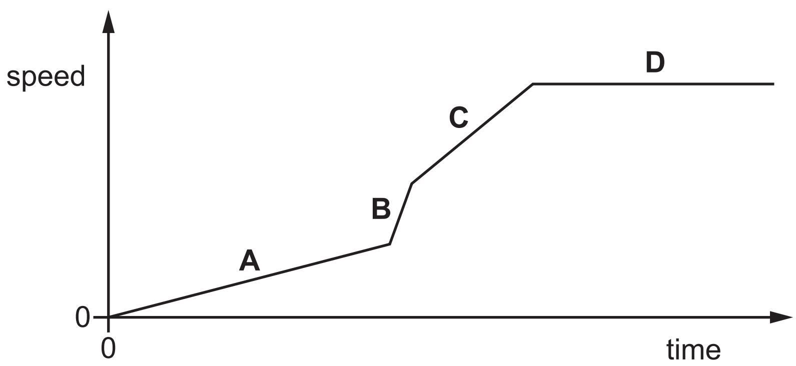


What is the area of the metal sheet?

A $700\,cm^2$ **B** $875\,cm^2$ **C** $900\,cm^2$ **D** $1125\,cm^2$

2 The diagram shows the speed–time graph of an object.

In which section does the object have the largest acceleration?



3 An object begins to fall close to the Earth's surface. Air resistance can be ignored.

Which statement about the object's acceleration is correct?

A The acceleration is constant.

B The acceleration decreases as the body falls.

C The acceleration increases as the body falls.

D The acceleration is zero.

4 Which quantity is equal to gravitational force?

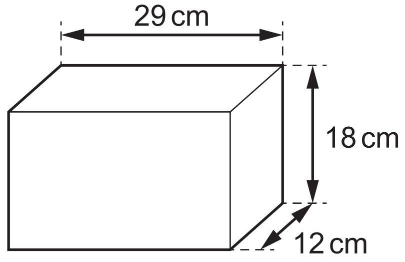
A gravitational field strength × mass

B gravitational field strength × weight

C mass per unit weight

D weight per unit mass

5 A concrete building block has the dimensions shown.



The mass of the block is 15 000 g.

What is the density of the block?

A 43 g / cm^3 **B** 2.4 g / cm^3 **C** 0.42 g / cm^3 **D** 0.023 g / cm^3

6 What is measured by the moment of a force?

A the acceleration produced by the force

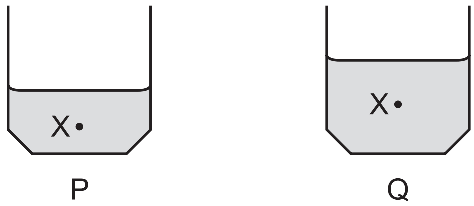
B the turning effect of the force

C the time for which the force acts

D the increase in energy caused by the force

7 Two identical containers, P and Q, are partly filled with different quantities of sand.

The position of the centre of gravity for each container is labelled X.



Which container is more stable and what is the reason why it is more stable?

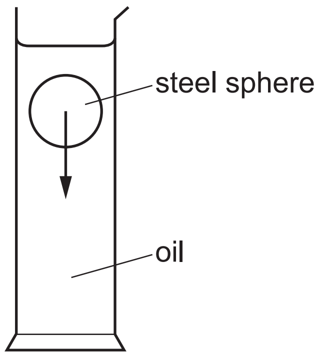
A P is more stable because it has a lower centre of gravity.

B P is more stable because it has a smaller mass.

C Q is more stable because it has a greater depth of sand.

D Q is more stable because it has a higher centre of gravity.

8 The diagram shows a steel sphere falling through a cylinder of oil.

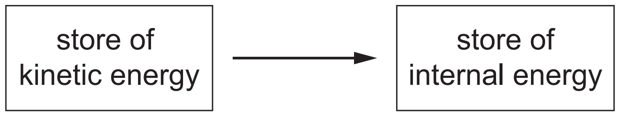


Which row indicates what happens to the steel sphere and what happens to the oil as the steel sphere falls?

	the steel sphere becomes warmer	the oil becomes warmer
A	yes	yes
B	yes	no
C	no	yes
D	no	no

9 A moving object is brought to rest by a frictional force of 30 N over a distance of 5.0 m.

The diagram shows the energy transferred between stores.



How much energy is transferred by this force and how is the energy transferred?

	energy transferred / J	how the energy is transferred
A	6.0	by electrical working
B	6.0	by mechanical working
C	150	by electrical working
D	150	by mechanical working

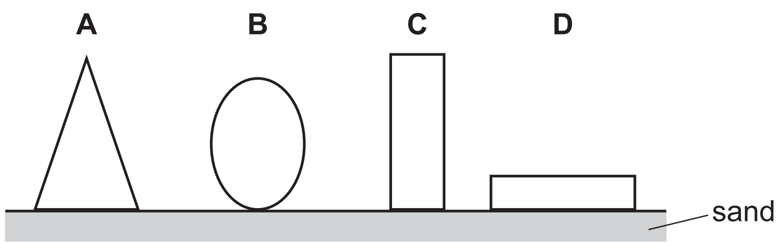
10 A motor transfers 24 J of energy in 60 seconds.

What is the power output of the motor?

A 0.40 W **B** 2.5 W **C** 24 W **D** 1400 W

11 An artist makes four sculptures with circular bases and places them on sand. All the sculptures are of equal weight and volume.

Which sculpture is least likely to sink into the sand?



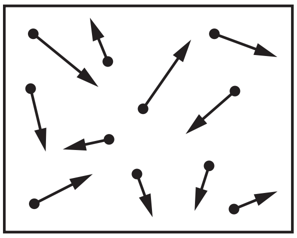
12 A quantity of water is boiled to form the same mass of steam.

Which row shows how the volume and density of the water change?

	volume	density
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

13 The diagram represents gas particles moving around in a sealed container.

The gas particles collide with the walls of the container.



The temperature of the gas is increased.

What happens to the average speed of the gas particles and what happens to the number of collisions by the gas particles with the walls of the container?

	average speed of gas particles	the number of collisions with the walls of the container
A	increases	less frequent collisions
B	increases	more frequent collisions
C	stays the same	less frequent collisions
D	stays the same	more frequent collisions

14 A metal object is heated strongly in an oven.

What happens to the volume of the object and the internal energy stored in the object?

	volume	internal energy
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

15 Which row describes the process of melting?

	initial state	final state	change in temperature
A	liquid	gas	yes
B	liquid	solid	no
C	solid	gas	yes
D	solid	liquid	no

16 A beaker of water is heated and thermal energy travels through the water by convection.

What happens to the density of the water when it is heated and how does the water move?

A The density decreases and the heated water moves downwards.

B The density decreases and the heated water moves upwards.

C The density increases and the heated water moves downwards.

D The density increases and the heated water moves upwards.

17 Waves travel across the surface of water.

What is meant by the amplitude of the wave?

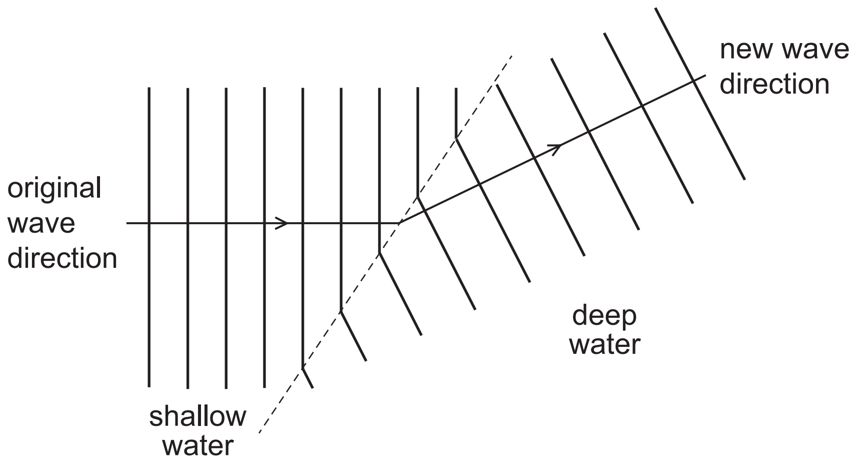
A the maximum distance of a water particle from its mean position

B how far the wave travels every second

C the number of waves passing a point every second

D the distance between the top of consecutive waves

18 Water waves change direction when they move from shallow water to deep water.



What is the name of this effect?

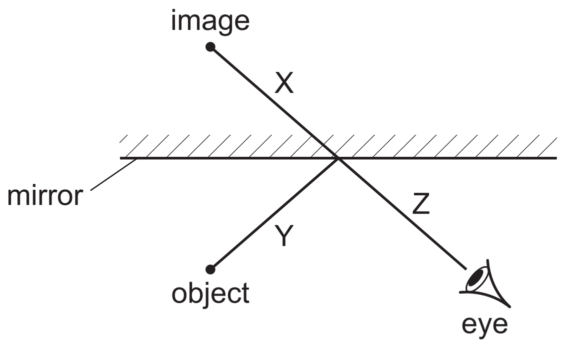
A diffraction

B dispersion

C reflection

D refraction

19 The ray diagram shows the formation of an image when light is reflected by a plane mirror.

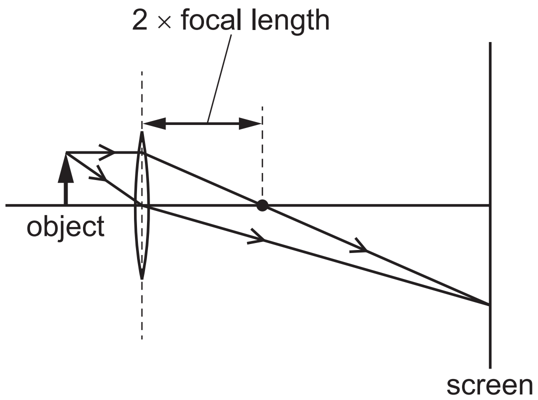


Which lines represent light rays?

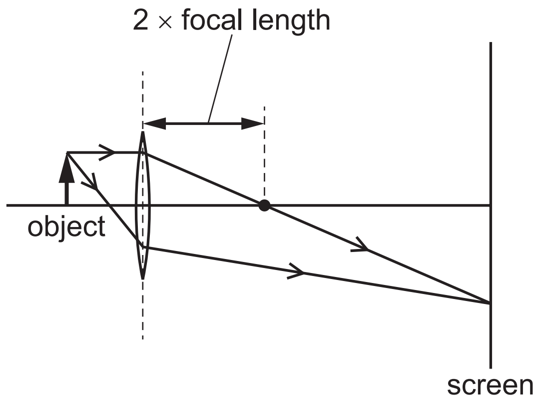
A X, Y and Z **B** X and Y only **C** X and Z only **D** Y and Z only

20 Which diagram shows how an image of an object is formed on a screen by a converging lens?

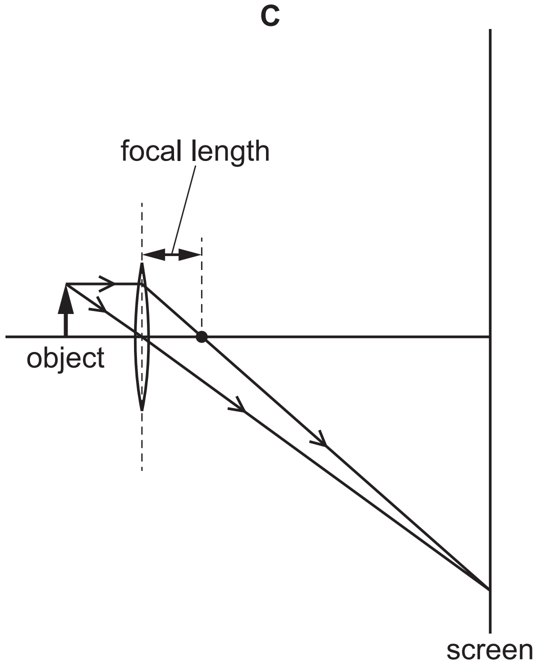
A



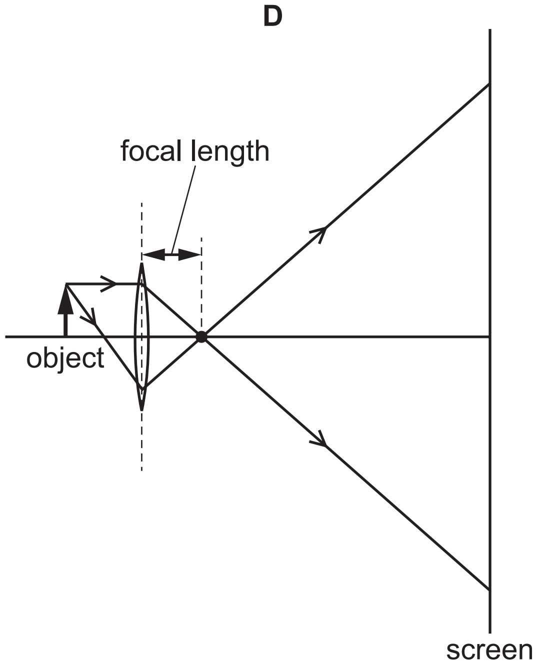
B



C

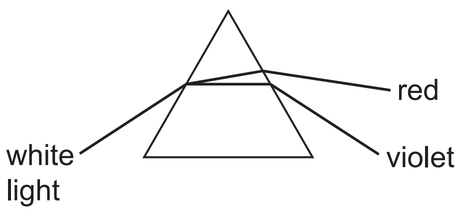


D

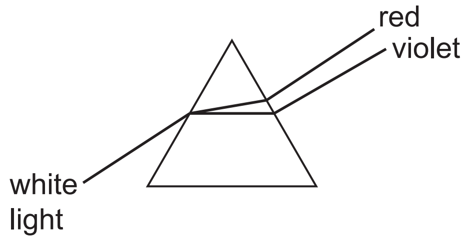


21 Which diagram shows the dispersion of white light by a glass prism?

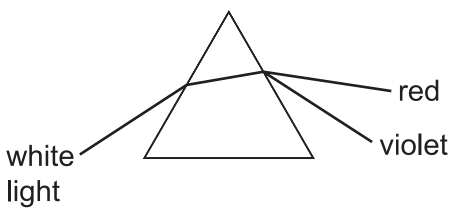
A



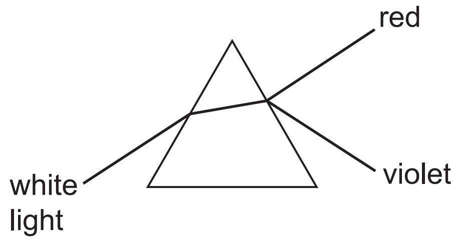
B



C



D



22 Which regions of the electromagnetic spectrum are used for satellite television and for security marking?

	satellite television	security marking
A	microwaves	infrared
B	microwaves	ultraviolet
C	X-rays	infrared
D	X-rays	ultraviolet

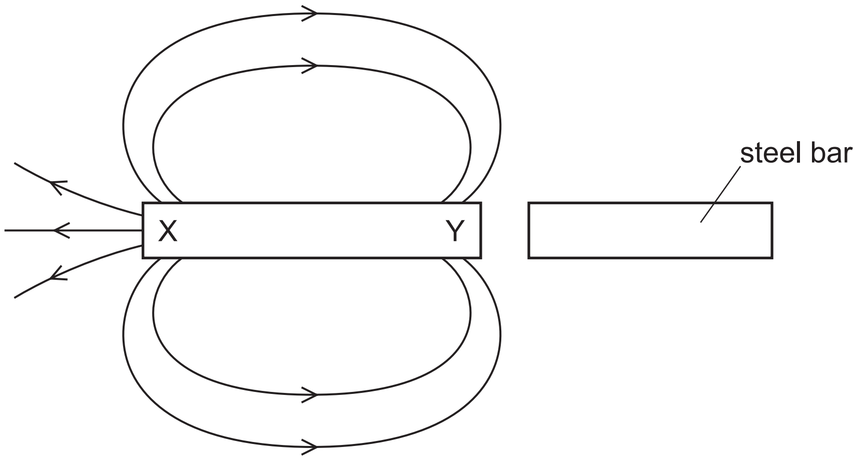
23 The horn on a ship makes a sound. The captain on the ship hears an echo from a cliff 4.0 s later.

The speed of sound is 340 m / s.

How far away is the cliff from the ship?

A 170 m **B** 340 m **C** 680 m **D** 1400 m

24 A bar magnet is next to a steel bar. Some of the magnetic field lines due to the magnet are shown.



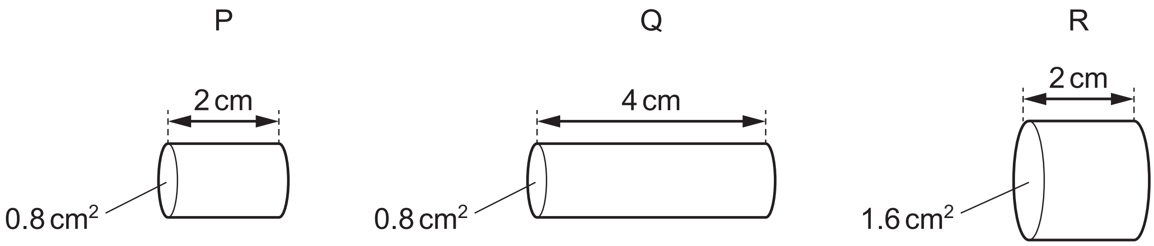
What are the correct poles of the magnet and would the steel bar attract or repel the bar magnet?

	X	Y	force
A	N pole	S pole	attraction
B	N pole	S pole	repulsion
C	S pole	N pole	attraction
D	S pole	N pole	repulsion

25 Which row about electrical conduction in metals is correct?

	statement 1	statement 2
A	It is the movement of free electrons.	Electric charge is a flow of current.
B	It is the movement of positive ions.	Electric charge is a flow of current.
C	It is the movement of free electrons.	Electric current is a flow of charge.
D	It is the movement of positive ions.	Electric current is a flow of charge.

26 The diagram shows three wires, P, Q and R. They are all made from the same metal.



Which list gives the wires in order of resistance, from lowest to highest?

A P → Q → R

B Q → R → P

C R → P → Q

D R → Q → P

27 A 60 W lamp operates at normal brightness for 2.0 minutes.

How much energy is transferred by the lamp?

A 0.50 J **B** 30 J **C** 120 J **D** 7200 J

28 What is a thermistor?

A a container to keep a hot liquid warm

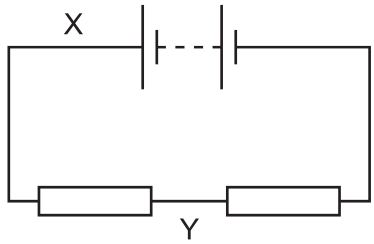
B a circuit to control room temperature

C a temperature-dependent resistor

D a heater for a room

29 The diagram shows a circuit containing two identical resistors connected to a battery.

The current is measured at X and at Y.



Which row is correct?

	arrangement of resistors	current at Y compared with current at X
A	parallel	less than X
B	parallel	equal to X
C	series	less than X
D	series	equal to X

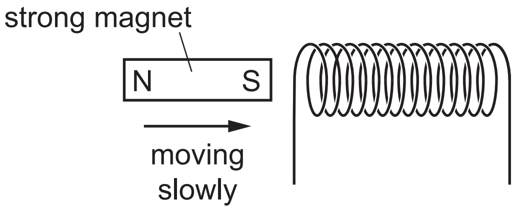
30 The mains voltage is 120 V.

Which fuse should be fitted to an electric kettle with a power of 1.5 kW?

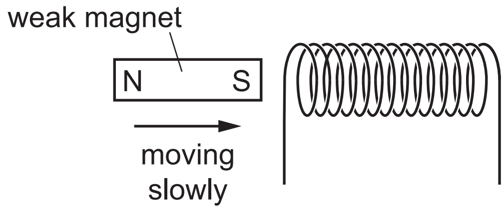
A 3 A **B** 5 A **C** 10 A **D** 13 A

31 The diagrams show a strong magnet and a weak magnet moving into the same coil of wire at different speeds.

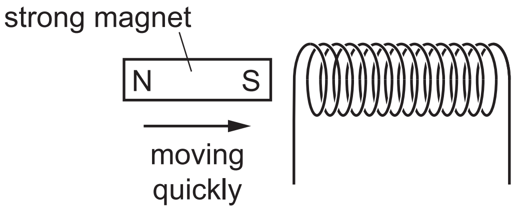
arrangement X



arrangement Y



arrangement Z

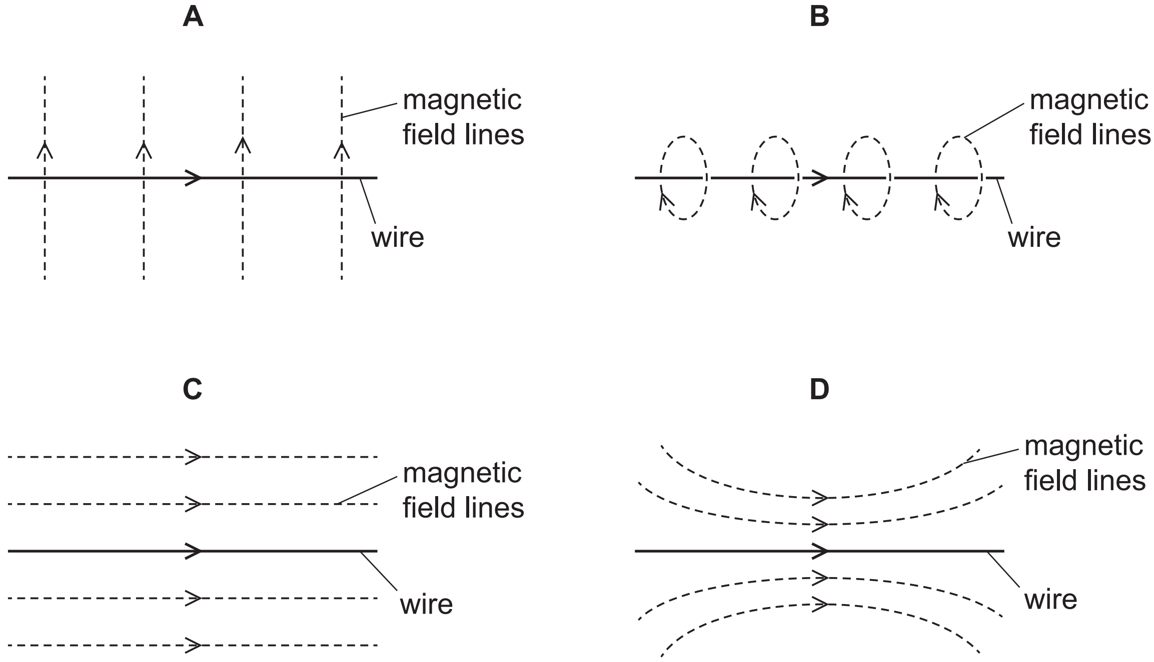


Which arrangement induces the largest electromotive force (e.m.f.) and which arrangement induces the smallest e.m.f.?

	largest e.m.f.	smallest e.m.f.
A	X	Y
B	Y	Z
C	Z	X
D	Z	Y

32 The diagrams show a current-carrying wire with an arrow in the direction of the current.

Which diagram shows the magnetic field produced by the current?



33 What are the best materials to use for the construction of a transformer?

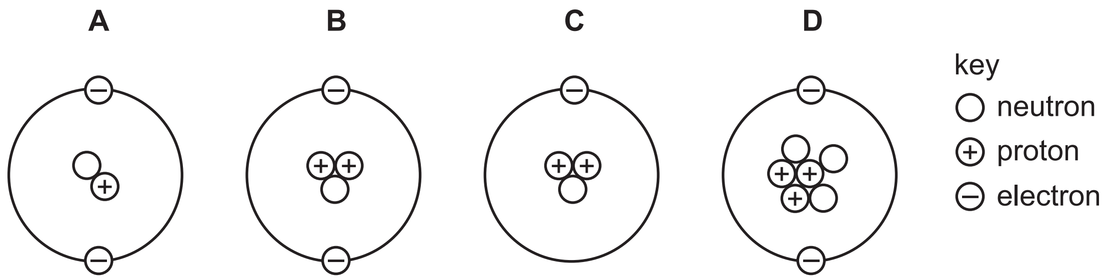
A copper for the core and steel for the coils wound around it

B copper for the core and copper for the coils wound around it

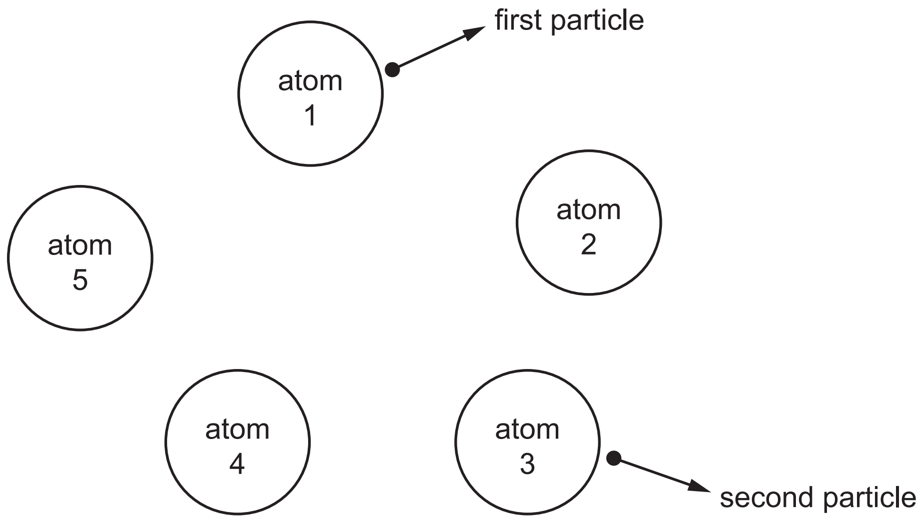
C soft iron for the core and copper for the coils wound around it

D steel for the core and copper for the coils wound around it

34 Which diagram represents the structure of a neutral atom?



35 The diagram shows five atoms in a radioactive substance. The atoms each give out an α-particle.



Atom 1 is the first to give out a particle. Atom 3 is the second to give out a particle.

Which atom will give out the next particle?

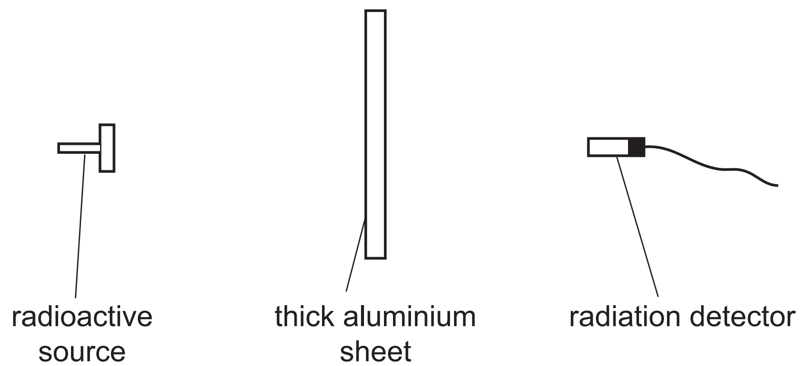
A atom 2

B atom 4

C atom 5

D impossible to tell

36 The diagram shows a radioactive source, a thick aluminium sheet and a radiation detector.



The radiation detector shows a reading greater than the background reading.

Which type of radiation is being emitted by the source and detected by the detector?

A α-radiation

B β-radiation

C γ-radiation

D infrared radiation

37 The half-life of the radioactive isotope caesium $^{137}_{55}Cs$ is 30 years.

Starting with 30 g of the isotope, which mass of the isotope remains after 90 years?

A 10.0 g **B** 7.50 g **C** 3.75 g **D** 1.25 g

38 Which list gives the names of the planets in the correct order?

A Mercury, Mars, Earth, Venus, Jupiter, Saturn, Uranus, Neptune

B Mercury, Venus, Earth, Mars, Saturn, Uranus, Jupiter, Neptune

C Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune

D Venus, Mars, Earth, Mercury, Jupiter, Saturn, Uranus, Neptune

39 Which row describes the Sun and the Milky Way?

	the Sun	the Milky Way
A	a galaxy	a galaxy
B	a galaxy	a star
C	a star	a galaxy
D	a star	a star

40 What is the approximate diameter of the Milky Way?

A 100 light-years

B 1000 light-years

C 10 000 light-years

D 100 000 light-years

Cambridge IGCSE™

PHYSICS **0625/13**

Paper 1 Multiple Choice (Core) **October/November 2024**

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	B	**1**
2	D	**1**
3	D	**1**
4	B	**1**
5	B	**1**
6	A	**1**
7	B	**1**
8	D	**1**
9	C	**1**
10	D	**1**
11	B	**1**
12	D	**1**
13	A	**1**
14	B	**1**
15	C	**1**
16	D	**1**
17	B	**1**
18	B	**1**
19	C	**1**
20	D	**1**
21	B	**1**
22	C	**1**
23	C	**1**
24	D	**1**
25	D	**1**
26	B	**1**
27	D	**1**
28	A	**1**

Question	Answer	Marks
29	B	**1**
30	B	**1**
31	D	**1**
32	A	**1**
33	D	**1**
34	A	**1**
35	A	**1**
36	D	**1**
37	A	**1**
38	C	**1**
39	C	**1**
40	A	**1**

Cambridge IGCSE™

PHYSICS **0625/12**

Paper 1 Multiple Choice (Core) **October/November 2024**

MARK SCHEME

Maximum Mark: 40

Published

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Question	Answer	Marks
1	C	**1**
2	B	**1**
3	A	**1**
4	D	**1**
5	C	**1**
6	D	**1**
7	A	**1**
8	A	**1**
9	A	**1**
10	C	**1**
11	B	**1**
12	B	**1**
13	C	**1**
14	A	**1**
15	A	**1**
16	B	**1**
17	B	**1**
18	C	**1**
19	A	**1**
20	D	**1**
21	D	**1**
22	D	**1**
23	A	**1**
24	D	**1**
25	D	**1**
26	D	**1**
27	D	**1**
28	C	**1**

Question	Answer	Marks
29	C	**1**
30	B	**1**
31	C	**1**
32	D	**1**
33	D	**1**
34	A	**1**
35	A	**1**
36	A	**1**
37	D	**1**
38	B	**1**
39	A	**1**
40	C	**1**

CAMBRIDGE
International Education

Cambridge IGCSE™

PHYSICS **0625/11**

Paper 1 Multiple Choice (Core) **October/November 2024**

MARK SCHEME

Maximum Mark: 40

Published

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Question	Answer	Marks
1	A	**1**
2	B	**1**
3	A	**1**
4	A	**1**
5	B	**1**
6	B	**1**
7	A	**1**
8	A	**1**
9	D	**1**
10	A	**1**
11	D	**1**
12	C	**1**
13	B	**1**
14	D	**1**
15	D	**1**
16	B	**1**
17	A	**1**
18	D	**1**
19	D	**1**
20	C	**1**
21	A	**1**
22	B	**1**
23	C	**1**
24	A	**1**
25	C	**1**
26	C	**1**
27	D	**1**
28	C	**1**

Question	Answer	Marks
29	D	**1**
30	D	**1**
31	D	**1**
32	B	**1**
33	C	**1**
34	B	**1**
35	D	**1**
36	C	**1**
37	C	**1**
38	C	**1**
39	C	**1**
40	D	**1**

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Cambridge IGCSE™

PHYSICS **0625/13**

Paper 1 Multiple Choice (Core) **October/November 2023**

45 minutes

You must answer on the multiple choice answer sheet.

5314735504

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

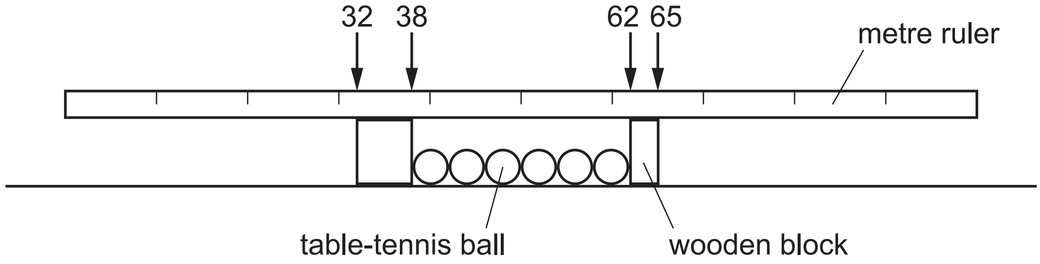
- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m / s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.

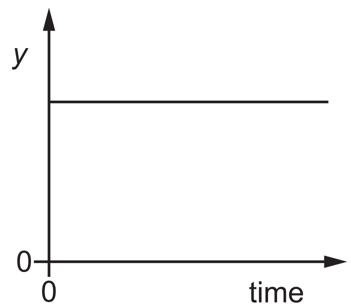
1 A student uses a metre ruler to measure the length of six identical table-tennis balls placed between two wooden blocks.



What is the diameter of one ball?

A 4 cm **B** 5 cm **C** 6 cm **D** 8 cm

2 A train is on a straight track. The graph shows how a quantity *y* varies with time.



Which statements can be true?

1 The train is stationary and *y* represents the distance from the last station.

2 The train is moving and *y* represents the distance from the last station.

3 The train is stationary and *y* represents the speed of the train.

4 The train is moving and *y* represents the speed of the train.

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

3 A vehicle sent to explore the surface of Mars has a mass of 200 kg.

The acceleration of free fall on Mars is $3.7\,m/s^2$.

What is the weight of the vehicle on Mars?

A 20 N **B** 54 N **C** 740 N **D** 2000 N

4 A student writes about mass and weight.

Which statement is correct?

A A ship which is floating has mass but no weight.

B Mass is a scientific word that means the same as weight.

C Mass is measured in newtons.

D The mass of an astronaut is the same on the Moon as on the Earth.

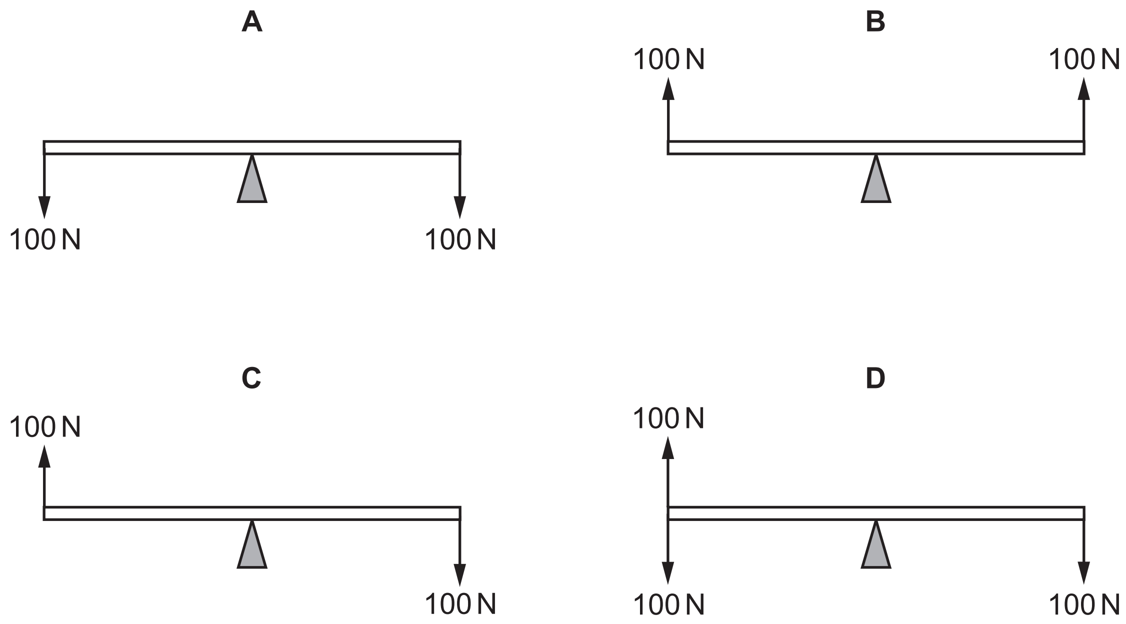
5 A student carries out an experiment to determine the density of an irregularly shaped solid. The solid is placed on a balance and a reading is taken. The solid is then immersed in a liquid in a measuring cylinder.

Which values should be used in the calculation?

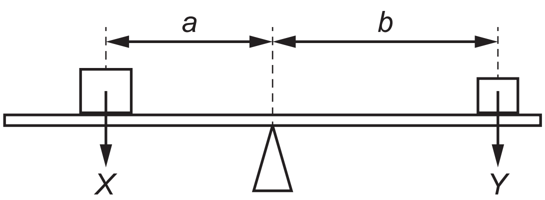
	value from measuring cylinder	value from balance
A	increase in reading after immersion of the solid	mass
B	increase in reading after immersion of the solid	weight
C	reading after immersion of the solid	mass
D	reading after immersion of the solid	weight

6 A uniform rod rests on a pivot at its centre. The rod is not attached to the pivot. Forces are then applied to the rod in four different ways, as shown. The weight of the rod can be ignored.

Which diagram shows the rod in equilibrium?



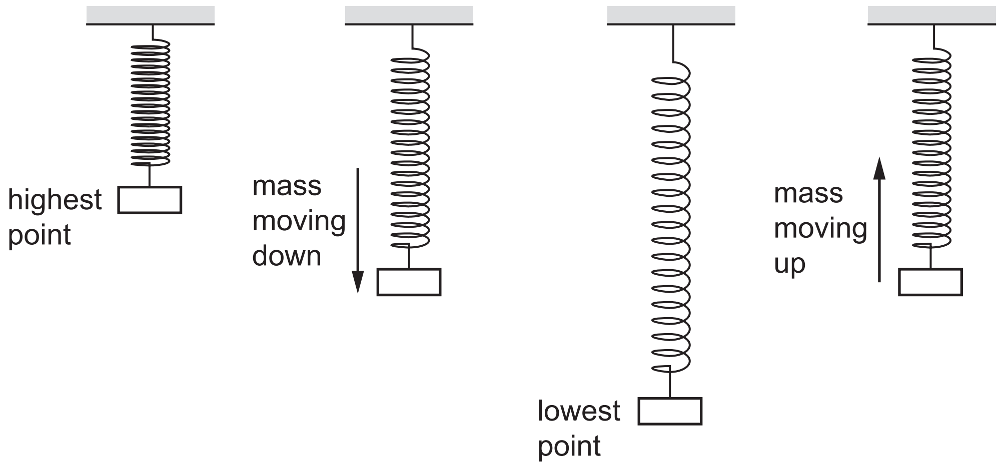
7 The diagram shows a beam balanced on a pivot. Two forces, *X* and *Y*, are acting on the beam.



Which calculation gives the moment of the force *Y* about the pivot?

A $Y \times (a + b)$ **B** $\dfrac{Y}{(a+b)}$ **C** $Y \times b$ **D** $\dfrac{Y}{b}$

8 A mass bounces up and down on a steel spring. The diagram shows the mass and the spring at different points during the motion.



At which point is the least energy in the gravitational potential store of the mass and at which point is the most energy in the elastic store of the spring?

	least energy in gravitational potential store of the mass	most energy in the elastic store of the spring
A	mass moving down	mass moving up
B	mass moving down	lowest point
C	lowest point	mass moving up
D	lowest point	lowest point

9 Electrical power is generated from different resources. Some of these resources are listed.

- chemical energy stored in biofuels
- chemical energy stored in fossil fuels
- energy stored in tides
- geothermal resources
- hydroelectric resources
- light from the Sun
- nuclear fuel

How many of the resources listed are classified as renewable?

A 3 **B** 4 **C** 5 **D** 6

10 A microwave oven is rated at 900 watts.

Which statement correctly describes the meaning of this value?

A 900 joules are transferred every second.

B 900 amperes are transferred every second.

C 900 volts are transferred every second.

D 900 ohms are transferred every second.

11 Why is it easier to push a sharp nail, rather than a blunt nail, into a piece of wood?

A The sharp nail exerts a larger force on the wood.

B The sharp nail exerts a smaller force on the wood.

C The sharp nail exerts a larger pressure on the wood.

D The sharp nail exerts a smaller pressure on the wood.

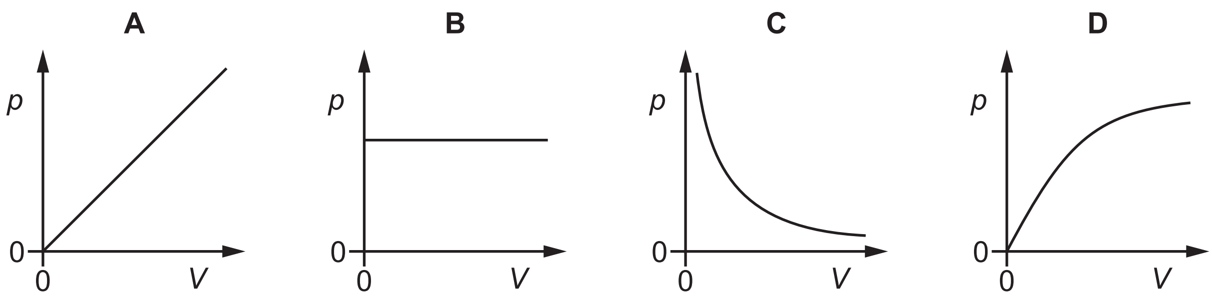
12 A sealed bottle of constant volume contains air.

The air in the bottle is heated by the Sun.

What is the effect on the average speed of the air particles in the bottle and the average distance between them?

	average speed of air particles	average distance between air particles
A	decreases	decreases
B	decreases	stays the same
C	increases	increases
D	increases	stays the same

13 Which graph shows the relationship between the pressure p of a fixed mass of gas and its volume V at a constant temperature?



14 What happens when a metal block is heated?

A Its width, height and length all increase.

B Its width increases only.

C Its height increases only.

D Its length increases only.

15 Which statement about the temperature of the solid describes what happens when a solid is melting?

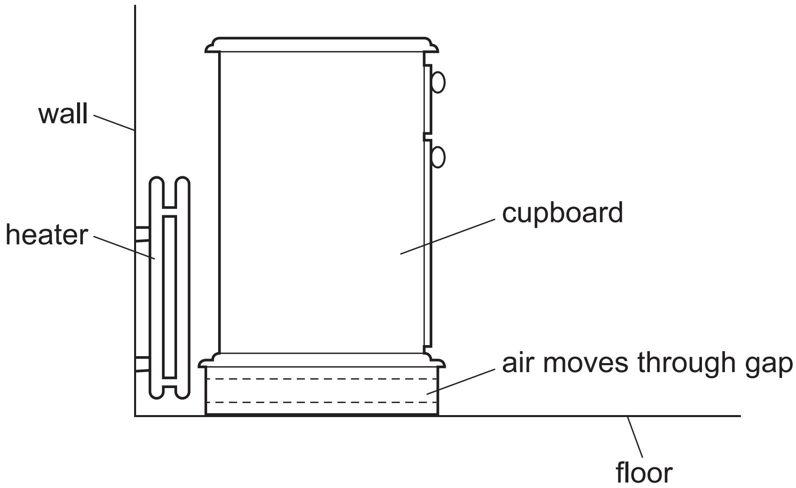
A The temperature increases and there is an input of energy.

B The temperature increases and there is no input of energy.

C The temperature remains constant and there is an input of energy.

D The temperature remains constant and there is no input of energy.

16 A cupboard is placed in front of a heater. Air can move through a gap under the cupboard.



Which row describes the temperature and the direction of movement of the air in the gap?

	air temperature	air direction
A	cool	away from the heater
B	cool	towards the heater
C	warm	away from the heater
D	warm	towards the heater

17 Which statement about waves is correct?

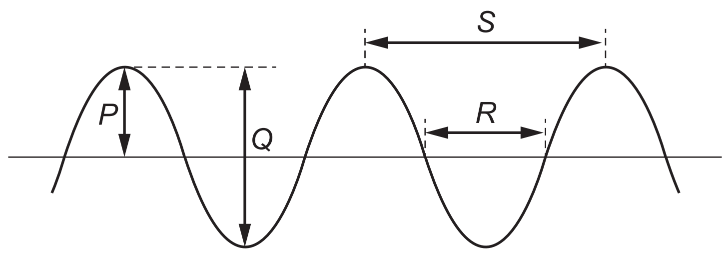
A All waves can travel through a vacuum.

B All waves travel at the same speed.

C Seismic S-waves can be modelled as longitudinal waves.

D Waves transfer energy without transferring matter.

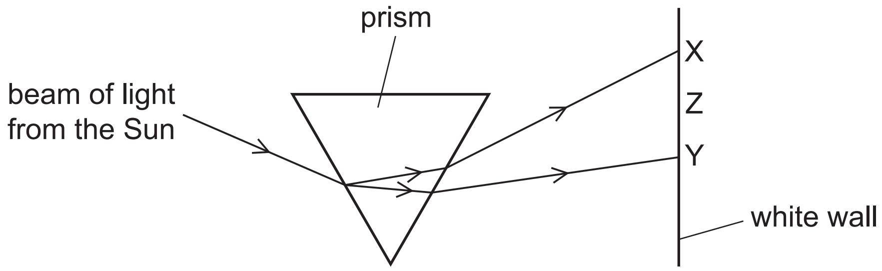
18 The diagram shows a transverse wave.



Which row identifies the amplitude and the wavelength of the wave?

	amplitude	wavelength
A	*P*	*R*
B	*P*	*S*
C	*Q*	*R*
D	*Q*	*S*

19 A beam of light from the Sun strikes a prism.

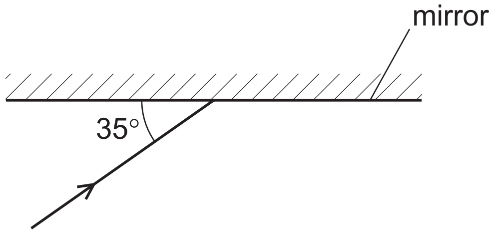


The dispersed beam is incident on a white wall between X and Y.

Which colours are seen at X, Z and Y?

	X	Z	Y
A	red	green	violet
B	red	violet	green
C	violet	green	red
D	violet	red	green

20 The diagram shows a ray of light incident on a plane mirror.



The angle between the ray and the mirror is 35°.

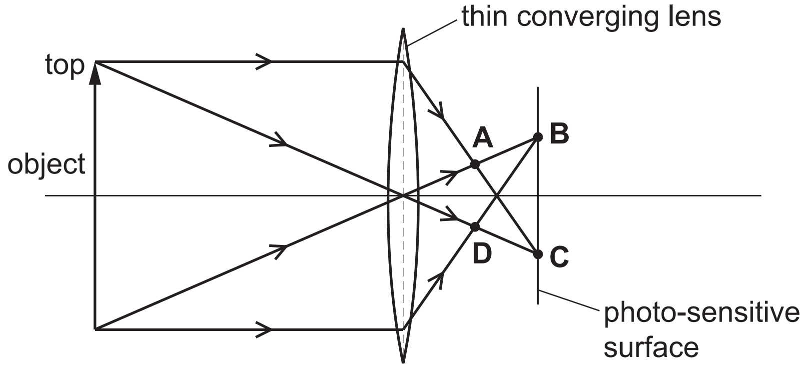
The ray is reflected by the mirror.

What is the angle of reflection?

A 35° **B** 55° **C** 70° **D** 110°

21 A thin converging lens in a camera produces a real image on a photo-sensitive surface, as shown.

At which position is the image of the top of the object formed?



22 The diagram shows the electromagnetic spectrum.

γ-rays	E	ultraviolet	F	infrared	microwaves	G

Which types of wave are E, F and G?

	E	F	G
A	radio	visible light	X-rays
B	radio	X-rays	ultrasound
C	X-rays	radio	ultrasound
D	X-rays	visible light	radio

23 A sound is produced and an echo is heard after the sound reflects off a wall.

How do the properties of the echo compare to the original sound wave?

	amplitude	frequency	speed
A	lower	lower	lower
B	lower	same	same
C	same	lower	lower
D	same	same	same

24 Which metal can be attracted by a magnet?

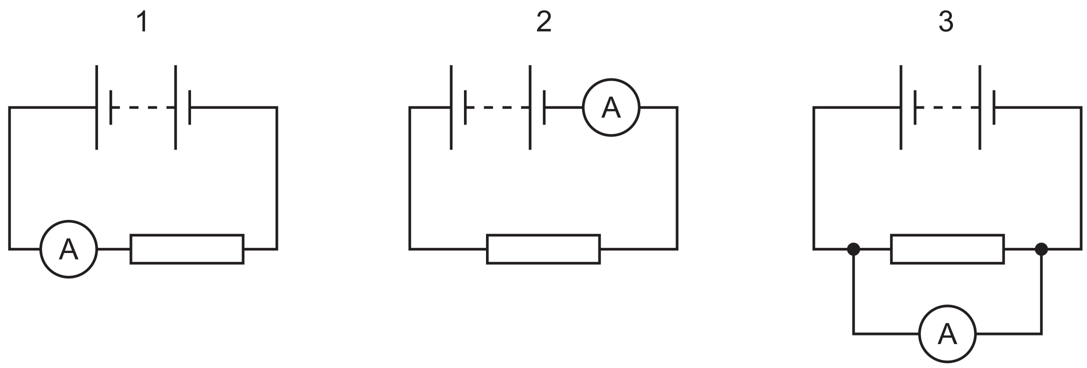
A zinc

B lead

C iron

D copper

25 A student uses an ammeter to measure the current in a resistor. He considers three different circuits, as shown.



In which of the circuits does the ammeter measure the current in the resistor?

A 1, 2 and 3 **B** 1 and 2 only **C** 1 only **D** 3 only

26 Which substances both contain large concentrations of free electrons?

A aluminium and glass

B copper and water

C copper and nylon

D silver and gold

27 What is the unit of resistance?

A ampere

B ohm

C volt

D watt

28 A teacher wishes to show the production of electrostatic charges.

She holds a rod and rubs it with a cotton cloth. A copper rod, a glass rod, a plastic rod and a steel rod are available.

Which two rods would both be suitable to use?

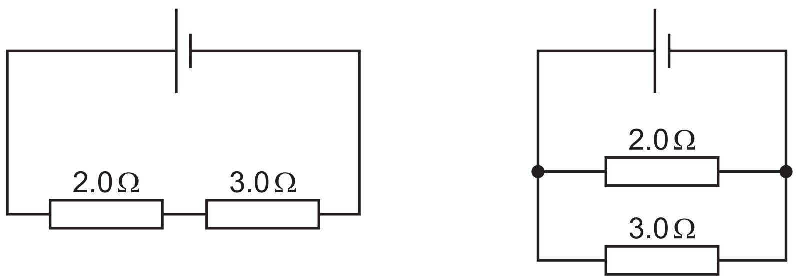
A a copper rod and a glass rod

B a glass rod and a plastic rod

C a plastic rod and a copper rod

D a plastic rod and a steel rod

29 Resistors of resistance 2.0 Ω and 3.0 Ω are connected in two different circuits.



What is the total resistance in each circuit?

	series	parallel
A	less than 2.0 Ω	less than 2.0 Ω
B	less than 2.0 Ω	greater than 3.0 Ω
C	greater than 3.0 Ω	less than 2.0 Ω
D	greater than 3.0 Ω	greater than 3.0 Ω

30 The current in an electrical heater is 5.0 A.

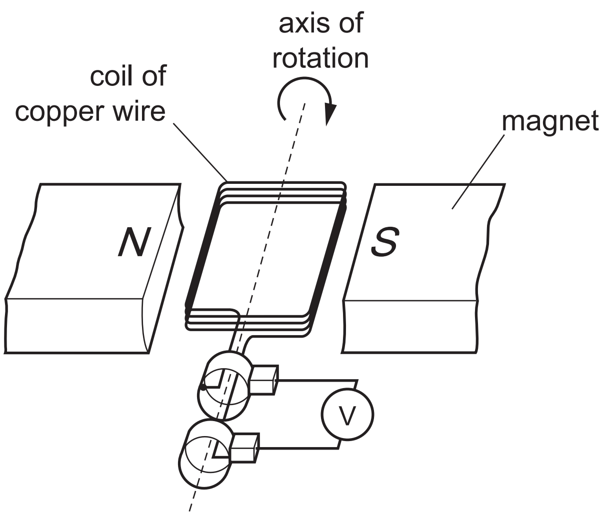
The heater is connected to the mains by a flexible cable that can carry a current of up to 15 A. The mains circuit can carry a current of up to 30 A.

Different fuses are available to protect the heater's cable.

Which fuse is the most suitable?

A 4.0 A **B** 10 A **C** 20 A **D** 40 A

31 A generator uses the principle of electromagnetic induction.



Which change would increase the induced electromotive force (e.m.f.) in the coil?

A increasing the number of turns in the coil

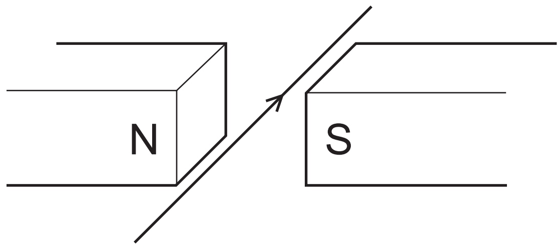
B placing the magnets further apart

C using a coil made from steel wire

D reversing one of the magnets

32 A current passes along a wire placed between the poles of a permanent magnet.

The wire experiences a force due to the magnetic field.



What will change the direction of this force?

A increasing the current

B reversing the current

C increasing the strength of the magnetic field

D using an electromagnet with the same polarity as the permanent magnet

33 What is a transformer used for?

A changing a direct current into an alternating current

B changing the magnitude of an alternating voltage

C reducing the frequency of an alternating current

D switching off the current in a circuit when there is a fault

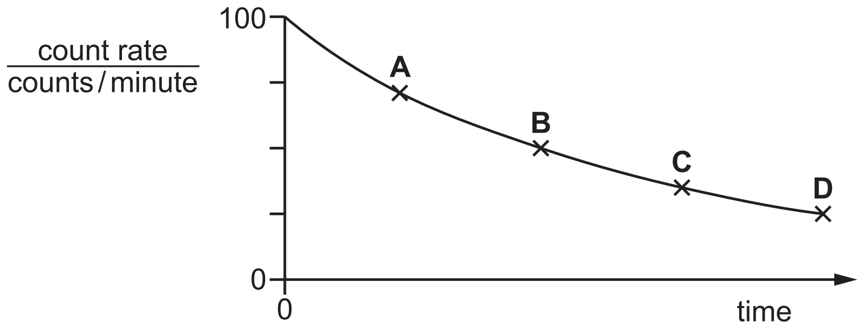
34 What are the relative charges on a proton, a neutron and an electron?

	proton	neutron	electron
A	0	–1	+1
B	0	–1	–1
C	+1	0	–1
D	+1	0	+1

35 The half-life of carbon-14 is 5700 years.

An object containing carbon-14 has a count rate of 100 counts / minute when it is first formed. The graph shows how the count rate decreases over time.

Which point on the graph corresponds to a time 11 400 years after the formation of the object?



36 Which type of radioactive decay causes the nucleus of one element to change into the nucleus of another element?

	emission of an alpha-particle	emission of a beta-particle	emission of a gamma ray
A	✓	✓	✗
B	✓	✗	✗
C	✗	✓	✓
D	✗	✓	✗

37 A scientist needs to use a source of γ-rays as safely as possible.

Which action will **not** reduce the total radiation that reaches the scientist?

A keeping the distance between the source and the scientist as large as possible

B keeping the temperature of the source as low as possible

C keeping the time for which the scientist uses the source as small as possible

D placing a lead screen between the scientist and the source

38 Which time period is approximately equal to 24 hours?

A the time for the Earth to complete one rotation on its axis

B the time for the Earth to orbit the Sun

C the time for the Moon to orbit the Earth

D the time for the Sun to orbit the Earth

39 The nearest star to the Sun is about four light-years away from the Earth.

A student makes three statements about the star.

1 Light from the star takes about four years to reach the Earth.

2 Light from the Sun takes about four years to travel to the star and back to the Earth.

3 The star is outside our galaxy.

Which statements are correct?

A 1, 2 and 3 **B** 1 and 3 only **C** 1 only **D** 2 and 3 only

40 The table shows some elements and some regions of the electromagnetic spectrum.

Which row shows one of the most common elements in the Sun and one of the regions in which the Sun radiates most of its energy?

	element	region
A	iron	gamma
B	iron	infrared
C	hydrogen	gamma
D	hydrogen	infrared

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PHYSICS **0625/12**

Paper 1 Multiple Choice (Core) **October/November 2023**

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS
- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m / s^2).

INFORMATION
- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.

1 A student investigates the oscillation of a mass suspended from a spring.

The student pulls the mass down from its rest position P and then releases it so that it oscillates vertically.

The student then follows the instructions listed to find the period of the oscillating mass.

1 Count 10 complete oscillations.

2 Divide the time on the stop-watch by 10.

3 Start the stop-watch as the mass passes upwards through point P.

4 Stop the stop-watch.

What is the correct order of these instructions?

A $1 \rightarrow 3 \rightarrow 4 \rightarrow 2$

B $3 \rightarrow 1 \rightarrow 4 \rightarrow 2$

C $3 \rightarrow 4 \rightarrow 1 \rightarrow 2$

D $4 \rightarrow 3 \rightarrow 2 \rightarrow 1$

2 A student measures the average speed of a cyclist in a race.

Which quantities must she measure?

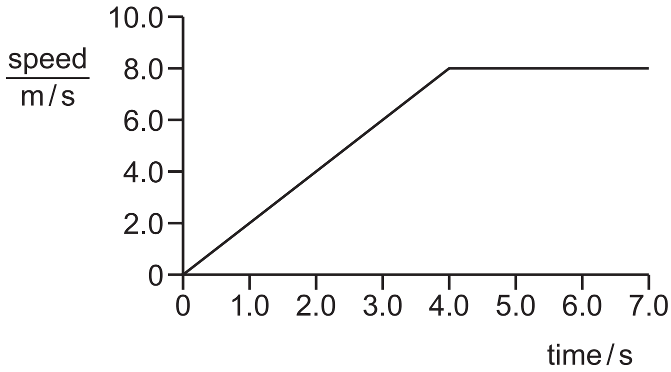
A the total time taken to complete the race and the time taken for the cyclist to reach her highest speed

B the total time taken to complete the race and the total distance travelled by the cyclist at her highest speed

C the total time taken to complete the race and the total distance travelled by the cyclist

D the time taken to reach her highest speed and the total distance travelled by the cyclist

3 The graph shows the motion of a sprinter.



She accelerates steadily from rest to 8.0 m / s in 4.0 s.

How far does she travel in the last three seconds of her acceleration?

A 9.0 m **B** 15 m **C** 16 m **D** 24 m

4 A person steps onto a bathroom scale.

The bathroom scale records both mass and weight.

Which row shows the readings on the bathroom scale?

	mass	weight
A	60 N	590 kg
B	60 kg	590 N
C	590 kg	60 N
D	590 N	60 kg

5 Which equation is correct?

A density = mass × volume

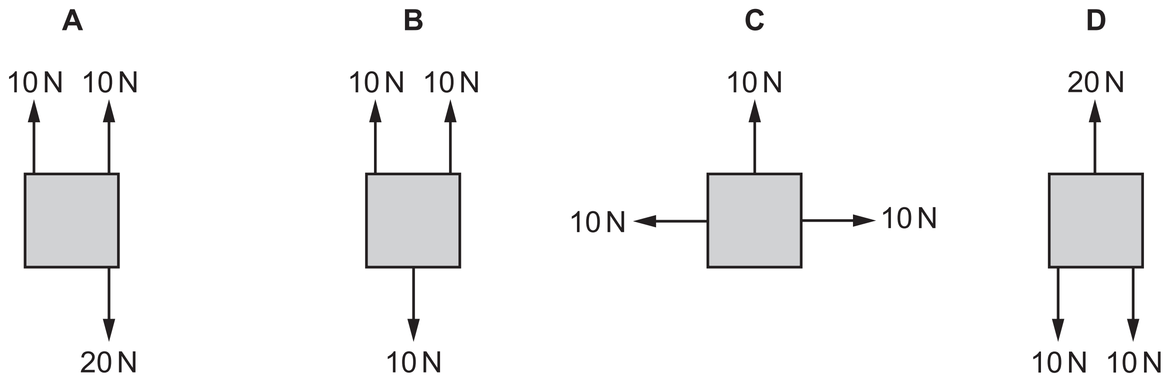
B density = weight × volume

C mass = density × volume

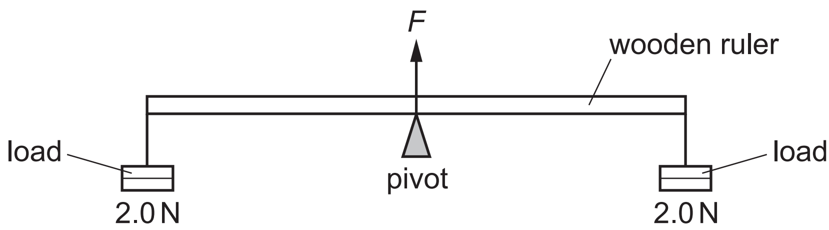
D weight = density × volume

6 The diagrams show four identical objects. Each object is acted on by only the forces shown.

Which diagram shows an object in equilibrium?



7 A uniform wooden ruler is pivoted at its centre. A load of 2.0 N is suspended from each end of the ruler.



The pivot exerts an upward force F on the ruler.

What is F equal to?

A 2.0 N

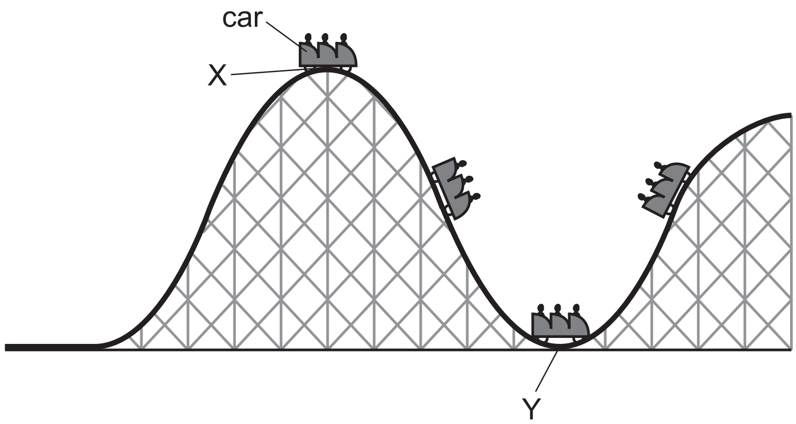
B the weight of the ruler

C 4.0 N

D 4.0 N plus the weight of the ruler

8 The diagram shows part of a rollercoaster ride with the car at different positions.

The car runs freely down from position X to position Y and up the hill on the other side.



What happens to the energy in the kinetic store and the gravitational potential store of the car as it moves from position X to position Y?

	energy in kinetic store	energy in gravitational potential store
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

9 In a small power station, biofuel is used to generate electricity.

Which energy store is reduced by this process?

A chemical

B kinetic

C nuclear

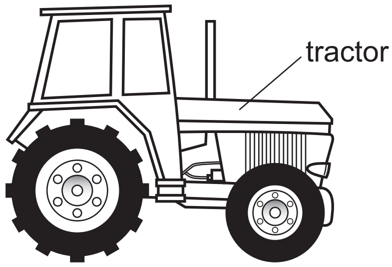
D thermal

10 An electric car is charged overnight. In 8.0 hours, 180 MJ of energy is transferred.

What is the power of the charger?

A 6.3 kW **B** 380 kW **C** 23 MW **D** 1400 MW

11 Tractors have large tyres. These help to prevent the wheels from sinking into soft ground.



Which statement explains this?

A Larger tyres exert a greater force on the ground.

B Larger tyres exert a greater pressure on the ground.

C Larger tyres exert a smaller force on the ground.

D Larger tyres exert a smaller pressure on the ground.

12 Brownian motion is the random motion of particles.

In which states of matter is Brownian motion observed?

A gases, liquids and solids

B gases and liquids only

C gases and solids only

D liquids and solids only

13 A student investigates the relationship between the pressure of a gas and its volume at constant temperature. He records his results in the table.

reading	$\frac{\text{pressure}}{\text{N/cm}^2}$	volume $/\text{cm}^3$
1	10.0	24
2	7.4	32
3	4.0	63
4	13.0	19

What is the correct conclusion from the experiment?

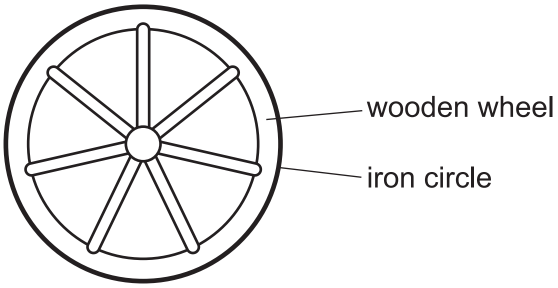
A The volume decreases when the pressure increases.

B The volume increases when the pressure increases.

C The volume initially increases when the pressure increases, but then decreases.

D The volume is independent of the pressure.

14 A wooden wheel can be strengthened by putting a tight circle of iron around it.



Which action would make it easier to fit the circle over the wood?

A cooling the iron circle only

B heating the iron circle

C heating the wooden wheel and cooling the iron circle

D heating the wooden wheel but not heating or cooling the iron circle

15 Which diagram shows the processes happening during changes of state?

A

$$\text{gas} \xrightarrow{\text{boiling}} \text{liquid} \xrightarrow{\text{solidification}} \text{solid}$$

B

$$\text{gas} \xrightarrow{\text{condensation}} \text{liquid} \xrightarrow{\text{melting}} \text{solid}$$

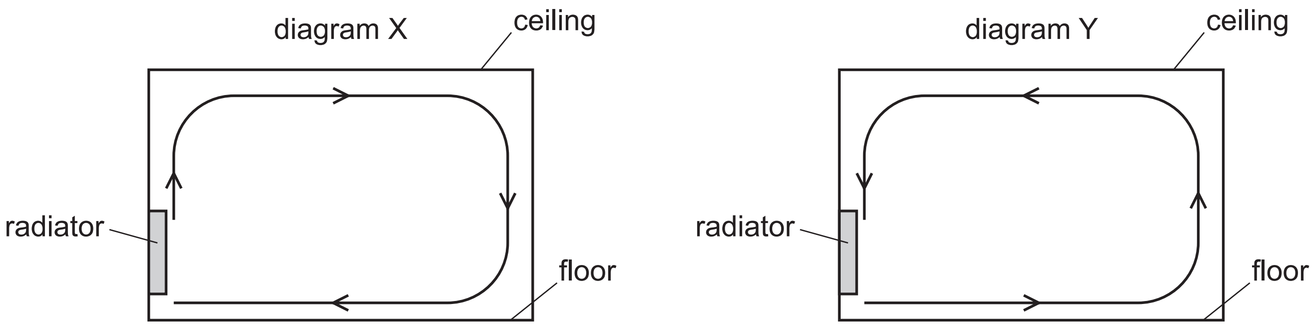
C

$$\text{solid} \xrightarrow{\text{melting}} \text{liquid} \xrightarrow{\text{boiling}} \text{gas}$$

D

$$\text{solid} \xrightarrow{\text{solidification}} \text{liquid} \xrightarrow{\text{boiling}} \text{gas}$$

16 A room is heated by a radiator. The diagrams X and Y show two possible circulations of hot air, which heat the room.



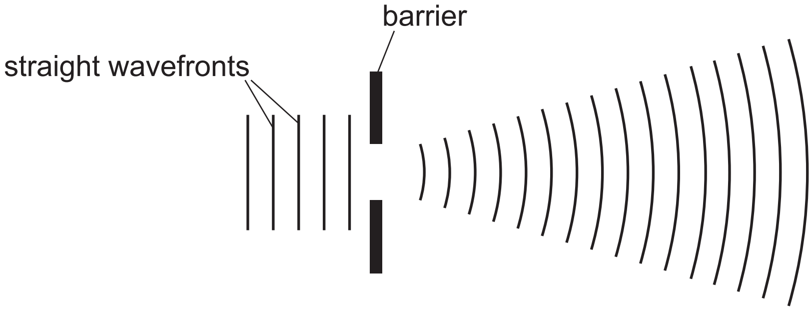
Which diagram and reason explain the heating of the room by convection?

	diagram	reason
A	X	air density decreases when air is heated
B	X	air density increases when air is heated
C	Y	air density decreases when air is heated
D	Y	air density increases when air is heated

17 Which description and example are correct for a transverse wave?

	description	example
A	The direction of vibration is parallel to the direction of propagation.	sound
B	The direction of vibration is parallel to the direction of propagation.	waves on a rope
C	The direction of vibration is at right angles to the direction of propagation.	sound
D	The direction of vibration is at right angles to the direction of propagation.	waves on a rope

18 Straight wavefronts on the surface of a ripple tank approach a gap in a barrier. The diagram shows how the wavefronts change shape as they pass through the gap.



What is the name of this effect?

A diffraction

B propagation

C reflection

D refraction

19 Red, green and violet lights are part of the visible spectrum of light.

What is the order of colours from shortest to longest wavelength?

A red → green → violet

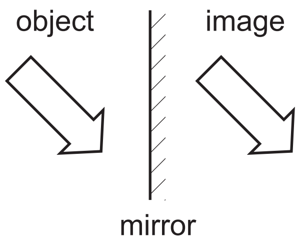
B red → violet → green

C violet → red → green

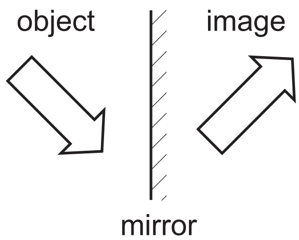
D violet → green → red

20 Which diagram shows the image correctly formed by reflection?

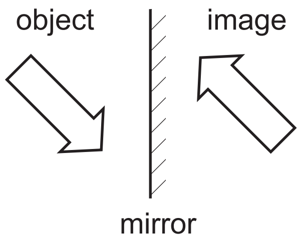
A



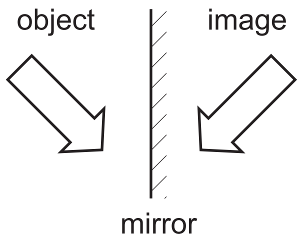
B



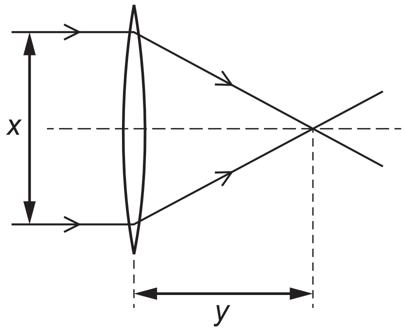
C



D



21 A student passes parallel rays of light through four different converging lenses. He measures the distance x and the distance y for each experiment.



Which lens has the longest focal length?

	x / cm	y / cm
A	4.6	2.0
B	5.1	3.1
C	5.9	2.3
D	6.1	2.4

22 The table shows different types of wave in the electromagnetic spectrum.

radio waves	microwaves	infrared waves	visible light	ultraviolet waves	X-rays	gamma rays

Where do all the waves travel at the same speed?

A in a vacuum

B in diamond

C in glass

D in water

23 Which statement about a sound that can be heard by a person with normal hearing is correct?

A The sound is a longitudinal wave with a frequency between 2.0 Hz and 20 Hz.

B The sound is a longitudinal wave with a frequency between 20 Hz and 20 000 Hz.

C The sound is a transverse wave with a frequency between 2.0 Hz and 2000 Hz.

D The sound is a transverse wave with a frequency between 2.0 Hz and 20 MHz.

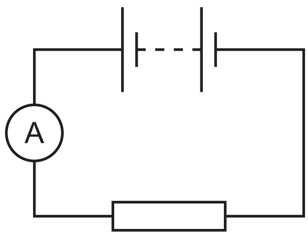
24 A hard magnetic material can be used to make a permanent magnet.

A soft magnetic material can be used to make a temporary magnet.

Which row shows whether iron and steel are hard or soft magnetic materials?

	iron	steel
A	hard	hard
B	hard	soft
C	soft	hard
D	soft	soft

25 A battery is connected to an ammeter and a resistor.



The ammeter reading is 0.20 A.

An electrical insulator is connected in parallel with the resistor.

What is the ammeter reading?

A 0 A

B between 0 A and 0.20 A

C 0.20 A

D greater than 0.20 A

26 Which unit is used to measure electromotive force (e.m.f.)?

A ampere

B joule

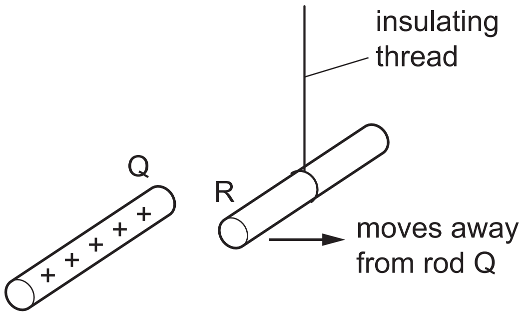
C volt

D watt

27 Which equation is correct for resistance R, potential difference (p.d.) V and current I?

A $R = \frac{V}{I}$ **B** $R = V + I$ **C** $R = \frac{I}{V}$ **D** $R = V \times I$

28 In the diagram, rod R is suspended from an insulating thread.



When the positively charged rod Q is brought close to rod R, rod R moves away from rod Q.

Which conclusion can be made from this observation?

A Rod R is charged, but it is not possible to identify the sign of the charge.

B Rod R must be positively charged.

C Rod R must be negatively charged.

D Rod R is uncharged.

29 In which heating system circuit would thermistors **not** be useful?

A to keep different rooms at different temperatures

B to turn an alarm on if the system overheats

C to turn a heating system off at a particular temperature

D to turn a heating system on when a sound is detected

30 Which statement is correct?

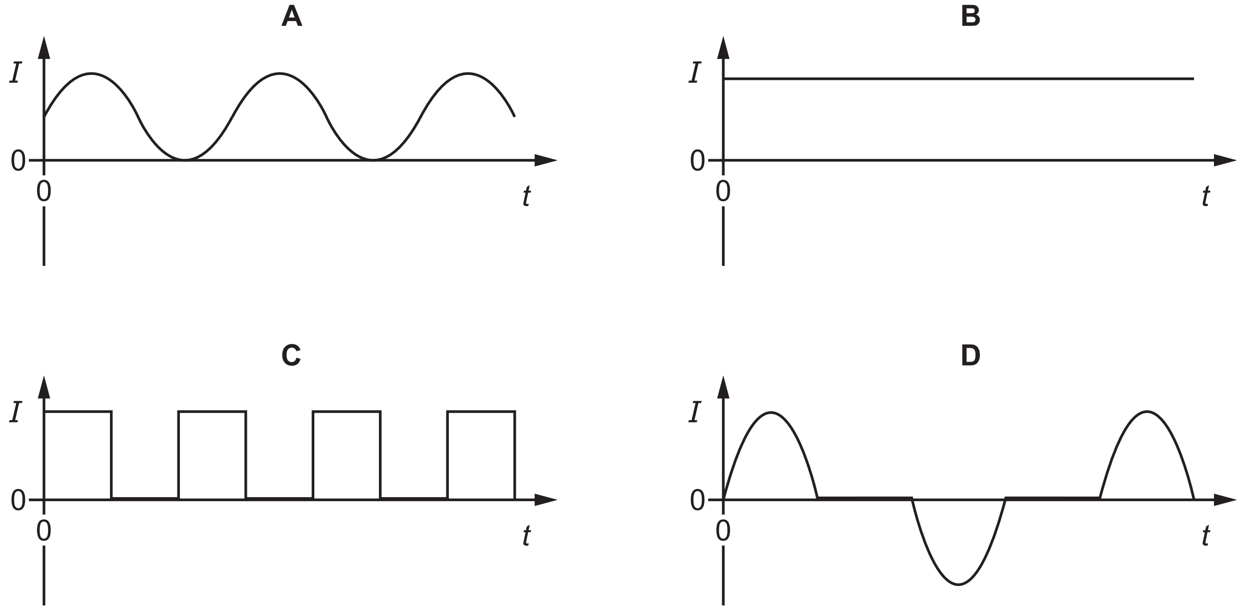
A A fuse is included in a circuit to prevent the current becoming too high.

B A fuse should be connected to the neutral wire in a plug.

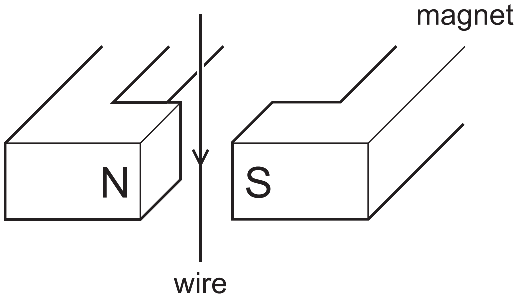
C An electric circuit will only work if it includes a fuse.

D An earth wire is needed to prevent the fuse blowing.

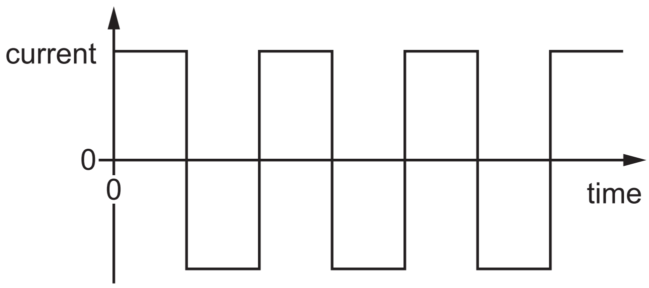
31 Which graph of current I against time t represents an alternating current (a.c.)?



32 The diagram shows a wire in the magnetic field between two poles of a magnet.



The current in the wire repeatedly changes between a constant value in one direction and a constant value in the opposite direction, as shown in the graph.



What is the effect on the wire?

A The force on the wire alternates between one direction and the opposite direction.

B The force on the wire is constant in size and direction.

C There is no force acting on the wire at any time.

D There is only a force on the wire when the current reverses.

33 A transformer has N_p turns on its primary coil and N_s turns on its secondary coil. The voltage across the primary coil is V_p and the voltage across the secondary coil is V_s.

What is the relationship between these four quantities?

A $V_p \times V_s = N_p \times N_s$

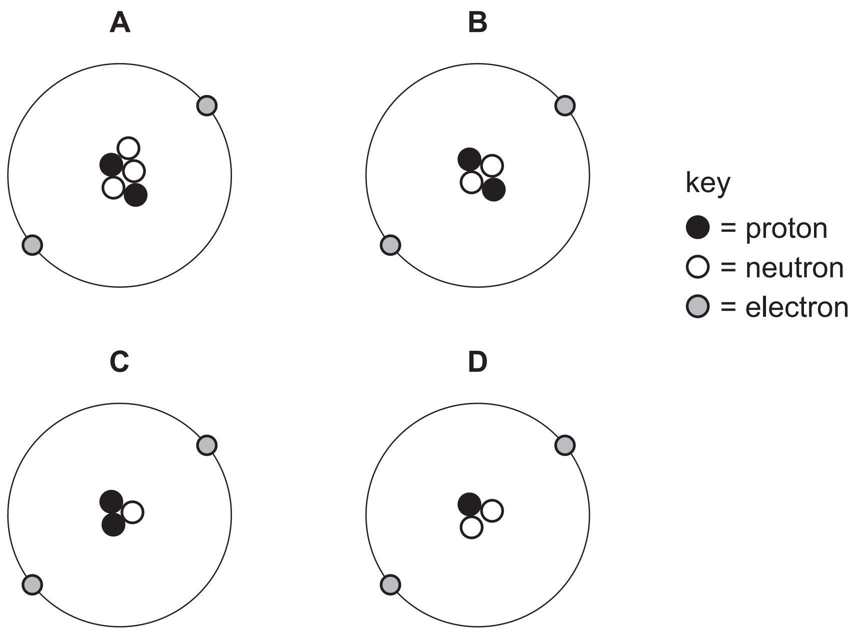
B $\frac{V_p}{V_s} = \frac{N_p}{N_s}$

C $\frac{V_p}{V_s} = \frac{N_s}{N_p}$

D $\frac{V_p}{V_s} = N_p \times N_s$

34 The diagrams represent the protons, neutrons and electrons in different atoms and ions.

Which diagram shows a negatively charged ion?



35 Which row correctly describes an example of radioactive decay?

	original nucleus	emission	change or no change of element
A	stable	γ	change of element
B	unstable	α	change of element
C	unstable	α	no change of element
D	unstable	β	no change of element

36 A detector is used to monitor the emissions from a radioactive source over several days.

The table shows the count rate from the source at different times.

time / days	count rate / counts / s
0	250
1	215
2	180
3	148
4	120
5	100

What is the half-life of the source?

A between 1 and 2 days

B between 2 and 3 days

C between 3 and 4 days

D between 4 and 5 days

37 What is the most effective precaution to reduce the risk when handling, storing or using a radioactive source that emits γ-rays?

A Handle the source for the least possible time.

B Have a fire extinguisher nearby when using the source.

C Store the source at a low temperature.

D Wear plastic safety goggles when handling the source.

38 Approximately how long does the Moon take to orbit the Earth?

A 1 day

B 7 days

C 28 days

D 365 days

39 The Sun transfers energy to the Earth through electromagnetic radiation.

What are two of the parts of the electromagnetic spectrum to which most of the energy belongs?

A gamma rays and X-rays

B infrared radiation and visible light

C microwaves and visible light

D radio waves and microwaves

40 What provides evidence that the Universe is expanding?

A Stars in galaxies outside the Milky Way are all red.

B The Andromeda galaxy is moving toward the Milky Way.

C Light from distant galaxies is shifted to longer wavelengths.

D The Universe is 14 billion years old.

Cambridge IGCSE™

PHYSICS **0625/11**

Paper 1 Multiple Choice (Core) **October/November 2023**

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS
- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = $9.8\,m/s^2$).

INFORMATION
- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.

IB23 11_0625_11/3RP

1 In order to determine the period of a pendulum, a student times one complete swing of the pendulum using an analogue stop-watch with a second hand.

Which change of method produces the greatest improvement in accuracy?

A asking a friend with a shorter reaction time to take the measurement

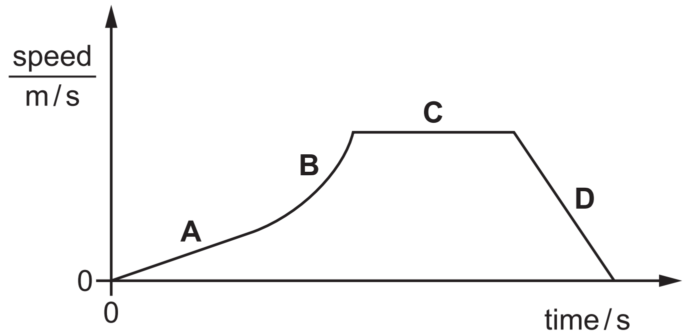
B measuring the time for 100 swings of the pendulum and dividing it by 100

C measuring the time for a half swing of the pendulum and doubling it

D using a digital timer

2 The graph shows the speed of a car travelling through a town.

Which section of the graph represents a period when the car is decelerating?

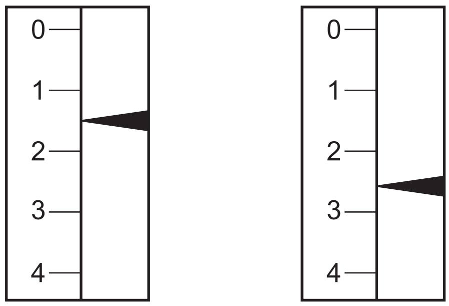


3 An object weighs 19 N on a planet where the acceleration of free fall is 3.8 m/s^2.

What is the mass of the object?

A 0.20 kg **B** 1.9 kg **C** 5.0 kg **D** 72 kg

4 An object is suspended from a spring balance on the Earth. The same object is suspended from the same spring balance on another planet.



Which statement explains the difference between the two readings?

A Both the mass and the weight of the object are greater on the other planet.

B The mass of the object is greater on the other planet than on the Earth, but the weight is unchanged.

C The spring stretches more easily when on the other planet.

D The weight of the object is greater on the other planet than on the Earth, but the mass is unchanged.

5 A student has a bottle of cooking oil.

She determines the density of the cooking oil.

Which apparatus does she need?

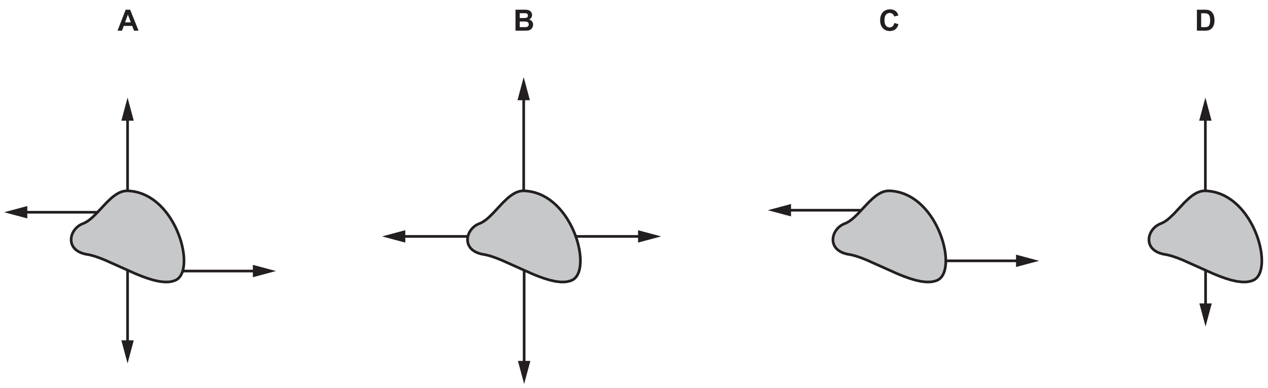
	balance	measuring cylinder	ruler	thermometer
A	✓	✓	✓	✓
B	✓	✓	✓	✗
C	✓	✓	✗	✗
D	✓	✗	✗	✗

key
✓ = needed
✗ = not needed

6 Forces are applied to four identical objects.

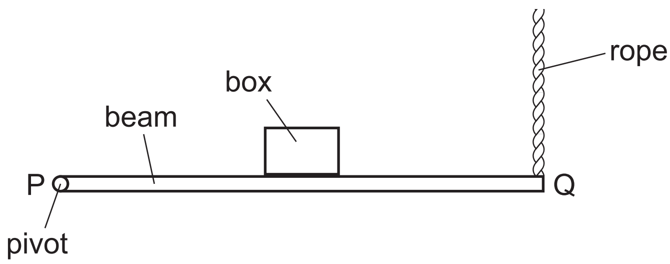
The length of each arrow indicates the magnitude of the force.

Which object is in equilibrium?



7 The diagram shows a wooden beam PQ which is attached to a wall by a pivot at P and kept in a horizontal position by a vertical rope attached at Q.

A box has been placed on the beam.



Which changes **must** reduce the tension in the rope at Q?

A Decrease the mass of the box and move it towards P.

B Decrease the mass of the box and move it towards Q.

C Increase the mass of the box and move it towards P.

D Increase the mass of the box and move it towards Q.

8 A cyclist travels down a hill from rest at point X without pedalling.

The cyclist applies his brakes and the cycle stops at point Y.

Which energy transfers have taken place between X and Y?

A gravitational potential $\rightarrow$ kinetic $\rightarrow$ internal (thermal)

B gravitational potential $\rightarrow$ internal (thermal) $\rightarrow$ kinetic

C kinetic $\rightarrow$ gravitational potential $\rightarrow$ internal (thermal)

D kinetic $\rightarrow$ internal (thermal) $\rightarrow$ gravitational potential

9 Which energy resource is non-renewable?

A geothermal

B natural gas

C solar

D wind

10 A pump does 460 000 J of work to raise water to fill a tank. It takes 7 minutes to fill the tank.

What is the power of the pump?

A 1.1 kW **B** 66 kW **C** 3200 kW **D** 190 000 kW

11 When a diver swims down from the surface of the water to a depth of 10 m, the pressure experienced increases from 100 000 N / m^2 to 200 000 N / m^2.

Which statement explains this increase in pressure?

A The density of the water increases with depth.

B The gravitational field strength increases with depth.

C The weight of water above the diver increases with depth.

D Water cannot be compressed.

12 Why can a gas be compressed easily into a smaller volume?

A The particles are far apart.

B The particles do not attract each other.

C The particles move randomly.

D The volume of each particle can be reduced.

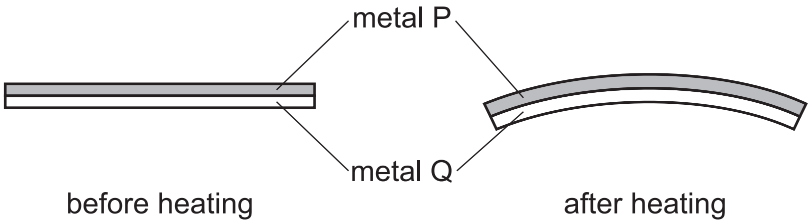
13 The melting point of mercury is –39 °C.

What is the melting point of mercury in kelvin?

A –234 K **B** 61 K **C** 234 K **D** 312 K

14 A bimetallic strip is used to control the temperature of an electrical appliance. It is made of two different metals fixed together.

The diagram shows the shape of the bimetallic strip before and after heating.



Which statement is correct?

A Metal P contracts more than metal Q on heating.

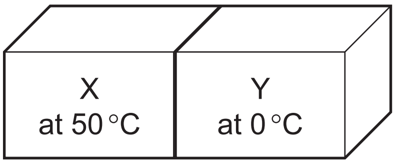
B Metal Q contracts more than metal P on heating.

C Metal P expands more than metal Q on heating.

D Metal Q expands more than metal P on heating.

15 A student has two blocks of metal, X and Y. The temperature of X is 50 °C and the temperature of Y is 0 °C.

The two blocks are placed in contact with each other, as shown.

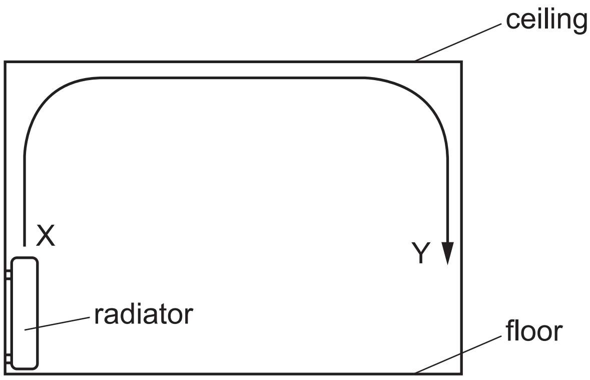


After some time, both blocks have a temperature of 25 °C.

What has happened to the internal energy of each block?

	internal energy of X	internal energy of Y
A	decreased	decreased
B	decreased	increased
C	increased	decreased
D	unchanged	unchanged

16 The diagram shows the view of a room heated by a radiator. The arrowed line from X to Y is the path of the convection current in the air.



Which row about the air temperature and the air density at X and at Y is correct?

	air temperature	air density
A	higher at X	higher at X
B	higher at X	higher at Y
C	higher at Y	higher at Y
D	higher at Y	higher at X

17 Which statements about waves are correct?

1 Only longitudinal waves can undergo diffraction.

2 All waves can undergo refraction.

3 All waves can undergo reflection.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

18 Which quantities relating to a wave on the surface of water can **both** be measured in metres?

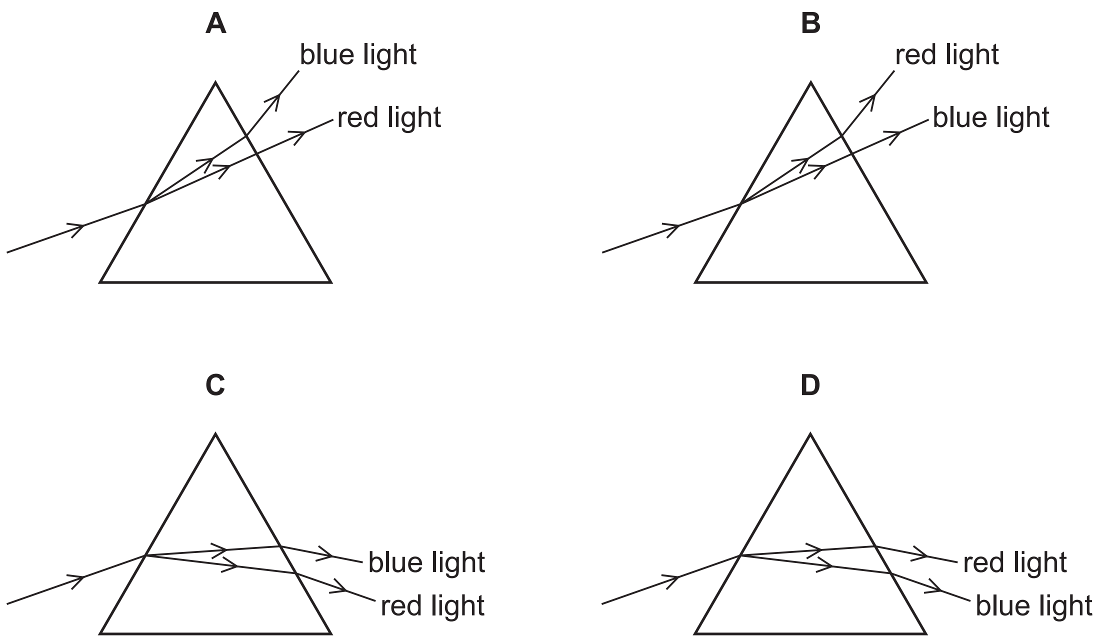
A amplitude and frequency

B amplitude and wavelength

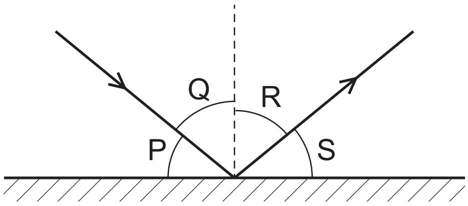
C amplitude and wave speed

D frequency and wavelength

19 Which diagram correctly shows the dispersion of white light through a glass prism?



20 A ray of light is reflected by a plane mirror.



Which row shows the angle of incidence and the angle of reflection?

	angle of incidence	angle of reflection
A	P	Q
B	P	S
C	Q	R
D	R	S

21 A thin converging lens is used to produce a real image of an object.

Which statement about the real image is always correct?

A It is nearer to the lens than the object.

B It is on the opposite side of the lens to the object.

C It is the same size as the object.

D It is upright.

22 Where do all types of electromagnetic waves travel at the same speed?

A air

B a vacuum

C glass

D water

23 Dogs can hear sounds in the range from 100 Hz to 45 kHz.

Which statement is correct?

A Any sound a dog can hear can also be heard by a human.

B Any sound a human can hear can also be heard by a dog.

C Dogs can hear some low frequency sounds that are silent for humans.

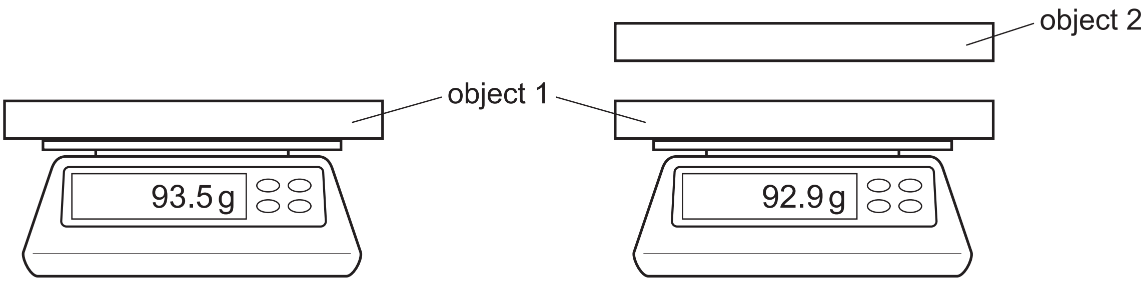
D Dogs can hear some high frequency sounds that are silent for humans.

24 Which metal could be used for a permanent magnet and which metal could be used for the core of an electromagnet?

	permanent magnet	core of electromagnet
A	iron	copper
B	iron	steel
C	steel	copper
D	steel	iron

25 Object 1 is given a negative charge and placed on a balance.

Object 2, which is also charged, is brought close to object 1 and the reading on the balance changes as shown.



Which action would further decrease the reading on the balance?

A Add the same number of electrons to both objects.

B Remove the same number of electrons from both objects.

C Transfer electrons from object 1 to object 2.

D Transfer electrons from object 2 to object 1.

26 Which statement about a voltmeter is correct?

A It has a scale which is marked in amperes (A).

B It must be connected in series in a circuit.

C It measures potential difference (p.d.).

D It must have three terminals.

27 How does the resistance of a metallic wire change:

- as its length increases
- as its cross-sectional area decreases?

	resistance as length increases	resistance as cross-sectional area decreases
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

28 A plastic rod is rubbed with a cloth. The rod becomes positively charged because of the movement of charged particles.

Which row gives the name of these charged particles and the direction in which they move?

	charged particles	direction of movement
A	electrons	from cloth to rod
B	electrons	from rod to cloth
C	protons	from cloth to rod
D	protons	from rod to cloth

29 Two $2.0\,\Omega$ resistors are connected in parallel.

What is the combined resistance of the resistors?

A less than $2.0\,\Omega$

B exactly $2.0\,\Omega$

C more than $2.0\,\Omega$, but less than $4.0\,\Omega$

D exactly $4.0\,\Omega$

30 An electric heater is plugged into the mains supply using a fused plug.

The current in the heater is 10 A.

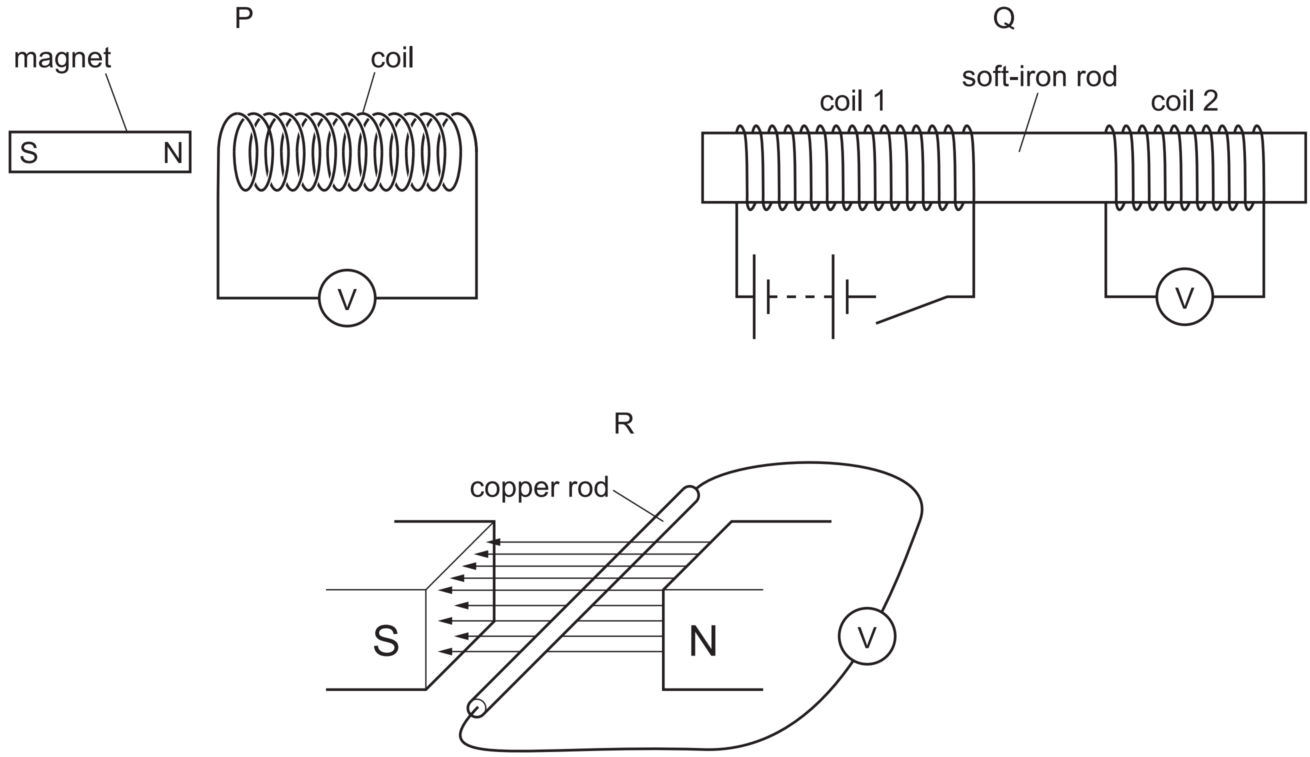
The cable attached to the heater is rated at 15 A.

The fuses available are rated at 1.0 A, 3.0 A, 5.0 A and 13 A.

Which fuse should be used?

A 1.0 A **B** 3.0 A **C** 5.0 A **D** 13 A

31 A teacher sets up the equipment for three demonstrations, P, Q and R.



In demonstration P, the magnet is moved to the right.

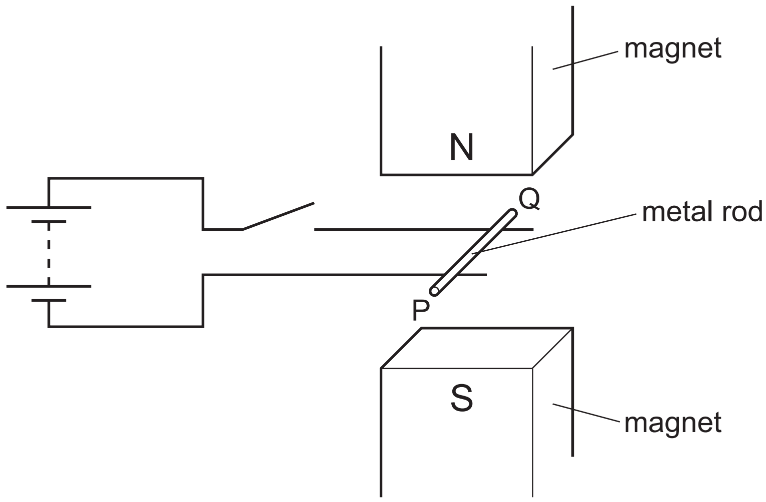
In demonstration Q, the switch is closed.

In demonstration R, the copper rod is moved vertically upwards.

Which demonstrations can be used to demonstrate electromagnetic induction?

A P and Q only **B** P and R only **C** Q and R only **D** P, Q and R

32 A metal rod PQ rests on two horizontal metal wires that are attached to a battery. The rod lies between the poles of a magnet.



When the switch is closed, the rod moves to the right.

What could be changed so that the rod moves to the left?

A Open the switch.

B Reverse the battery terminals and exchange the poles of the magnet.

C Reverse the battery terminals but without exchanging the poles of the magnet.

D Turn the metal rod around (P and Q exchanged).

33 A transformer in a computer is used to transform the mains voltage of 240 V to 12 V.

The number of turns on the secondary coil is 2000.

Which statement about the transformer is correct?

A It is a step-down transformer and has 100 turns on its primary coil.

B It is a step-down transformer and has 40 000 turns on its primary coil.

C It is a step-up transformer and has 100 turns on its primary coil.

D It is a step-up transformer and has 40 000 turns on its primary coil.

34 Atom P has 6 electrons, 6 protons and 6 neutrons.

Atom Q has 6 electrons, 6 protons and 7 neutrons.

Which statement is correct?

A Atom P has atomic number 12.

B Atom P has nucleon number 6.

C Atom Q has nucleon number 13.

D Atoms P and Q are different chemical elements.

35 α-particles, β-particles and γ-rays are emitted by radioactive nuclei when they decay.

Which emissions can be deflected by an electric field?

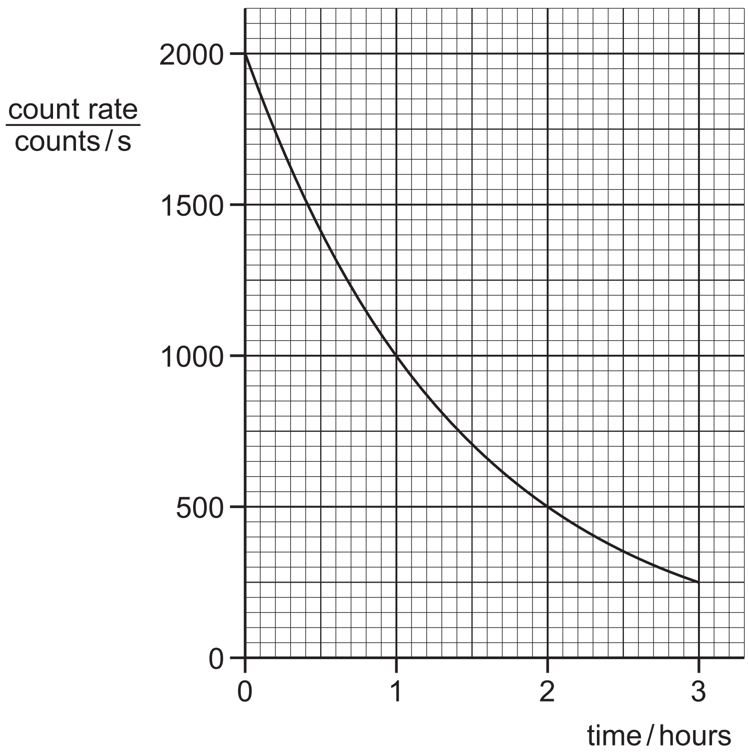
A α-particles, β-particles and γ-rays

B α-particles and β-particles only

C β-particles and γ-rays only

D γ-rays and α-particles only

36 The graph shows the count rate from a radioactive source over a period of time.



What is the half-life of the source?

A 0.5 hour **B** 1.0 hour **C** 1.5 hours **D** 3.0 hours

37 A scientist carries out an experiment using a sealed source which emits β-particles. The range of the β-particles in the air is about 30 cm.

Which precaution is the most effective to protect the scientist from the radiation?

A handling the source with long tongs

B keeping the temperature of the source low

C opening all windows in the laboratory

D washing his hands before leaving the laboratory

38 Which planet orbits the Sun between Mars and Saturn?

A Earth

B Jupiter

C Mercury

D Neptune

39 Which statement about the Sun is correct?

A The Sun is a dwarf star consisting mostly of hydrogen and oxygen.

B The Sun is a giant star consisting mostly of helium and carbon dioxide.

C The Sun is a medium-sized star consisting mostly of hydrogen and helium.

D The Sun is a medium-sized star consisting mostly of nitrogen and oxygen.

40 Which quantity does a light-year measure?

A an angle

B a distance

C a speed

D a time

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Cambridge IGCSE™

PHYSICS **0625/13**

Paper 1 Multiple Choice (Core) **October/November 2023**

MARK SCHEME

Maximum Mark: 40

Published

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Question	Answer	Marks
1	A	**1**
2	B	**1**
3	C	**1**
4	D	**1**
5	A	**1**
6	A	**1**
7	C	**1**
8	D	**1**
9	C	**1**
10	A	**1**
11	C	**1**
12	D	**1**
13	C	**1**
14	A	**1**
15	C	**1**
16	B	**1**
17	D	**1**
18	B	**1**
19	C	**1**
20	B	**1**
21	C	**1**
22	D	**1**
23	B	**1**
24	C	**1**
25	B	**1**
26	D	**1**
27	B	**1**
28	B	**1**

Question	Answer	Marks
29	C	**1**
30	B	**1**
31	A	**1**
32	B	**1**
33	B	**1**
34	C	**1**
35	D	**1**
36	A	**1**
37	B	**1**
38	A	**1**
39	C	**1**
40	D	**1**

Cambridge IGCSE™

PHYSICS **0625/12**

Paper 1 Multiple Choice (Core) **October/November 2023**

MARK SCHEME

Maximum Mark: 40

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Question	Answer	Marks
1	B	**1**
2	C	**1**
3	B	**1**
4	B	**1**
5	C	**1**
6	D	**1**
7	D	**1**
8	C	**1**
9	A	**1**
10	A	**1**
11	D	**1**
12	B	**1**
13	A	**1**
14	B	**1**
15	C	**1**
16	A	**1**
17	D	**1**
18	A	**1**
19	D	**1**
20	D	**1**
21	B	**1**
22	A	**1**
23	B	**1**
24	C	**1**
25	C	**1**
26	C	**1**
27	A	**1**
28	B	**1**

Question	Answer	Marks
29	D	**1**
30	A	**1**
31	D	**1**
32	A	**1**
33	B	**1**
34	D	**1**
35	B	**1**
36	C	**1**
37	A	**1**
38	C	**1**
39	B	**1**
40	C	**1**

Cambridge Assessment International Education

Cambridge IGCSE™

PHYSICS **0625/11**

Paper 1 Multiple Choice (Core) **October/November 2023**

MARK SCHEME

Maximum Mark: 40

Published

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This document consists of **3** printed pages.

Question	Answer	Marks
1	B	**1**
2	D	**1**
3	C	**1**
4	D	**1**
5	C	**1**
6	B	**1**
7	A	**1**
8	A	**1**
9	B	**1**
10	A	**1**
11	C	**1**
12	A	**1**
13	C	**1**
14	C	**1**
15	B	**1**
16	B	**1**
17	D	**1**
18	B	**1**
19	D	**1**
20	C	**1**
21	B	**1**
22	B	**1**
23	D	**1**
24	D	**1**
25	D	**1**
26	C	**1**
27	D	**1**
28	B	**1**

Question	Answer	Marks
29	A	**1**
30	D	**1**
31	D	**1**
32	C	**1**
33	B	**1**
34	C	**1**
35	B	**1**
36	B	**1**
37	A	**1**
38	B	**1**
39	C	**1**
40	B	**1**

Cambridge Assessment International Education

Cambridge IGCSE™

PHYSICS **0625/13**

Paper 1 Multiple Choice (Core) **October/November 2022**

45 minutes

You must answer on the multiple choice answer sheet.

* 7 0 4 2 4 2 0 3 8 2 *

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m / s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.

1 A student uses a ruler to measure the length of a spring.

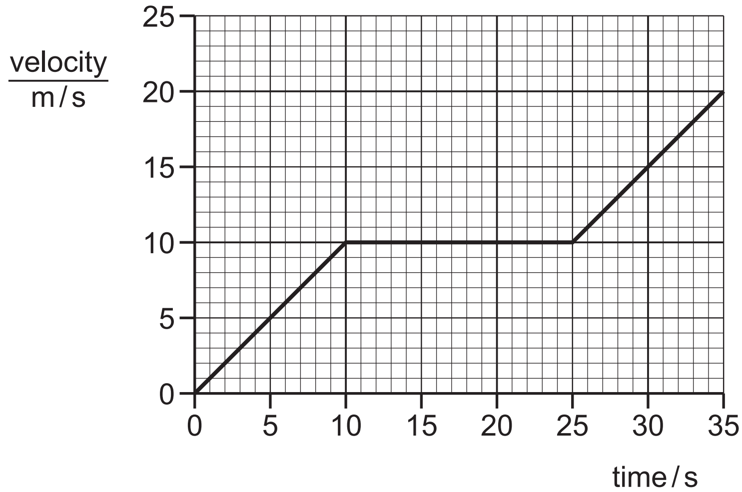
His results are shown.

14.9 cm   14.8 cm   14.8 cm   14.7 cm

What is the average length of the spring to three significant figures?

A 14.8 cm   **B** 14.9 cm   **C** 15.0 cm   **D** 15 cm

2 The velocity–time graph for a car is shown.



What is the distance travelled by the car in 35 s?

A 250 m   **B** 350 m   **C** 450 m   **D** 700 m

3 Which statements about weight are correct?

1 Weight is the quantity of matter in an object.

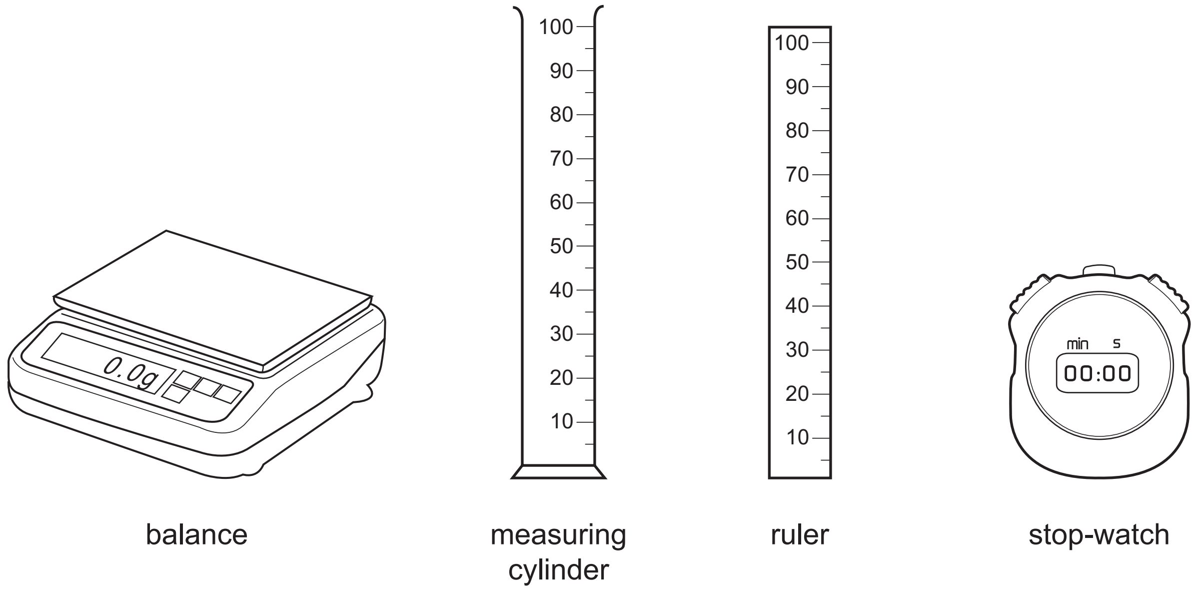
2 Weight is the force due to gravity acting on an object.

3 Weight is measured in kilograms.

4 Weight is measured in newtons.

A 1 and 2   **B** 1 and 4   **C** 2 and 3   **D** 2 and 4

4 The diagram shows four pieces of laboratory apparatus.



Which pieces of apparatus are used to find the density of a liquid?

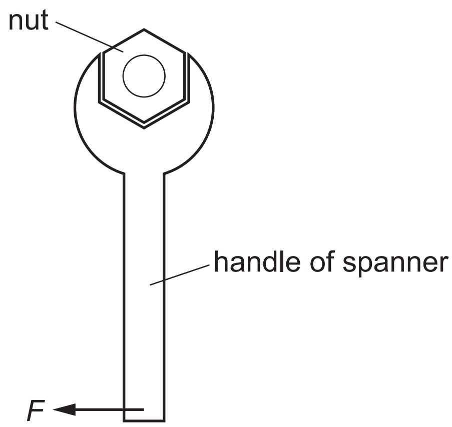
A balance and stop-watch

B balance and measuring cylinder

C measuring cylinder and ruler

D stop-watch and ruler

5 A spanner is used to tighten a nut on a bicycle wheel. A force *F* is needed.



How can the force *F* be reduced?

A Use a longer handle.

B Use a shorter handle.

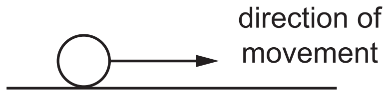
C Use a thinner handle.

D Use a thicker handle.

6 On which ball is a non-zero resultant force acting?

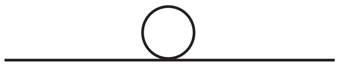
A

a ball moving at constant speed on a smooth surface



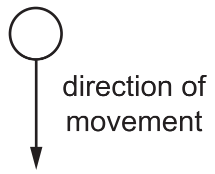
B

a ball at rest on a bench



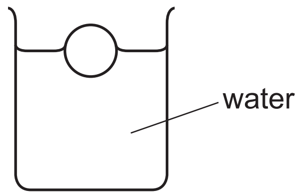
C

a free-falling ball which has just been released

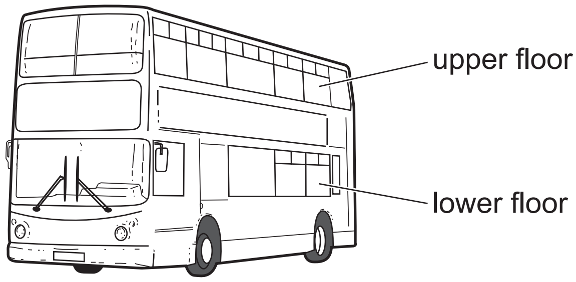


D

a ball floating on water



7 The diagram shows a bus with an upper floor and a lower floor.



Passengers on the upper floor are not allowed to stand while the bus is moving. Standing passengers make the bus less stable.

Why is this?

A The total weight is greater when they stand.

B There is too much pressure on the floor.

C The centre of mass is higher up.

D The density is greater.

8 An object falls towards the Earth's surface.

What happens to the gravitational potential energy and to the kinetic energy of the object?

	gravitational potential energy	kinetic energy
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

9 What is the unit of work?

A J **B** N **C** N / kg **D** W

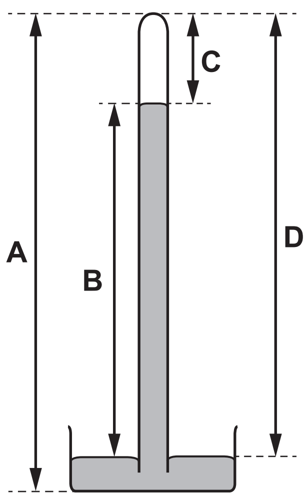
10 In some situations, a force does work.

Which set of conditions increases the quantity of work done by the force?

	magnitude of force	distance moved by the force
A	decreases	decreases
B	decreases	stays the same
C	increases	increases
D	stays the same	decreases

11 The diagram shows a simple mercury barometer.

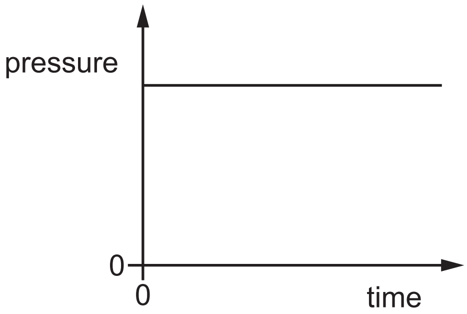
Which labelled distance decreases when atmospheric pressure increases?



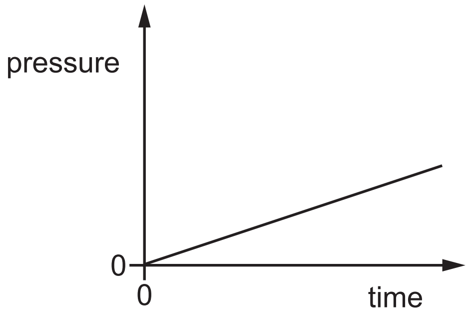
12 The pressure of a fixed mass of gas in a cylinder is measured. The volume of the gas in the cylinder is slowly decreased. The temperature of the gas does not change.

Which graph shows how the pressure of the gas changes during this process?

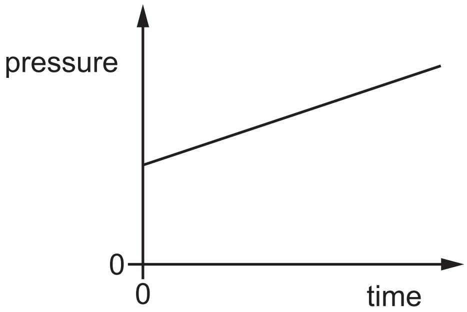
A



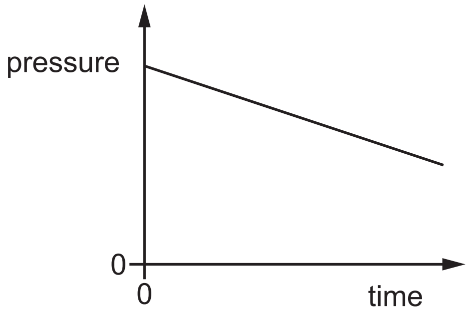
B



C



D



13 A liquid is evaporating.

What does it lose from its surface?

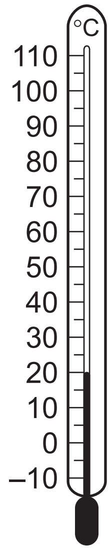
A less energetic ions

B less energetic molecules

C more energetic ions

D more energetic molecules

14 The diagram shows a thermometer calibrated in degrees Celsius.



What are the values of the lower fixed point and of the upper fixed point on the Celsius scale?

	lower fixed point / °C	upper fixed point / °C
A	–10	110
B	0	20
C	0	100
D	20	100

15 Which statements does the term 'melting point' refer to?

1 the temperature at which a liquid changes to solid without a change in temperature

2 the temperature at which a solid changes to liquid without a change in temperature

3 the temperature at which a liquid changes to vapour without a change in temperature

4 the temperature at which a vapour changes to liquid without a change in temperature

A 1 and 2 **B** 2 only **C** 3 and 4 **D** 4 only

16 Which piece of equipment is designed to produce a type of electromagnetic wave?

A electric fire

B electric generator

C electric motor

D electromagnet

17 How is energy transferred from the Sun to the Earth?

A by α-particles

B by conduction

C by convection

D by radiation

18 Wavefronts are incident on a boundary.

What is needed for the wave to refract at the boundary?

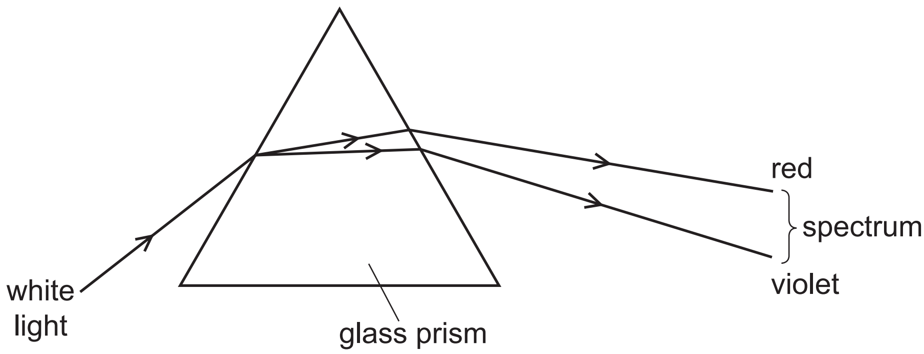
A a shiny surface at the boundary

B a small gap in the boundary

C different mediums either side of the boundary in which the frequency of the wave is different

D different mediums either side of the boundary in which the speed of the wave is different

19 A 60° glass prism disperses white light as shown.



The spectrum can be seen emerging from the prism.

Which spectrum shows the colours in the correct order?

A violet, green, blue, yellow, orange, red

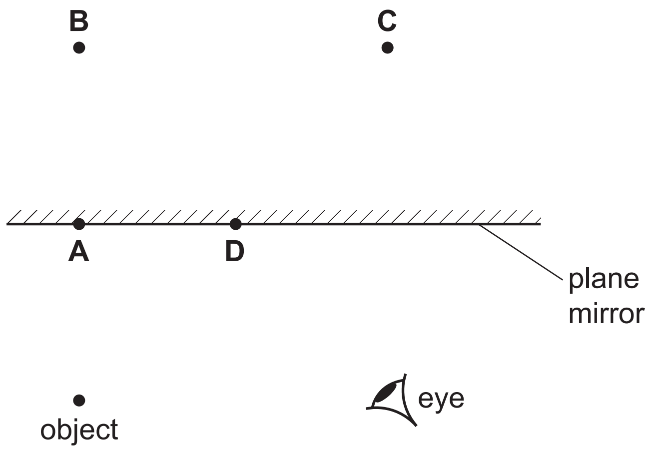
B violet, blue, green, orange, yellow, red

C violet, blue, green, yellow, orange, red

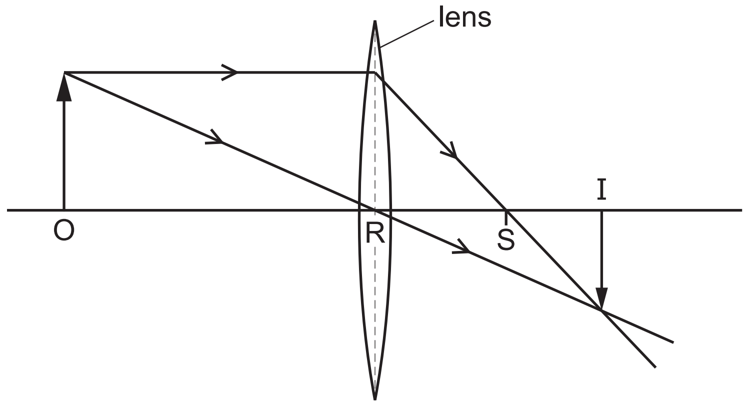
D violet, green, blue, orange, yellow, red

20 The diagram shows an object in front of a plane mirror.

At which labelled position is the image of the object formed?



21 The diagram shows the action of a thin converging lens on two rays of light. The rays are from the top of an object O. An inverted image I is formed.



Which name is given to the distance RS?

A principal axis

B principal focus

C focal length

D real length

22 Visible light has a frequency of approximately 5.0×10^{14} Hz.

M and N are two other types of electromagnetic radiation.

The frequency of M is 5.0×10^{6} Hz.

The frequency of N is 5.0×10^{15} Hz.

Which types of radiation are M and N?

	M	N
A	radio waves	infrared
B	radio waves	ultraviolet
C	ultraviolet	X-rays
D	X-rays	infrared

23 Which wave is **not** an electromagnetic wave?

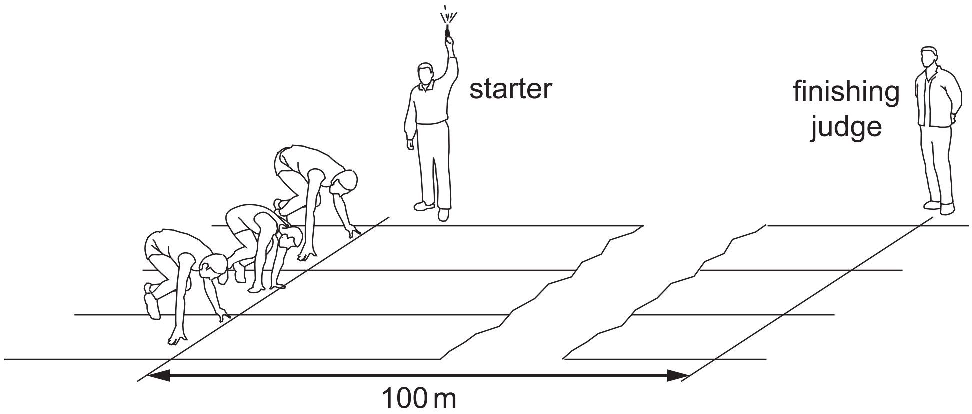
A microwaves

B radio

C sound

D ultraviolet

24 A 100 m race is started by firing a gun. The gun makes a bang and a puff of smoke at the same time.



When does the finishing judge see the smoke and when does he hear the bang?

	sees the smoke	hears the bang
A	almost immediately	almost immediately
B	almost immediately	after about 0.3 s
C	after about 0.3 s	almost immediately
D	after about 0.3 s	after about 0.3 s

25 Copper wires are used to connect an electric circuit.

Which particles flow in the wires when the circuit is switched on?

A atoms

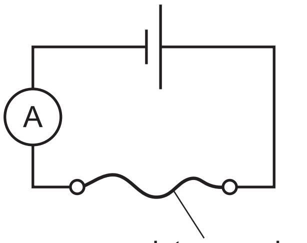
B electrons

C ions

D neutrons

26 A student is investigating a resistance wire.

She measures the current in a 50 cm length of resistance wire.



The student repeats the experiment using a 100 cm length of the same resistance wire.

What is the effect of this change on the current in the circuit and on the resistance of the wire?

	effect on current	effect on resistance
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

27 What is the full name of the term e.m.f.?

A electromotive field

B electromotive force

C electromotive frequency

D electromotive friction

28 A plastic rod is rubbed with a dry woollen cloth. The rod becomes positively charged.

Which statement is correct?

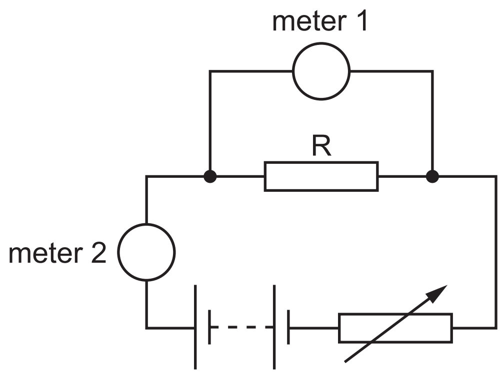
A Electrons move from the cloth to the rod.

B Electrons move from the rod to the cloth.

C Protons move from the cloth to the rod.

D Protons move from the rod to the cloth.

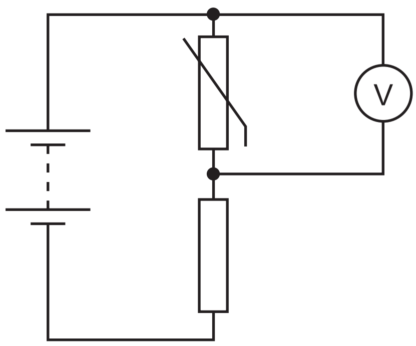
29 The diagram shows the circuit that a student uses to determine the resistance of resistor R.



Which row describes meter 1 and meter 2?

	meter 1	meter 2
A	ammeter	ammeter
B	ammeter	voltmeter
C	voltmeter	ammeter
D	voltmeter	voltmeter

30 The diagram shows a potential divider circuit.

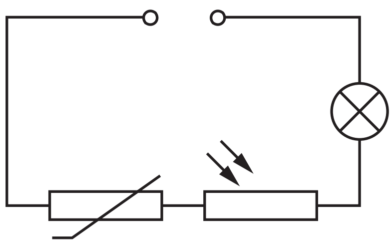


The temperature of the thermistor increases.

What happens to the resistance of the thermistor, and what happens to the reading on the voltmeter?

	resistance of thermistor	voltmeter reading
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

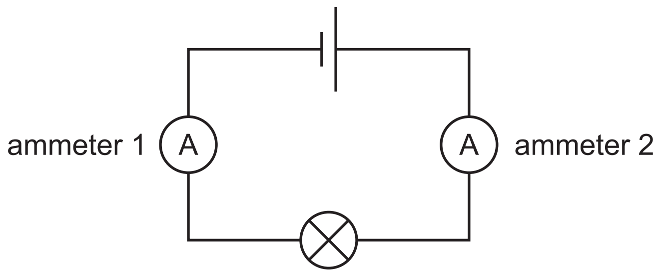
31 The diagram shows a series circuit containing a fixed voltage power supply, a thermistor, a light-dependent resistor (LDR) and a lamp.



Which set of conditions makes the lamp brightest?

	light intensity shining on LDR	temperature of thermistor
A	bright	high
B	bright	low
C	dim	high
D	dim	low

32 The diagram shows a circuit containing a cell, a lamp and two ammeters.



The current reading on ammeter 2 is 0.20 A.

What is the name for this type of circuit and what is the reading on ammeter 1?

	type of circuit	reading on ammeter 1
A	series	0.20 A
B	series	greater than 0.20 A
C	parallel	0.20 A
D	parallel	greater than 0.20 A

33 The circuit connecting a room heater to the mains electricity supply has a fuse in it.

The fuse melts and switches off the circuit.

Why does the wire in the fuse melt?

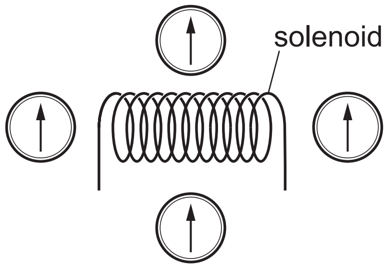
A The air in the room becomes too damp.

B The air in the room reaches its required temperature.

C The insulation around the circuit wire becomes damaged.

D The current in the circuit becomes too large.

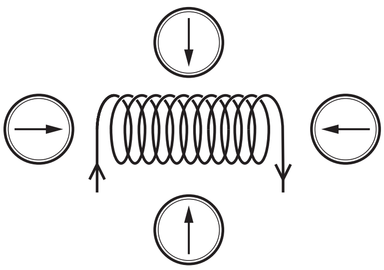
34 Four small compasses are placed around a solenoid.



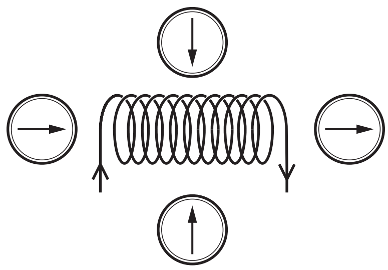
A current is now switched on in the solenoid.

Which diagram shows possible new directions of the compass needles?

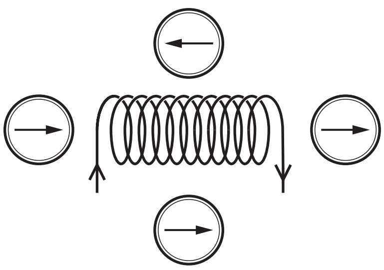
A



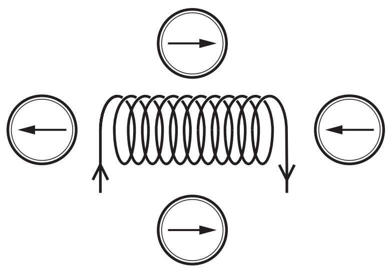
B



C



D



35 Which metal is used for the core of a transformer?

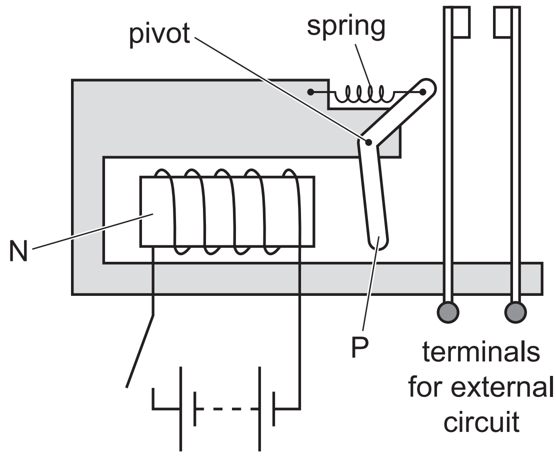
A aluminium

B copper

C soft iron

D steel

36 The diagram shows the design of a type of relay.



Two parts are labelled N and P.

From which metal is N made and from which metal is P made?

	N	P
A	soft iron	soft iron
B	soft iron	steel
C	steel	soft iron
D	steel	steel

37 A nuclide of cobalt contains 27 protons and 32 neutrons.

Which symbol represents this nuclide?

A $^{27}_{59}\mathrm{Co}$ **B** $^{32}_{27}\mathrm{Co}$ **C** $^{32}_{59}\mathrm{Co}$ **D** $^{59}_{27}\mathrm{Co}$

38 In the processes shown, X and Y are elements.

Which process describes α-decay?

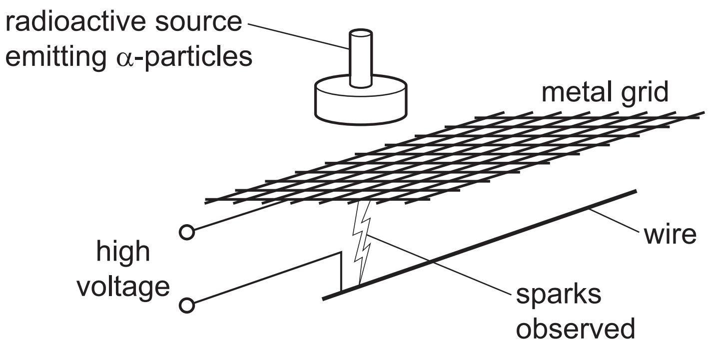
A Atoms of X collide with atoms of Y.

B Atoms of X emit atoms of Y.

C Atoms of X change into ions of Y.

D Atoms of X absorb α-particles.

39 A high-voltage power supply is connected to a metal grid and a wire, as shown.



When the radioactive source is placed close to the grid, sparks are observed in the position indicated.

Which statement explains why the sparks are formed?

A α-particles have a long range.

B α-particles have no charge.

C α-particles have no mass.

D α-particles are strongly ionising.

40 A radioactive substance has a half-life of 6 hours.

It has an initial rate of emission of 120 counts per second.

How long will it take for this rate of emission to fall to 30 counts per second?

A 1.5 hours **B** 12 hours **C** 30 hours **D** 240 hours

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PHYSICS **0625/12**

Paper 1 Multiple Choice (Core) **October/November 2022**

45 minutes

You must answer on the multiple choice answer sheet.

2354292637

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.

1 Which measuring instrument can be used to find the volume of a small stone?

A measuring cylinder partly filled with water

B measuring tape

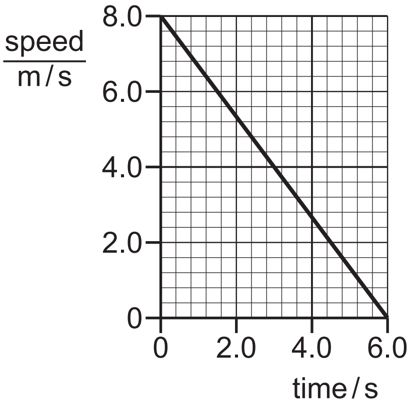
C metre rule

D protractor

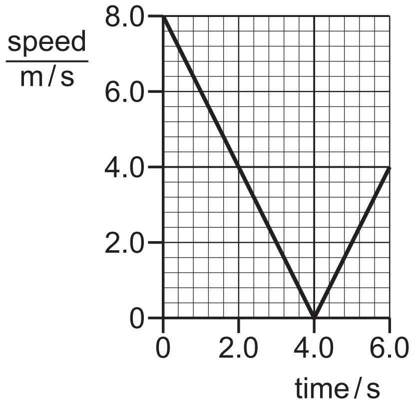
2 The diagrams show speed–time graphs for four different bodies moving for 6.0 s.

Which body travelled the least distance?

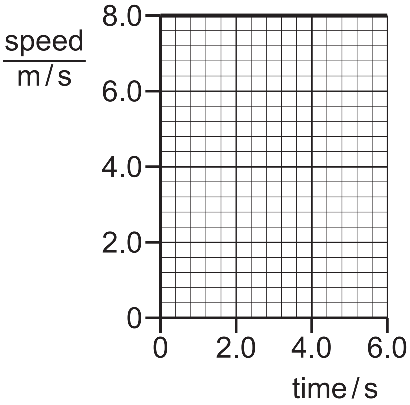
A



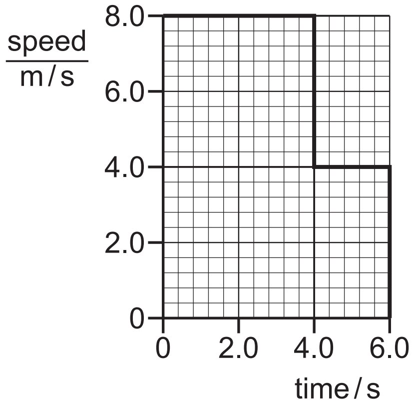
B



C



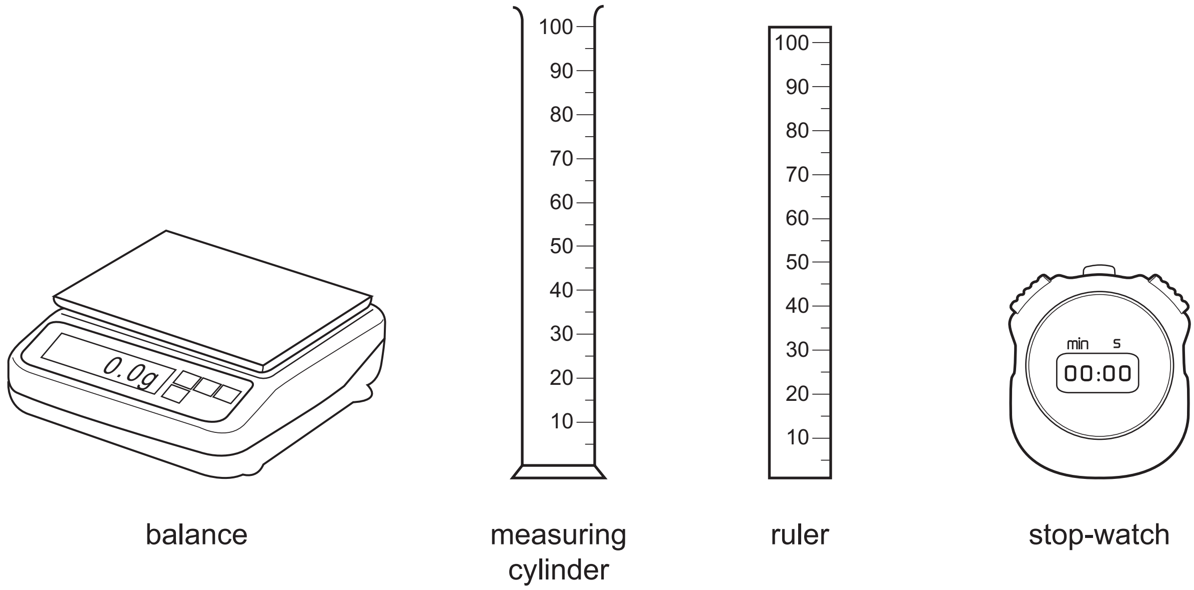
D



3 Which row shows the mass and the weight of an object near the Earth's surface?

	mass / kg	weight / N
A	0.2	0.2
B	2	0.2
C	2	20
D	20	10

4 The diagram shows four pieces of laboratory apparatus.



Which pieces of apparatus are used to find the density of a liquid?

A balance and stop-watch

B balance and measuring cylinder

C measuring cylinder and ruler

D stop-watch and ruler

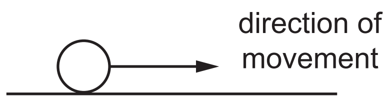
5 What is the unit for the moment of a force about a point?

A W **B** N s **C** N / m **D** N m

6 On which ball is a non-zero resultant force acting?

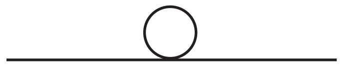
A

a ball moving at constant speed on a smooth surface



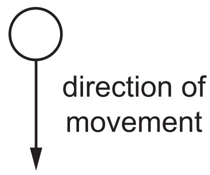
B

a ball at rest on a bench



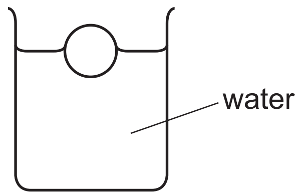
C

a free-falling ball which has just been released



D

a ball floating on water



7 Which statements **must** be correct for an object to be in equilibrium?

1 The object is moving in a straight line.

2 There is no resultant force on the object.

3 There is no resultant moment on the object.

A 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

8 An object falls towards the Earth's surface.

What happens to the gravitational potential energy and to the kinetic energy of the object?

	gravitational potential energy	kinetic energy
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

9 Which source of energy is **not** currently used to generate electrical energy?

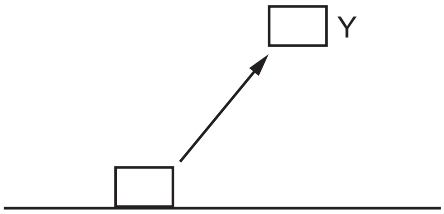
A nuclear fusion

B solar

C tidal

D waves

10 A mass is lifted from rest on the ground to Y. There is no air resistance.



P is the increase in gravitational energy of the mass.

Q is the kinetic energy of the mass at Y.

Which expression is equal to the mechanical work done on the mass?

A P + Q **B** P – Q **C** Q – P **D** P × Q

11 A pressure gauge is lowered into the sea. Measurements of the pressure and depth are taken as the pressure gauge is lowered.

Which statement describes how and why the pressure changes as the gauge is lowered?

A The density of the sea water decreases so the pressure increases.

B The depth of the gauge below the surface of the sea increases so the pressure increases.

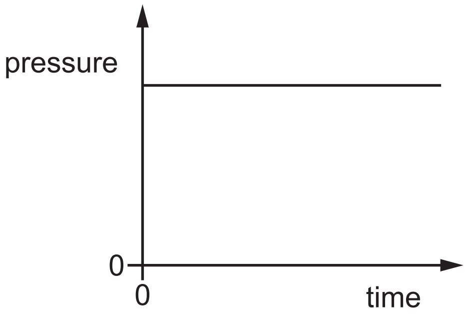
C The height of the gauge above the sea bed decreases so the pressure decreases.

D The temperature of the sea water decreases so the pressure decreases.

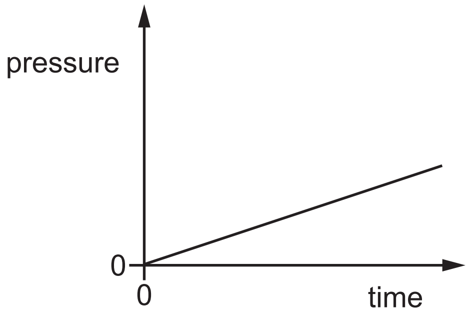
12 The pressure of a fixed mass of gas in a cylinder is measured. The volume of the gas in the cylinder is slowly decreased. The temperature of the gas does not change.

Which graph shows how the pressure of the gas changes during this process?

A



B



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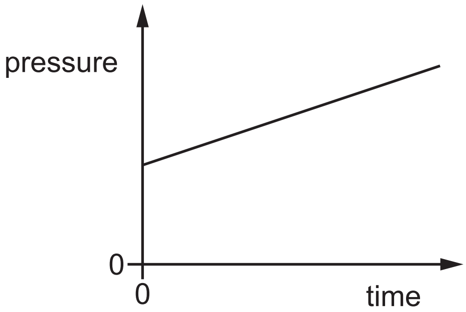
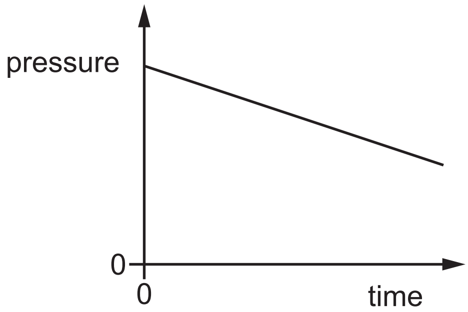


D



13 Which statement about the motion of molecules describes the process of evaporation?

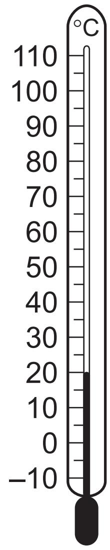
A Molecules break free from their fixed positions.

B Freely moving molecules collide and join together.

C Molecules escape from the surface of a liquid.

D Freely moving molecules gain energy and move further apart.

14 The diagram shows a thermometer calibrated in degrees Celsius.



What are the values of the lower fixed point and of the upper fixed point on the Celsius scale?

	lower fixed point / °C	upper fixed point / °C
A	–10	110
B	0	20
C	0	100
D	20	100

15 What happens to a solid when its temperature increases?

A It contracts.

B Its density increases.

C Its internal energy increases.

D Its molecules move freely.

16 Which piece of equipment is designed to produce a type of electromagnetic wave?

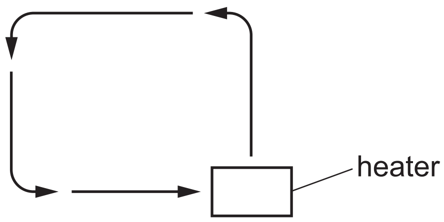
A electric fire

B electric generator

C electric motor

D electromagnet

17 Particles can move, transferring thermal energy, as shown.



In which states of matter does this movement occur?

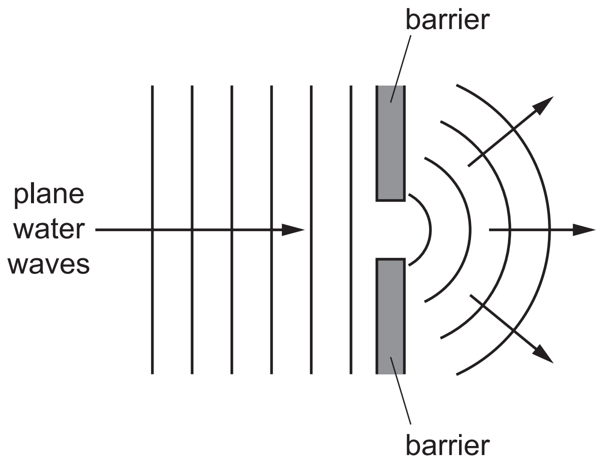
A gas and liquid only

B gas and solid only

C gas, liquid and solid

D liquid and solid only

18 The diagram shows plane water waves in a ripple tank passing through a gap between two barriers and spreading out.



Which name is given to this effect?

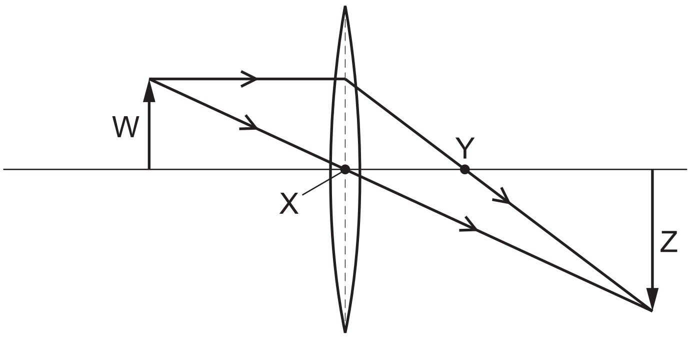
A diffraction

B reflection

C refraction

D total internal reflection

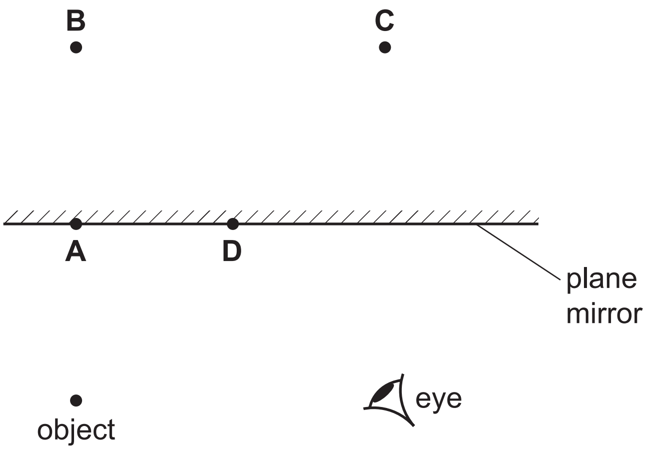
19 What are the correct labels for the ray diagram?



	object	image	principal focus
A	W	X	Y
B	W	Z	Y
C	X	Y	Z
D	X	Z	W

20 The diagram shows an object in front of a plane mirror.

At which labelled position is the image of the object formed?



21 The angle between an incident ray and the surface of a plane mirror reflecting the ray is 70°.

What is the angle of incidence?

A 20° **B** 40° **C** 70° **D** 140°

22 Visible light has a frequency of approximately 5.0×10^{14} Hz.

M and N are two other types of electromagnetic radiation.

The frequency of M is 5.0×10^{6} Hz.

The frequency of N is 5.0×10^{15} Hz.

Which types of radiation are M and N?

	M	N
A	radio waves	infrared
B	radio waves	ultraviolet
C	ultraviolet	X-rays
D	X-rays	infrared

23 Two students are describing different types of electromagnetic radiation.

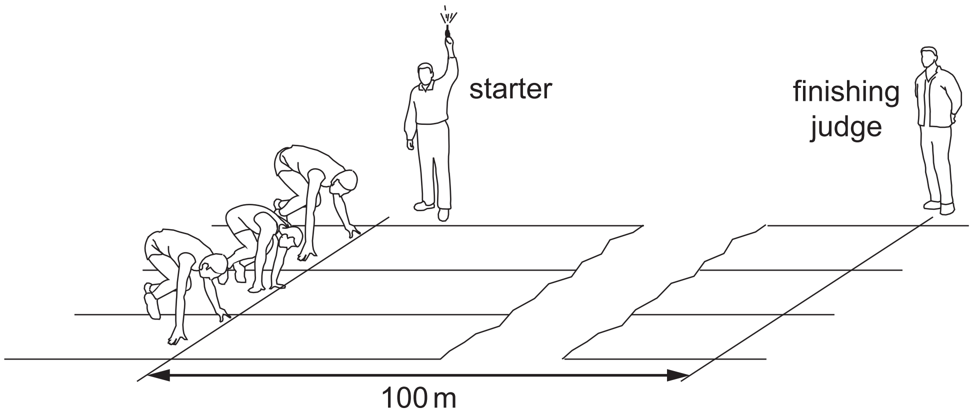
student 1 This radiation is used in communications.

student 2 This radiation is used in remote controllers.

Which row shows the possible type of radiation that each student is describing?

	student 1	student 2
A	microwave	infrared
B	radio	ultraviolet
C	sound waves	visible light
D	X-rays	gamma rays

24 A 100 m race is started by firing a gun. The gun makes a bang and a puff of smoke at the same time.



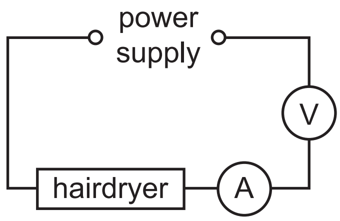
When does the finishing judge see the smoke and when does he hear the bang?

	sees the smoke	hears the bang
A	almost immediately	almost immediately
B	almost immediately	after about 0.3 s
C	after about 0.3 s	almost immediately
D	after about 0.3 s	after about 0.3 s

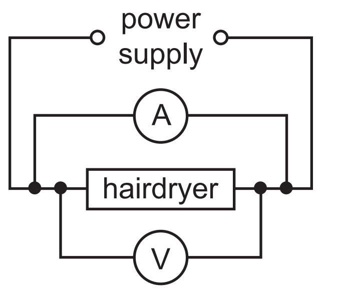
25 An electric hairdryer is rated 230 V, 2 A.

Which circuit could be used to check that these ratings are correct?

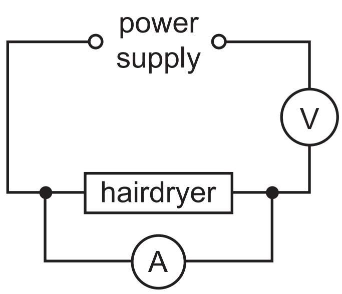
A



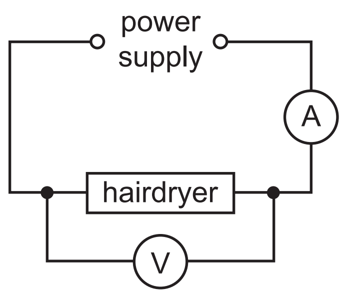
B



C



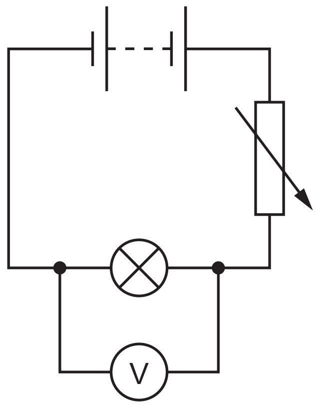
D



26 The diagram shows a circuit used to control the potential difference (p.d.) across a lamp.

The variable resistor is adjusted until the p.d. across the lamp is 6.0 V.

The current in the lamp is 0.5 A.



What is the resistance of the lamp?

A 0.083 Ω **B** 3.0 Ω **C** 6.5 Ω **D** 12.0 Ω

27 What are the units of electromotive force (e.m.f.)?

A amperes

B watts

C ohms

D volts

28 A plastic rod is rubbed with a dry woollen cloth. The rod becomes positively charged.

Which statement is correct?

A Electrons move from the cloth to the rod.

B Electrons move from the rod to the cloth.

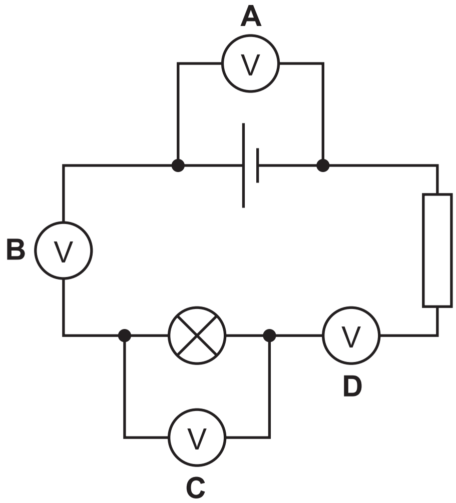
C Protons move from the cloth to the rod.

D Protons move from the rod to the cloth.

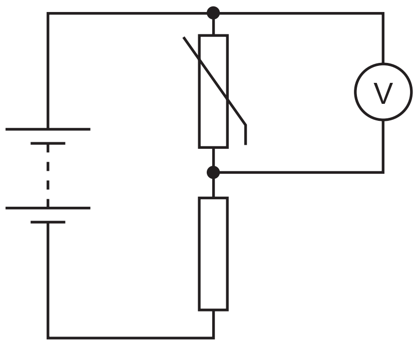
29 A circuit containing a cell, a resistor and a lamp is set up as shown.

A student connects a voltmeter to the circuit in one of the positions shown.

In which position does the voltmeter measure the potential difference (p.d.) across the lamp?



30 The diagram shows a potential divider circuit.



The temperature of the thermistor increases.

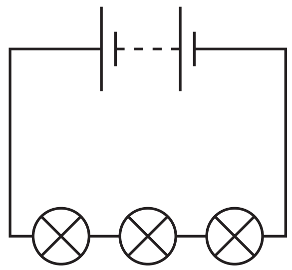
What happens to the resistance of the thermistor, and what happens to the reading on the voltmeter?

	resistance of thermistor	voltmeter reading
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

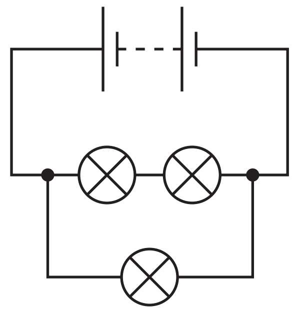
31 A student sets up four circuits using identical batteries and three identical lamps.

In which circuit will all the lamps be the brightest?

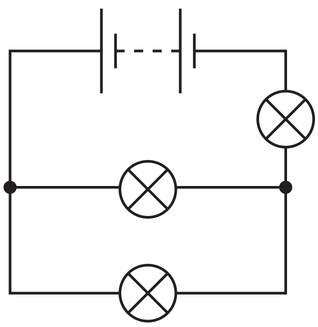
A



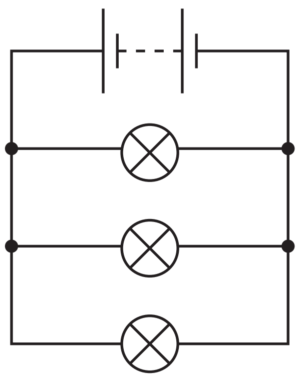
B



C



D



32 Which diagram shows the circuit symbol for a fuse?

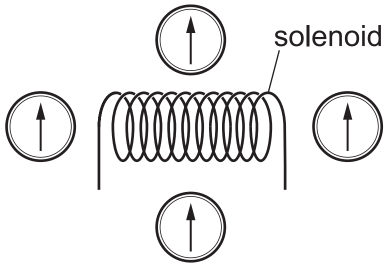
A **B** **C** **D**

33 If the insulation within a mains cable becomes damaged, two of the wires in it may touch and cause a short circuit.

Which row is correct?

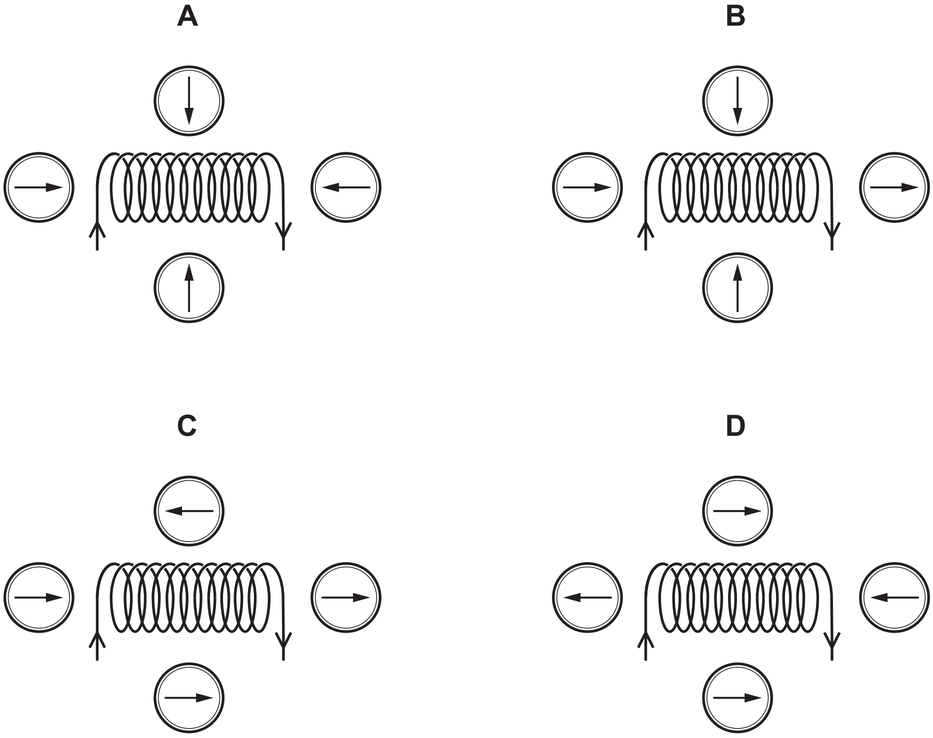
	the danger when this happens	safety device which avoids this danger
A	a large current will overheat the wiring and lead to a fire	a fuse
B	a large current will overheat the wiring and lead to a fire	a relay
C	the appliance at the end of the cable will be damaged	a fuse
D	the appliance at the end of the cable will be damaged	a relay

34 Four small compasses are placed around a solenoid.



A current is now switched on in the solenoid.

Which diagram shows possible new directions of the compass needles?

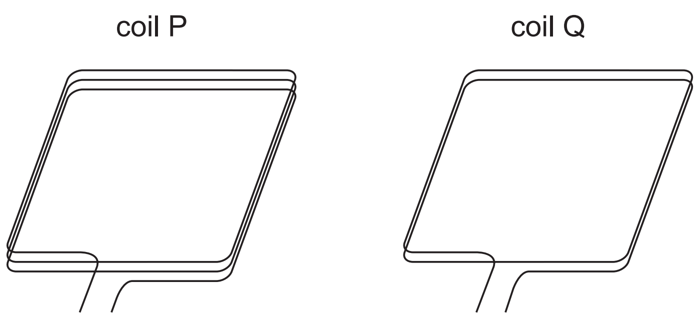


35 Transformers are used in the transmission of electrical power to houses.

Which type of transformer is used at the power station prior to connection to the power lines and which type is used prior to delivery to the houses?

	power station	before houses
A	step-down	step-down
B	step-down	step-up
C	step-up	step-down
D	step-up	step-up

36 The diagrams show the coils of two simple direct current (d.c.) motors. Coil P has three turns of wire and coil Q has two turns. Coil P has the same dimensions as coil Q. The coils are in identical magnet fields.



What produces the greatest turning effect?

	coil	current / A
A	P	2
B	P	4
C	Q	2
D	Q	4

37 A nuclide of cobalt contains 27 protons and 32 neutrons.

Which symbol represents this nuclide?

A $^{27}_{59}\text{Co}$ **B** $^{32}_{27}\text{Co}$ **C** $^{32}_{59}\text{Co}$ **D** $^{59}_{27}\text{Co}$

38 Everyone is exposed to background radiation.

What are sources of background radiation?

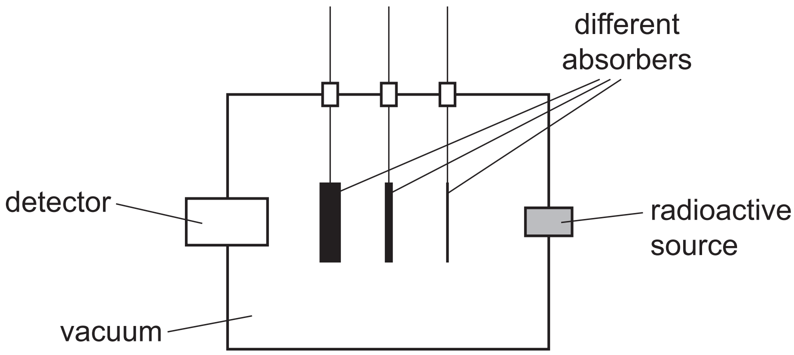
A food and drink only

B rocks only

C cosmic rays only

D food and drink, rocks and cosmic rays

39 The diagram shows a piece of apparatus used to determine the nature of the emissions from a radioactive source. The absorbers can be raised out of or lowered into the path of the radiation from the source to the detector. The apparatus is evacuated.



The table gives a set of results for a particular radioactive source.

absorber in use	count rate on detector (counts per second)
none	350
thin paper	350
1.0 mm aluminium	180
1.0 cm lead	23

Which types of radiation are being emitted by the radioactive source?

A α-particles and β-particles

B α-particles only

C β-particles and γ-rays

D β-particles only

40 The half-life of a sample of radioactive material is 400 years.

How long will it take until only $\frac{1}{4}$ of this sample remains undecayed?

A 100 years

B 400 years

C 800 years

D 1600 years

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PHYSICS **0625/11**

Paper 1 Multiple Choice (Core) **October/November 2022**

45 minutes

You must answer on the multiple choice answer sheet.

* 3 7 4 3 8 3 8 2 2 4 *

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m / s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.

1 The times for 10 swings of a pendulum are measured.

measurement	time for 10 swings / s
1	10.12
2	10.48
3	10.24

What is the average time for **one** swing?

A 1.028 s **B** 1.036 s **C** 1.042 s **D** 10.28 s

2 A car starts from rest.

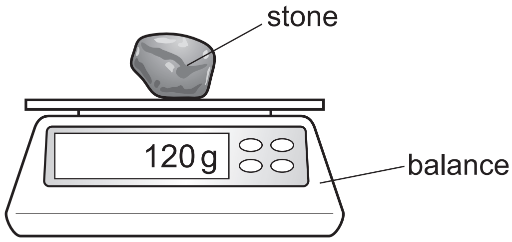
The table shows the readings from its speedometer every 10 s.

time / s	0	10	20	30	40	50	60
$\frac{\text{speed}}{\text{m/s}}$	0	4	8	12	12	12	12

Which row describes the car's motion in the first 30 seconds and in the last 30 seconds?

	motion during first 30 s	motion during last 30 s
A	non-zero acceleration	at rest
B	zero acceleration	constant speed
C	zero acceleration	at rest
D	non-zero acceleration	constant speed

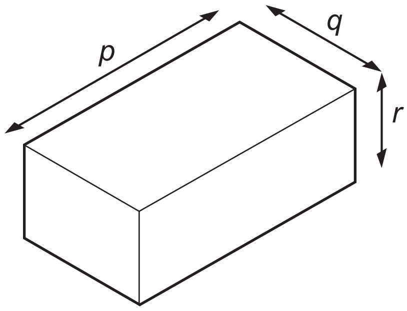
3 A stone is placed on a balance as shown.



Which row gives the mass and weight of the stone?

	mass	weight
A	120 g	1.2 N
B	120 g	1200 N
C	1.2 N	120 g
D	1200 N	120 g

4 The diagram shows the dimensions of a solid rectangular block of metal of mass *m*.



Which expression is used to calculate the density of the metal?

A $\dfrac{m}{(p \times q)}$

B $\dfrac{m}{(p \times q \times r)}$

C $m \times p \times q$

D $m \times p \times q \times r$

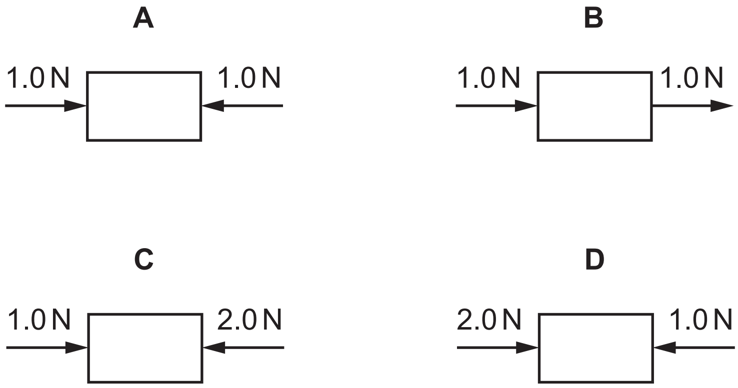
5 A force F acts on an object at a distance x from a pivot.

Which two changes both increase the moment of the force about the pivot?

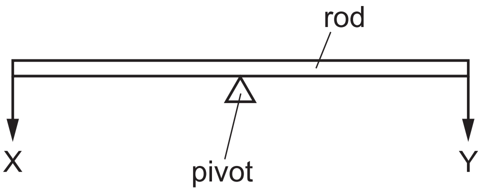
	change 1	change 2
A	decrease F	decrease x
B	decrease F	increase x
C	increase F	decrease x
D	increase F	increase x

6 The diagrams represent the only two forces acting on an object.

Which object could be moving to the right at constant speed?



7 The diagram shows a uniform rod with its midpoint on a pivot. Two weights X and Y are hung from the rod. The rod is in equilibrium.



Which statement is correct?

A The forces at X and Y have different values.

B There is a resultant turning effect on the rod.

C The resultant force on the rod is zero.

D The rod does not have a centre of mass.

8 Brakes are used to slow down a moving car.

Into which form of energy is most of the kinetic energy converted as the car slows down?

A chemical

B elastic

C thermal

D sound

9 What is a disadvantage of nuclear fission as a source of energy?

A Nuclear power stations are expensive to build.

B Nuclear power stations are unreliable.

C Nuclear power stations can only provide small quantities of energy.

D Nuclear power stations release large quantities of carbon dioxide into the atmosphere.

10 The statements describe what happens when the power of a machine is increased.

1 The work done in a given time decreases.

2 The work done in a given time increases.

3 The time taken to do a given quantity of work decreases.

4 The time taken to do a given quantity of work increases.

Which statements are correct?

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

11 Which quantities, in addition to the acceleration of free fall *g*, affect the pressure at the bottom of a pond of water?

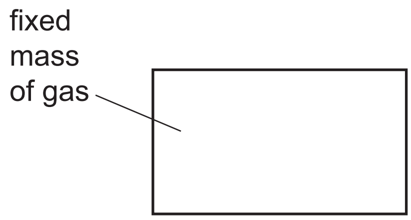
A the density of the water and the depth of the pond only

B the density of the water and the surface area of the pond only

C the depth of the pond and the surface area of the pond only

D the depth of the pond, the density of the water and the surface area of the pond

12 A fixed mass of gas is trapped in a container. The temperature of the gas is increased but the volume of the gas is kept constant.



How does this change affect the average kinetic energy of the molecules and the pressure on the walls of the container?

	average kinetic energy	pressure
A	increases	increases
B	stays the same	increases
C	increases	decreases
D	decreases	increases

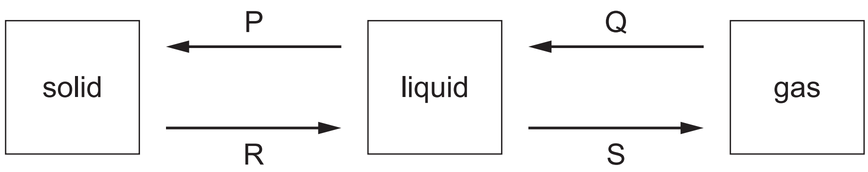
13 Which row correctly describes the arrangement and the motion of particles in a solid?

	arrangement of particles	motion of particles
A	far apart	moving randomly from place to place
B	far apart	vibrating about one position
C	tightly packed	moving randomly from place to place
D	tightly packed	vibrating about one position

14 What is the temperature difference between the fixed points on the °C temperature scale?

A 10 °C **B** 100 °C **C** 110 °C **D** 120 °C

15 In the diagram, each box represents a state of matter and each arrow represents a change of state.



Which row correctly identifies the changes of state?

	P	Q	R	S
A	freezing	condensation	boiling	melting
B	boiling	melting	condensation	freezing
C	freezing	condensation	melting	boiling
D	condensation	freezing	melting	boiling

16 Four thermometers, with their bulbs painted different colours, are placed at equal distances from a radiant heater.

Which thermometer shows the slowest temperature rise when the heater is first switched on?

A dull black

B dull white

C shiny black

D shiny white

17 Which method of transfer of thermal energy is caused by changes in density?

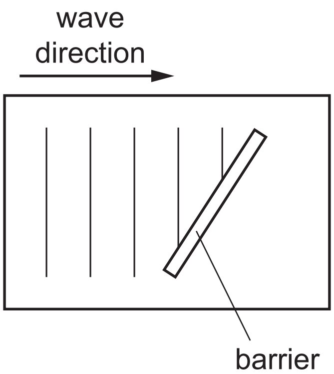
A conduction

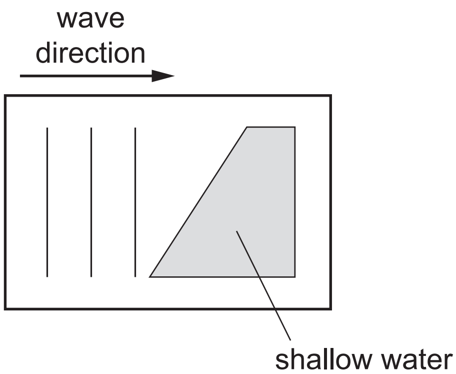
B convection

C evaporation

D radiation

18 The diagrams show two sets of wavefronts in a ripple tank.





A student makes two statements about the waves.

1. When the waves reflect from the barrier the direction changes but the wavelength remains the same.
2. When the waves refract as they enter the shallow water the direction remains the same, but the wavelength changes.

Which statements are correct?

A statement 1 and statement 2

B statement 1 only

C statement 2 only

D neither statement 1 nor statement 2

19 What is the correct order of the colours in a spectrum of white light?

A blue → green → yellow

B blue → yellow → green

C yellow → blue → green

D green → blue → yellow

20 An object is placed 30 cm in front of a plane mirror.

Which statement describes the image of the object?

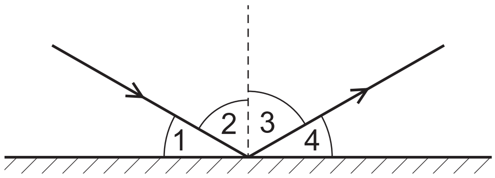
A The image is the same size and 30 cm from the object.

B The image is the same size and 60 cm from the object.

C The image is smaller and 30 cm from the object.

D The image is smaller and 60 cm from the object.

21 A ray of light is shone onto the surface of a mirror.



Which two angles represent the angle of incidence and the angle of reflection?

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

22 The diagram shows the electromagnetic spectrum. The numbers indicate the approximate wavelength at the boundaries between the various regions of the spectrum.

For a device to be able to make use of electromagnetic radiation, it needs an aerial of approximately the same size as the wavelength of the radiation it is designed to work with.

P	Q	R	S	T	U	V
1 m	10^{-3} m	7×10^{-7} m	4×10^{-7} m	10^{-8} m	10^{-11} m	

Which statement is correct?

A A mobile phone uses radiation from region P.

B A television satellite dish uses radiation from region Q.

C The receptor cells in an eye use radiation from region R.

D The remote controller for a television uses radiation from region U.

23 Microwaves and X-rays are regions of the electromagnetic spectrum.

Which statement about microwaves and X-rays is correct?

A Microwaves and X-rays have the same frequency.

B Microwaves and X-rays travel at the same speed in a vacuum.

C Microwaves have a shorter wavelength than X-rays.

D Microwaves travel at a lower speed than X-rays in a vacuum.

24 An observer stands at the finish line of a 100 m race. He wants to time the winner's run. He starts his stop-watch as soon as he sees the smoke from the starting gun instead of when he hears the bang.

What is the reason for doing this?

A Light travels much faster than sound.

B There is a risk he might respond to an echo from a wall.

C Humans react slower to sound than to light.

D Humans react more quickly to sound than to light.

25 Which description of a current in a metal is correct?

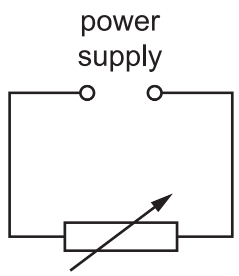
A a flow of electrons

B a flow of molecules

C a flow of positive atoms

D a flow of protons

26 The diagram shows a circuit containing a variable resistor connected to a variable power supply.



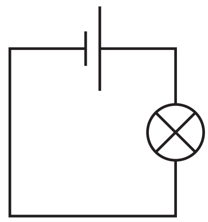
The table shows the currents for different values of the potential difference (p.d.) and the resistance.

p.d. / V	resistance / Ω	current
3.6	12	I_1
1.2	12	I_2
3.6	6	I_3

What is the order of the currents from smallest to largest?

A $I_1 \rightarrow I_2 \rightarrow I_3$ **B** $I_1 \rightarrow I_3 \rightarrow I_2$ **C** $I_2 \rightarrow I_1 \rightarrow I_3$ **D** $I_3 \rightarrow I_1 \rightarrow I_2$

27 The e.m.f. of the cell in this circuit is 1.5 V.



What does e.m.f. stand for?

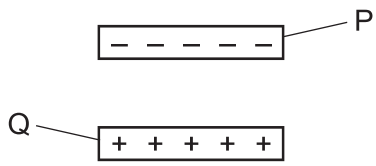
A electromagnetic field

B electromagnetic force

C electromotive field

D electromotive force

28 A negatively charged plastic rod P is placed above a positively charged plastic rod Q.

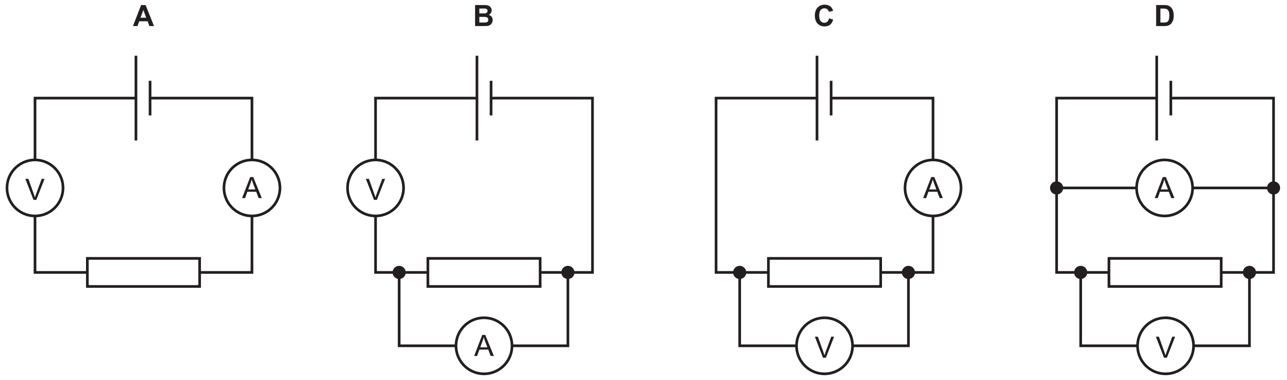


What are the directions of the electrostatic forces on rod P and on rod Q?

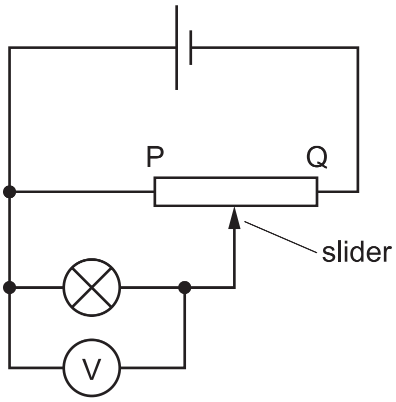
	electrostatic force on rod P	electrostatic force on rod Q
A	downwards	downwards
B	downwards	upwards
C	upwards	downwards
D	upwards	upwards

29 A student finds the resistance of a resistor.

Which circuit is used to measure the potential difference (p.d.) across the resistor and the current in it?



30 The circuit diagram shows a variable potential divider.

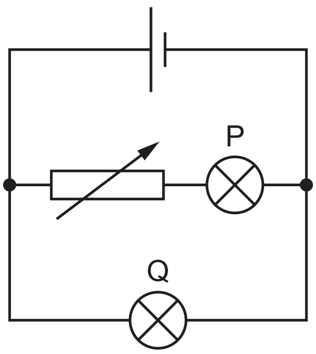


The slider is moved from P towards Q.

What happens to the reading on the voltmeter and to the brightness of the lamp?

	reading on voltmeter	brightness of lamp
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

31 The diagram shows a circuit containing a cell, a variable resistor and two bulbs, P and Q.

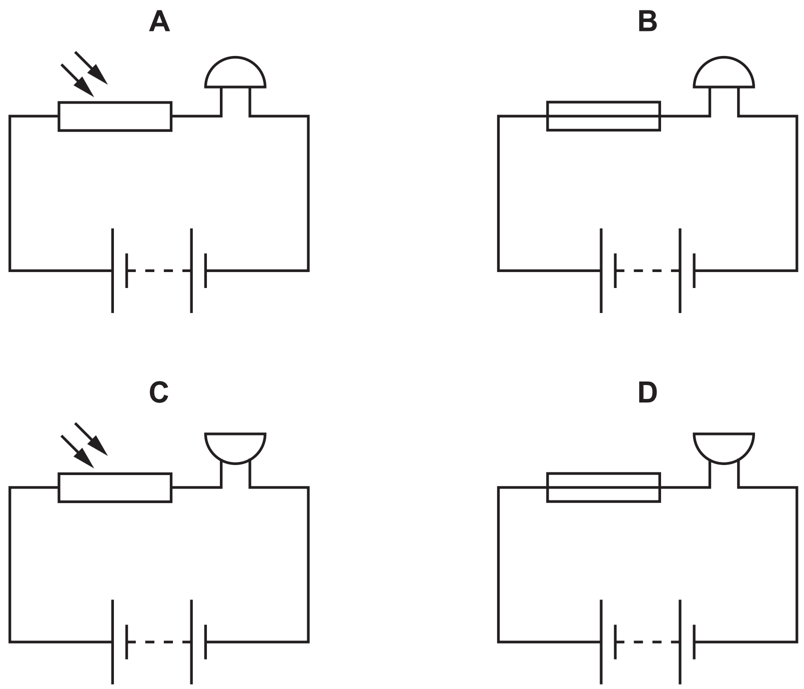


The resistance of the variable resistor is increased.

What happens to the brightness of each bulb?

	brightness of bulb P	brightness of bulb Q
A	dimmer	dimmer
B	dimmer	remains the same
C	remains the same	dimmer
D	remains the same	remains the same

32 Which diagram shows a circuit containing a battery, a fuse and a buzzer?



33 Why is a fuse used in an electric circuit?

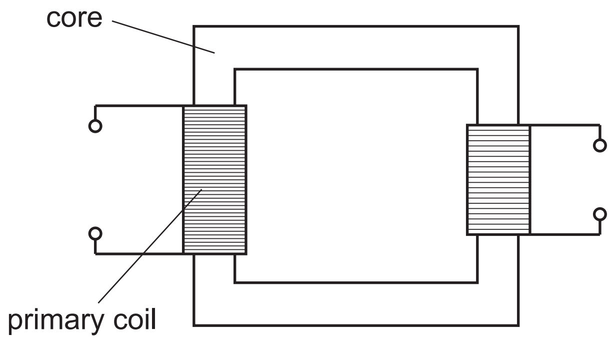
A to increase the circuit resistance

B to prevent short circuits

C to reduce the power loss

D to stop the cables from overheating

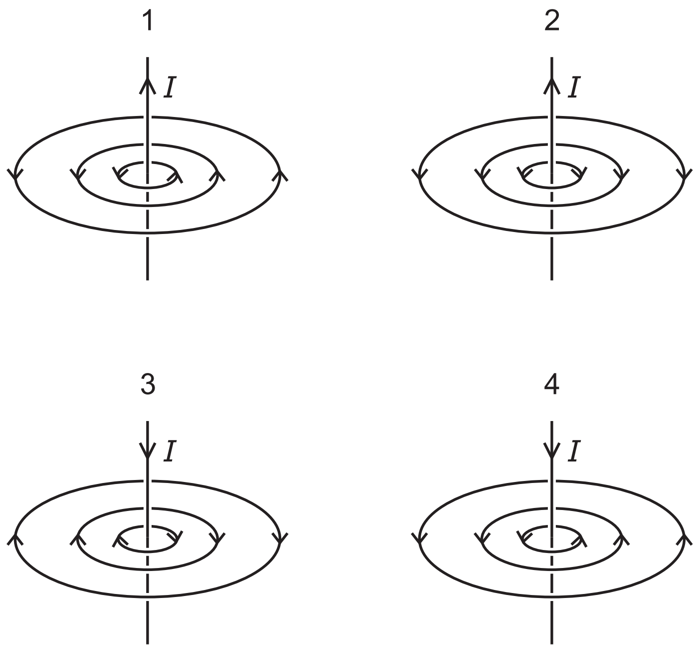
34 The diagram represents a transformer.



Which row shows materials suitable for making the core and the primary coil?

	core	primary coil
A	iron	copper
B	iron	plastic
C	steel	copper
D	steel	plastic

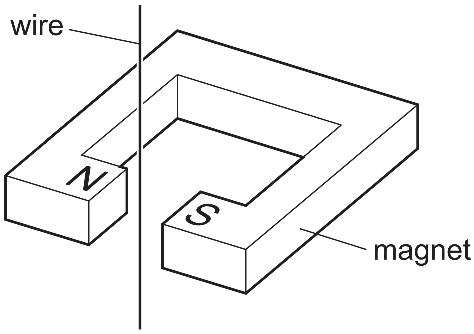
35 The diagrams show the magnetic field lines around a wire carrying a current, I.



Which diagrams are correct?

A 1 only **B** 2 and 3 **C** 4 only **D** 1 and 3

36 A wire is placed between the poles of a magnet.



Which statements are correct?

1 An a.c. current in the wire causes a changing force on it.

2 A downward d.c. current in the wire causes a constant force on it.

3 An upward d.c. current in the wire causes a constant force on it.

A 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

37 The nuclide notation for the isotope lithium-7 is ${}^{7}_{3}\mathrm{Li}$.

How many neutrons are there in an atom of lithium-7?

A 3 **B** 4 **C** 7 **D** 10

38 A radioactive material is emitting α-particles. The radioactive material is used in a demonstration in a school laboratory experiment.

Which safety precaution must be taken by the person carrying out the experiment?

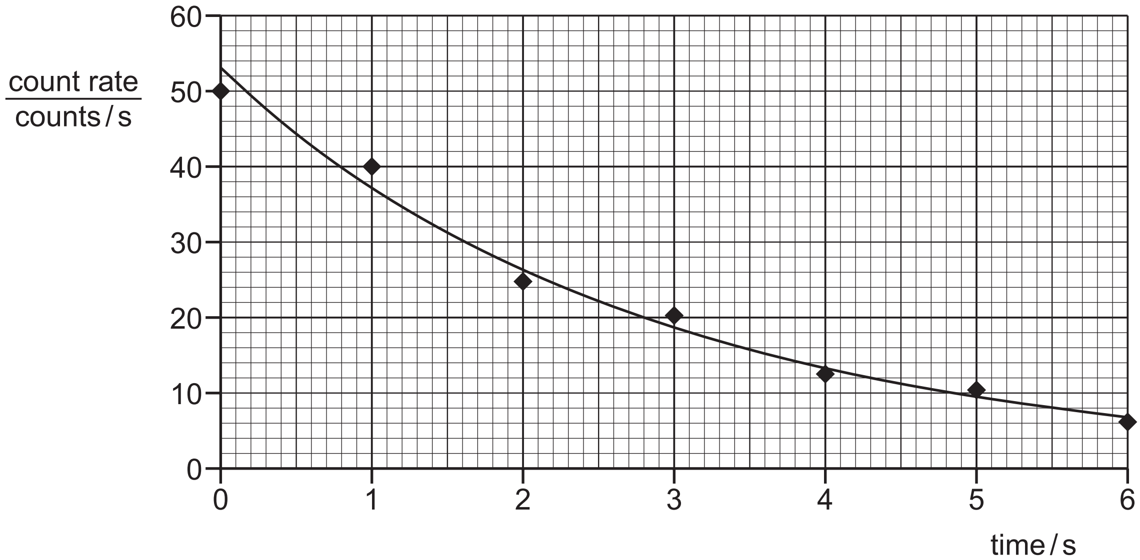
A Handle the source with tongs.

B Place the source on a heat-proof mat.

C Surround the experiment with a lead screen.

D Wear goggles.

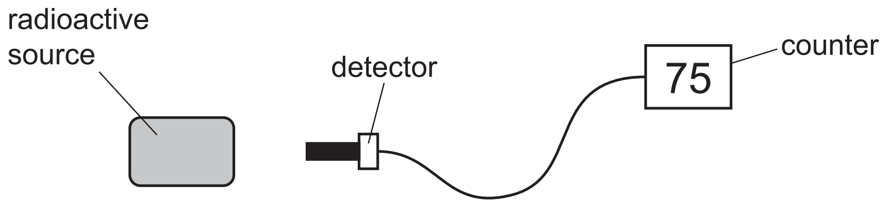
39 The graph shows how the count rate from a radioactive isotope changes with time.



What is the half-life of this isotope?

A 2.0 s **B** 6.0 s **C** 12 s **D** 53 s

40 A student measures the count rate near a radioactive source using a detector of ionising radiation. The diagram shows the arrangement.



The counter reads 75 counts per minute.

When the source is taken away, the reading on the counter decreases to 5 counts per minute.

What was the rate of emission from the radioactive source when the counter reading is corrected for background radiation?

A 5 counts per minute

B 15 counts per minute

C 70 counts per minute

D 80 counts per minute

Cambridge IGCSE™

PHYSICS | **0625/13**

Paper 1 Multiple Choice (Core) | **October/November 2022**

MARK SCHEME

Maximum Mark: 40

Published

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This document consists of **3** printed pages.

[Turn over

Question	Answer	Marks
1	A	**1**
2	B	**1**
3	D	**1**
4	B	**1**
5	A	**1**
6	C	**1**
7	C	**1**
8	B	**1**
9	A	**1**
10	C	**1**
11	C	**1**
12	C	**1**
13	D	**1**
14	C	**1**
15	A	**1**
16	A	**1**
17	D	**1**
18	D	**1**
19	C	**1**
20	B	**1**
21	C	**1**
22	B	**1**
23	C	**1**
24	B	**1**
25	B	**1**
26	B	**1**
27	B	**1**
28	B	**1**

Question	Answer	Marks
29	C	**1**
30	A	**1**
31	A	**1**
32	A	**1**
33	D	**1**
34	D	**1**
35	C	**1**
36	A	**1**
37	D	**1**
38	C	**1**
39	D	**1**
40	B	**1**

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PHYSICS **0625/12**

Paper 1 Multiple Choice (Core) **October/November 2022**

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Question	Answer	Marks
1	A	**1**
2	B	**1**
3	C	**1**
4	B	**1**
5	D	**1**
6	C	**1**
7	C	**1**
8	B	**1**
9	A	**1**
10	A	**1**
11	B	**1**
12	C	**1**
13	C	**1**
14	C	**1**
15	C	**1**
16	A	**1**
17	A	**1**
18	A	**1**
19	B	**1**
20	B	**1**
21	A	**1**
22	B	**1**
23	A	**1**
24	B	**1**
25	D	**1**
26	D	**1**
27	D	**1**
28	B	**1**

Question	Answer	Marks
29	C	**1**
30	A	**1**
31	D	**1**
32	D	**1**
33	A	**1**
34	D	**1**
35	C	**1**
36	B	**1**
37	D	**1**
38	D	**1**
39	C	**1**
40	C	**1**

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PHYSICS **0625/11**

Paper 1 Multiple Choice (Core) **October/November 2022**

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Question	Answer	Marks
1	A	**1**
2	D	**1**
3	A	**1**
4	B	**1**
5	D	**1**
6	A	**1**
7	C	**1**
8	C	**1**
9	A	**1**
10	C	**1**
11	A	**1**
12	A	**1**
13	D	**1**
14	B	**1**
15	C	**1**
16	D	**1**
17	B	**1**
18	B	**1**
19	A	**1**
20	B	**1**
21	C	**1**
22	B	**1**
23	B	**1**
24	A	**1**
25	A	**1**
26	C	**1**
27	D	**1**
28	B	**1**

Question	Answer	Marks
29	C	**1**
30	D	**1**
31	B	**1**
32	D	**1**
33	D	**1**
34	A	**1**
35	D	**1**
36	D	**1**
37	B	**1**
38	A	**1**
39	A	**1**
40	C	**1**

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PHYSICS **0625/13**

Paper 1 Multiple Choice (Core) **October/November 2021**

45 minutes

You must answer on the multiple choice answer sheet.

* 2 6 4 2 5 3 7 2 2 8 *

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = $10\,m/s^2$).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.

1 A teacher asks a student to measure the volume of a pencil sharpener.

Which piece of apparatus would **not** be useful?

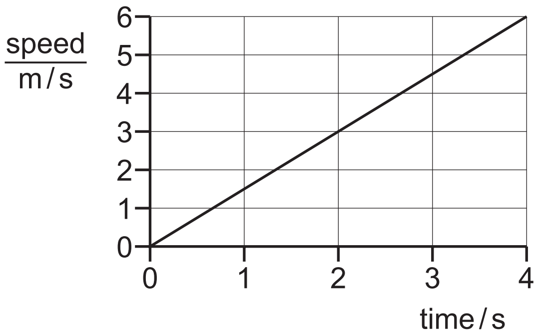
A beaker

B displacement can

C balance

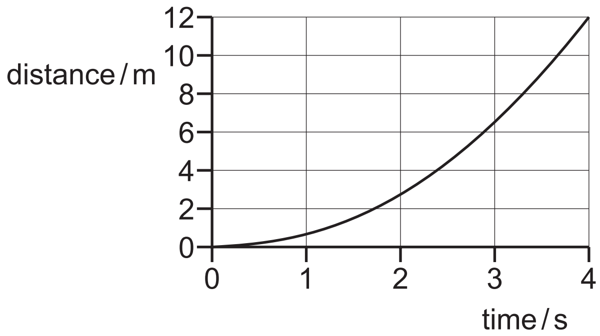
D measuring cylinder

2 The graph shows how the speed of a car varies with time at the start of a journey.

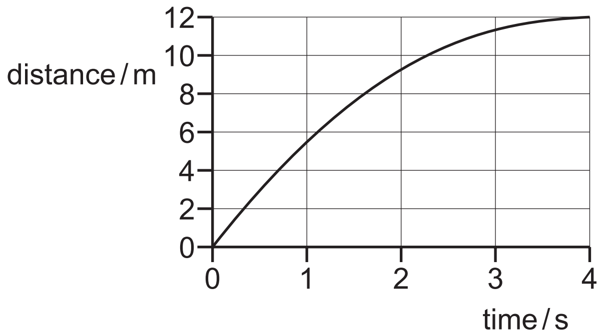


Which distance–time graph represents the motion of the car over the same time period?

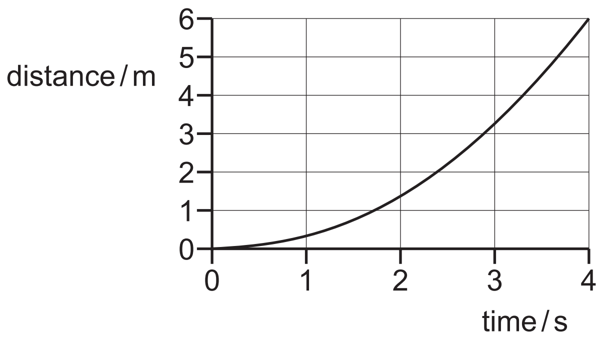
A



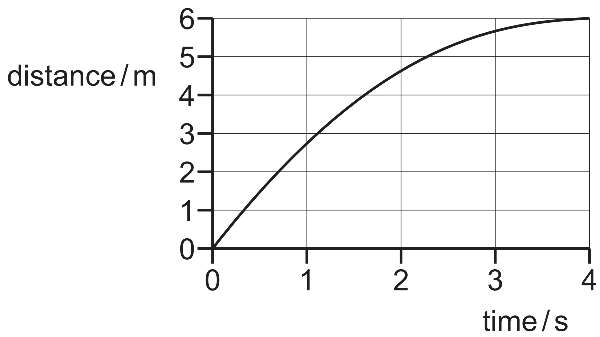
B



C



D



3 A student uses a force meter to measure the weights and a balance to measure the masses of four different objects.

He puts his measurements in a table.

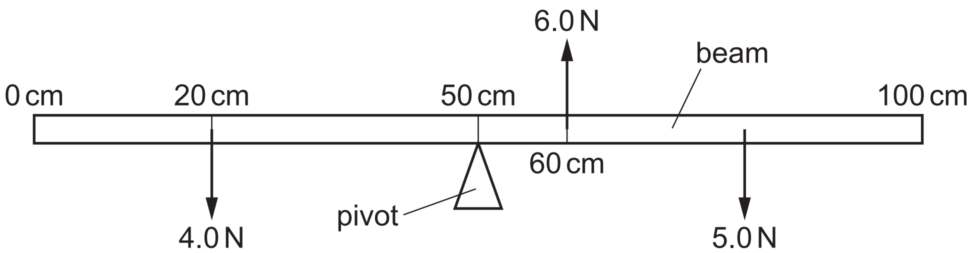
Which row is correctly recorded?

	mass	weight
A	1 kg	10 N
B	5 g	50 N
C	10 N	100 kg
D	20 kg	2 N

4 Which substance in the table has the lowest density?

	substance	mass / g	volume / cm^3
A	nylon	1.2	1.0
B	cotton	1.5	1.0
C	olive oil	1.8	2.0
D	water	2.0	2.0

5 The diagram shows a uniform beam 100 cm long. The diagram is not drawn to scale.



The beam remains balanced on a pivot at the 50 cm mark under the action of the forces of 4.0 N, 5.0 N and 6.0 N.

The 4.0 N force is at the 20 cm mark and the 6.0 N force is at the 60 cm mark.

At which point on the beam is the 5.0 N force acting?

A at the 62 cm mark

B at the 74 cm mark

C at the 86 cm mark

D at the 88 cm mark

6 A spacecraft is travelling in space with no resultant force and no resultant moment acting on it.

Which statement about the spacecraft is correct?

A Its direction is changing.

B It is in equilibrium.

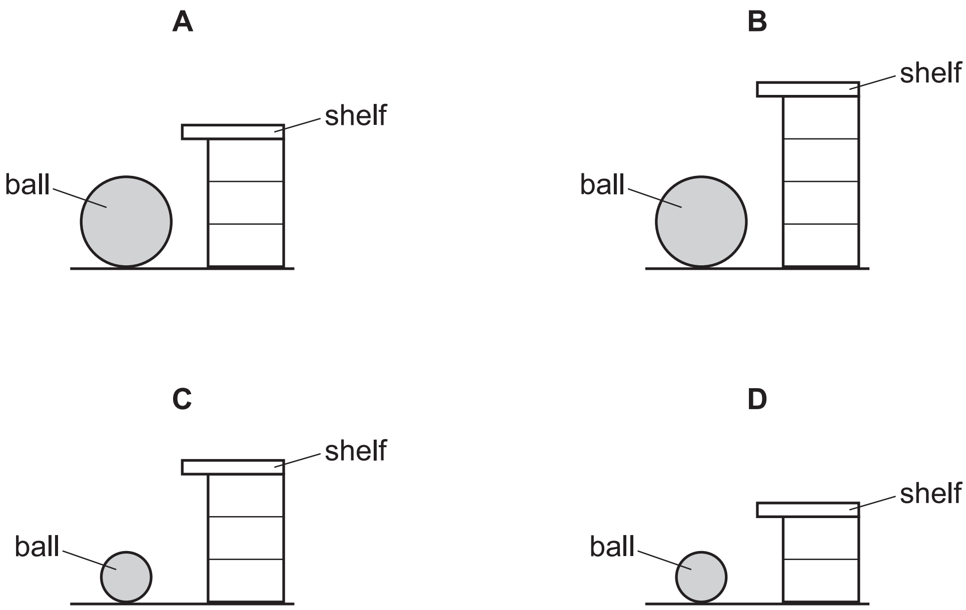
C Its speed is decreasing.

D Its speed is increasing.

7 A weightlifter picks up a stone ball and places it on a shelf.

Each lift takes the same time.

Which situation requires the greatest power?



8 Electrical energy may be obtained from nuclear fission.

In which order is the energy transferred in this process?

A nuclear fuel $\rightarrow$ generator $\rightarrow$ reactor and boiler $\rightarrow$ turbines

B nuclear fuel $\rightarrow$ generator $\rightarrow$ turbines $\rightarrow$ reactor and boiler

C nuclear fuel $\rightarrow$ reactor and boiler $\rightarrow$ generator $\rightarrow$ turbines

D nuclear fuel $\rightarrow$ reactor and boiler $\rightarrow$ turbines $\rightarrow$ generator

9 A person lifts a book from floor level to a shelf. It falls to the floor and a second person lifts it back up to the shelf.

Which statement **must** be correct?

A The second person does the same work on the book as the first person.

B The second person takes the same time as the first person.

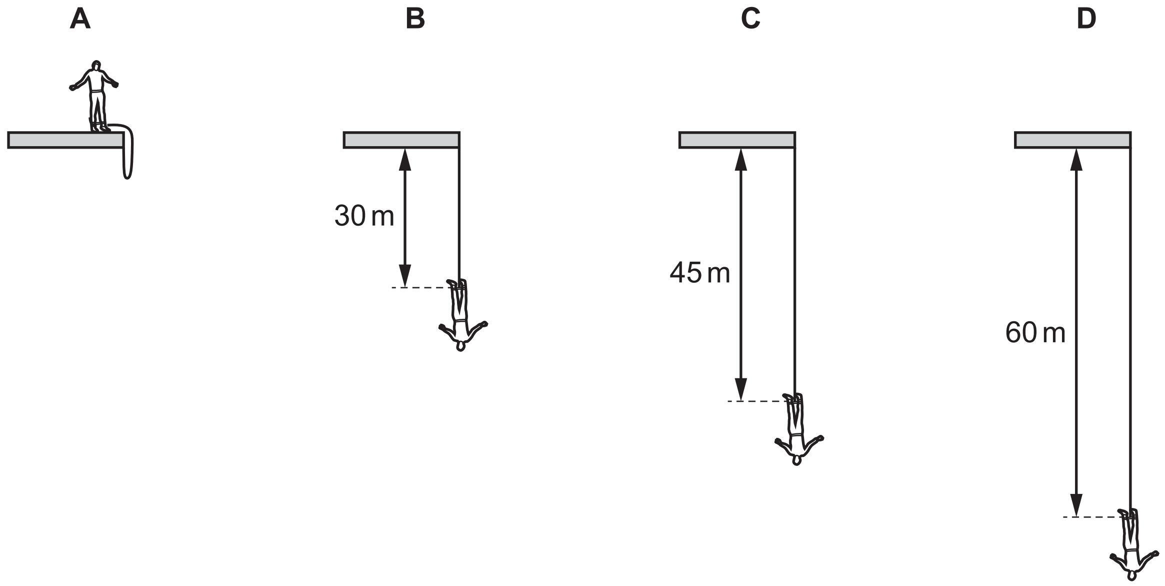
C The second person develops the same power as the first person.

D The second person does the same work on the book as the first person, develops the same power and takes the same time.

10 A man, attached to an elastic cord, jumps from a platform. He falls 60 m before starting to rise. The length of the unextended cord is 30 m.

The diagrams show four successive stages in his fall.

In which position is elastic (strain) energy and kinetic energy present?



11 Liquid is stored in a tank. The area of the base of the tank is 2.2 m^2 and the pressure at the base due to the liquid is 15 000 Pa.

What is the weight of the liquid?

A 3300 N **B** 6800 N **C** 15 000 N **D** 33 000 N

12 A liquid is evaporating. The liquid is not boiling.

Which statement about the liquid is correct at an instant in time?

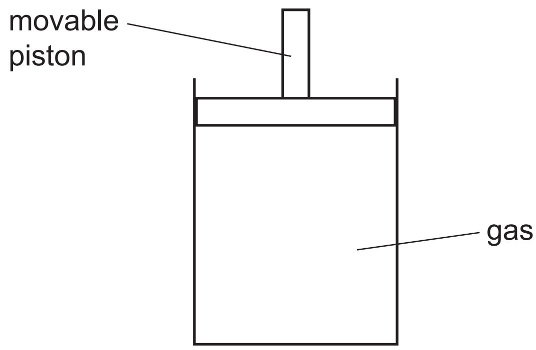
A Any molecule can escape, and from any part of the liquid.

B Any molecule can escape, but only from the liquid's surface.

C Only molecules with enough energy can escape, and only from the liquid's surface.

D Only molecules with enough energy can escape, but from any part of the liquid.

13 The diagram shows a sealed gas cylinder with a movable piston.

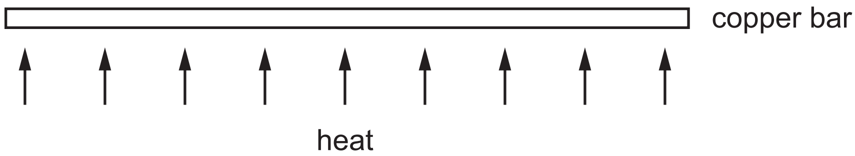


The piston is moved slowly downwards and there is no change in temperature of the gas.

What happens to the average distance between the gas molecules and to the pressure of the gas?

	average distance between gas molecules	pressure of gas in cylinder
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

14 A long, thin bar of copper is heated gently and evenly along its length.



What happens to the bar?

A It becomes less heavy.

B It becomes longer.

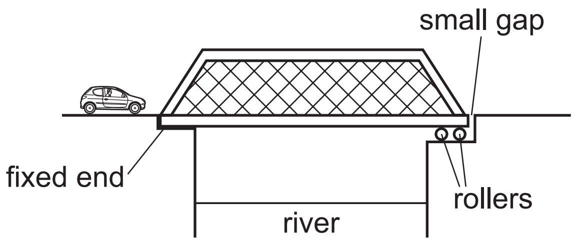
C It becomes shorter.

D It bends at the ends.

15 The diagrams show four examples of thermal expansion. In three of the examples, thermal expansion is useful. In one of the examples, expansion is unwanted and has to be allowed for.

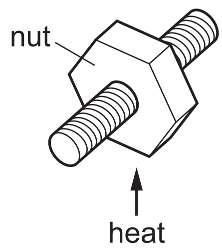
In which example is thermal expansion unwanted?

A



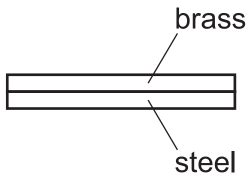
a bridge getting longer in hot weather

B



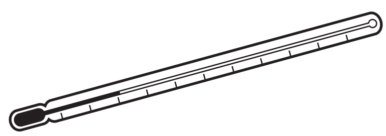
loosening a very tight nut by heating it

C



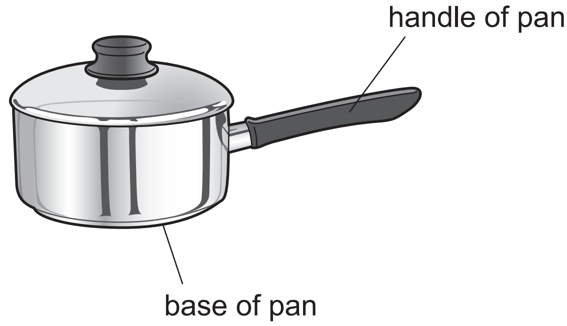
a bimetallic strip to indicate temperature

D



liquid-in-glass thermometer

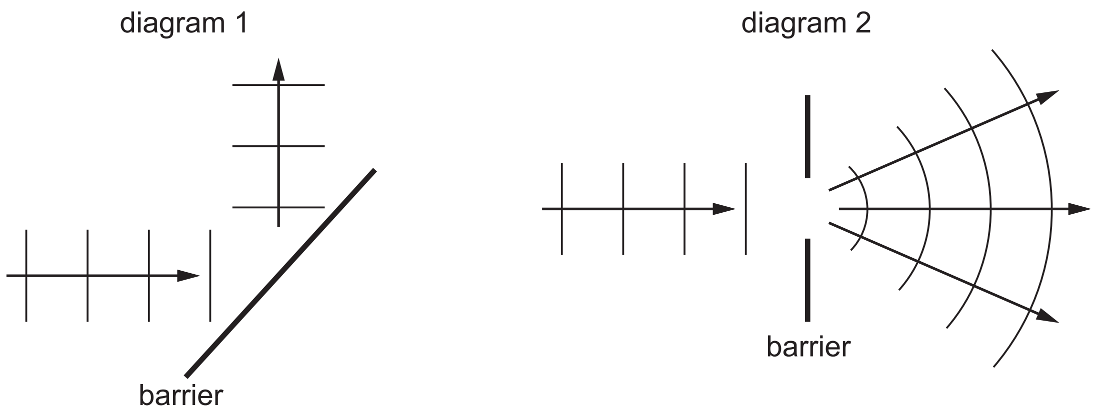
16 The diagram shows a pan used for cooking food.



Which row is correct for the materials used to make the base and the handle of the pan?

	base of pan	handle of pan
A	good thermal conductor	good thermal conductor
B	good thermal conductor	poor thermal conductor
C	poor thermal conductor	good thermal conductor
D	poor thermal conductor	poor thermal conductor

17 The diagrams show two patterns produced by water waves.



Which two effects are shown in the diagrams?

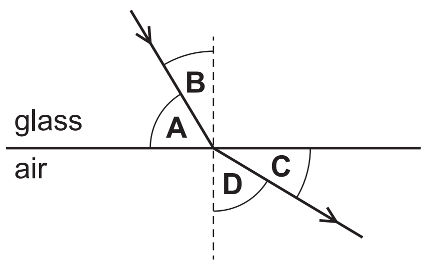
	diagram 1	diagram 2
A	reflection	diffraction
B	reflection	refraction
C	refraction	diffraction
D	refraction	reflection

18 Which row correctly defines the frequency and the speed of a wave?

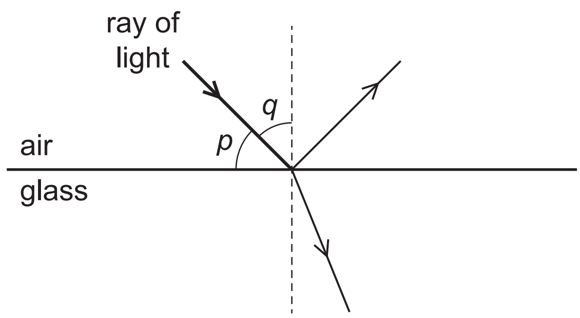
	frequency	speed
A	number of waves	distance travelled per unit time
B	number of waves	time taken for one complete wave to pass a point
C	number of waves passing per unit time	distance travelled per unit time
D	number of waves passing per unit time	time taken for one complete wave to pass a point

19 A narrow beam of light travels through glass. It reaches the edge of the glass and refracts into the air.

What is the angle of refraction?



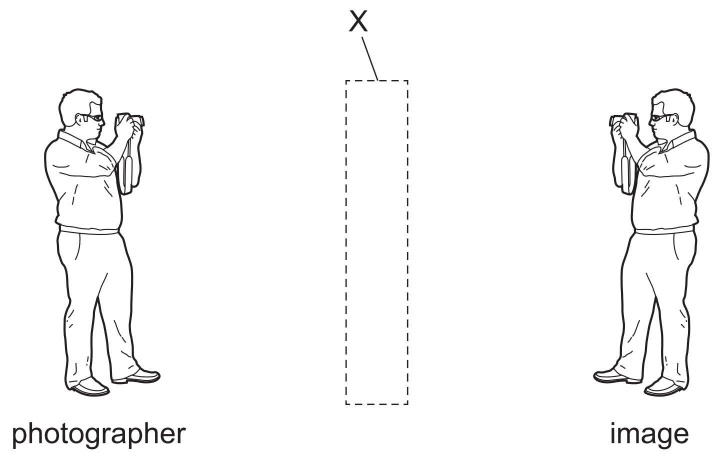
20 The diagram shows a ray of light in air incident on a glass block. Some of the light is refracted and some of the light is reflected. Two angles, *p* and *q*, are marked on the diagram.



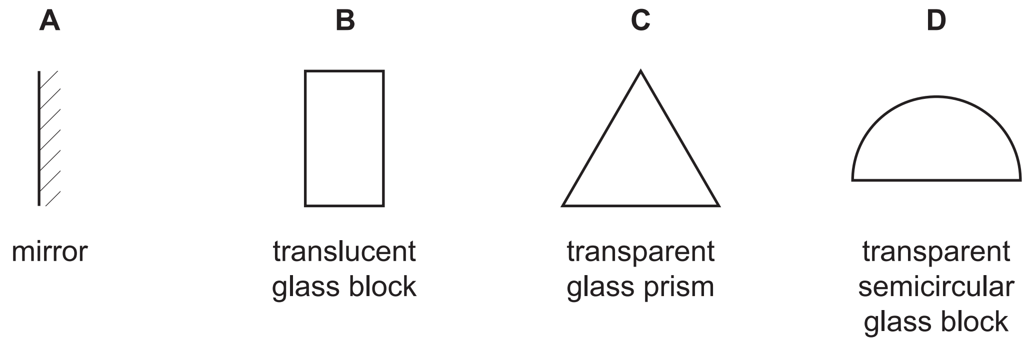
Which row gives the angle of incidence and states whether total internal reflection occurs?

	angle of incidence	total internal reflection
A	*p*	no
B	*p*	yes
C	*q*	no
D	*q*	yes

21 A photographer sees his image as shown.



What could X be?



22 Visible light, X-rays and microwaves are all components of the electromagnetic spectrum.

Which statement about the waves is correct?

A In a vacuum, microwaves travel faster than visible light and have a shorter wavelength.

B In a vacuum, microwaves travel at the same speed as visible light and have a shorter wavelength.

C In a vacuum, X-rays travel faster than visible light and have a shorter wavelength.

D In a vacuum, X-rays travel at the same speed as visible light and have a shorter wavelength.

23 What can transmit some types of transverse waves but **not** longitudinal waves?

A air

B a steel bar

C a vacuum

D sea water

24 What is ultrasound?

A sound waves that are so loud that they damage human hearing

B sound waves that are too high-pitched for humans to hear

C sound waves that are too low-pitched for humans to hear

D sound waves that are too quiet for humans to hear

25 What would be the least successful method of magnetising a steel bar?

A Insert the bar in a coil with a large direct current (d.c.).

B Place the bar at right angles to a weak magnetic field and hit it with a hammer.

C Place the bar parallel to a strong magnetic field, heat it and let it cool.

D Stroke the bar with a magnet.

26 An electric current in a copper wire is due to the flow of charge.

Which particles are moving along the wire?

A α-particles

B copper nuclei

C electrons

D protons

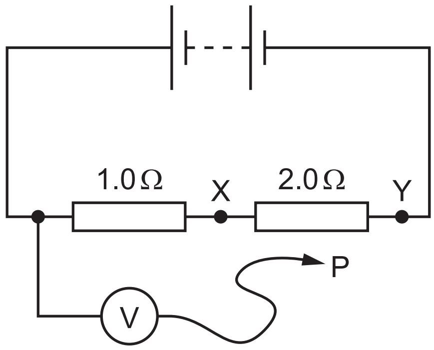
27 Which row correctly shows a conductor and an insulator?

	conductor	insulator
A	rubber	plastic
B	iron	nylon
C	air	wood
D	copper	steel

28 The diagram shows a circuit containing two resistors of resistance $1.0\,\Omega$ and $2.0\,\Omega$.

A voltmeter is connected across the $1.0\,\Omega$ resistor by connecting P to X.

The reading on the voltmeter is 6.0 V.



P is moved to point Y in the circuit.

What is the new reading on the voltmeter?

A 3.0 V **B** 6.0 V **C** 12 V **D** 18 V

29 What is the unit of electromotive force (e.m.f.)?

A ampere

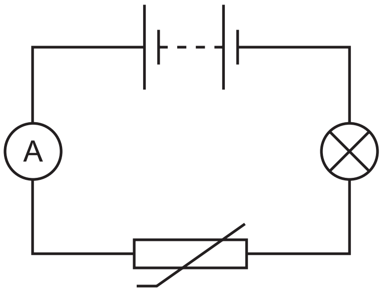
B newton

C ohm

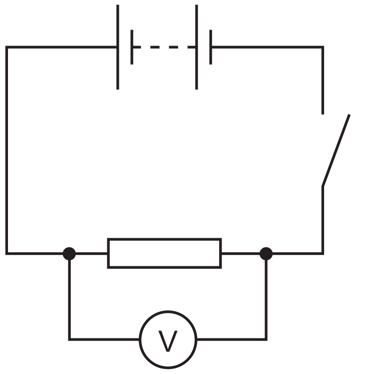
D volt

30 Which circuit contains a fuse?

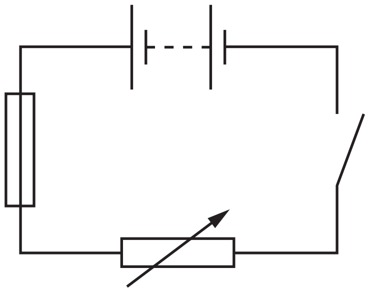
A



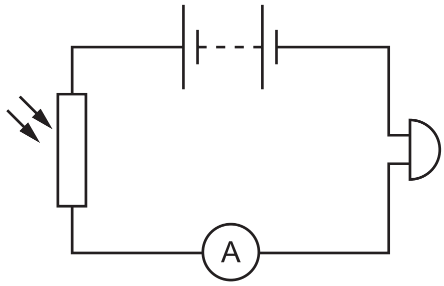
B



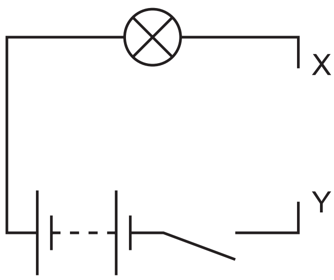
C



D



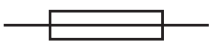
31 A student sets up the circuit shown, with a gap XY.



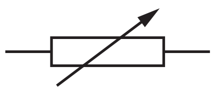
The student wishes to connect a component between X and Y to enable her to vary the brightness of the lamp.

Which component should be used?

A



B

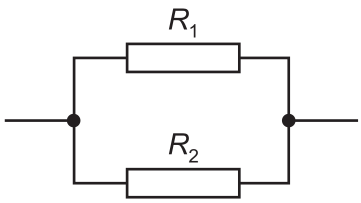


C

D

32 Two resistors, with resistances R_1 and R_2, are connected in parallel.

The resistance R_1 is greater than the resistance R_2.



What is the resistance of the parallel combination?

A less than either R_1 or R_2

B equal to R_1

C equal to R_2

D the average of R_1 and R_2

33 Circuit breakers and fuses are devices used to protect a circuit from overloading.

Which statement correctly describes the difference between a circuit breaker and a fuse?

A Circuit breakers can be reset if they operate but fuses need to be replaced.

B Circuit breakers need to be replaced if they operate but fuses can be reset.

C Circuit breakers can be used in an a.c. circuit but fuses cannot.

D Circuit breakers cannot be used in an a.c. circuit but fuses can.

34 Why might it be dangerous to use an electrical appliance in damp conditions?

A It might lead to the fuse blowing.

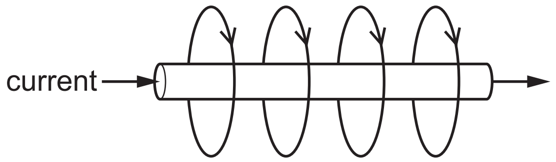
B It might lead to the insulation on the supply cable becoming damaged.

C It might lead to an electric shock.

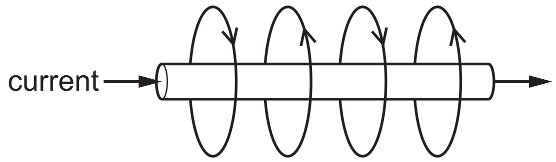
D It might lead to the supply cable overheating.

35 Which diagram shows the magnetic field around a straight, current-carrying wire?

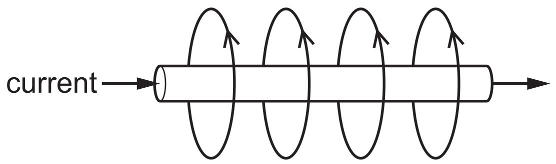
A



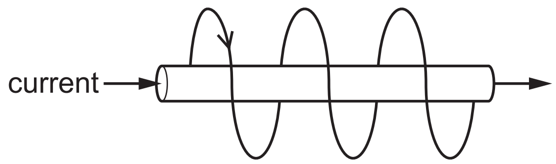
B



C



D



36 When wiring a house, it is important to use the correct cables.

The current ratings of four different cables are listed.

All four cables are used in series in the same circuit.

Which cable is most likely to overheat?

A 6 A **B** 10 A **C** 15 A **D** 30 A

37 A very important experiment improved scientists' understanding of the structure of matter.

The experiment involved α-particles being fired at a thin, gold foil.

What happened?

A All the α-particles were absorbed by the nuclei of the gold atoms.

B All the α-particles were unaffected by the gold atoms.

C Some of the α-particles were attracted by the neutrons in the nuclei of the gold atoms.

D Some of the α-particles were repelled by the protons in the nuclei of the gold atoms.

38 A nuclide has the symbol $^{23}_{11}Na$.

Which statement about all atoms of this nuclide is correct?

A There are 11 protons in the nucleus.

B There are 23 neutrons in the nucleus.

C There are 11 electrons in the nucleus.

D There are 34 nucleons in the nucleus.

39 The half-life for lead-202 is 52 500 years.

A sample of lead-202 produces 800 counts / s.

How long will it take for the count rate to drop to 100 counts / s?

A 105 000 years

B 157 500 years

C 210 000 years

D 420 000 years

40 Why is a thick shield made of lead needed to protect people from a source of γ-rays?

A Gamma radiation is strongly ionising and so is not very penetrating.

B Gamma radiation is strongly ionising and so is very penetrating.

C Gamma radiation is weakly ionising and so is not very penetrating.

D Gamma radiation is weakly ionising and so is very penetrating.

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PHYSICS **0625/12**

Paper 1 Multiple Choice (Core) **October/November 2021**

45 minutes

You must answer on the multiple choice answer sheet.

4547524422

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m / s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.

1 Which list places units of length in increasing order of magnitude (size)?

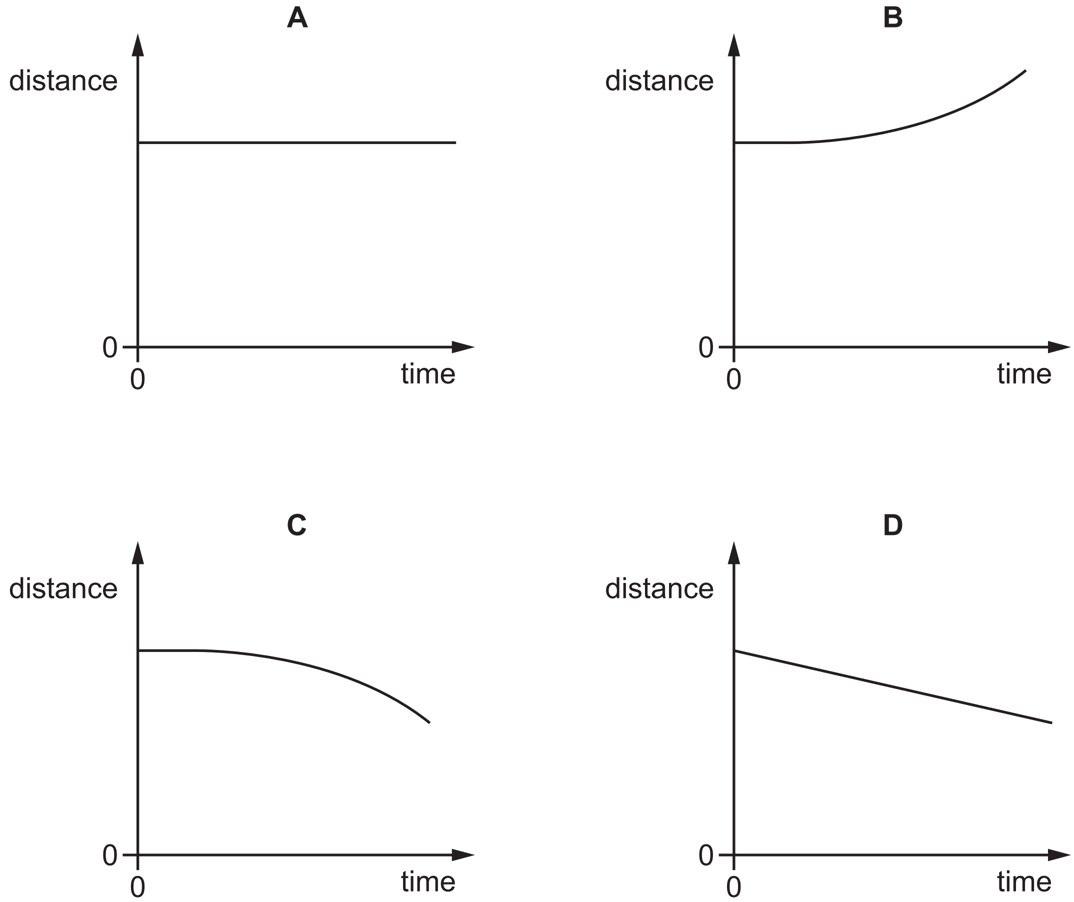
A cm → mm → m

B mm → cm → m

C mm → m → cm

D m → mm → cm

2 Which graph represents an object that is moving at constant speed?



3 Which statement about the equation shown is correct?

$$W = mg$$

A g is a force, m and W are not forces.

B m is a force, g and W are not forces.

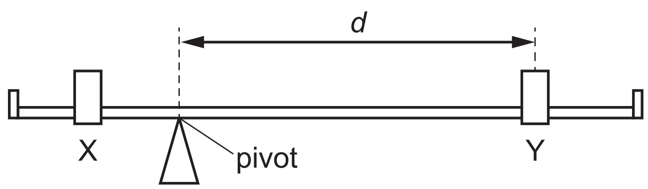
C W is a force, g and m are not forces.

D None of g, m and W are forces.

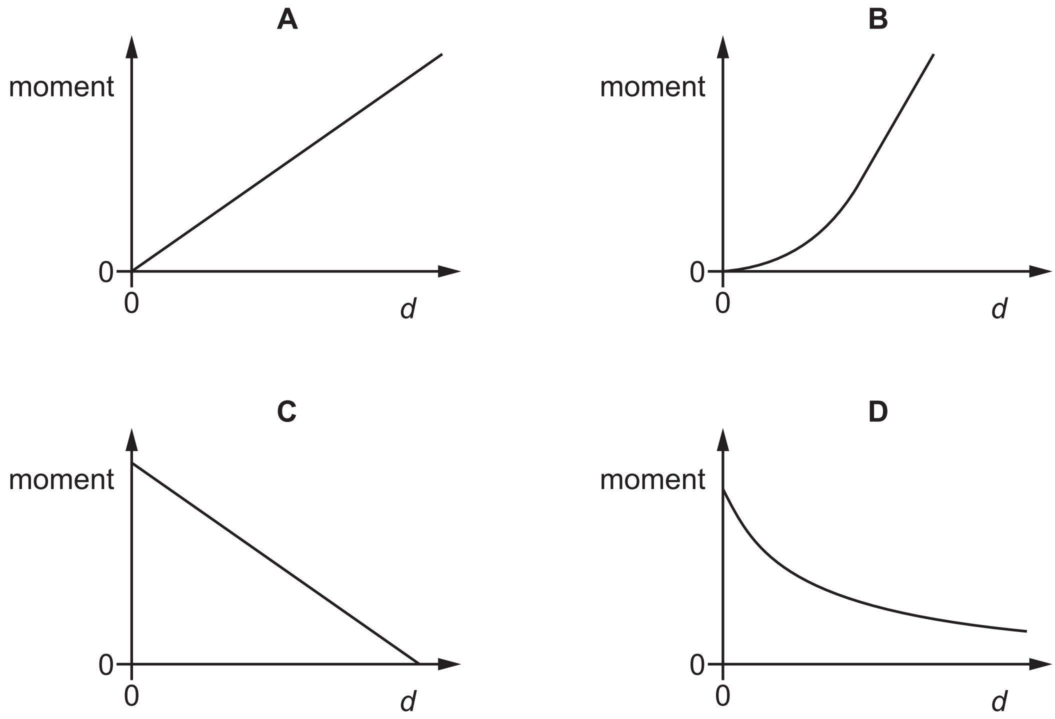
4 Which substance in the table has the lowest density?

	substance	mass / g	volume / cm^3
A	nylon	1.2	1.0
B	cotton	1.5	1.0
C	olive oil	1.8	2.0
D	water	2.0	2.0

5 The diagram shows a simple balance. The two loads, X and Y, can be moved along the beam.



Which graph shows how the moment produced by load Y varies as the perpendicular distance *d* from the pivot changes?



6 A spacecraft is travelling in space with no resultant force and no resultant moment acting on it.

Which statement about the spacecraft is correct?

A Its direction is changing.

B It is in equilibrium.

C Its speed is decreasing.

D Its speed is increasing.

7 A student carries out an investigation by pulling four different boxes across the floor.

The results are shown in the table.

On which box is the most work done?

	frictional force needed to pull the box / N	distance moved across the floor / m
A	5	4
B	10	2
C	15	2
D	20	4

8 Electrical energy may be obtained from nuclear fission.

In which order is the energy transferred in this process?

A nuclear fuel → generator → reactor and boiler → turbines

B nuclear fuel → generator → turbines → reactor and boiler

C nuclear fuel → reactor and boiler → generator → turbines

D nuclear fuel → reactor and boiler → turbines → generator

9 A stone falls.

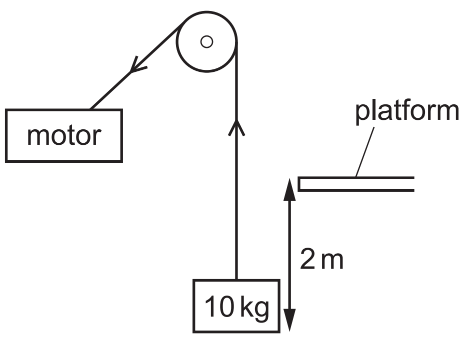
Which row gives the energy changes?

	gravitational potential energy	kinetic energy
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

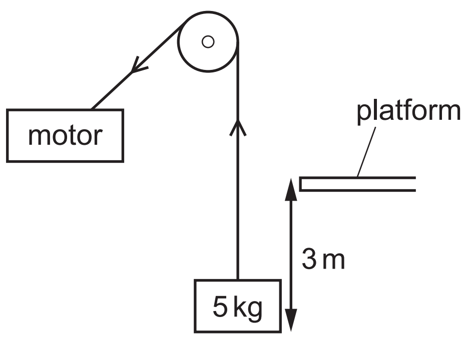
10 A rope, connected to a pulley system and motor, is used to lift different objects through different distances. The time taken to lift each object is the same. The diagrams are not to scale.

Which motor requires the greatest power?

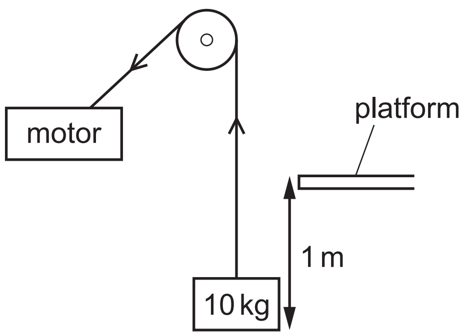
A



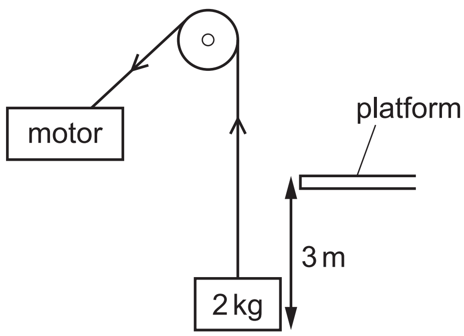
B



C

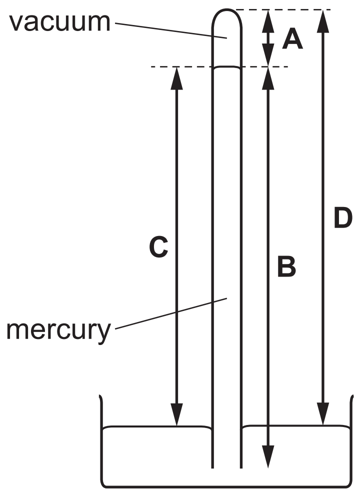


D



11 The diagram shows a simple mercury barometer.

Which length is used to indicate atmospheric pressure in a simple mercury barometer?



12 A liquid is evaporating. The liquid is not boiling.

Which statement about the liquid is correct at an instant in time?

A Any molecule can escape, and from any part of the liquid.

B Any molecule can escape, but only from the liquid's surface.

C Only molecules with enough energy can escape, and only from the liquid's surface.

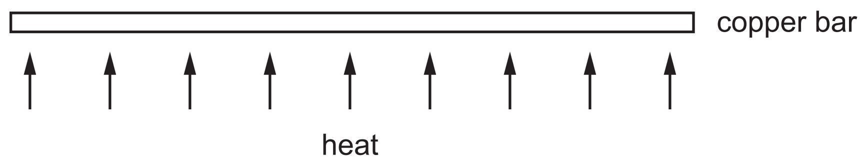
D Only molecules with enough energy can escape, but from any part of the liquid.

13 The temperature of the gas in a sealed container of constant volume decreases from 20 °C to 12 °C.

Which row is correct?

	pressure of the gas in the container	average speed of the molecules of gas
A	decreases	decreases
B	stays the same	increases
C	increases	stays the same
D	stays the same	decreases

14 A long, thin bar of copper is heated gently and evenly along its length.



What happens to the bar?

A It becomes less heavy.

B It becomes longer.

C It becomes shorter.

D It bends at the ends.

15 A teacher makes the statement, ‘Object P has a higher thermal capacity than object Q.’

What does this statement mean?

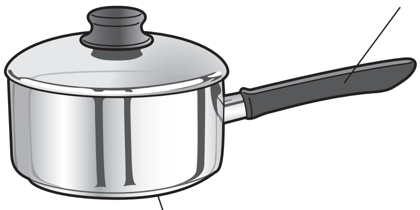
A Object P has a higher melting point than object Q.

B Object P has a lower melting point than object Q.

C The increase in temperature of object P is greater than that of object Q for the same increase in internal energy.

D The increase in temperature of object P is smaller than that of object Q for the same increase in internal energy.

16 The diagram shows a pan used for cooking food.



Which row is correct for the materials used to make the base and the handle of the pan?

	base of pan	handle of pan
A	good thermal conductor	good thermal conductor
B	good thermal conductor	poor thermal conductor
C	poor thermal conductor	good thermal conductor
D	poor thermal conductor	poor thermal conductor

17 The diagrams show two patterns produced by water waves.

diagram 1

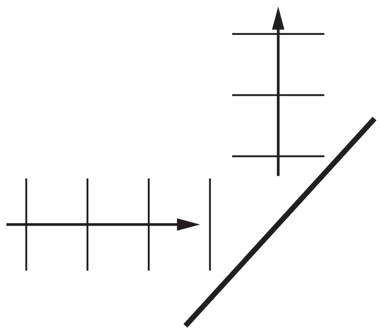
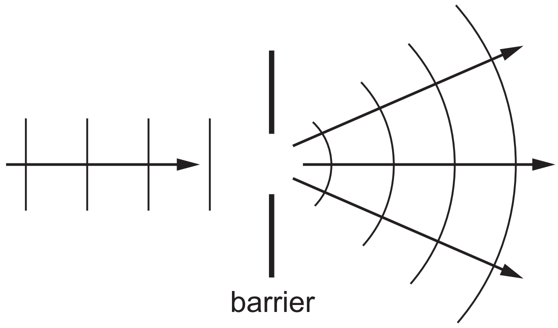


diagram 2



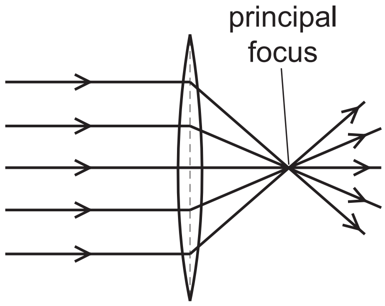
Which two effects are shown in the diagrams?

	diagram 1	diagram 2
A	reflection	diffraction
B	reflection	refraction
C	refraction	diffraction
D	refraction	reflection

18 Which row is **not** correct for a wave on the surface of water?

	quantity	usual unit
A	amplitude	m
B	frequency	Hz
C	wavelength	λ
D	speed	m / s

19 A thin, converging lens causes parallel rays of light to converge to a single point known as the principal focus.



Which statement explains this?

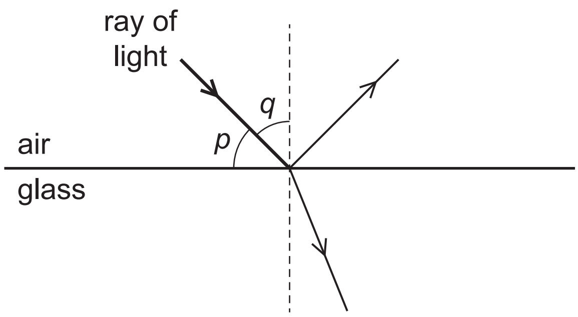
A The light diffracts.

B The light disperses.

C The light reflects.

D The light refracts.

20 The diagram shows a ray of light in air incident on a glass block. Some of the light is refracted and some of the light is reflected. Two angles, *p* and *q*, are marked on the diagram.



Which row gives the angle of incidence and states whether total internal reflection occurs?

	angle of incidence	total internal reflection
A	*p*	no
B	*p*	yes
C	*q*	no
D	*q*	yes

21 The letter F is reflected in a mirror.

What does the optical image look like?

22 Visible light, X-rays and microwaves are all components of the electromagnetic spectrum.

Which statement about the waves is correct?

A In a vacuum, microwaves travel faster than visible light and have a shorter wavelength.

B In a vacuum, microwaves travel at the same speed as visible light and have a shorter wavelength.

C In a vacuum, X-rays travel faster than visible light and have a shorter wavelength.

D In a vacuum, X-rays travel at the same speed as visible light and have a shorter wavelength.

23 Which radiation has a higher frequency than red light?

A ultraviolet

B radio waves

C microwaves

D infrared

24 What is ultrasound?

A sound waves that are so loud that they damage human hearing

B sound waves that are too high-pitched for humans to hear

C sound waves that are too low-pitched for humans to hear

D sound waves that are too quiet for humans to hear

25 Which material is magnetic?

A aluminium

B copper

C iron

D silver

26 Which circuit symbol represents a component used to measure electric current?

27 Four wires made of the same metal have different lengths and different diameters.

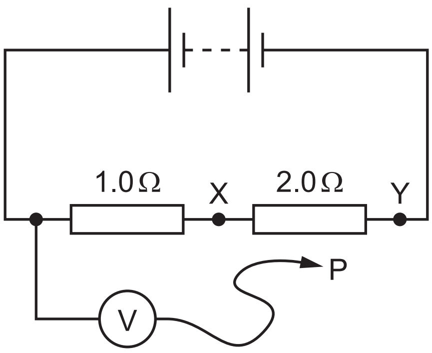
Which wire has the lowest resistance?

	length	diameter
A	long	large
B	long	small
C	short	large
D	short	small

28 The diagram shows a circuit containing two resistors of resistance 1.0 Ω and 2.0 Ω.

A voltmeter is connected across the 1.0 Ω resistor by connecting P to X.

The reading on the voltmeter is 6.0 V.



P is moved to point Y in the circuit.

What is the new reading on the voltmeter?

A 3.0 V **B** 6.0 V **C** 12 V **D** 18 V

29 Which device is used to measure the flow of charge in an electrical circuit?

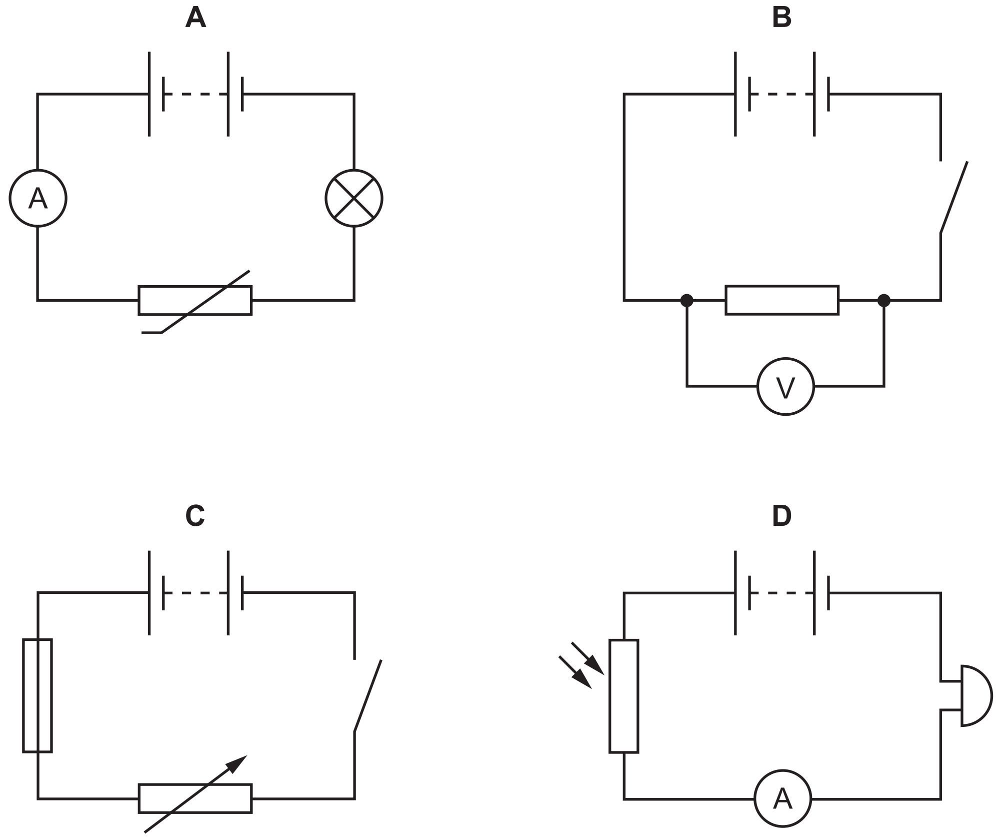
A ammeter

B voltmeter

C battery

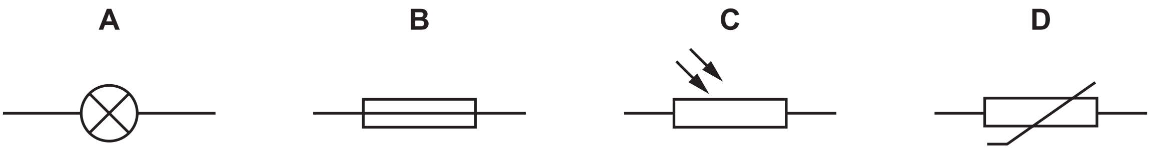
D newton meter

30 Which circuit contains a fuse?



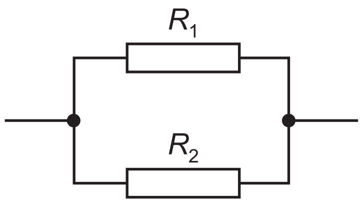
31 A student designs a circuit to turn on a fan when the temperature increases.

Which component does the student need to use in her circuit?



32 Two resistors, with resistances R_1 and R_2, are connected in parallel.

The resistance R_1 is greater than the resistance R_2.



What is the resistance of the parallel combination?

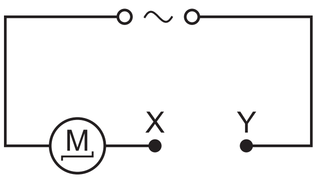
A less than either R_1 or R_2

B equal to R_1

C equal to R_2

D the average of R_1 and R_2

33 The diagram shows a motor connected to an a.c. supply. The circuit is incomplete.



Which device needs to be connected between point X and point Y to prevent the wires from overheating if a fault in the motor causes the current to get too high?

A an ammeter

B a fuse

C a transformer

D a length of thick copper wire

34 A hairdresser is using a hairdryer with a plastic casing. He notices that there is no wire attached to the earth pin of the plug.

Why is an earth connection **not** needed?

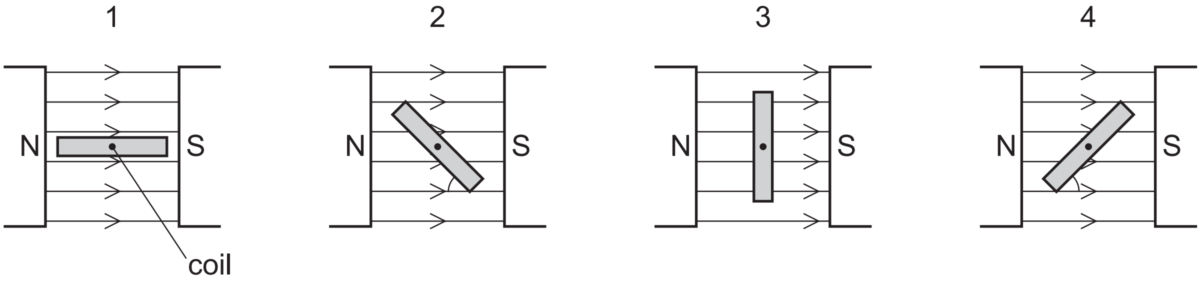
A Plastic is an insulator.

B The hairdresser only touches the handle of the dryer.

C The hairdryer uses a.c. so cannot give the hairdresser a shock.

D Wet hands do not conduct electricity.

35 Four positions of a current-carrying coil in a magnetic field, as in a d.c. motor, are shown. In diagrams 2 and 4, the coil is at an angle of 45° to the field lines.



Which row is correct?

	turning effect of the forces in positions 1 and 3	turning effect of the forces in positions 2 and 4
A	different	different
B	different	same
C	same	different
D	same	same

36 An electric drill, operating from a supply voltage of 240 V, uses a current of 3.5 A.

Which rating of fuse should be used to protect the drill's cable?

A 250 V **B** 200 V **C** 5 A **D** 3 A

37 A very important experiment improved scientists' understanding of the structure of matter.

The experiment involved α-particles being fired at a thin, gold foil.

What happened?

A All the α-particles were absorbed by the nuclei of the gold atoms.

B All the α-particles were unaffected by the gold atoms.

C Some of the α-particles were attracted by the neutrons in the nuclei of the gold atoms.

D Some of the α-particles were repelled by the protons in the nuclei of the gold atoms.

38 A sample contains 0.0016 g of a radioactive isotope.

After 4.0 hours the mass of the radioactive isotope in the sample falls to 0.00080 g.

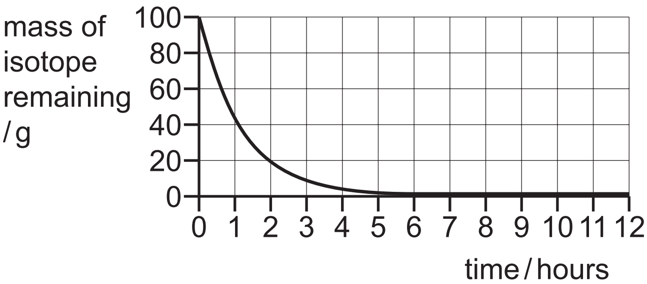
What is the half-life of the radioactive isotope?

A 2.0 hours **B** 4.0 hours **C** 8.0 hours **D** 16 hours

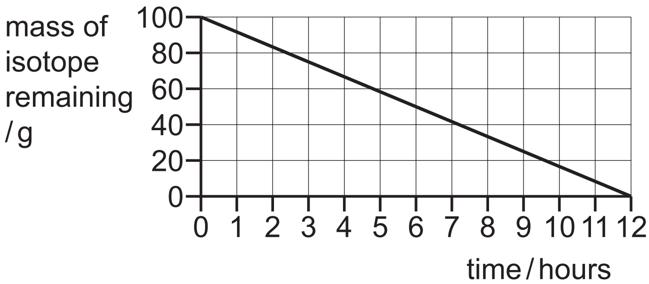
39 A sample of a radioactive isotope has a mass of 100 g. The half-life of the radioactive isotope is 6.0 hours.

Which graph shows the decay for this isotope?

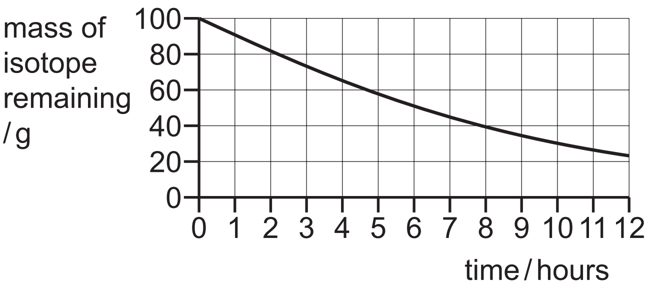
A



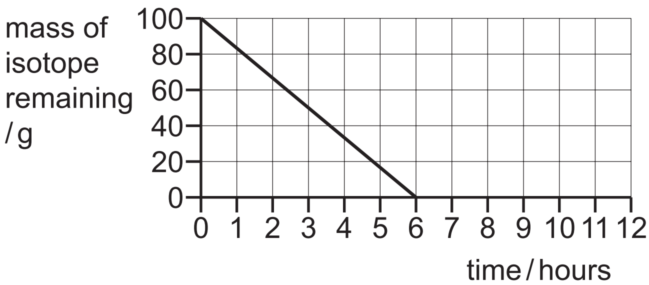
B



C



D



40 Which statement best describes background radiation?

A any harmful level of radiation

B radiation that is only found in space

C radiation from natural sources

D radiation that is absorbed by rocks

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PHYSICS **0625/11**

Paper 1 Multiple Choice (Core) **October/November 2021**

45 minutes

You must answer on the multiple choice answer sheet.

7987287000

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = $10\,m/s^2$).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.

1 A student measures the volume of a quantity of water.

Which apparatus is suitable?

A a balance

B a measuring cylinder

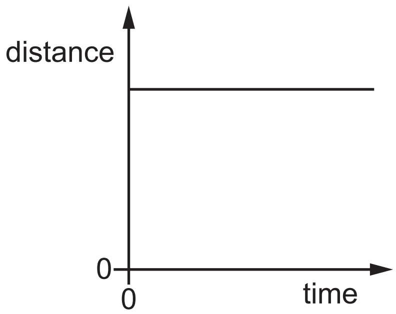
C a ruler

D a thermometer

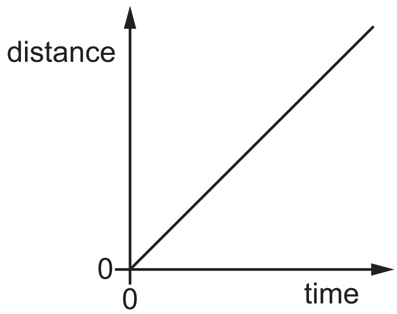
2 The diagrams show distance–time graphs for four objects.

Which graph represents an object moving with an increasing speed?

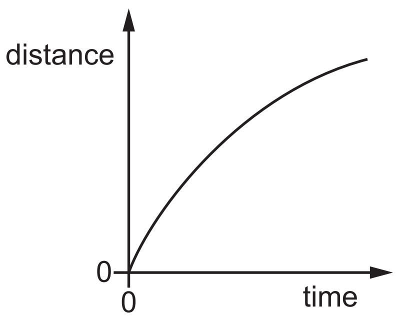
A



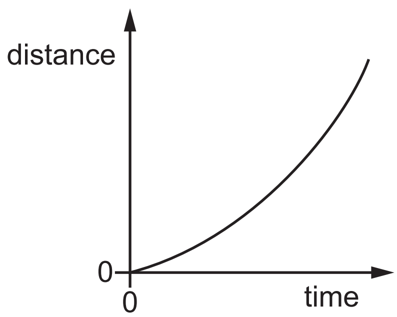
B



C



D



3 The gravitational field strength on the Moon is smaller than that on the Earth.

A scientist examines a rock which has been brought back from the Moon.

He measures three quantities.

1 the density of the rock

2 the mass of the rock

3 the weight of the rock

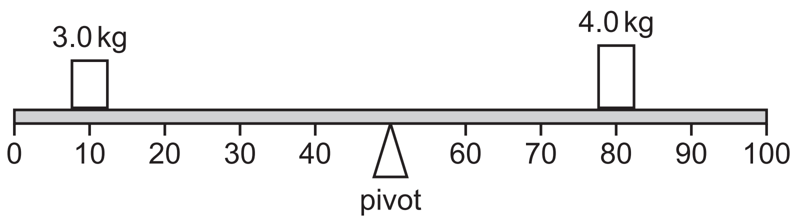
Which quantities are the same size on the surface of the Earth and on the surface of the Moon?

A 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

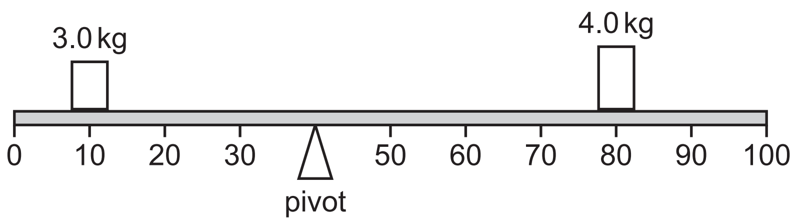
4 Which substance in the table has the lowest density?

	substance	mass / g	volume / cm^3
A	nylon	1.2	1.0
B	cotton	1.5	1.0
C	olive oil	1.8	2.0
D	water	2.0	2.0

5 A 100 cm beam balances as shown.



The pivot is moved 10 cm to the left.



What will be the effect of this change on the anticlockwise and clockwise moments about the pivot?

	anticlockwise moment	clockwise moment
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

6 A spacecraft is travelling in space with no resultant force and no resultant moment acting on it.

Which statement about the spacecraft is correct?

A Its direction is changing.

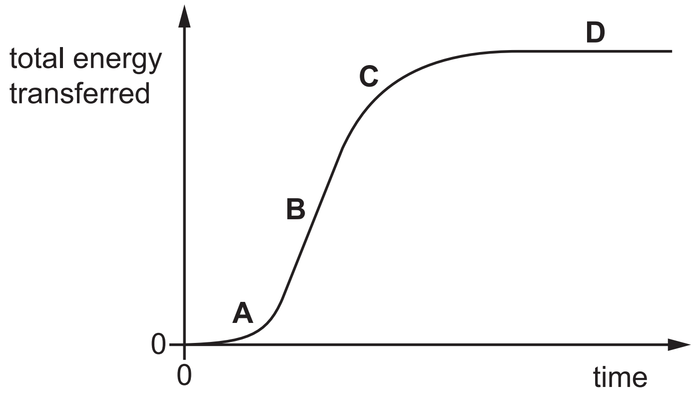
B It is in equilibrium.

C Its speed is decreasing.

D Its speed is increasing.

7 The graph shows the total energy transferred by an electric motor over a period of time.

In which region of the graph is the greatest power being developed by the motor?



8 Electrical energy may be obtained from nuclear fission.

In which order is the energy transferred in this process?

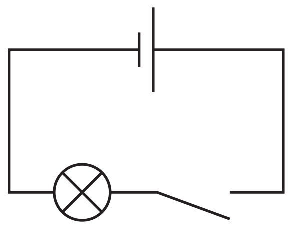
A nuclear fuel → generator → reactor and boiler → turbines

B nuclear fuel → generator → turbines → reactor and boiler

C nuclear fuel → reactor and boiler → generator → turbines

D nuclear fuel → reactor and boiler → turbines → generator

9 The diagram shows an electric circuit. When the switch is closed, the lamp is lit.



Which row states the type of energy stored in the cell and how this energy is usefully transferred to the lamp?

	type of energy stored in the cell	how this energy is usefully transferred to the lamp
A	chemical	by electric current
B	chemical	by light
C	electrical	by electric current
D	electrical	by light

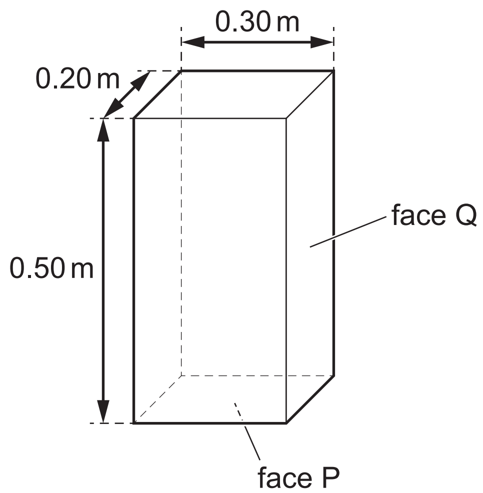
10 A scientist uses an electric motor to lift a load through a vertical distance of 2.0 m.

He then increases the input power to the motor and repeats the experiment. The efficiency of the motor does not change.

Which row correctly describes the effect that this has on the useful work done lifting the load and the time taken to lift it?

	work done	time taken
A	decreases	decreases
B	stays the same	decreases
C	decreases	stays the same
D	stays the same	stays the same

11 The box shown has a weight of 15 N.



The box is resting on a horizontal surface with face P in contact with the surface.

What is the change in pressure on the surface if the box falls over onto face Q?

A $0.0040\,\mathrm{m^2/N}$ **B** $0.0067\,\mathrm{m^2/N}$ **C** $100\,\mathrm{N/m^2}$ **D** $250\,\mathrm{N/m^2}$

12 A liquid is evaporating. The liquid is not boiling.

Which statement about the liquid is correct at an instant in time?

A Any molecule can escape, and from any part of the liquid.

B Any molecule can escape, but only from the liquid's surface.

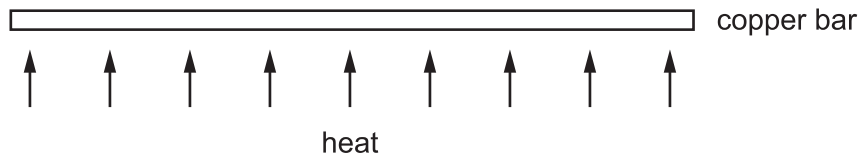
C Only molecules with enough energy can escape, and only from the liquid's surface.

D Only molecules with enough energy can escape, but from any part of the liquid.

13 Which row correctly describes the movement of particles in solids and liquids?

	solids	liquids
A	no movement	move around each other
B	no movement	vibration only
C	vibration only	move around each other
D	vibration only	vibration only

14 A long, thin bar of copper is heated gently and evenly along its length.



What happens to the bar?

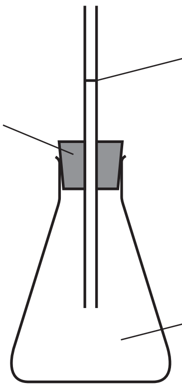
A It becomes less heavy.

B It becomes longer.

C It becomes shorter.

D It bends at the ends.

15 The diagram shows a flask which has been filled with liquid X at room temperature.



When the flask is placed in warm water, the liquid rises higher up the tube. When the flask is put in cold water, the liquid drops below the original level in the tube.

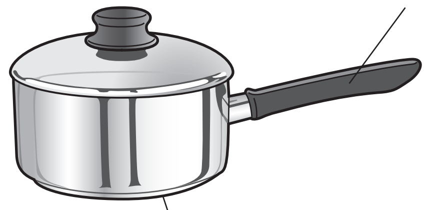
The experiment is repeated using an identical flask but a different liquid Y. The initial level of the liquid in the tube is the same as that in the original experiment.

Liquid Y expands more, per degree increase in temperature, than liquid X.

Which row is correct for the level of the liquid?

	rises most in warm water	falls most in cold water
A	X	X
B	X	Y
C	Y	X
D	Y	Y

16 The diagram shows a pan used for cooking food.



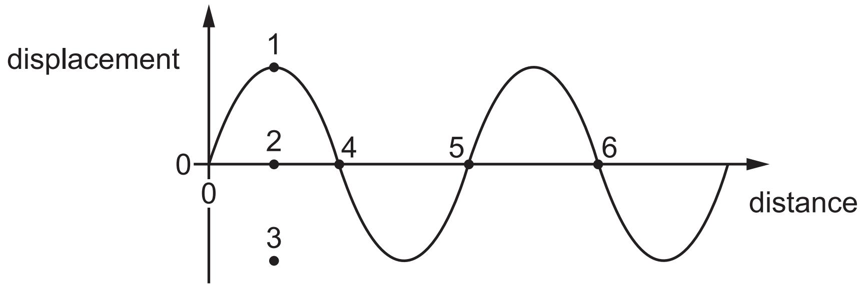
Which row is correct for the materials used to make the base and the handle of the pan?

	base of pan	handle of pan
A	good thermal conductor	good thermal conductor
B	good thermal conductor	poor thermal conductor
C	poor thermal conductor	good thermal conductor
D	poor thermal conductor	poor thermal conductor

17 Which row gives the correct description of a sound wave?

	action of wave	type of wave
A	transfers energy without transferring matter	longitudinal
B	transfers energy without transferring matter	transverse
C	transfers matter without transferring energy	longitudinal
D	transfers matter without transferring energy	transverse

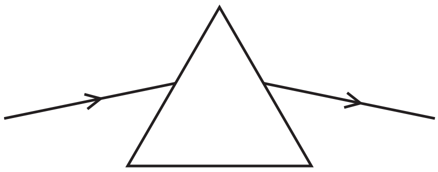
18 The diagram shows a wave.



Which row correctly indicates the amplitude and the wavelength of the wave?

	amplitude	wavelength
A	the distance between 1 and 2	the distance between 4 and 5
B	the distance between 1 and 2	the distance between 4 and 6
C	the distance between 1 and 3	the distance between 4 and 5
D	the distance between 1 and 3	the distance between 4 and 6

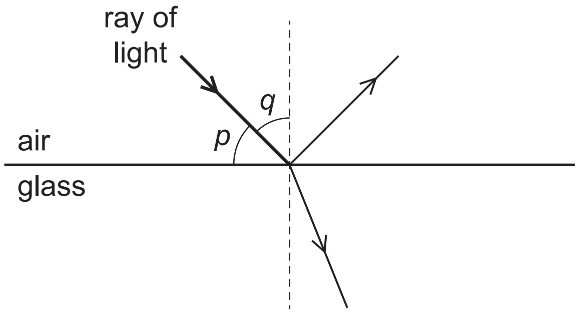
19 A ray of green light passes through a glass prism as shown.



Which colours of light refract as shown in the table?

	refract more than green	refract less than green
A	red	blue
B	red	yellow
C	violet	blue
D	violet	yellow

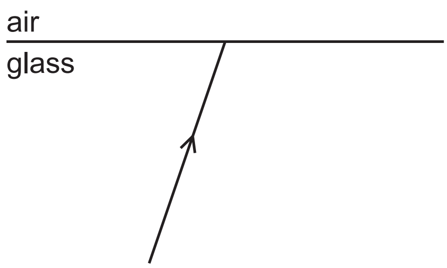
20 The diagram shows a ray of light in air incident on a glass block. Some of the light is refracted and some of the light is reflected. Two angles, *p* and *q*, are marked on the diagram.



Which row gives the angle of incidence and states whether total internal reflection occurs?

	angle of incidence	total internal reflection
A	*p*	no
B	*p*	yes
C	*q*	no
D	*q*	yes

21 The diagram shows a ray of light in glass incident on the surface between the glass and air.



What happens if the angle of incidence is made larger than the critical angle for the glass?

A The angle of refraction becomes equal to 90°.

B There is a refracted ray and a ray reflected inside the glass.

C There is a refracted ray only.

D There is only a ray reflected inside the glass.

22 Radiation from which part of the electromagnetic spectrum is used in the remote controller for a television?

A infrared waves

B microwaves

C radio waves

D ultraviolet waves

23 Two rays of light are different colours.

Which row is correct?

	speed of the two colours in a vacuum	wavelengths of the two colours in a vacuum
A	different	different
B	different	the same
C	the same	different
D	the same	the same

24 What is ultrasound?

A sound waves that are so loud that they damage human hearing

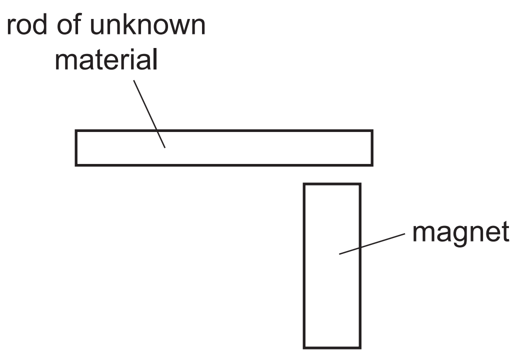
B sound waves that are too high-pitched for humans to hear

C sound waves that are too low-pitched for humans to hear

D sound waves that are too quiet for humans to hear

25 A bar magnet is brought near to a rod of unknown material.

The unknown rod is attracted to both ends of the bar magnet.



Which material is the rod made from?

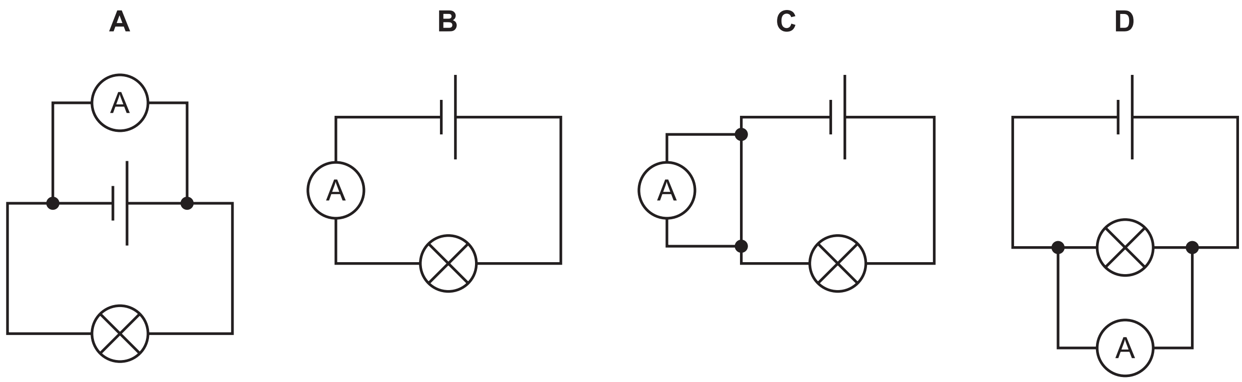
A aluminium

B magnetised steel

C unmagnetised iron

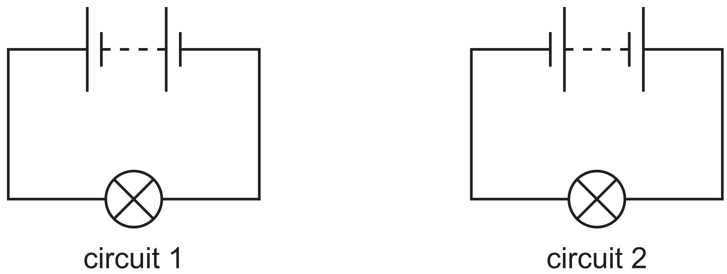
D plastic

26 In which circuit is the ammeter measuring the flow of charge through the lamp?



27 In circuit 1, a negative charge flows in a clockwise direction. The bulb is bright.

In circuit 2, the battery is reversed as shown. The bulb is equally bright.



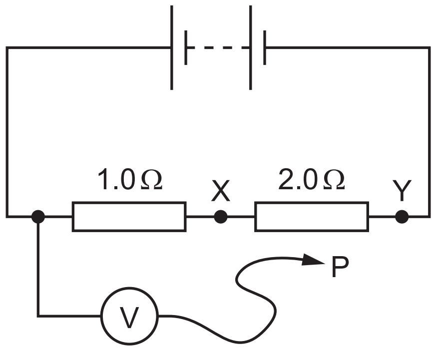
Which charge flows in circuit 2 and in which direction?

	charge	direction
A	negative	anticlockwise
B	negative	clockwise
C	positive	anticlockwise
D	positive	clockwise

28 The diagram shows a circuit containing two resistors of resistance 1.0 Ω and 2.0 Ω.

A voltmeter is connected across the 1.0 Ω resistor by connecting P to X.

The reading on the voltmeter is 6.0 V.

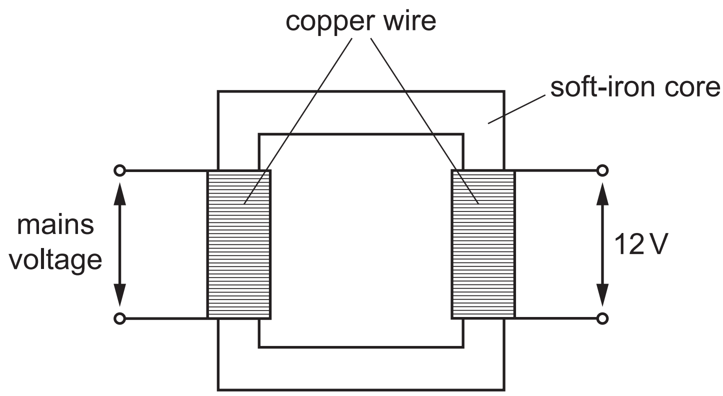


P is moved to point Y in the circuit.

What is the new reading on the voltmeter?

A 3.0 V **B** 6.0 V **C** 12 V **D** 18 V

29 The step-down transformer shown reduces mains voltage to 12 V.



When the transformer is used, some energy is transferred to the surroundings.

Which type of energy is transferred to the surroundings?

A chemical energy

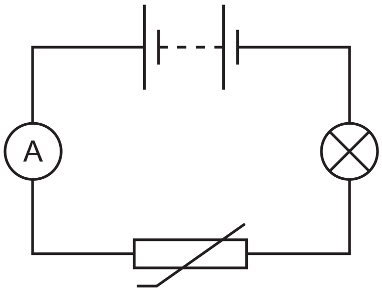
B light energy

C thermal energy

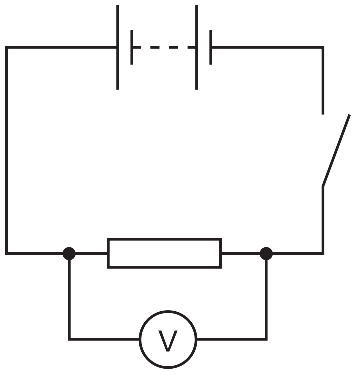
D elastic energy

30 Which circuit contains a fuse?

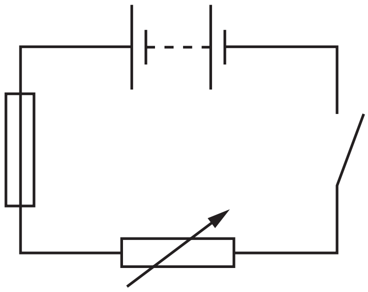
A



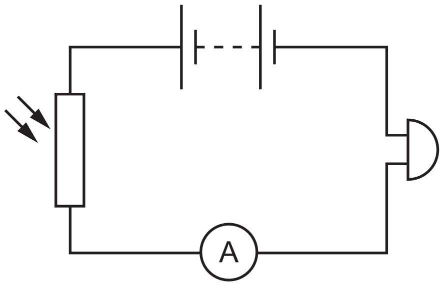
B



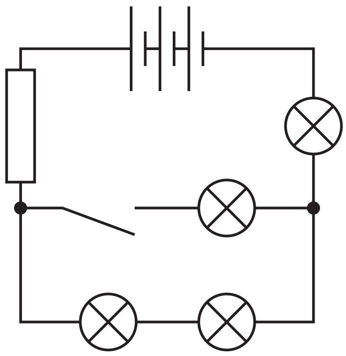
C



D



31 The diagram shows an electric circuit.

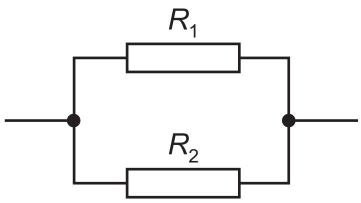


Which row correctly shows the numbers of different components in the circuit?

	cells	lamps	switches
A	1	1	4
B	1	4	1
C	3	1	4
D	3	4	1

32 Two resistors, with resistances R_1 and R_2, are connected in parallel.

The resistance R_1 is greater than the resistance R_2.



What is the resistance of the parallel combination?

A less than either R_1 or R_2

B equal to R_1

C equal to R_2

D the average of R_1 and R_2

33 The metal cases of electrical appliances are connected to an earth wire.

Which statement is **not** correct?

A The live wire may become loose and touch the metal case.

B If the metal case becomes live, the earth wire conducts current to the ground.

C The earth wire needs to have a high resistance.

D Earthing metal cases helps prevent a person from receiving an electric shock.

34 A teacher asks, ‘Why do we put a fuse in a mains circuit?’

Student 1 says, ‘It protects the wiring from overheating.’ Student 2 says, ‘It protects us from getting a shock if we touch the live wire.’

Who is correct?

A both students

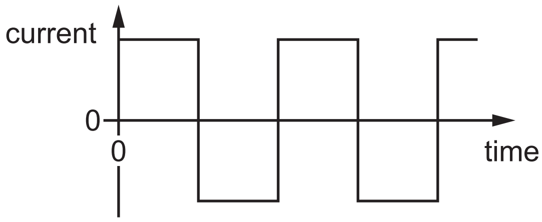
B neither student

C student 1 only

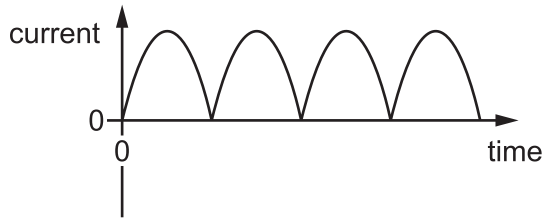
D student 2 only

35 Which graph represents an alternating current (a.c.)?

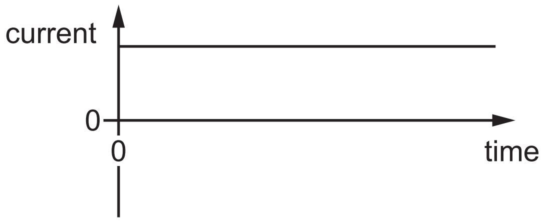
A



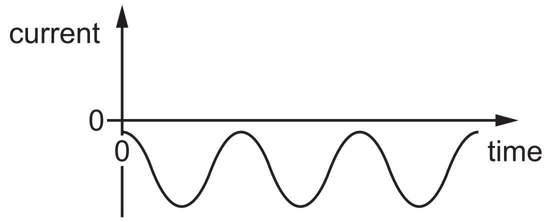
B



C



D



36 The current in a small electric heater is 4.0 A.

The cable connected to the heater is able to carry currents up to 10 A.

Fuses rated 1 A, 3 A, 5 A and 13 A are available.

Which fuse should be used?

A 1 A **B** 3 A **C** 5 A **D** 13 A

37 A very important experiment improved scientists' understanding of the structure of matter.

The experiment involved α-particles being fired at a thin, gold foil.

What happened?

A All the α-particles were absorbed by the nuclei of the gold atoms.

B All the α-particles were unaffected by the gold atoms.

C Some of the α-particles were attracted by the neutrons in the nuclei of the gold atoms.

D Some of the α-particles were repelled by the protons in the nuclei of the gold atoms.

38 Some sources of background radiation are natural and others are due to human activity.

Which source is natural?

A medical X-rays

B nuclear weapons testing

C radioactive waste from power stations

D radon gas from rocks

39 A radioactive material is placed near a detector.

The detector shows a count rate of 28 000 counts / min.

When a piece of card is put between the material and the counter, the reading decreases to 25 000 counts / min.

When an aluminium sheet is put between the material and the counter, the reading remains at 25 000 counts / min.

When a sheet of lead is put between the material and the counter, the reading decreases to 19 000 counts / min.

What is being emitted by the radioactive material?

A α, β and γ-radiation

B α and β-radiation only

C α and γ-radiation only

D β and γ-radiation only

40 A radioactive isotope has a half-life of 3 years.

A sample gives a count rate of 100 counts / min on a detector.

Which calculation is used to predict the count rate after 12 years?

A $100 \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$

B $100 \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$

C $100 \times \frac{3}{12}$

D $100 \times \frac{12}{3} \times \frac{1}{2}$

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Cambridge IGCSE™

PHYSICS **0625/13**

Paper 1 Multiple Choice (Core) **October/November 2021**

MARK SCHEME

Maximum Mark: 40

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This document consists of **3** printed pages.

[Turn over

Question	Answer	Marks
1	C	**1**
2	A	**1**
3	A	**1**
4	C	**1**
5	C	**1**
6	B	**1**
7	B	**1**
8	D	**1**
9	A	**1**
10	C	**1**
11	D	**1**
12	C	**1**
13	B	**1**
14	B	**1**
15	A	**1**
16	B	**1**
17	A	**1**
18	C	**1**
19	D	**1**
20	C	**1**
21	A	**1**
22	D	**1**
23	C	**1**
24	B	**1**
25	B	**1**
26	C	**1**
27	B	**1**
28	D	**1**

Question	Answer	Marks
29	D	**1**
30	C	**1**
31	B	**1**
32	A	**1**
33	A	**1**
34	C	**1**
35	A	**1**
36	A	**1**
37	D	**1**
38	A	**1**
39	B	**1**
40	D	**1**

Cambridge IGCSE™

PHYSICS **0625/12**

Paper 1 Multiple Choice (Core) **October/November 2021**

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Question	Answer	Marks
1	B	**1**
2	D	**1**
3	C	**1**
4	C	**1**
5	A	**1**
6	B	**1**
7	D	**1**
8	D	**1**
9	B	**1**
10	A	**1**
11	C	**1**
12	C	**1**
13	A	**1**
14	B	**1**
15	D	**1**
16	B	**1**
17	A	**1**
18	C	**1**
19	D	**1**
20	C	**1**
21	B	**1**
22	D	**1**
23	A	**1**
24	B	**1**
25	C	**1**
26	A	**1**
27	C	**1**
28	D	**1**

Question	Answer	Marks
29	A	**1**
30	C	**1**
31	D	**1**
32	A	**1**
33	B	**1**
34	A	**1**
35	B	**1**
36	C	**1**
37	D	**1**
38	B	**1**
39	C	**1**
40	C	**1**

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Cambridge IGCSE™

PHYSICS	**0625/11**
Paper 1 Multiple Choice (Core)	**October/November 2021**
MARK SCHEME	
Maximum Mark: 40	

Published

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Question	Answer	Marks
1	B	**1**
2	D	**1**
3	A	**1**
4	C	**1**
5	B	**1**
6	B	**1**
7	B	**1**
8	D	**1**
9	A	**1**
10	B	**1**
11	C	**1**
12	C	**1**
13	C	**1**
14	B	**1**
15	D	**1**
16	B	**1**
17	A	**1**
18	B	**1**
19	D	**1**
20	C	**1**
21	D	**1**
22	A	**1**
23	C	**1**
24	B	**1**
25	C	**1**
26	B	**1**
27	A	**1**
28	D	**1**

Question	Answer	Marks
29	C	**1**
30	C	**1**
31	D	**1**
32	A	**1**
33	C	**1**
34	C	**1**
35	A	**1**
36	C	**1**
37	D	**1**
38	D	**1**
39	C	**1**
40	A	**1**

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PHYSICS **0625/13**

Paper 1 Multiple Choice (Core) **October/November 2020**

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m / s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
- Any rough working should be done on this question paper.

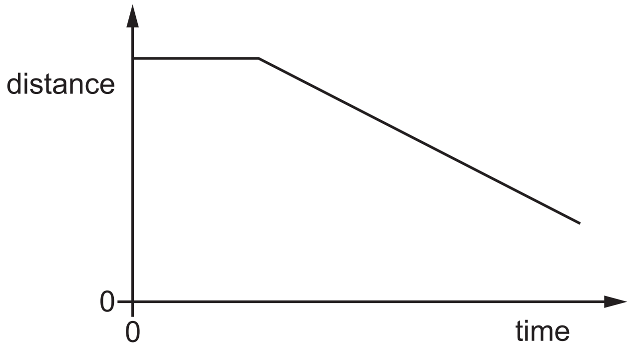
This document has **16** pages. Blank pages are indicated.

1 A student uses a metre rule to measure the length of a sheet of paper.

Which measurement is shown to the nearest millimetre?

A 0.3 m **B** 0.29 m **C** 0.293 m **D** 0.2932 m

2 The diagram shows the distance–time graph for the motion of an object.



How can the motion of the object be described?

A at rest, then constant deceleration

B at rest, then constant speed

C constant speed, then constant acceleration

D constant speed, then constant deceleration

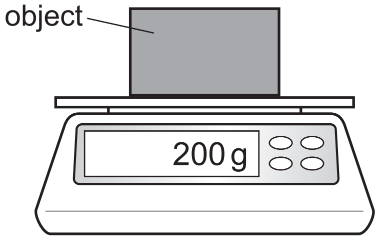
3 An athlete runs 300 m up a hill in 100 s.

She then runs down the same hill in 50 s.

What is her average speed for the whole run?

A 2.0 m/s **B** 4.0 m/s **C** 8.0 m/s **D** 9.0 m/s

4 The diagram shows an object on a balance. The reading on the balance is shown.



Which quantity is shown?

A the mass of the object

B the pressure exerted by the object

C the volume of the object

D the weight of the object

5 Which quantity is weight an example of?

A acceleration

B force

C mass

D pressure

6 A student is asked to predict whether a solid floats in a liquid.

Which information does the student require?

A the density of the liquid and the mass of the solid

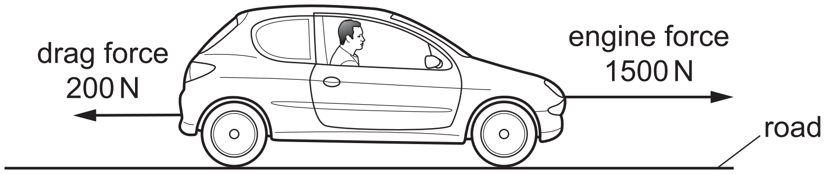
B the density of the solid and the density of the liquid

C the density of the solid and the mass of the liquid

D the mass of the solid and the mass of the liquid

7 The diagram shows a car moving along a road.

The force due to the engine is 1500 N and the total drag force is 200 N.



What is the motion of the car?

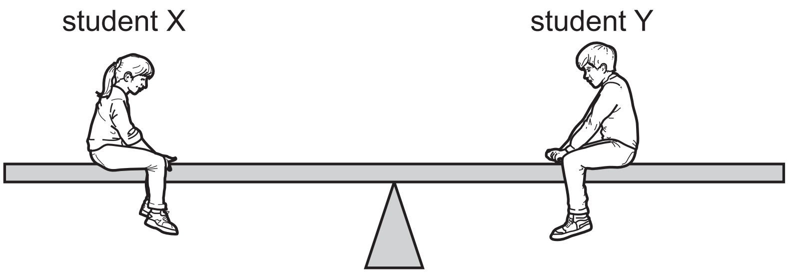
A constant speed

B decreasing speed

C increasing speed

D reversing

8 Students X and Y are sitting on a seesaw. Student X has a weight of 400 N and student Y has a weight of 600 N. The seesaw is in equilibrium.



Which statement correctly describes why the seesaw is in equilibrium?

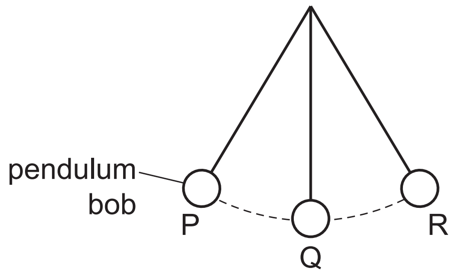
A The resultant force is zero and the resultant moment is zero.

B The resultant force is 200 N and the resultant moment is zero.

C The sum of the downward forces is zero and the resultant moment is zero.

D The total downward force is 1000 N and the resultant moment is 200 N m.

9 The diagram shows a frictionless pendulum.



The pendulum bob swings from point P through point Q to point R then back to P.

At which point is the energy of the pendulum bob greatest?

A at point P

B at point Q

C at point R

D it is the same at points P, Q and R

10 Which statement correctly compares the production of electricity using wind and the production of electricity using nuclear fission?

A Wind is less reliable than nuclear fission; both wind and nuclear fission are renewable.

B Wind is less reliable than nuclear fission; wind is renewable, but nuclear fission is not.

C Wind is more reliable than nuclear fission; both wind and nuclear fission are renewable.

D Wind is more reliable than nuclear fission; wind is renewable, but nuclear fission is not.

11 To calculate the power produced by a force, the size of the force must be known.

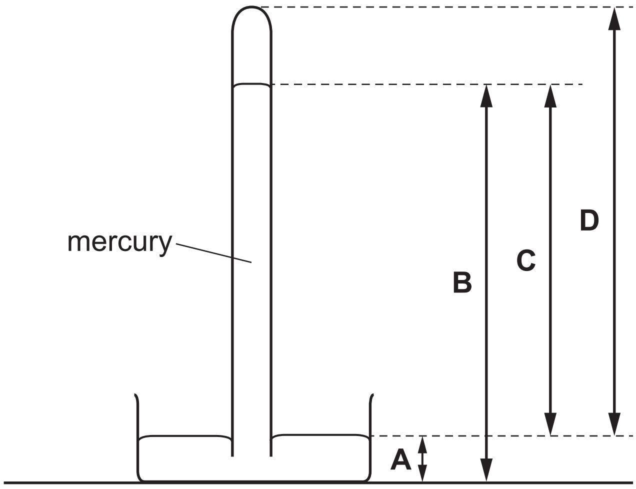
What else needs to be known to calculate the power?

	the distance that the force moves the object	the time for which the force acts on the object
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

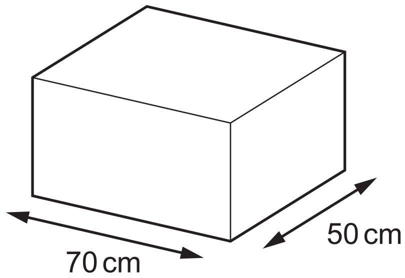
key
✓ = needed
✗ = not needed

12 The diagram shows a mercury barometer.

Which height is used as a measurement of atmospheric pressure?



13 A large box has a weight of 700 N. The box is placed on the floor.



What is the pressure on the floor due to the box?

A $0.20\,N/m^2$ **B** $10\,N/m^2$ **C** $245\,N/m^2$ **D** $2000\,N/m^2$

14 On a warm day, a driver checks the air pressure in a car tyre. At night, the temperature drops and the air pressure in the tyre decreases. There are no air leaks in the tyre.

Why does the pressure decrease?

A The air molecules in the tyre move more slowly.

B The air molecules in the tyre stop moving.

C The volume of the air in the tyre decreases.

D The volume of the air in the tyre increases.

15 A textbook gives the description of a thermal process as 'more-energetic molecules escape from the surface of a liquid which causes the liquid to cool'.

Which process is being described?

A boiling

B Brownian motion

C condensation

D evaporation

16 Which physical property changes when temperature is measured with a liquid-in-glass thermometer?

A electromotive force

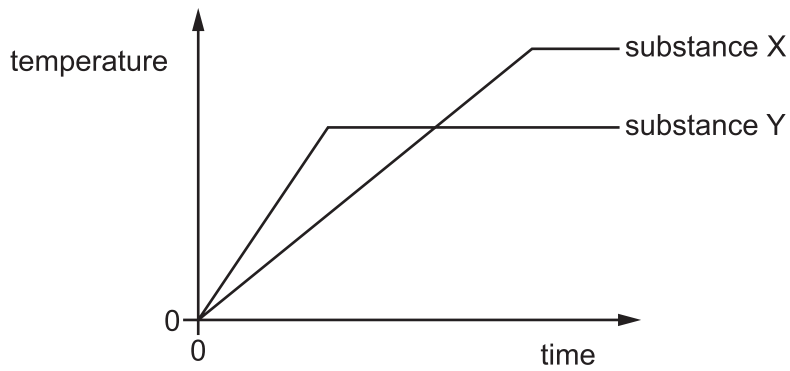
B pressure

C resistance

D volume

17 Two different pure substances X and Y are heated. Both substance X and substance Y are initially in the solid state.

The graph shows how the temperature of each substance changes with time.



What does the graph tell us about the substances?

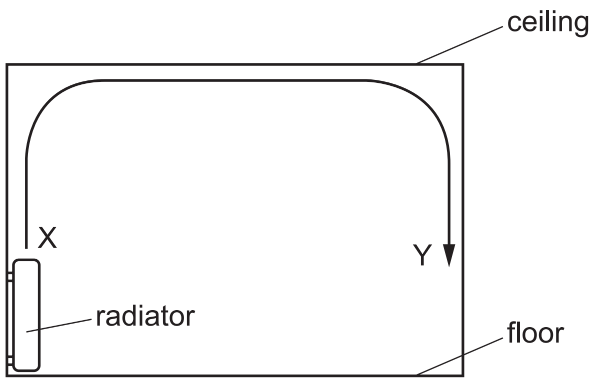
A Substance X has a lower boiling point than substance Y.

B Substance X has a lower melting point than substance Y.

C Substance Y has a lower boiling point than substance X.

D Substance Y has a lower melting point than substance X.

18 The diagram shows the view of a room heated by a radiator. The arrowed line from X to Y is the path of the convection current in the air.



Which row about the air temperature and the air density at X and at Y is correct?

	air temperature	air density
A	higher at X	higher at X
B	higher at X	higher at Y
C	higher at Y	higher at Y
D	higher at Y	higher at X

19 Which thermal process can transfer energy through a vacuum?

A conduction

B convection

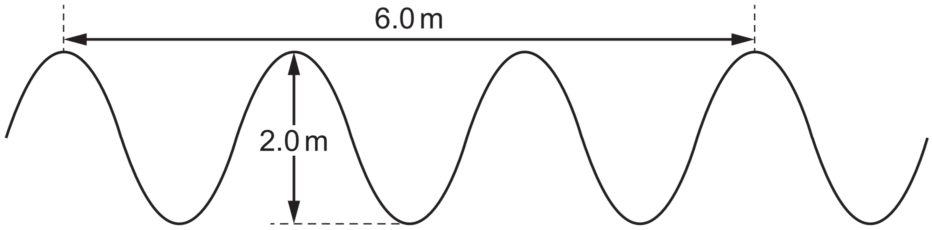
C evaporation

D radiation

20 Which row correctly describes light waves?

	wave type	direction of vibrations
A	longitudinal	parallel to direction of wave travel
B	longitudinal	perpendicular to direction of wave travel
C	transverse	parallel to direction of wave travel
D	transverse	perpendicular to direction of wave travel

21 The diagram shows a wave. It is not drawn to scale.



What are the amplitude and the wavelength of the wave?

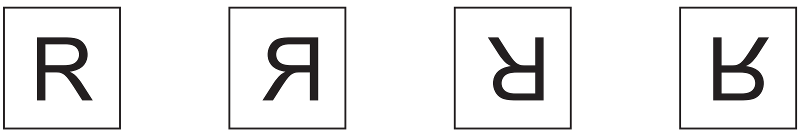
	amplitude / m	wavelength / m
A	1.0	1.0
B	1.0	2.0
C	2.0	2.0
D	2.0	3.0

22 A boy is having his eyes tested. A letter is printed on a card placed over his head. He sees the card in a plane mirror placed on the far wall of the room.

He sees the letter ‘R’ in the mirror.

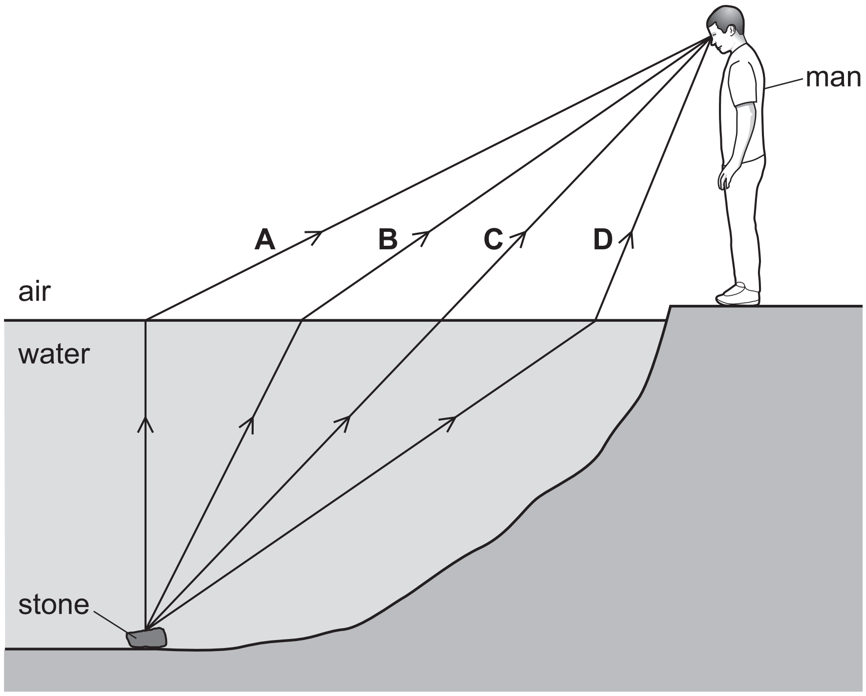
How is it printed on the card?

A **B** **C** **D**

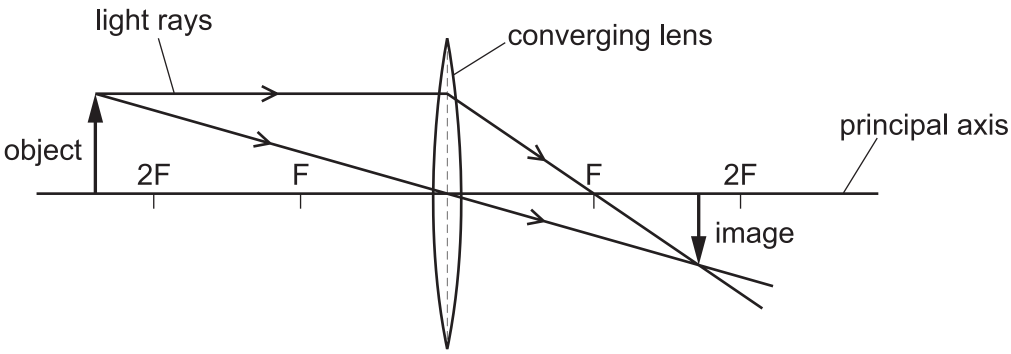


23 A man sees a stone at the bottom of a pool of water.

Which path could be taken by light from the stone to the man?



24 The diagram shows the image made by a converging lens.



Which statement describes the image formed?

A diminished and inverted

B diminished and upright

C enlarged and inverted

D enlarged and upright

25 The diagram shows three types of electromagnetic radiation listed in a particular order. The electromagnetic radiation is travelling in a vacuum.

microwaves infrared X-rays

Which quantities increase in magnitude going from left to right across the list?

A frequency only

B neither speed nor frequency

C speed and frequency

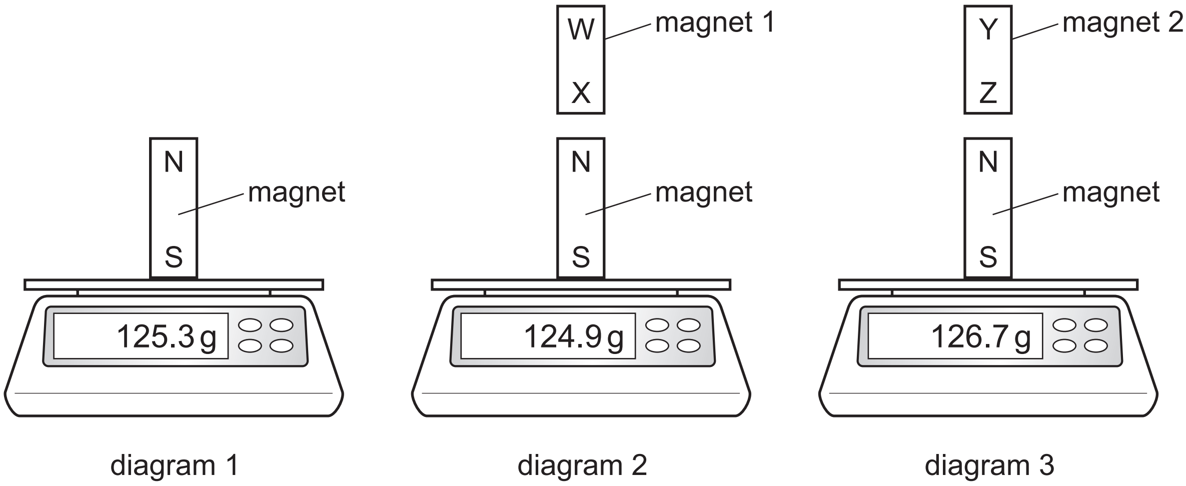
D speed only

26 A police car with its siren sounding is stationary in heavy traffic. A pedestrian notices that, although the loudness of the sound produced does not change, the pitch varies.

Which row describes the amplitude and the frequency of the sound?

	amplitude	frequency
A	constant	constant
B	constant	varying
C	varying	constant
D	varying	varying

27 Diagrams 1, 2 and 3 show an experiment to compare two magnets 1 and 2.

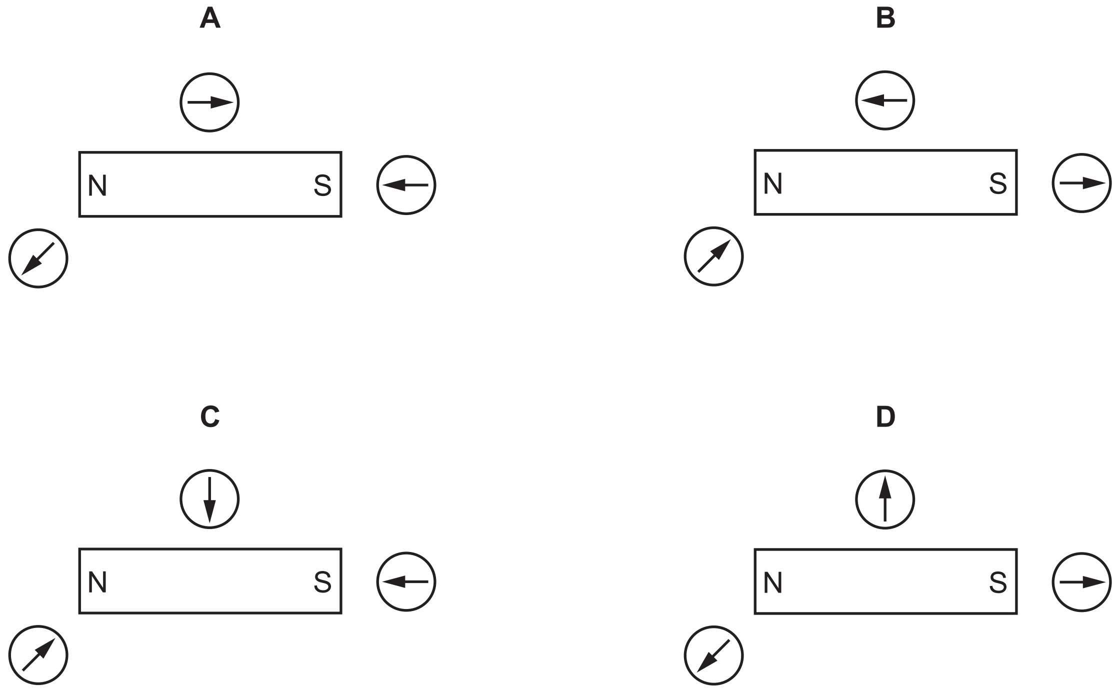


Which row explains the readings on the balances?

	polarity of magnet 1	polarity of magnet 2
A	X is an N pole	Z is an N pole
B	X is an N pole	Z is an S pole
C	X is an S pole	Z is an N pole
D	X is an S pole	Z is an S pole

28 A student uses three small plotting compasses to investigate the magnetic field around a bar magnet.

Which diagram shows the directions in which the compass needles point?



29 Which statement about charging an insulator is correct?

A It can be given an electrostatic charge by placing it in a magnetic field.

B It can be given an electrostatic charge by connecting it to a battery.

C It can be given an electrostatic charge by friction.

D It cannot be given an electrostatic charge.

30 Each diagram shows part of a circuit. The circle represents an instrument used to measure the potential difference (p.d.) across the resistor.

diagram 1

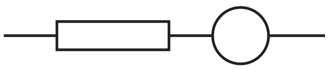
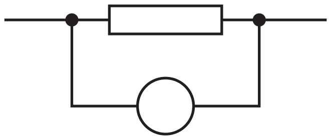


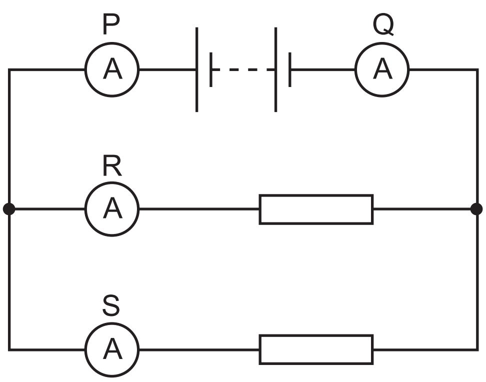
diagram 2



Which row is correct?

	the unit of p.d.	diagram which shows the meter correctly connected
A	amperes	diagram 1
B	amperes	diagram 2
C	volts	diagram 1
D	volts	diagram 2

31 A student uses four ammeters P, Q, R and S to measure the current in different parts of the circuit shown.



Which two ammeters read the largest current?

A P and Q **B** P and R **C** R and Q **D** R and S

32 Three statements about a relay are given.

1. A relay has a coil that becomes a temporary magnet when in operation.
2. A large current in a relay coil is used to switch off a smaller current.
3. A small current in a relay coil is used to switch on a larger current.

Which statements are correct?

A 1 and 2 only **B** 2 and 3 only **C** 1 and 3 only **D** 1, 2 and 3

33 An electrical appliance is powered from a mains supply.

The appliance normally uses a current of 3 A, but the current briefly rises to 4 A at the instant the appliance is switched on. The cable to the appliance is designed for currents up to 6 A.

A fuse is used to protect the circuit.

What should be the rating of the fuse?

A 1 A **B** 3 A **C** 5 A **D** 13 A

34 When a metal wire moves up, cutting a magnetic field, an electromotive force (e.m.f.) is induced across the wire.

Which change affects the magnitude of the induced e.m.f.?

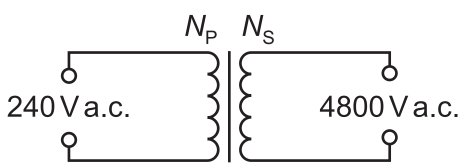
A moving the wire down at the same speed

B moving the wire up at a faster speed

C using a thicker wire

D using a wire made from a different metal

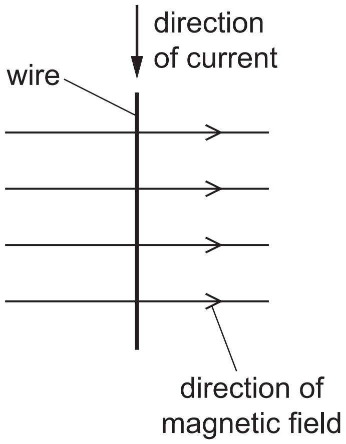
35 A transformer is needed to convert a supply of 240 V a.c. into 4800 V a.c..



Which pair of coils would be suitable for this transformer?

	number of turns on primary coil N_P	number of turns on secondary coil N_S
A	50	1 000
B	240	48 000
C	480	24
D	2000	100

36 The diagram shows a wire carrying a current in the direction shown. There is a magnetic field acting from left to right. The wire experiences a force acting out of the page.



The current is now reversed.

In which direction does the force on the wire now act?

A into the page

B out of the page

C to the left

D to the right

37 Which statement is correct for the nucleus of **any** atom?

A The nucleus contains electrons, neutrons and protons.

B The nucleus contains the same number of protons as neutrons.

C The nucleus has a total charge of zero.

D The nucleus is very small compared with the size of the atom.

38 How many protons and how many neutrons are in a nucleus of $^{234}_{90}\mathrm{Th}$?

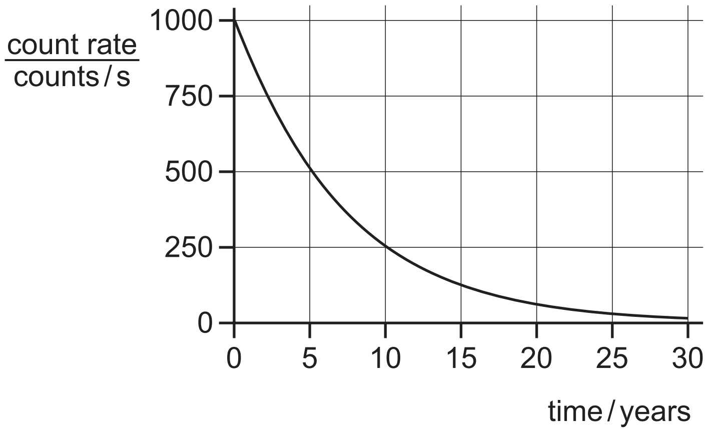
	protons	neutrons
A	90	144
B	90	234
C	144	90
D	234	90

39 A radiation detector in a laboratory is measuring background radiation.

Which row describes the readings and the cause?

	readings	cause
A	vary with no pattern	background radiation is random
B	vary with no pattern	radiation detectors are unstable
C	slowly increase during the day	background radiation increases as temperature increases
D	slowly reduce during the day	background radiation decreases as temperature increases

40 The graph shows the radioactive decay curve of a substance.



What is the half-life of this substance?

A 0.5 years **B** 5 years **C** 15 years **D** 30 years